

Harnessing Individual Participant Data from Clinical Trials to Predict Relapse in Visceral Leishmaniasis



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Abstract

This needs completing.

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List of Abbreviations

ABD	Amphotericin B deoxycholate
ABLE	Amphotericin B lipid emulsion
ALT	Alanine aminotransferase
AST	Aspartate aminotransferase
AUC	Area under the receiver operating characteristic curve
BPKIHS	B.P. Koirala Institute of Health Sciences
BMGF	Bill & Melinda Gates Foundation
BMI	Body mass index
CD	Cluster of differentiation
CDISC	Clinical Data Interchange Standards Consortium
CI	Confidence interval
CL	Cutaneous leishmaniasis
CRP	C-reactive protein
Crt	Creatinine
DAG	Directed acyclic graph
DAT	Direct agglutination test
DND<i>i</i>	Drugs for Neglected Diseases initiative
EC	European Commission
EoT	End of treatment
ESR	Erythrocyte sedimentation rate
GDPR	General Data Protection Regulation
HAART	Highly active antiretroviral therapy
Hb	Haemoglobin
HR	Hazard ratio
HIV	Human immunodeficiency virus
ICT	Immunochromatographic test

IDDO		Infectious Diseases Data Observatory
IFNγ		Interferon gamma
IM		Intramuscular
IPD		Individual participant data
IQR		Interquartile range
IRS		Indoor residual spraying
ISC		Indian subcontinent
ITN		Insecticide treated nets
IV		Intravenous
KAEP		Kala-azar Elimination Programme
KAMIS		Kala-azar Management Information System
KAMRC		Kala-azar Medical Research Centre
kDNA		Kinetoplast deoxyribonucleic acid
LAMB		Liposomal amphotericin B (AmBisome®; Gilead Sciences)
LEAP		Leishmaniasis East Africa Platform
LST		Leishmanin skin test
MA		Meglumine antimoniate
MCL		Mucocutaneous leishmaniasis
MF		Miltefosine
MPS		Mononuclear phagocytic system
MSF		Médecins Sans Frontières
NCVBDC	..	National Centre for Vector Borne Diseases Control
NTD		Neglected tropical disease
PCR		Polymerase chain reaction
PD1		Programmed cell death protein-1
PICOTS		Population, Index Model, Comparator Model, Outcome, Timing, Setting
PK-PD		Pharmacokinetic-pharmacodynamic
Plt		Platelets
PM		Paromomycin
qPCR		Quantitative polymerase chain reaction

RBC		Red blood cell
RDT		Rapid diagnostic test
RMRIMS		Rajendra Memorial Research Institute of Medical Sciences
SDA		Single dose liposomal amphotericin B
SDTM		Study Data Tabulation Model
SINAN		Sistema de Informação de Agravos de Notificação
SSG		Sodium stibogluconate
TDR		UNICEF/UNDP/World Bank/WHO Special Programme for Research and Training in Tropical Diseases
TGFβ		Transforming growth factor beta
UN		United Nations
UNDP		United Nations Development Programme
UNICEF		United Nations International Children's Emergency Fund
VL		Visceral leishmaniasis
WBC		White blood cells
WHO		World Health Organization

1

Background

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1.1 Introduction

This chapter provides important context for the development and validation of prognostic models predicting VL relapse in the Indian subcontinent (ISC) and East Africa. It begins with an overview of VL, outlining its epidemiology, pathophysiology, clinical features, and management. Emphasis is placed on the epidemiology of VL and the ongoing WHO-supported elimination programmes in the ISC and East Africa, defining the public health landscape in which a VL prognostic model would be implemented.

The section on relapse is presented as a narrative synthesis informed by a systematic review of the literature (full methods in Appendix A). Using pre-specified inclusion criteria, the review identifies and evaluates studies describing the burden, mechanisms, and determinants of relapse. The discussion highlights how relapse constitutes a high-risk infection reservoir that, together with PKDL and VL/HIV co-infection, poses a major threat to elimination efforts — a threat that could be mitigated through early identification of patients at high risk of relapse, enabling timely diagnosis and treatment.

1.2 Visceral Leishmaniasis

The leishmaniasis are a diverse group of neglected tropical diseases (NTDs) caused by protozoan parasites of the *Leishmania* genus and transmitted between susceptible mammalian hosts via the bite of infected female sandflies[1]. At least 20 *Leishmania* species infect humans, causing four principal disease forms: cutaneous leishmaniasis (CL), mucocutaneous leishmaniasis (MCL), visceral leishmaniasis (VL), and post kala-azar dermal leishmaniasis (PKDL) (see Box 1.1)[1–4]. Disease form is largely determined by parasite species and strain, resulting in a spectrum of clinical presentations ranging in severity from the relatively common and usually self-healing skin lesions seen in CL, to the often disfiguring mucosal destruction of MCL, and life-threatening systemic illness of VL. PKDL is a disseminated dermal eruption that may occur following VL recovery, and while patients are systemically well, they have been shown to be infective to sandflies[3].

VL represents the most severe manifestation of leishmaniasis, accounting for the vast majority of its morbidity and mortality. The disease overwhelmingly affects impoverished rural populations with poor access to healthcare. VL is also recognised as an opportunistic infection in patients living with human immunodeficiency virus (HIV), in whom co-infection leads to particularly high rates of treatment failure and death.

1.2.1 Epidemiology

This section provides an overview of the current global distribution, transmission dynamics, and risk factors for developing visceral leishmaniasis, highlighting major regional differences in disease burden and risk.

VL is endemic¹ in at least 80 countries across tropical, semi-tropical and temperate climates. The disease is caused by two closely related *Leishmania* species whose

¹Defined by the WHO as the occurrence of at least one autochthonous case with demonstrated local transmission within a country[5].

Box 1.1: Principal disease forms of Leishmaniasis[2, 4]

Cutaneous leishmaniasis (CL) Results in lesions on exposed skin that can lead to ulceration and life-long scarring. Often self-healing within a year, but can manifest atypical and disseminated forms, especially in immunocompromised patients. Up to 1 million new cases per year with most cases occurring in the Americas, Mediterranean basin, Middle East and Central Asia.

Mucocutaneous leishmaniasis (MCL) Rare complication of CL seen especially in the Americas with most cases reported in Bolivia, Perú, and Brazil. Results in destructive ulceration of the oral and nasal mucosa. Highly stigmatising and challenging to treat.

Visceral leishmaniasis (VL) Also known as kala-azar, the most severe form of leishmaniasis caused by *L. donovani* in the Old World and *L. infantum* in the New World. With an estimated 50,000–90,000 cases/year, VL presents with progressive weight loss, splenomegaly and fever.

Post kala-azar dermal leishmaniasis (PKDL) Benign macular and/or papular rash often including the face, arms, and trunk. Affects 5–20% of patients months to years after successful initial treatment for VL in the Indian subcontinent and East Africa. Rarely seen with *L. infantum* infections. Often self-limiting, although known to be infective to sandflies and therefore acts as a disease reservoir.

distribution defines the four principal global regions of high endemicity (see Figure 1.1): *L. donovani*, responsible for anthroponotic transmission (human→sandfly→human) in the ISC and East Africa, and *L. infantum*², responsible for zoonotic transmission (mammal→sandfly→human) in the Americas (limited to Central and South America) and the Mediterranean basin, extending into the Middle East and Central Asia[5, 6]. Together, these two species comprise the *L. donovani complex*.

²Previously referred to as *L. chagasi* in the Americas.

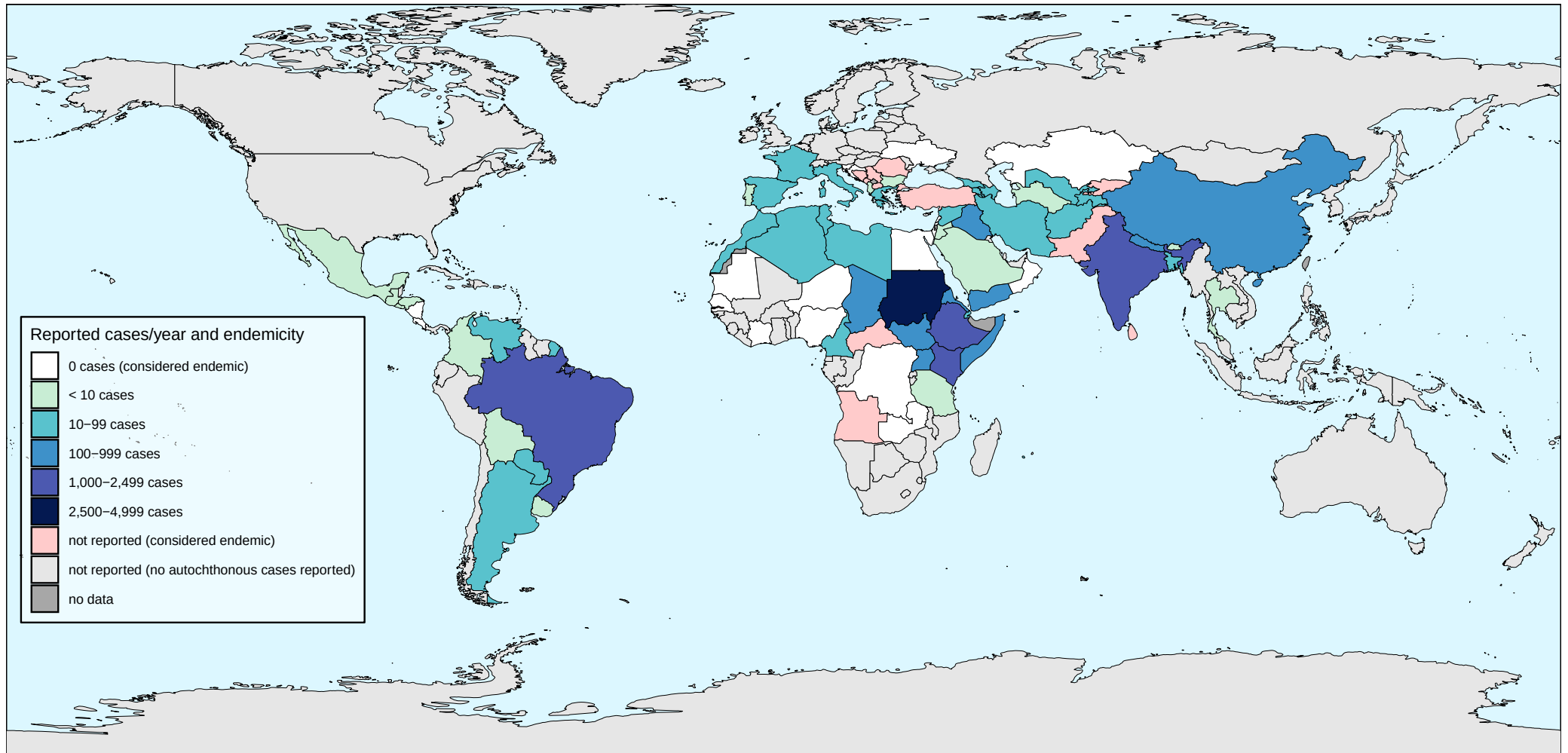


Figure 1.1: Geographical distribution of VL reported per country. Incidence is derived by averaging the reported cases/year over the last 6 reporting years (2019–2024). ‘Considered endemic’: at least one autochthonous case has been reported, with or without the entire cycle of transmission being demonstrated. Data source: World Health Organization Global Health Observatory, accessed 18 January 2026[6].

Disease Burden

Estimating the global burden of VL is problematic. True case numbers are obscured by significant underreporting due to limited access to healthcare, inadequate diagnostic facilities, misdiagnosis, and poor surveillance systems in many endemic countries[7–10]. In 2012, Alvar et al published the results of a WHO-led update to the global incidence of leishmaniasis using country-level reporting from the mid-late 2000s[11]. Underreporting rates were estimated through consultation with country representatives and disease experts. The global incidence was estimated at 200,000–400,000 cases/year, with approximately 80% of the burden originating from the ISC and 15% from East Africa. Compared to official reporting over the same period, this reflected a global underreporting rate of 3.5–7-fold.

Since the publication of Alvar et al, the number of cases has undisputedly fallen, driven largely by reductions in the ISC following the launch of the Kala-Azar Elimination Programme (KAEP) in 2005. Reported incidence dropped from > 50,000 cases/year prior to 2012, to approximately 22,500 cases/year in 2017, and remaining between 11,000 and 13,000 cases/year from 2020–2024 (inclusive)[6] (high incidence country breakdowns presented in Figure 1.2). Notably, this downward trend has persisted despite improvements in surveillance and reporting systems in many endemic countries[5]. Reflecting these changes, the WHO revised its estimated annual incidence in 2017 to 50,000–90,000 new cases/year³[4, 12].

Based on the most recent reporting data from 2024[6], the five countries with the highest case numbers are now, in descending order, Sudan, Kenya, Ethiopia, Brazil, and South Sudan, collectively comprising 78.1% of the global total. In stark contrast to the situation 20 years ago — when the majority of the global burden originated from the ISC — these countries now contribute only 5.5% of all reported cases: India with 464 cases (3.6%), Nepal with 216 cases (1.7%), and Bangladesh with just 23 cases (0.2%).

Mortality

The disease is widely described as ‘fatal without treatment’[2, 3]. Supporting this statement are the high mortality figures recorded during conflict-related epidemics in East Africa over the last 40 years, and prior to effective therapy, 19th century accounts of outbreaks devastating communities across the Gangetic plains[13–15]. Despite

³According to the online WHO Leishmaniasis Fact Sheet: <https://www.who.int/news-room/fact-sheets/detail/leishmaniasis>. Alternate WHO online content estimates 30,000 new cases/year since 2020: <https://www.who.int/health-topics/leishmaniasis> (online material accessed 18 January 2026).

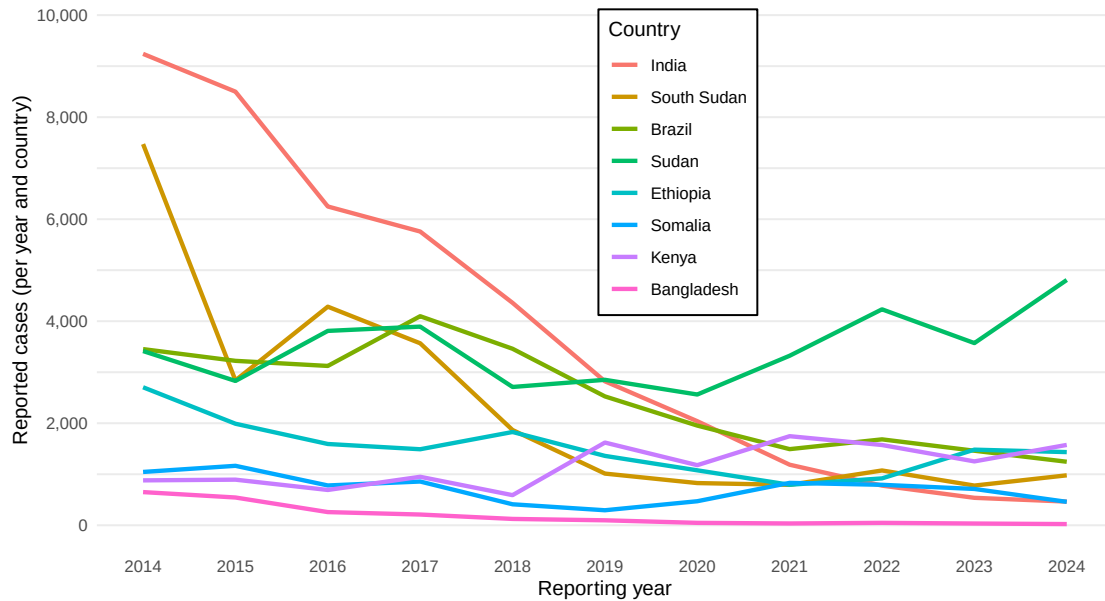


Figure 1.2: Temporal trends of visceral leishmaniasis incidence by country. Comparison between the top 8 countries with the highest average case numbers reported between 2014 and 2024. Data sourced from the World Health Organization Global Health Observatory (accessed 18 January 2026)[6].

this, subclinical disease forms have been reported with spontaneous resolution, although their extent and epidemiological significance remains elusive[16, 17].

With treatment, ~5–15% of cases result in death, although accurate estimates are challenged by a lack of reporting. Where deaths are reported, they frequently only reflect hospital deaths and omit those where a definite diagnosis was missed. In the 2000s, it was often cited that VL was responsible for up to 50,000 deaths/year — the second highest mortality from a parasitic disease after malaria[7] — although more recent estimates tentatively place the figure at a more modest 4,627 deaths/year with a wide uncertainty range of 1,853–8,725 deaths/year[18].

Vector

Measuring 2–4 mm and covered in dense hairs, phlebotomine sandflies (Diptera: Psychodidae) appear distinctly fuzzy under magnification. Females from an estimated 31 species across two genera are known to transmit the parasite between human hosts: *Lutzomyia* in the New World and *Phlebotomine* in the Old World[19]. Sandflies occupy a wide range of ecological niches, found on every continent except Antarctica. Biting occurs from dusk to dawn, with females requiring a blood meal for larval development. During the day they are found in cool and sheltered locations, such as in cracks and crevices in walls as seen with *Ph. argentipes*, responsible for

transmission in the ISC. In East Africa, three sandfly vectors have been implicated in *L. donovani* transmission, defining two distinct and non-overlapping ecological settings: (i) the *Acacia-Balanites* and black cotton soil savannah regions in northern focus, incorporating Sudan, northern South Sudan, and northern Ethiopia, where *Ph. orientalis* thrives, and (ii) the savannah and forested areas in the southern focus, incorporating southern Ethiopia, Kenya, and Uganda, where *Ph. martini* and *Ph. celiae* are seen in association with *Macrotermes* termite mounds.

In addition to sandflies, needle sharing among people who inject drugs was considered an important route of transmission in the southern Mediterranean region during the 1990s and 2000s, particularly among people living with HIV [20]. Exceptionally, transmission can also result from blood transfusion, organ transplantation, laboratory accidents[2], mother-to-child transmission[21], and possibly even sexual contact[22, 23].

Reservoirs

Similar to the majority of *Leishmania* spp. causing CL and MCL, *L. infantum* demonstrates zoonotic transmission, with domestic dogs being the main reservoir host in both the Americas and the Old World. This being said, an ever-increasing list of wild and domestic animals are known to harbour the parasite, including cats, foxes, horses, rodents, bats and opossums, although their relevance to human infection is unclear[24, 25]. An outbreak near Madrid (2009–2012) was attributed to hares[26].

In contrast to *L. infantum*, and importantly from an elimination perspective, *L. donovani* transmission in the ISC and East Africa is predominantly anthroponotic. Xenodiagnosis studies confirm that patients with active VL and those with PKDL are competent human reservoirs, in contrast to individuals with asymptomatic infection[27, 28]. Although *L. donovani* infections have been detected in several animal species in both regions—including cattle, dogs, and rats—their relevance as sources of human transmission remains unproven[29, 30].

Risk Factors

From population prevalence studies we know that only a minority of people with detectable parasites develop symptoms[3, 31]. Risk factors for acquiring an initial asymptomatic infection and subsequent progression to symptomatic disease reflect a tangled ecology of determinants linking host factors (sandfly exposure, immunity, genetics) and parasite factors (strain, virulence, inoculum). A common theme, woven into many of these determinants, is poverty.

In both the ISC and East Africa the median age of infection during stable transmission is similar at 15–20 years. More men than women are infected and develop disease, likely reflecting their increased occupational exposure to sandflies (for example, cattle herding and other outdoor activities)[5].

In the ISC⁴, VL endemicity is centred on the fertile and low-lying alluvial plains of the Ganges river, where high humidity, heavy monsoon rains, and abundant vegetation provide ideal conditions for sustained transmission between sandflies and humans[32]. Significant clustering of cases is seen across the rural farming communities of Bihar, Jharkhand, Uttar Pradesh, and West Bengal States in northeastern India, central and western Bangladesh, and the southeastern plains of Nepal neighbouring India. In a systematic review by Bern et al, determinants of VL transmission in the ISC included living in mud houses, proximity to prior cases (in the same or nearby household), presence of vegetation and standing water surrounding the house, sleeping on the floor or outside, malnutrition, and a lack of bed net use[33].

The greatest concentration of cases in East Africa is reported in the northern focus, specifically between the eastern Sudanese State of Gederaf and the bordering northern States of Ethiopia. Many of the epidemiological determinants of VL in East Africa are shared with the ISC, with poverty remaining the central overarching factor. Notable determinants include living in rural settings near sandfly breeding and resting sites (living near termite mounds in the southern focus, sleeping under *Acacia* trees in the northern focus), living in proximity to other VL infected (or recently infected) people, and malnutrition[34].

VL/HIV co-infection has reshaped VL epidemiology in many endemic regions, and remains the most important risk factor for asymptomatic infection, disease progression, and poor treatment outcomes. In the mid-1980s in Spain and other southern European countries, VL shifted from a rare childhood disease to one predominantly affecting HIV-positive adults[35, 36]. VL/HIV co-infection rates are currently increasing in Brazil (now reported in 20% of new cases[37]) and India (> 10% of VL episodes in 2023 and 2024[38]). In northern Ethiopia, VL/HIV co-infection presents a significant challenge to elimination, affecting between 20 and 50% of cases predominantly in male migrant workers[39].

Host immunity — particularly cell-mediated immunity — is a major determinant of disease onset and treatment outcomes, and thought to be a key driver of several devastating epidemics associated with war and natural disasters. A striking example is Sudan in the 1980s-2000s, where population displacement driven by conflict and

⁴Despite sporadic VL cases reported from Pakistan, Bhutan and Sri Lanka, in this thesis ISC refers to India, Nepal and Bangladesh.

famine brought immune-naïve populations into endemic areas, and spread infection into previously unaffected areas. Combined with severe malnutrition and the collapse of health infrastructure, the resulting mortality was catastrophic[13, 40, 41].

The geographic and demographic heterogeneity of VL reflects fundamental differences in host-parasite interactions and immune responses. To understand how these epidemiological patterns arise and why certain groups experience more severe disease, it is necessary to examine the pathophysiology of human *Leishmania* infection.

1.2.2 Pathophysiology

Parasites of the genus *Leishmania* belong to the family *Trypanosomatidae* (class *Kinetoplastea*, phylum *Euglenozoa*) and share many features with two other human pathogens of the same family; *Trypanosoma brucei* — the causative agent of human African Trypanosomiasis (African sleeping sickness), and *Trypanosoma cruzi* — the agent of Chagas disease in the Americas. These vector-borne NTDs are all single-celled protozoa characterized by a flagellum and a kinetoplast — a dense network of mitochondrial DNA giving them a distinctive microscopic appearance.

Life Cycle

In the sandfly gut, *Leishmania* parasites exist in their extracellular flagellated promastigote forms. During a blood meal, the parasite is regurgitated into the bite wound and rapidly phagocytosed by macrophages. Intracellularly, in the human host, the protozoa lose their flagellum and transform into amastigotes: oval bodies measuring $\sim 2\text{--}6\mu\text{m}$ in length with their characteristic nucleus and kinetoplast clearly visible on nucleic acid staining. Intracellular amastigotes multiply through binary fission, eventually leading to cell lysis and subsequent uptake by neighbouring macrophages to perpetuate infection. In viscerotropic disease caused by *L. donovani complex*, this cycle drives widespread dissemination with macrophage proliferation and granuloma formation across the mononuclear phagocytic system, including in the spleen, liver, bone marrow, and lymph nodes. The cycle completes when sandflies ingest parasitised macrophages during a subsequent blood meal, initiating parasite development within the vector (Figure 1.3).

Immune Evasion

Remarkably, intracellular amastigotes have evolved to survive and replicate within the hostile phagolysosomal environment of the macrophage. Although the full immunological mechanisms remain incompletely understood and beyond the scope of this chapter, key strategies include the presence of a lipophosphoglycan surface

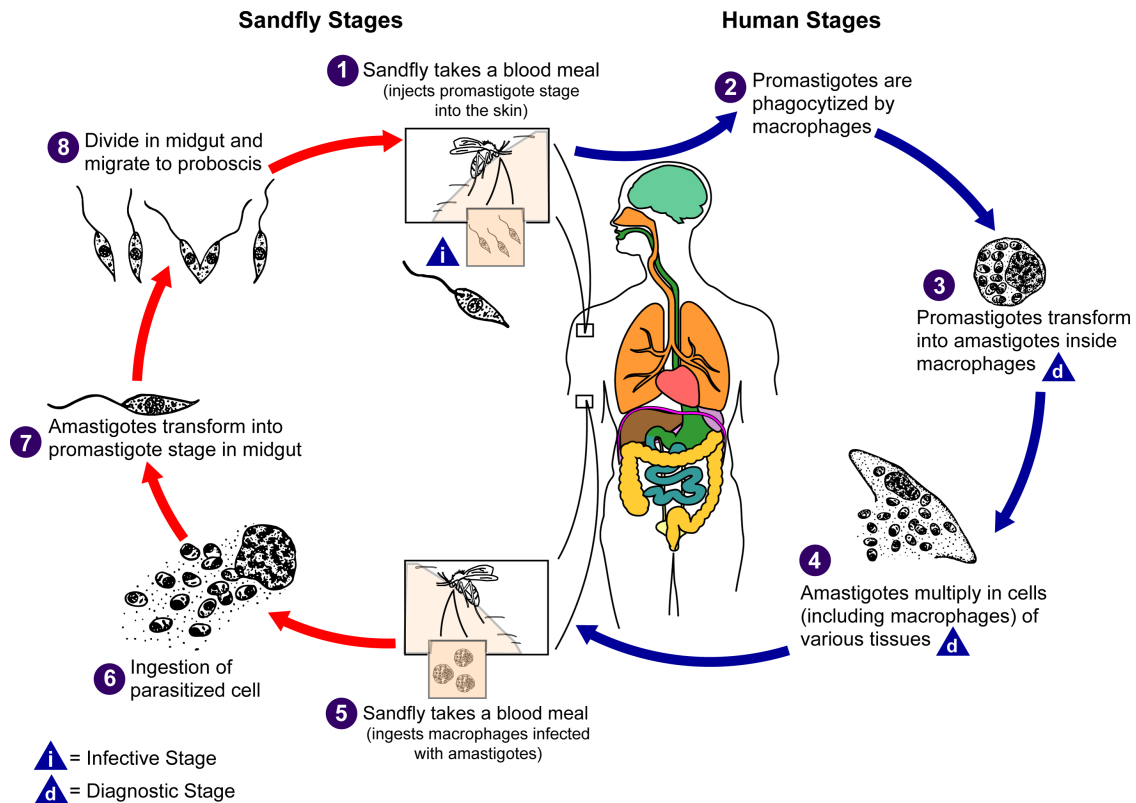


Figure 1.3: Life cycle of the *Leishmania* parasite consists of a vector (sandfly) stage and reservoir (animal) stage. While vector transmission is by far the most common route, it is not essential — direct human-to-human transmission has been reported via needle sharing, organ transplantation, blood transfusion, and mother-to-child transmission. Illustration credit: Centers for Disease Control and Prevention/Alexander J. da Silva/Melanie Moser.

coat that protects promastigotes from complement-mediated lysis, and suppression of macrophage activation by inducing functional impairment of cytotoxic T-cell responses. Despite the presence of a pronounced humoral response, with a polyclonal hypergammaglobulinaemia typically seen, the resulting antibodies are predominantly non-neutralising and may even play a role in maintaining the immunosuppressive milieu required for amastigote survival[42, 43]. As the disease progresses, so too does the host's immunosuppressive state, driven by rising levels of immunoregulatory cytokines (especially IL-10 and TGF β) and the destructive remodelling of lymphoid architecture[43, 44].

1.2.3 Clinical Features

The clinical spectrum of *Leishmania* infection is remarkably broad, with asymptomatic carriers outnumbering symptomatic cases by a factor of 5–200 in endemic

settings[3, 31, 45]. If symptomatic, disease onset is typically insidious, following an incubation period ranging from several weeks, to months, and occasionally years[46].

Classic symptoms result from a persistent systemic inflammatory response syndrome with infiltration of the mononuclear phagocytic system, resulting in intermittent fever, weight loss, splenomegaly, often with accompanying hepatomegaly and, in parts of East Africa, lymphadenopathy. The spleen is usually non-tender on palpation, but massive splenomegaly $\pm$ hepatomegaly can result in severe abdominal pain. The greyish skin hyperpigmentation (‘kala-azar’ — literally ‘black disease’ or ‘black fever’ in Hindi), described in historical accounts of chronic disease in the ISC before effective therapy became available, is now uncommon[2, 7].

Characteristic laboratory abnormalities include raised inflammatory markers (CRP, ESR), pancytopenia, a polyclonal hypergammaglobulinaemia, and hypoalbuminaemia. Anaemia is often severe, normocytic, and normochromic, and likely multifactorial in origin, with splenic sequestration and anaemia of chronic disease thought to play important roles[47, 48]. Haemoglobin levels at presentation are frequently between 6–10 g/dL[47], resulting in pallor and contributing to weakness.

Without treatment, disease severity correlates with increasing parasite burden[49, 50]. Patients may develop vomiting, diarrhoea, cough, and dyspnoea, reflecting a combination of direct mucosal invasion, and opportunistic infections arising from the host’s immunosuppressed state. The presence of oedema, jaundice, severe co-infection, and bleeding with disseminated intravascular coagulation indicates a poor prognosis[51].

1.2.4 Diagnosis

The gold standard for diagnosing VL combines compatible clinical features with direct visualisation of *Leishmania* amastigotes in tissue aspirates obtained from the spleen, bone marrow, or lymph node. Splenic aspiration offers the highest sensitivity (93–99%) but carries a small risk of fatal haemorrhage. Bone marrow and lymph node aspirates are safer but considered less sensitive (~50–80%)[2, 3]. Parasite density is usually expressed using a logarithmic semiquantitative grading system (0–6+) originally described by Chulay and Bryceson (Table 1.1)[52].

Culture of tissue aspirates can improve diagnostic sensitivity but is limited by cost, technical complexity, and a slow turnaround of up to four weeks. Molecular assays have been developed but remain infrequently available outside research settings and regions of high endemicity. Notably, peripheral blood quantitative polymerase chain reaction (qPCR) assays amplifying kinetoplast DNA (kDNA) have shown high sensitivities in patients with both VL and VL/HIV co-infection[53–55],

Table 1.1: Logarithmic grading system for parasite load on microscopy of tissue aspirates described by Chulay and Bryceson in the early 1980s[52]. Smears are stained with Giemsa and examined using a 10× eyepiece and 100× objective lens.

Grade	Average parasite density
0	0 amastigotes/1,000 fields
1+	1–10 amastigotes/1,000 fields
2+	1–10 amastigotes/100 fields
3+	1–10 amastigotes/10 fields
4+	1–10 amastigotes/field
5+	10–100 amastigotes/field
6+	> 100 amastigotes/field

obviating the need for invasive sampling where available. Urine antigen detection has demonstrated specificity but suffers from low diagnostic sensitivity[56, 57].

A wide gamut of serological (antibody-detecting) tests is available for the diagnosis of VL, including enzyme-linked immunosorbent assays (ELISA), indirect fluorescent antibody (IFA) assays, immunoblots, and rapid diagnostic tests (RDTs)[2]. The rK39 immunochromatographic test (ICT) is the most widely used RDT, providing a binary result within 10–20 minutes from a finger-prick blood sample. In the ISC, rK39 RDTs demonstrate excellent sensitivity (97.0%, 95% CI 90.0–99.5%)[58] and have served as the first-line diagnostic test since the mid-2000s. In East Africa, however, sensitivity is lower (85.3%, 95% CI 74.5–93.2%), and a second serological test — the direct agglutination test (DAT) — is recommended to confirm negative rK39 results[58]. Key limitations of serological tests include their inability to distinguish between active and past infection, with antibodies persisting for months to years following successful treatment and their reduced sensitivity in patients with severe immunosuppression, particularly those with VL/HIV co-infection[3].

1.2.5 Treatment

Intramuscular (IM) pentavalent antimonials have been the workhorse of VL treatment since the 1940s,⁵ and despite their relatively toxic adverse effects (pancreatitis, cardiac arrhythmias, hepatitis), they remain first-line therapy in both East Africa (sodium stibogluconate [SSG], as part of combination therapy) and in Brazil (as meglumine antimoniate [MA])[59, 60]. From the early 1980s in Bihar, India, treatment failure rates exceeding 50% were observed with SSG despite dose escalation

⁵Previously, since the 1910s, tartar emetic was used — a toxic *trivalent* antimonial compound also used as an emetic.

(10→20 mg/kg/day) and duration (6→40 days). Blame was attributed to poor stewardship, with subtherapeutic dosing practices driving resistance[61]. As efficacy declined, second-line agents were used — initially IM pentamidine (diabetogenic, and limited by severe toxicity), and later intravenous (IV) amphotericin B deoxycholate (ABD) — effective but constrained by infusion reactions, nephrotoxicity, and the need for prolonged hospitalisation.

Over the past two decades, the introduction of three new agents — oral miltefosine, IV liposomal amphotericin B (initially registered in 1997 as AmBisome®; Gilead Sciences), and IM paromomycin — has transformed the previously limited and often toxic treatment arsenal[62].

Miltefosine (MF), a repurposed anticancer drug, is the only effective oral agent for VL. Introduced as the first-line treatment in the ISC in 2005, MF played a pivotal role in the early KAEP[63–65]. Its use has since declined. Concerns about teratogenicity, gastrointestinal toxicity, and decreasing efficacy[66, 67] led to its replacement in the KAEP by single-dose liposomal amphotericin B (LAMB) (AmBisome®; Gilead Sciences) in 2014–15[68]. Outside the ISC, MF performs poorly as monotherapy in East Africa and Brazil[62, 69].

Supported by a WHO-Gilead donation programme, a single intravenous IV dose of 10 mg/kg LAMB remains the first-line regimen in the ISC owing to its efficacy and safety[70]. At higher cumulative doses of 21–30 mg/kg given over 5–10 days, LAMB is also used as first-line therapy in Europe and as a second-line option in Brazil and East Africa, particularly in patients intolerant to pentavalent antimonials (severe disease, extremes of age, pregnancy, renal or hepatic dysfunction, relapse, VL/HIV co-infection).

Paromomycin (PM), an injectable aminoglycoside, is effective both as monotherapy and in combination regimens in the ISC[71]. Despite its low cost and favourable safety profile, uptake in the ISC has been limited. In East Africa, by contrast, SSG/PM combination therapy has been first-line since 2010[59, 72], with evidence also supporting its use alongside MF in both East Africa and the ISC[73, 74].

1.2.6 Elimination Efforts

In areas of anthroponotic transmission, control of VL centres on two interdependent strategies: (i) reduction of the human infection reservoir through early diagnosis and treatment, and (ii) interruption of transmission events through vector control measures. Effective implementation of these approaches requires evidence-based decision-making and sustained political will.

Guided by these principles, in 2005 the governments of Nepal, Bangladesh, and India, with WHO support, signed a Memorandum of Understanding to reduce VL incidence to fewer than 1 case per 10,000 population at district or sub-district levels by 2015. This deadline was later extended and is now embedded within the 2021–2030 WHO NTD roadmap[65, 75–78]. The initial success of the KAEP was attributed to the introduction of highly sensitive rK39 RDTs, oral MF, and subsequently single-dose LAMB. These advances were complemented by active case detection strengthened through community engagement and public awareness, including the use of frontline community health workers such as Accredited Social Health Activists (ASHAs) in India[79]. Vector control measures, including indoor residual spraying (IRS) and insecticide-treated nets (ITN), further supported transmission reduction[80]. In October 2023, Bangladesh became the first country to eliminate VL as a public health problem[81]. Nepal and India are currently working to sustain their targets for three consecutive years to achieve elimination status[82].

As the ISC enters the consolidation and maintenance phases of the KAEP, concerns regarding sustainability are increasingly voiced[68]. Waning political momentum and financial support threaten the viability of costly IRS and active case detection. Moreover, as the proportion of immunologically naïve individuals increases, so too does the risk of future outbreaks[83].

Buoyed by the KAEP experience, calls for VL elimination in East Africa[84] culminated in the 2023 *Nairobi Declaration*, endorsed by the health ministries of nine endemic East African countries⁶ and aiming to reduce VL incidence by 90% by 2030. Similar to the KAEP, a phased approach has been proposed — consisting of planning, attack, consolidation, and maintenance phases — with emphasis placed on equitable access to diagnosis and treatment, integrated vector management adapted to diverse ecological settings, and strong cross-border collaboration[85, 86].

However, in contrast to the KAEP, the East Africa initiative faces greater operational complexity. Challenges include heterogeneous transmission ecologies, weaker health systems, reliance on multi-dose treatments, limited access to sensitive diagnostic tools, and perhaps more importantly, political instability and the consequences of population displacement due to civil wars and famine[86].

Early diagnosis and treatment of all human infection reservoirs remains a cornerstone of any successful elimination campaign where transmission is largely anthroponotic. The next section examines the burden, mechanisms, and determinants of VL relapse — a reservoir of particular concern and of critical relevance to elimination.

⁶Chad, Djibouti, Eritrea, Ethiopia, Kenya, Somalia, South Sudan, Sudan and Uganda

1.3 Relapse

This section is guided by a systematic review of the literature and presented as a narrative review (see Box 1.2). Search terms were constructed to identify the burden, timing, mechanisms, and determinants of VL relapse, with a focus on immunocompetent patients in the ISC and East Africa. Further details are available in Appendix A.

VL relapse is defined as the reappearance of VL signs and symptoms following an initial treatment response[87, 88]. In both research and routine clinical settings, relapse is typically confirmed by direct visualisation of the parasite on a tissue aspirate smear[89].

Relapse can only occur once an initial treatment response is achieved — termed initial cure in clinical efficacy studies. In research settings, this requires both clinical improvement (e.g., defervescence, reduction in spleen size, weight gain, improvement of anaemia) and confirmation of parasite clearance by microscopy. This is referred to as *test-of-cure*, and usually occurs within a month of the end of treatment (EoT), although considerable heterogeneity exists in precisely how and when test-of-cure is assessed[90].

Box 1.2: Literature search

Literature search performed from database inception to 11th August 2025 in PubMed (below), Embase and Web of Science. Articles in English were reviewed. Full search details available in Appendix A.

```
("Leishmaniasis, Visceral"[Mesh] OR "visceral leishmaniasis" OR
"leishmaniasis, visceral" OR "kala azar" OR "kala-azar") AND
("Recurrence"[Mesh] OR relapse* OR recurrent OR recurrence OR
recrudescence OR "treatment failure")
```

1.3.1 Burden of Relapse

Reported relapse rates among immunocompetent patients vary widely across studies, reflecting differences in host, treatment, and parasite factors[89]. As a ‘rule of thumb’, relapse rates with current first-line regimens in patients without significant immunosuppression range between 2.5% to 10%[89]. According to a meta-analysis of VL clinical efficacy studies by Chhajed et al, the overall proportion of HIV-negative patients relapsing in the ISC within 6 months of treatment was estimated at 3.5% (95% confidence interval [CI]: 2.8–4.5%) following first-line treatment with single-dose LAMB[89]. Under pragmatic conditions in East Africa, slightly higher

6-month relapse rates of approximately 5% are seen with the first-line combination therapy of PM and SSG[91–93].

These estimates, however, understate the true relapse burden: (i) relapses occurring beyond 6-months of follow-up are missed, and (ii) clinical trial populations do not reflect the broader patient population seen in routine care. Chhajer et al also showed that, across 21 studies reporting both 6 and 12 month relapse rates, one third of all relapses by 12 months occurred during the second half of the 12 month period[89]. Similarly, several large studies from the ISC have shown comparable or even higher relapses counts occurring between 6–12 months than in the initial 6 month period, prompting calls to extend routine follow-up from 6 to 12 months[66, 94–97]. Furthermore, trials often exclude patients at increased risk of relapse, such as those at the extremes of age, with more severe forms of the disease, and with comorbidities. Trial patients are also managed under controlled conditions that incentivise adherence. These factors collectively bias relapse estimates downward compared with outcomes observed under routine conditions.

1.3.2 Relapse vs. Reinfection

Distinguishing relapse resulting from parasite recrudescence, from re-infection with a new strain, is important when assessing drug efficacy and relapse determinants. However, unlike malaria[98], VL lacks validated molecular targets for PCR-based confirmation. Nevertheless, accumulating evidence indicates that most relapses result from recrudescence rather than reinfection. For example, Rijal et al. performed kDNA fingerprinting in 8 pairs of bone marrow samples in HIV-negative patients prior to primary infection treatment and at the time of relapse. No evidence of reinfection was found, with 8 distinct kDNA fingerprints identified across the 16 samples that matched at the individual patient level[66]. Similar studies have been performed in VL/HIV co-infection patients in East Africa and Europe employing a variety of molecular techniques, and showing that while reinfection does occur, it is considerably less common than recrudescence[99–102].

1.3.3 Relapse Determinants

When selecting candidate predictors for inclusion in a prognostic model, it is important to draw on previously described predictors[103]. To provide important context for the subsequent chapters on relapse model development, this section summarises the determinants of relapse identified through the literature review.

For immunocompetent patients, the review aims to be exhaustive. Studies were excluded if they (i) did not adopt a longitudinal design in which index VL cases were

linked to subsequent relapse episodes in the same patients, (ii) included more than 5% VL/HIV co-infection without reporting relapse predictors separately by HIV status; or (iii) focused solely on treatment regimens and/or biomarkers (including molecular targets or cytokines) that are not routinely measured in clinical practice. Although not intended to be comprehensive, key studies conducted in patients with VL/HIV co-infection, as well as those reviewing biomarkers or treatments predictive of relapse, are also discussed.

Four systematic reviews describing predictors of relapse were identified: two including patients with and without VL/HIV co-infection[104, 105] and two focussing specifically on patients with VL/HIV co-infection [106, 107]. Where appropriate, references from these reviews inform the narrative synthesis.

It is important to recognise that whether a study identifies a significant association between a predictor and relapse does not necessarily correspond to whether or not the predictor is important. For example, (i) not all predictors are considered by all studies, (ii) many studies have small sample sizes and are therefore underpowered to detect certain associations, and (iii) a wide range of study designs, statistical tests, predictor transformations, and modelling strategies are employed, affecting what is considered ‘significant’. Furthermore, the significance of an association will also depend on the other variables adjusted for in the prediction model.

Immunocompetent Patients

A total of 11 studies reporting relapse determinants in immunocompetent patients were identified, and are presented in Table 1.2. Eight studies were conducted in the ISC, including India[94, 96, 97, 108, 109], Bangladesh[110, 111], and Nepal[66], two in East Africa, including South Sudan[112] and Kenya[113], and one in Georgia[114]. The median study size was 1,143 patients (range 115[66] to 8,537[96]). Two studies presented outcomes from the same trial[97, 108], although they are reported separately here due to differences in sample size and methodology. Follow-up ranged from 6 months to 4 years, and relapse proportions from 2.6%[110] to 20.8%[66]. Further details, including study design and methodologies, are provided in the [Supplementary Material](#).

Age, treated categorically (≥ 2 levels), was the most frequently reported relapse determinant. All 11 studies considered age as a candidate predictor, with eight demonstrating statistically significant ($p < 0.05$) or borderline significant ($0.05 \leq p < 0.1$) associations in unadjusted models[66, 94, 96, 97, 108–110, 114]. When adjusted models were considered, age remained significant in all but one study[94]. Two ISC studies that modelled ≥ 3 age categories showed a ‘U’-shaped relationship,

with increased relapse risk in young children (< 5 years) and older adults (≥ 40 or ≥ 45 years)[96, 110]. Other studies identified increased risk only in younger groups; < 12 or < 15 years in the ISC[66, 94, 97, 108] and < 1 year in Georgia[114]. Notably, the two East African studies did not identify age as a significant predictor[112, 113], although this may reflect limited power (17 relapses in Kennedy et al.[113]), or selection bias due to incomplete linking of relapse and index cases[112]. No study explicitly modelled age as a continuous variable.

Spleen size was the next most frequently identified determinant, considered in seven studies. Five reported significant associations in unadjusted analyses, and four in adjusted models[94, 96, 110, 112, 114]. Definitions and timing varied. Lucero et al. reported that larger spleens at discharge predicted relapse (OR 1.27, 95% CI 1.10–1.47 per cm below the costal margin). Sundar et al. found approximately twice the odds of relapse in patients with admission spleen size > 4 cm[94]. Gorski et al. identified increased relapse odds with larger spleens at discharge *and* admission, when measured with Hackett grade[112]. Kajaia et al. reported increased relapse risk with larger spleens at admission when measured by the Kandelaki splenometric method[114, 115]. Instead of absolute spleen size, Burza et al. modelled change in spleen size during admission: patients with a reduction ≤ 0.5 cm/day had 1.7-fold higher odds of relapse (95% CI 1.1–2.5) compared with those with greater reductions[96].

Symptom duration prior to treatment was considered in six studies, with four identifying significant associations in both unadjusted and adjusted models[96, 97, 108, 114]. In the ISC, Burza et al. reported that patients with ≤ 4 weeks of symptoms had higher relapse odds (1.6-fold vs 4–8 weeks; 2.3-fold vs > 8 weeks), with similar effects after adjusting for age, sex, and change in spleen size[96]. Goyal et al. found that ≤ 8 weeks of symptoms was associated with 3.3-fold (95% CI 1.3–8.4) higher relapse odds[97] and a 3.6-fold (95% CI 1.4–9.1) higher relapse hazard in time-to-event analysis[108]. In contrast, Kajaia et al. reported that ≥ 90 days of symptoms was associated with higher relapse risk in Georgia (OR 3.9; 95% CI 1.8–8.5) after adjusting for haemoglobin and age[114]. Kennedy et al. included symptom duration but did not identify a significant association, perhaps due to low statistical power[113].

Although sex was considered in all studies, only three ISC studies found significant associations with relapse[94, 96, 109]. Each reported an approximate doubling in relapse odds[94, 96] or rates[109], with similar effect sizes seen in adjusted analyses.

Admission haemoglobin (Hb) was included in eight studies[66, 94, 96, 97, 110, 113, 114] and was significant in three[110, 113, 114]. Lucero et al. observed strong

associations between lower Hb and relapse in partially adjusted models, although these were not retained in the fully adjusted model (OR 1.40, 95% CI 1.09–1.72 and 1.45, 95% CI 1.15–1.85, per 10 g/L decrease, for admission and discharge Hb, respectively). Kajaia et al. reported a strong association, with relapse 12-fold (95% CI 4.1–34.8) more likely in patients with Hb < 60 g/L vs $\geq$ 80 g/L, persisting after adjustment for age and symptom duration.

Markers of malnutrition were inconsistently defined across studies, and reporting was often unclear. Gorski et al. and Burza et al. used age-specific definitions combining BMI, BMI-for-age z-scores, and weight-for-height z-scores[96, 112]. Lucero et al. used a similar approach, combining weight-for-height z-scores and BMI[110]. Goyal et al. and Kennedy et al. described ‘severe wasting’ and ‘malnutrition’, respectively, without further definitions[97, 113]. Rijal et al. and Sundar et al. relied on single indicators (BMI and weight, respectively)[66, 94]. Only Sundar et al. observed a significant association, with patients $\leq$ 30 kg at higher risk, although the causal interpretation is likely confounded by age, given that the study recruited participants of all ages[94].

Table 1.2: Summary of studies presenting significant predictors of VL relapse in immunocompetent patients. Excluding: studies only looking at treatment regimen or biomarkers. BPKIHS: B.P. Koirala Institute of Health Sciences; ISC: Indian subcontinent; KAMRC: Kala-azar Medical Research Centre; MA: meglumine antimoniate; MSF: Médecins Sans Frontières; Hb: haemoglobin; LAMB: liposomal amphotericin B; MF: miltefosine; PM: paromomycin; RBC: red blood cell; SDA: single-dose liposomal amphotericin B 10mg/kg; SSG: sodium stibogluconate; VL: visceral leishmaniasis; wks: weeks; yrs: years.

Study	Location	Treatment	n (relapse) %	Significant predictors — unadjusted model ¹	Significant predictors – adjusted model ²
ISC					
Burza 2014[96]	MSF clinics, Bihar, India	20 mg/kg LAMB (4 × 5 mg/kg)	8,537 (119) 1.4%	age < 5 ^{****} , age ≥ 45 ^{**} vs. 15–30 yrs; male sex ^{***} ; other backwards caste [*] vs. general category; symptom duration ³ > 8 wks ^{***} , > 4 to ≤ 8 wks ^{**} vs. ≤ 4 wks; spleen size change ≤ 0.5 cm/day vs. > 0.5 cm/day	age < 5 ^{****} , age ≥ 45 ^{**} vs. 15–30 yrs; male sex ^{***} ; symptom duration ³ > 8 wks ^{**} , > 4 to ≤ 8 wks ^{**} vs. ≤ 4 wks; spleen size change ≤ 0.5 cm/day vs. > 0.5 cm/day
Goyal 2019[97]	Public health facilities, Bihar, India	SDA; LAMB+MF; MF+PM	1,353 (75) 5.5%	age 2–12 yrs ^{**} vs. > 12 yrs; symptom duration ≤ 8 wks ^{***} vs. > 8 wks; treatment ^{****}	age 2–12 yrs [†] vs. > 12 yrs; symptom duration ≤ 8 wks [†] vs. > 8 wks; treatment MF+PM [†] vs. SDA
Goyal 2020[108]	[as above]	[as above]	1,750 (79) 4.5%	age 2–12 yrs ^{††} vs. > 12 yrs; treatment with MF+PM ^{††}	age 2–12 yrs ^{***} vs > 12 yrs; symptom duration ≤ 8 wks ^{***} vs. > 8 wks; treatment MF+PM ^{****} vs. SDA
Lucero 2015[110]	MSF clinic, Fulbaria, Bangladesh	15 mg/kg LAMB (3 × 5 mg/kg)	1,521 (39) 2.6%	age < 5 yrs ^{***} , age ≥ 40 yrs ^{**} vs. 18–39 yrs; lower admission Hb ^{***} ; lower discharge Hb ^{***} ; larger discharge spleen size ^{***}	age < 5 yrs ^{***} , age ≥ 40 yrs ^{**} vs. 18–39 yrs; larger discharge spleen size ^{**}
Mondal 2019[28]	Hospital in Mymensingh, Bangladesh.	8 different regimens	984 (69) 7.0%	treatments compared only: MD-LAMB [*] , MF ^{**} , LAMB+PM ^{**} , LAMB+MF ^{***} , SDA ^{****} , PM ^{****} , MF+PM ^{****} vs. SSG	treatment: MD-LAMB [*] , MF ^{**} , LAMB+PM ^{**} , SDA ^{***} , LAMB+MF ^{****} , PM ^{****} , MF+PM ^{****} vs. SSG
Rijal 2013[66]	BPKIHS, Eastern Nepal	MF	115 (24) 20.9%	age ≤ 12 yrs ^{**} vs. > 12 yrs	[not performed]
Sundar 2019[94]	KAMRC, Bihar, India	SDA	1,143 (66) 5.8%	age ≤ 15 yrs [*] vs. > 15 yrs; male sex ^{**} ; weight ≤ 30 kg ^{**} vs. > 30 kg; spleen size > 4 cm ^{***} vs. ≤ 4 cm	male sex ^{***} ; weight ≤ 30 kg ^{**} vs. > 30 kg; spleen size > 4 cm ^{***} vs. ≤ 4 cm

continued on next page

Table 1.2: continued

Study	Location	Treatment	n (relapse) %	Significant predictors — unadjusted model ¹	Significant predictors – adjusted model ²
East Africa					
Gorski 2010[112]	MSF clinics, South Sudan	SSG, SSG/PM, LAMB	8,090 (166) ⁴	treatment centre: Lankien ^{****} vs other; admission spleen size ^{5**} ; discharge spleen size ^{***} ; treatment ^{****} ; lower % gain in body weight [*]	admission spleen size ^{5†} , 2 [†] , ≥ 3 [†] vs 0; discharge spleen size 2 [†] , ≥ 3 [†] vs 0; treatment SSG/PM [†] vs SSG
Kennedy 2024[113]	Chemolingot Sub-county Hospital, Baringo County, Kenya	SSG/PM, LAMB	248 (17) 6.9%	anaemia [*] ; shorter hospital stay [*]	[not performed]
Europe					
Kajaia 2011[114]	Institute of Parasitology and Tropical Medicine, Tbilisi, Georgia	MA	300 (21) 7.0%	age ^{**} : higher risk in < 1 yrs; symptom duration ^{***} : higher risk with longer duration; Hb ^{****} : higher risk with Hb < 60 g/L; RBC [*] : higher risk < 2.0×10 ¹² /L; lymphocytes ^{**} : higher risk with ≥ 80%; admission spleen size ^{**} : higher risk with larger spleens	age < 1 yr ^{**} vs. > 1 yr; symptom duration ≥ 90 days ^{***} vs < 90 days; Hb < 60 g/L ^{****} vs. Hb ≥ 80 g/L

¹ Significant predictors in Lucero et al. are in fact adjusted for sex and age. Kajaia et al present the overall p-value across all predictor categories and 95% CIs for each category. Overall p-values are presented, and the trend commented on.

² 3 studies adjusted the multivariable model for other non-significant variables, including Lucero et al. (sex, admission and discharge haemoglobin, admission spleen size), Mondal et al. (sex, age, treatment, and various others), Gorski et al. (age, sex, calendar year, treatment centre)

³ Shorter symptom duration associated with higher relapse risk in Burza 2014.

⁴ 166 relapses cases could be matched with primary case records, and does not reflect total relapse number.

⁵ Spleen size recorded as Hackett grade in Gorski 2010.

^{*} 0.05 ≤ p < 0.1.

^{**} 0.01 ≤ p < 0.05;

^{***} 0.001 ≤ p < 0.01

^{****} p < 0.001.

[†] p < 0.05 (inferred as confidence interval does not cover null).

^{††} Significance level not stated.

VL/HIV Co-infection

Relapse rates in patients with VL/HIV co-infection often exceed 20–50%[\[106, 116, 117\]](#), highlighting the importance of host immunity in achieving lasting cure.

Two systematic reviews were identified that summarised relapse determinants in this immunosuppressed population, drawing on 18 studies (Cota et al., 2011)[\[106\]](#) and 15 studies (Alemayehu et al., 2016)[\[107\]](#), respectively. Significant predictors of relapse included low baseline CD4⁺ T-cell counts (particularly < 100 cells/ μ L), previous history of VL relapse, lack of improvement in CD4⁺ counts during follow-up, absence of secondary prophylaxis, and the late initiation of highly active antiretroviral therapy (HAART) initiation.

Since the publication of these reviews, several important studies have shed further light on relapse determinants in VL/HIV co-infected patients[\[116, 118, 119\]](#). In 2017, Abongomera et al. reported outcomes from a retrospective study of 146 co-infected patients treated at an MSF-supported clinic in Northwest Ethiopia between 2008 and 2013, in which 44 (30.1%) relapsed during follow-up. In addition to the timing of HAART initiation, further predictors included high baseline tissue parasite load (adjusted hazard ratio [aHR] 6.63, 95% CI 2.64–16.63, 6+ vs $\leq$ 6+) and the presence of splenomegaly on admission (borderline significant in unadjusted analysis)[\[118\]](#).

More recently, Costa et al. (2023) published a prospective study of 169 co-infected patients from Maranhão, Brazil (2013–2020), in which 70 (41.1%) relapsed during 12 months of follow-up. Baseline splenomegaly, lymphadenopathy, previous VL relapse, HAART regimen, HAART duration, elevated creatinine, and elevated urea were all statistically significant predictors of relapse in unadjusted analyses[\[119\]](#).

In contrast to findings in immunocompetent patients, only one study identified age as a statistically significant predictor of relapse. Burza et al. demonstrated an approximate doubling of relapse hazard among patients $\geq$ 40 years, using routinely collected data from MSF treatment centres in Bihar, India (2007–2012)[\[117\]](#).

Treatments

After VL/HIV co-infection, treatment is arguably the second most important predictor of relapse. This is evident across numerous studies, where the most common reason for failing to achieve the primary outcome of definite cure — defined as initial cure and relapse-free survival for at least 6 months — is relapse[\[62, 72, 73, 94, 108\]](#). Increases in relapse rates have been suggested as a sensitive indicator of declining treatment efficacy and the emergence of drug resistance, often preceding overt increases in primary treatment failure[\[120\]](#). Summarising

outcomes from all VL efficacy studies is beyond the scope of this thesis, although several observations merit discussion.

Firstly, as summarised by Chhajed et al., although relapse risk varies considerably across regimens, combination therapies are generally associated with lower relapse risk than monotherapies[89]. After adjusting for age and symptom duration, Goyal et al. showed that in India, treatment with paromomycin and miltefosine combination therapy was associated with significantly reduced odds of relapse compared with single-dose LAMB 10 mg/kg (aOR 0.21, 95% CI 0.08–0.55)[97]. Similarly, in a large cohort study from Bangladesh, Mondal et al. found that, with the exception of SSG and multiple-dose LAMB, combination regimens were associated with lower relapse rates compared with paromomycin and miltefosine monotherapies[111]. A possible exception to this pattern is SSG/PM combination therapy. Using programmatic MSF data from South Sudan, 1999–2007, Gorski et al. demonstrated an approximate doubling of relapse odds in the SSG/PM group compared to SSG monotherapy (OR 2.08, 95% CI 1.21–3.58)[112].

Secondly, drug dose and treatment duration are also important predictors of treatment failure and relapse. Chhajed et al. showed this at the aggregate level by performing a meta-regression of relapse risk in the ISC against the dose of single-dose LAMB, demonstrating that higher LAMB doses were associated with lower relapse risk (OR 0.81 per 1 mg/kg increase, 95% CI 0.72–0.91)[89]. Similar relationships between dose and overall treatment failure (lack of initial cure or subsequent relapse) have been demonstrated under trial conditions for both LAMB[121] and paromomycin[122, 123]. In East Africa, Dorlo et al. used a paediatric pharmacokinetic-pharmacodynamic (PK-PD) model that demonstrated an inverse relationship between miltefosine exposure and relapse hazard, with similar findings reported in the ISC[124]. These results supported the development of allometric dosing strategies for paediatric miltefosine[125, 126].

Lastly, both the intrinsic infectivity of the parasite strain and the development of drug resistance contribute to increasing rates of initial treatment failure and, when initial cure is achieved, subsequent relapse. A well-known example is the emergence of antimony resistance in Bihar, India, where treatment failure and relapse rates following treatment with SSG increased from < 5% to > 50% during the 1980s and 1990s, despite incremental escalation in SSG dose and duration[61, 127]. These failures have been attributed to both antimony resistance[128], and also to the selective survival and transmission of more infectious parasite strains[129]. Similar concerns arose following the widespread introduction of miltefosine in the ISC in the late 2000s, when increasing relapse rates were observed[66, 67, 130].

Interestingly, comparisons of relapsing versus non-relapsing strains revealed no evidence of drug resistance, although relapsing strains were found to be more infectious[131]. In East Africa, while relapse rates following SSG have never reached the scale seen in India, parasite strains from relapsed patients have been found to harbour increased *in vitro* SSG resistance[132]. More recently, Naylor-Leyland et al. reported increasing relapse rates between 2001 and 2018 that could not be explained by patient-level factors, prompting concerns about declining SSG/PM efficacy, potentially due to emerging resistance[133].

Biomarkers

Since tissue aspirate sampling is invasive, substantial effort has been directed toward developing biomarkers that can serve both as a test-of-cure following treatment and as a diagnostic test of relapse[88]. In fact, a 2015 systematic review identified 53 biomarkers from 170 studies with potential to be used for monitoring post-treatment outcomes in Leishmaniasis[134]. Biomarkers can be broadly categorised into two categories: (1) direct markers of parasite burden, including DNA and antigen detection; and (2) indirect markers of host immunity, such as antibodies, cytokines and acute-phase proteins.

A large portion of the VL biomarker literature focuses on the direct detection of *Leishmania* DNA/RNA in both immunosuppressed[135–140] and immunocompetent patients[55, 141–144]. In particular, real-time quantitative PCR (qPCR) targeting kDNA, performed on the buffy coat of peripheral blood, has been confirmed as a strong predictor of relapse[55, 141, 143, 144], although routine use is limited by the need for specialised laboratory expertise and high costs. In a landmark 2021 study, Verrest et al. used qPCR measurements from 177 immunocompetent patients enrolled in DNDi trials in East Africa⁷ and demonstrated that qPCR levels measured between treatment initiation and day 56 significantly predicted relapse, with areas under the receiver operating curve (AUC) of 0.71, 0.74, and 0.92 on days 14, 28, and 56, respectively[144]. Using the same qPCR data, Verrest et al. (2024) developed a semi-mechanistic population PK-PD model describing parasite replication, drug action, and post-treatment parasite clearance. The model successfully predicted relapse based on modelled day 28 and 56 parasite loads and provided the first direct evidence that relapse risk depends not only on initial parasite clearance but also on subsequent host immune responses[55].

⁷LEAP 0714 (30 children treated with allometric miltefosine), LEAP 0208 (151 patients receiving miltefosine plus LAMB combination regimens), FEXI VL 001 (14 patients receiving fexinidazole)

Interestingly, inclusion of haematological and biochemical parameters,⁸ did not account for variation in post-treatment qPCR values.

Beyond molecular testing, several studies have shown an association between relapse and the degree of *Leishmania* antigenuria in patients with VL/HIV co-infection[56, 57]. In Ethiopian patients, higher levels of urinary antigen at test-of-cure were strongly associated with relapse over the following 12 months (HR 9.8, 95% CI 1.8–82.1, comparing 1–10 parasites/10 fields with no parasites detected using the KATex assay).

Post-treatment IgG subclass titres have also been linked to relapse in both immunocompetent and immunosuppressed individuals[145–148]. However, because of the long half-life of immunoglobulins, their ability to predict relapse at the time of initial cure assessment remains uncertain. In asymptomatic patients, however, serological studies have shown that individuals with higher antibody titres are more likely to progress to symptomatic disease[149, 150].

A variety of non-serological and indirect biomarker panels have been evaluated for assessing disease activity during asymptomatic infection, active disease, and post-treatment recovery[151–156]. In East African patients, Kip et al. reported that the change in neopterin, a marker of macrophage activation, between EoT and day 60 of follow-up was predictive of relapse, with an AUC of 0.84[155]. In a subsequent study of 34 Ethiopian VL/HIV co-infected patients, half of whom relapsed, Takele et al. showed that relapse was associated with (i) failure to restore antigen-specific IFN γ production, (ii) persistently low CD4⁺ T-cell counts, and (iii) high T-cell expression of PD1. These results underscore the importance of effective cell-mediated immunity in achieving sustained cure.

Summary of Relapse Determinants

A substantial and rapidly growing evidence base has been identified describing the determinants of VL relapse, although methodological heterogeneity complicates synthesis.

Among immunocompetent patients — the focus of this thesis — the most consistently identified predictors are clear. Extremes of age (young children and older adults), larger spleen size, whether measured at admission or at EoT, male sex, and the severity of anaemia (at admission and EoT) are repeatedly associated with higher relapse risk. In contrast, evidence for malnutrition is weak, perhaps in part due to the varying definitions, and nuances in how measures are adjusted for age. Symptom duration shows conflicting associations. Shorter pre-treatment

⁸including Hb, white blood cells (WBC), platelets, and creatinine

symptom intervals were strongly linked to higher relapse risk in two ISC studies, yet an opposite association was reported in the study from Georgia — highlighting potential contextual or methodological modifiers.

In patients with VL/HIV co-infection, predictors align more with immunological plausibility: markers of profound immunosuppression (notably low baseline CD4⁺ counts) and indicators of inadequate immune reconstitution (including timing and duration of HAART, absence of secondary prophylaxis) were the dominant determinants of relapse identified.

As anticipated, treatment-related factors are also highly influential. Treatment regimen, dose, and duration modify both initial cure and subsequent relapse risk, and therefore any prognostic model should be accounting for treatment variables where possible.

Finally, biomarker research (particularly peripheral blood qPCR) shows promising predictive performance, but current approaches are constrained by cost, laboratory requirements and limited routine availability, restricting their immediate applicability in many endemic settings.

1.4 Summary

VL remains a highly neglected tropical disease, disproportionately affecting the poorest and most marginalised communities. Despite substantial gains over the last two decades — most notably in the ISC, where coordinated elimination efforts have driven steep reductions in incidence — VL continues to impose a considerable clinical and economic burden. Sustaining these achievements is challenging, and translating success to high-burden regions such as East Africa remains an urgent global health priority[77].

Relapse is a critical, yet often understated, component of this challenge, with consequences both at the individual patient level and for public health more broadly.

For the patient, relapse is associated with heightened morbidity and mortality, exposure to prolonged and often toxic second-line therapies, and considerable direct and indirect financial costs for already vulnerable households[157].

From a public health and elimination perspective, relapse carries further implications. In anthroponotic transmission settings, early identification and effective treatment of relapse cases are essential to interrupt transmission. Moreover, parasite strains isolated from relapse cases can harbour resistance mutations and demonstrate greater fitness, underscoring their potential role as a consequential parasite reservoir.

Given the clinical, economic, and epidemiological significance of relapse, there is a clear and globally recognised need to improve the identification of individuals

at greatest risk[86, 88]. Motivated by this, the overarching aim of this thesis is to determine whether clinical trial data contained within the IDDO VL data platform can be leveraged to support the risk stratification of patients who are likely to progress to relapse following initial cure. In particular, this thesis explores whether such data can inform the development of a robust prognostic model for relapse.

If a prognostic model for VL relapse already exists, data from the IDDO VL data platform could be used to validate and, where necessary, update the model[103]. Therefore, to establish the current evidence base, the next chapter presents a systematic review of prognostic models developed for VL, encompassing both relapse-specific models and those predicting broader clinical outcomes.

2

Prognostic Models Predicting Clinical Outcomes in Patients with Visceral Leishmaniasis: A Systematic Review

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Context

This systematic review was submitted for publication to *BMJ Public Health* in early 2024. Following editorial and peer review feedback received in December 2025, the literature search was updated and substantial revisions made. The version presented in this chapter is adapted from the final [published article](#)[158].

A protocol for this systematic review has also been published[159] and prospectively registered (PROSPERO registration number: [CRD42023417226](#)). Preliminary findings were disseminated through oral presentations at MSF Scientific Day (London, 2024) and the ASTMH 2024 Annual Meeting.

Authorship and Acknowledgements

The following co-authors contributed to this work (listed in publication order):

Forhad Chowdhury;¹ Shermarke Hassan;¹ Elinor Harriss;² Fabiana Alves;³ Ahmed Musa;⁴ Prabin Dahal;¹ Kasia Stepniewska;¹ Philippe Guérin.¹

PG, JW (myself), KS, and PD conceptualised the study. The final manuscript was written by JW, with review and input from all co-authors. Disease-specific expertise was provided by FA, AM, and PG. The search strategy was developed and executed by EH, with the literature search conducted by JW and FH. The data collection tool was designed by JW, and data extraction was performed by JW and SH.

In contrast to other chapters in this thesis, and to reflect co-author contributions, the plural first-person pronoun “we” is used throughout.

2.1 Introduction

In endemic settings, accurately identifying patients at high risk of adverse outcomes is paramount when prioritising limited resources, including admission, treatment selection, and follow-up intensity. Prognostic models — most commonly developed using multivariable regression techniques — estimate an individual patient’s probability of experiencing a future clinical event[160]. Often presented as simplified risk scores, such models abound in the medical literature and inform clinical decision-making and guideline development[161]. In infectious diseases alone, systematic

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reviews have identified over 600 prognostic models for COVID-19[162], 37 models for tuberculosis[163], and 27 models for malaria[164]. However, concerns have been raised on the methodological quality and reporting of prediction models. Biased models may overestimate performance, generate misleading predictions, and ultimately contribute to suboptimal or inequitable clinical decisions[161, 165].

Several prognostic models have been developed and implemented for patients with VL[166, 167]. In Brazil, national guidelines introduced in 2011 recommended the use of four related risk scores, based on combinations of clinical and/or laboratory factors, to guide hospital admission and the use of liposomal amphotericin B[167–169]. Similarly, since 2003, Médecins Sans Frontières (MSF) Holland has used simple VL risk scores in South Sudan to support clinical decision-making regarding liposomal amphotericin B therapy, broad-spectrum antibiotics, blood transfusions, and nutritional support[13, 112, 166]. Additional models exist[170, 171], although the full range of models, including their characteristics, comparative performance, and inherent biases, have yet to be systematically described.

We therefore conducted a systematic review to identify, summarise, and critically appraise prognostic models predicting future clinical outcomes in patients with VL. This review aims to support policymakers in evaluating the incorporation of prognostic models into treatment guidelines, and to help clinicians assess the applicability of existing models to their own patient populations. In addition, researchers may use this review to identify evidence gaps and to determine whether available data are better suited to the development of new models, or to the external validation (evaluation) and/or updating of existing models[172].

A glossary of key terms relating to model development and evaluation is presented in Table 2.1.

2.2 Methodology

We adhered to Transparent Reporting of Multivariable Prediction Models for Individual Prognosis or Diagnosis: Checklist for Systematic Reviews and Meta-Analyses (TRIPOD-SRMA) when reporting this systematic review[173]. Data extraction was guided by the Checklist for Critical Appraisal and Data Extraction for Systematic Reviews of Prediction Modelling Studies (CHARMS)[174] and the Prediction Model Risk of Bias Assessment Tool (PROBAST)[175, 176]. Risk of bias assessment is performed with PROBAST[175].

Box 2.1: Key Messages of the Systematic Review**What is already known on this topic**

- Visceral leishmaniasis (VL) is a neglected tropical disease associated with high mortality, predominantly affecting populations in resource-constrained settings.
- Identification of patients at high risk of adverse outcomes is critical for prioritising limited healthcare resources, including hospital admission, treatment selection, and follow-up care.
- Risk stratification of patients can be performed using prognostic models, however, the range of available models and their key methodological characteristics have not been systematically evaluated.

What this study adds

- Using existing reporting guidelines for systematic reviews of prediction model studies, this review provides the first comprehensive synthesis of prognostic models predicting future clinical outcomes in patients with VL.
- We identified 12 prognostic models, all of which predict mortality and were developed in Brazil or East Africa.
- All identified models and model evaluations were judged to be at high risk of bias; therefore model predictions and performance measures should be interpreted with caution.

How this study might affect research, practice or policy

- This review highlights key evidence gaps in the VL prediction model landscape and supports researchers in identifying candidate models for external validation or updating using local patient data.
- By summarising and appraising available models, this review enables clinicians and policymakers to assess the applicability of existing models to their own patient populations.
- By identifying common methodological limitations, this review encourages researchers to review contemporary guidance on the reporting of prediction model development and evaluation.

2.2.1 Eligibility Criteria

We followed a Population, Index model, Comparator model, Outcomes, Timing, Setting (PICOTS) approach to frame our review question and define our eligibility criteria[174, 177].

The population consisted of all human patients with a confirmed or suspected diagnosis of VL, as defined by the study authors. Index models included all prognostic models developed in patients with VL, including model development studies, external validation studies and/or model updating studies. No individual comparator model was defined, given the aim of the review was to summarise and

critically appraise all identified model developments and evaluations. All clinical outcomes were considered that occur following the intended time of model use, with no upper limit on the prediction horizon. Timing of model use was either at the time of or following diagnosis. We imposed no restriction on the setting of model development or evaluation.

In accordance with best practice in prediction modelling research[175, 178, 179], we defined a prognostic model as a multivariable model (including two or more predictors) developed with the intention of predicting future outcomes at the individual patient level. Prediction model studies were distinguished from predictor finding or prognostic factor studies, where the aim is to investigate the effect of a single or group of factors on an outcome of interest[180]. We excluded unpublished studies (including conference abstracts, educational theses), studies that only report diagnostic prediction models, and animal studies.

2.2.2 Information Sources and Search Strategy

An information specialist (EH) created the search strategy to retrieve relevant records from the following databases: Ovid Embase; Ovid MEDLINE; the Web of Science Core Collection, SciELO and LILACS. No language restrictions were imposed. The databases were initially searched from database inception to 1 March 2023. Using the same strategy, the search was updated to include additional records from 1 March 2023 to 18 December 2025. The search strategy used text words and relevant indexing terms to retrieve studies describing eligible prognostic models. The Ingui search filter was augmented with an additional search string as described by Geersing et al[181, 182], and combined with VL-specific keywords. Google Scholar was used to identify any complementary grey literature (full search strategy presented in the [Supplementary Material](#)).

2.2.3 Study Selection

Deduplication and screening of references were performed in Covidence[183]. Screening was performed independently by two reviewers (JW, FC); initially at title and abstract level, and subsequently at full-text level. Where discordance existed, a third expert reviewer (PD) was consulted to make the final judgement. Subsequent forward and backward citation searching were performed to identify records missed by the initial search.

2.2.4 Data Collection

Study information was captured using a REDCap server hosted at the University of Oxford[184]. A data extraction form was created and piloted as per the CHARMS checklist and PROBAST ([Supplementary Material](#))[174–176]. Two reviewers (JW, SH) independently extracted the study information. Where discordance remained after discussion, a final decision was made by a third expert reviewer (PD). Study authors were not contacted in the event of unclear or missing information.

2.2.5 Risk of Bias Assessment

Risk of bias was assessed using PROBAST[175, 176]. Two reviewers (JW, SH) independently assessed each model development (including updating) and external validation, by answering 20 signalling questions across four domains (participants, predictors, outcome, and analysis). Responses were used to judge the overall risk of bias as either ‘low’, ‘high’ or ‘unclear’. Any discordance was resolved through discussion.

As part of the bias assessment, we established whether the model predictors and their corresponding risk scores were consistent with the reported multivariable regression coefficients (PROBAST signalling question 4.9). Consistency was assessed by referring to the study’s reported methodology, and expert guidance on the presentation of clinical risk scores[185].

2.2.6 Applicability

Applicability was not formally assessed given the broad remit of the research question. Instead, we summarise a broad range of information to facilitate comparison of each model’s predictors, participants, and outcomes, with those of the intended target setting[176].

2.2.7 Synthesis of Results

Key characteristics of the identified models and their evaluations are summarised at the aggregate study and model levels, with complementary information presented in accompanying tables and [Supplementary Material](#). As all identified models predict mortality, a figure is included to facilitate comparison of candidate and final predictors across models.

A complementary narrative summary is also provided in the [Supplementary Material](#), where each study is described individually to provide additional contextual detail.

All identified models reported discrimination using the c-statistic. Accordingly, median and range values are presented for models evaluated in the development dataset (apparent performance) and in new patient data (external validation). Where c-statistics or other measures were not explicitly reported, no attempt was made to derive them from other performance measures. Owing to sparse and heterogeneous reporting across studies, other reported performance measures were not suitable for quantitative synthesis and are instead reported separately for individual models where available.

Meta-analysis was not performed due to substantial heterogeneity across models, including differences in predictors, populations, and outcome definitions, precluding meaningful pooling of c-statistics.

Patient and Public Involvement

No patients or members of the public were involved in this systematic review.

2.3 Results

2.3.1 Study and Model Selection

After deduplication, 3,313 records were identified from the combined original and updated literature searches (Figure 2.1). Title and abstract screening yielded 71 records for full-text review, of which eight prognostic model studies were identified[166, 167, 170, 171, 186–189]. In total, 12 prognostic models were described, of which 10 underwent one or more evaluations in patient data from either different settings and/or time periods (19 external validations presented in four studies)[166, 167, 188, 189].

2.3.2 Model Developments

Key characteristics of the identified models are summarised in Table 2.2 (with further predictor, participant, and outcome information presented in the [Supplementary Material](#)). All models use multivariable logistic regression and predict mortality as a binary outcome, reported either as in-hospital mortality (10 models)[166, 167, 171, 187–189], or registry-reported mortality (two models)[170, 186].

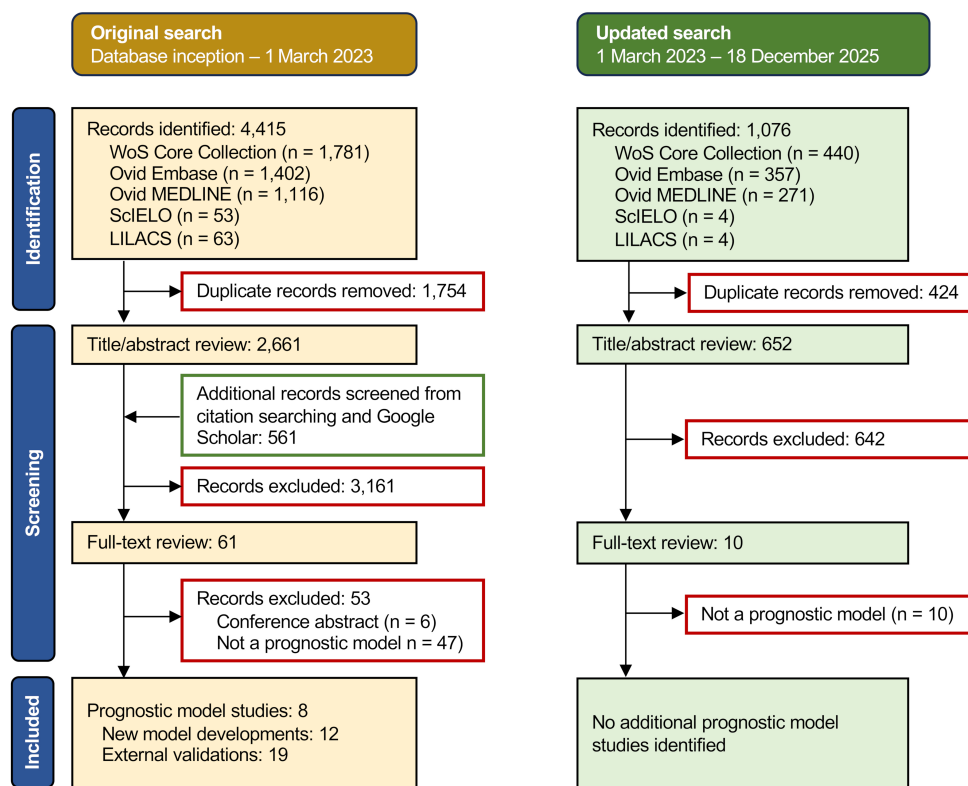


Figure 2.1: PRISMA-like flow diagram depicting the record screening process, performed initially on 1 March 2023 with subsequent updating on 18 December 2025.

Table 2.1: Glossary of key terms.

Term	Description
Prediction model	An equation or set of rules for estimating an individual's probability of an outcome based on two or more predictors. Traditionally developed using multivariable regression, although machine learning methods are increasingly used.
Outcome	The event being predicted. Also termed the response or dependent variable. Models are described as <i>prognostic</i> when outcomes occur after the time of model use, and <i>diagnostic</i> when outcomes are present at the time of model use.
Predictors	Patient or group characteristics used to estimate an outcome, also termed covariates, inputs, determinants, or independent variables. Predictors may be <i>candidate predictors</i> (considered for inclusion during model development) or <i>final predictors</i> (retained in the final model).
Overall performance	An overall summary measure of how well a model fits the data. Commonly presented measures include explained variation (R^2) and the Brier score.
Discrimination	The ability of a model to distinguish between individuals with and without the outcome. For binary outcomes this is often quantified using the concordance (c-)statistic, also termed the AUC, defined as the probability that the model assigns a higher predicted risk to an individual with the outcome than to one without. Values range from 0.5 (no better than chance) to 1.0 (perfect discrimination).
Calibration	The agreement between predicted risks and observed outcomes. For binary outcomes, calibration is best assessed using a calibration plot comparing predicted risks with observed outcome frequencies across the range of predictions.
Apparent performance	Model performance evaluated using the same dataset in which the model was developed. Performance can be optimistically biased due to overfitting, particularly in small samples or when data-driven predictor selection is used.
Internal validation	Model performance evaluated in the population represented by the development dataset, ideally using resampling techniques (e.g. cross-validation or bootstrapping) to account for overfitting. Split-sample approaches are generally considered inefficient.
External validation	Model performance evaluated in new data that were not used for model development and providing an assessment of model generalisability to new populations or settings.

Table 2.2: Key characteristics on the 12 prognostic model developments, ordered by outcome and year published. Each row corresponds to a different model developed by the referenced study. -: not reported; +ve: positive; Cand: candidate; clin: clinical; EPP: events per predictor parameter; HIV: human immunodeficiency virus; lab: laboratory; y: years

Study	Model described	Data source	Location Period	% Male	% HIV +ve	Events ¹ Sample size (%)	Predictors ² Final, Cand.	EPP ²
Outcome: Registry-reported mortality								
de Araújo 2012[186]	-	Registry	Brazil (Belo Horizonte) 2007–09	-	-	49 376 (13.0)	4, 48	1.0
Coura-Vital 2014[170]	-	Registry	Brazil (Nationwide) 2007–11	61.7	7.0	770 12,333 (6.2)	12, (29)	(26.6)
Outcome: In-hospital mortality								
Werneck 2003[171]	-	Case-control	Brazil (Teresina) - (< 2003)	68.9	-	12 90 (13.3)	4, (15)	(0.8)
Sampaio 2010[187]	-	Retrospective	Brazil (Recife) 1996–2006	50.4	-	57 546 (10.4)	6, (15)	(3.8)
Costa 2016[167] ³	< 2 y, clin only	Prospective	Brazil (Teresina) 2005–08	-	-	- 314 (-)	6, (25)	(0.9)
	< 2 y, clin + lab	Prospective	Brazil (Teresina) 2005–08	-	-	- 291 (-)	6, (31)	(0.7)
	≥ 2 y, clin only	Prospective	Brazil (Teresina) 2005–08	-	-	- 569 (-)	9, (27)	(1.6)
	≥ 2 y, clin + lab	Prospective	Brazil (Teresina) 2005–08	-	-	- 538 (-)	9, (33)	(1.2)
Abongomera 2017[166]	-	Retrospective	Ethiopia (Abdurafi) 2008–13	95.9	19.3	99 1,686 (5.9)	8, 16	6.2
Kämink 2017[188]	< 19 y	Retrospective	South Sudan (Lankien) 2013–15	54.2	excl.	116 4,931 (2.4)	8, 20	5.8
	≥ 19 y	Retrospective	South Sudan (Lankien) 2013–15	56.2	excl.	70 1,702 (4.1)	8, 21	3.3
Foinquinos 2021[189]	Sampaio updating	Retrospective	Brazil (Recife) 2008–18	48.7	-	10 156 (6.4) ⁴	1, 1	10.0

¹ Including patients with missing predictor information and excluding patients with missing outcomes.

² Figures presented in brackets are estimated from incomplete reporting.

³ Total events in combined development dataset reported as 7.5% (66/883) and not disaggregated by age group. EPP is calculated based on extrapolating the 7.5% event rate to each model development dataset.

⁴ Sample size excludes both participants with missing predictors and missing/excluded outcomes.

Two studies (three models) were developed in East African MSF treatment centres (one model from Ethiopia that included patients with VL/HIV co-infection[12], and two models from South Sudan, for patients ≥ 19 and < 19 years, and excluding VL/HIV co-infection[188]). The remaining six studies (nine models) were performed in Brazil. Two Brazilian studies (two models) were developed using registry data; either at a national level[170] or for residents of Belo Horizonte, state of Minas Gerais[186]. The remaining Brazilian studies were developed in hospital settings, including two studies (five models) developed from patients admitted to a hospital in Teresina[167, 171], and two studies (two models) that were developed for children < 15 years and admitted to a hospital in Recife, state of Pernambuco[187, 189]. No models were developed in South Asia or the Mediterranean region.

Most studies employed a retrospective cohort design, using hospital records (four studies, five models)[166, 187–189] or registry data (two studies, two models)[170, 186]. One study used a prospective cohort design (four models)[167] and one study (one model) used a case-control design[171]. The median number of patients used for model development was 542 (range 90–12,333).

Participant age formed the inclusion criteria of eight models[167, 187–189], with five models limiting inclusion to adolescents and younger[167, 187–189]. Where reported, the median proportion of male participants was 56.2% (range 48.7–95.9%, seven models). No model excluded participants based on sex. Of the models developed in adults (≥ 15 years), patients living with HIV were either excluded (two models)[188], not reported (two models)[171, 186], and where reported, ranged from 7.0–19.3% of the model development datasets (four models)[166, 167, 170].

2.3.3 External Validations

Both East African studies performed external validations of their model developments (two studies, three models)[166, 188]. The model developed in Ethiopia (Abdurafi health centre, Abdurafi, Amhara region) was validated using data from a nearby treatment centre (Leishmaniasis Research and Treatment Centre, Gondar, Gondar, Amhara region)[166], and the two models developed in South Sudan were validated using retrospectively collected data from the same treatment centre (Lankien hospital, Jonglei state) and a treatment centre from a neighbouring state in South Sudan (Malakal hospital, Upper Nile state) (three external validations per model)[188].

Two Brazilian studies reported the external validations of eight models[167, 189]. One study, conducted in a prospective hospital cohort (Teresina, state of Piauí), both developed and evaluated four models. All four models were evaluated

in patients attending the same hospital over the following five years[167]. The same study used their prospective cohort to evaluate three further models: one developed from historical cohort from the same hospital[171], one developed from a retrospective hospital cohort (Recife, state of Pernambuco)[187], and one using national registry data[170]. The second Brazilian study[189] used a retrospective hospital cohort (Recife, state of Pernambuco) to both evaluate and update a model previously developed in the same hospital[187].

Further details on validation datasets are presented in the [Supplementary Material](#).

2.3.4 Model Performance

Model discrimination measures were reported as c-statistics for all risk scores, and presented in Table 2.3. Further performance measures, where reported, are presented in the [Supplementary Material](#). For external validations, the median c-statistic was 0.78 (range 0.62–0.92, 10 models, 19 external validations). When evaluated in the same patients used for model development (apparent performance), the median c-statistic was 0.86 (range 0.56–0.93, 12 models). No studies presented overall measures of performance or calibration plots. One model’s calibration plot could be reproduced from an internal (split-sample) validation of the risk score[170].

Table 2.3: Summary of model discrimination and risk of bias assessments, ordered by outcome and year published. Each row corresponds to a individual model that was either developed or externally validated in the referenced study. Unless otherwise stated, the reported c-statistics represent performance of the model risk score and not the full model equation. -: not reported; dev: development; eval: evaluation; n/a, not applicable (external validations only); perf: performance; val: validation; **Risk of bias (RoB)**: A: analysis, O: outcome; OA: overall assessment; P: participants; Pr: predictors; +: high RoB; -: low RoB; ?: unclear RoB

Study	Model described	Model discrimination (c-statistic) ¹			Risk of bias ²					
		Apparent perf.	Internal val.	External val.	Eval type	P	Pr	O	A	OA
Outcome: Registry-reported mortality										
de Araújo 2012 ^[186]	-	0.756	-	-	dev	+	+	?	+	+
Coura-Vital 2014 ^[170]	-	0.80 (0.78–0.82)	0.78 (0.75–0.82) ³	-	dev	+	+	?	+	+
Outcome: In-hospital mortality										
Werneck 2003 ^[171]	-	0.882	-	-	dev	+	+	-	+	+
Sampaio 2010 ^[187]	-	0.895	-	-	dev	+	+	-	+	+
Costa 2016 ^[167]	< 2 y, clin only	0.90 (0.84–0.97)	-	0.83 (0.64–1)	dev	-	-	-	+	+
				0.86 (0.74–0.98)	val (×2)	-	-	-	+	
	< 2 y, clin + lab	0.93 (0.88–0.98)	-	0.80 (0.57–1)	dev	-	-	-	+	+
				0.92 (0.84–1)	val (×2)	-	-	-	+	
	≥ 2 y, clin only	0.89 (0.84–0.93)	-	0.75 (0.68–0.83)	dev	-	-	-	+	+
				0.88 (0.83–0.93)	val (×2)	-	-	-	+	
	≥ 2 y, clin + lab	0.92 (0.88–0.96)	-	0.79 (0.62–0.96)	dev	-	-	-	+	+
				0.71 (0.34–1)	val (×2)	-	-	-	+	
	Werneck 2003	n/a	n/a	0.75	val (×1)	?	-	-	+	+
	Sampaio 2010	n/a	n/a	0.87	val (×1)	?	-	-	+	+
	Coura-Vital 2014	n/a	n/a	0.77	val (×1)	?	-	-	+	+
Abongomera 2017 ^[166]	-	0.83 (0.79–0.87)	0.82 (0.77–0.88) ⁴	0.78 (0.72–0.83)	dev	+	-	-	+	+
					val (×1)	+	-	-	+	
Kämink 2017 ^[188]	< 19 y	0.83 (0.78–0.87)	-	0.72; 0.83; 0.77	dev	+	+	-	+	+
					val (×3)	+	+	?	+	
	≥ 19 y	0.74 (0.68–0.81)	-	0.72; 0.80; 0.71	dev	+	+	-	+	+
					val (×3)	+	+	?	+	
Foinquinos 2021 ^[189]	Sampaio updating	0.556	-	-	dev	+	+	-	+	+
		0.762 (0.662–0.901) ⁵								
	Sampaio 2010	n/a	n/a	0.618 ⁵	val (×1)	+	+	-	+	+

¹ Significant figures and 95% confidence intervals are reproduced as reported in the publication.

² For studies presenting multiple external validations of the same model, risk of bias assessments were the same and therefore grouped together.

³ Split-sample (random split with 2:1 development:validation).

⁴ Cross-validation (5-fold).

⁵ C-statistic from evaluation of full model equation.

2.3.5 Predictors

A visual comparison of the candidate and final predictors is presented in Figure 2.2. Where authors provide further predictors definitions, these are reported in the [Supplementary Material](#). The four most common candidate predictors were jaundice (12 models), age (11 models), sex (10 models) and bleeding (10 models). Initial VL treatment was included as a candidate predictor in two models, although not retained in the final models [12,35]. Predictors most frequently retained in the models were jaundice (11 models), bleeding (eight models), and age (seven models). No models included sex as a final predictor. One model did not include HIV status as a candidate predictor, despite being conducted in adults and not pre-specifying the exclusion of patients with HIV [20]. Apart from HIV testing, four models did not consider laboratory tests as a predictor[167, 170, 186]. The remaining eight models included laboratory tests both as candidate and final predictors[166, 167, 171, 187–189].

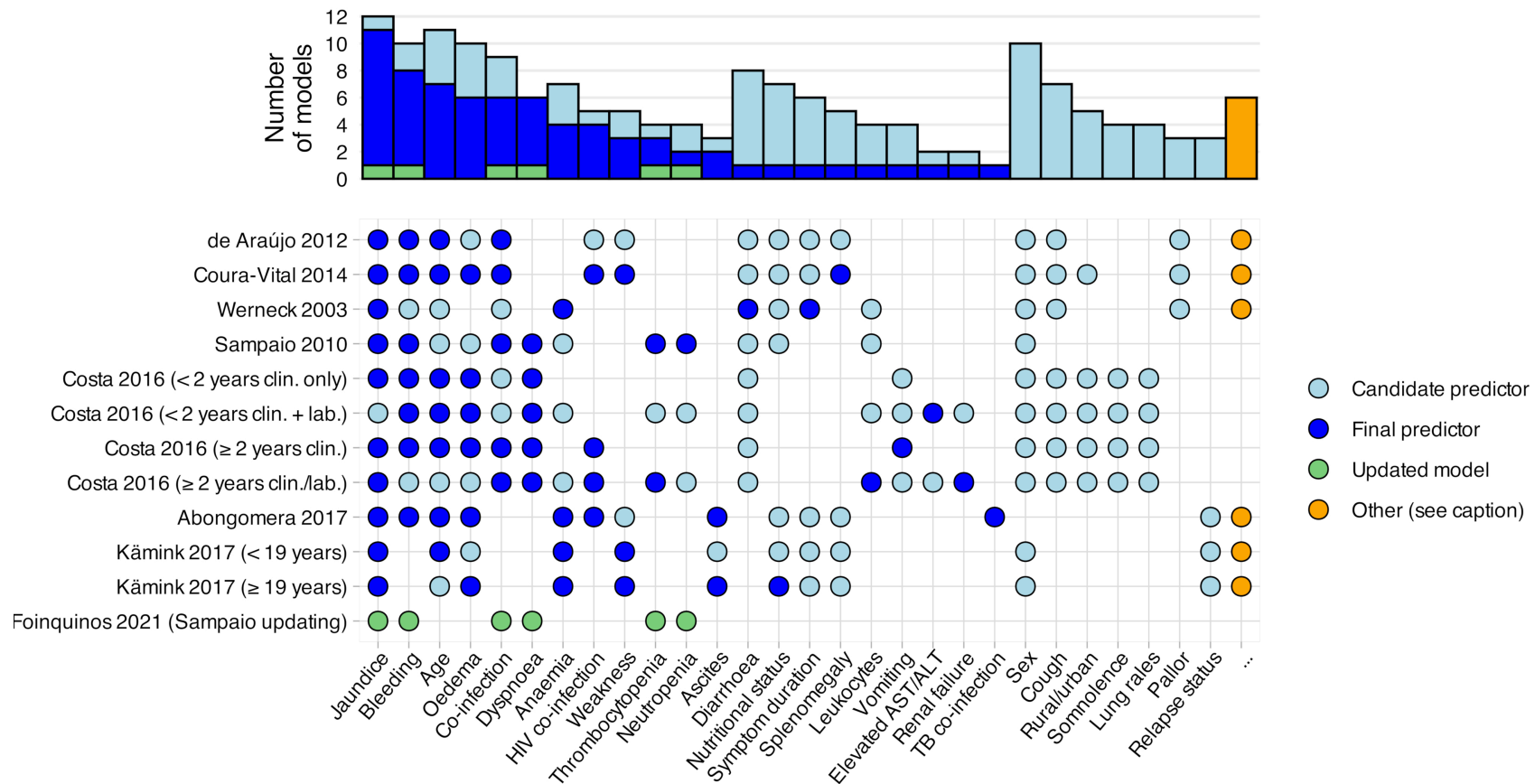


Figure 2.2: Candidate (considered) and final (retained) predictors included in prognostic models of mortality in visceral leishmaniasis. Bars indicate the number of models incorporating each predictor. Models are labelled by first author, publication year, and model name (in brackets where multiple models were reported). Conceptually similar predictors were grouped and renamed; definitions and groupings as reported by the study authors are provided in Supplemental Table 4. For models by de Araújo et al. and Coura-Vital et al., cough and/or diarrhoea were treated as a single predictor in accordance with national registry reporting. For models by Kämink et al., oedema and ascites were combined as a single predictor, whereas Abongomera et al. treated these as separate variables. Predictors shown for the updated model correspond to the predetermined final predictors of the model being updated (Sampaio et al.). Abbreviations: ALT, alanine transaminase; AST, aspartate transaminase; clin, clinical; HIV, human immunodeficiency virus; lab, laboratory; TB, tuberculosis. ‘...’ predictors assessed in ≤ 2 models and not included in the final model: [Werneck 2003]: ‘abdominal distension’, episodes of blood transfusion; [de Araújo 2012]: fever, hepatomegaly, “other clinical manifestations”, initial VL regimen, VL drugs following initial regimen, antimony treatment duration; [Coura-Vital 2014]: fever, hepatomegaly, ‘other clinical manifestations’, race, education; [Abongomera 2017]: initial VL regimen; [Kämink 2017, both models]: lymphadenopathy.

2.3.6 Model Presentation

All 12 models were presented as simplified risk scores. Score ranges, suggested risk groupings, and the authors' opinions on how different risk groups should inform clinical decision-making are presented in the [Supplementary Material](#). Outcome (mortality) probabilities corresponding to the risk scores were presented either in tabular format (four models)[166, 170, 188], graphically and through a web application (four models)[167], or not presented (four models)[171, 186, 187, 189]. The full model equation, including model intercept, was reproducible for three models in total — presented either in the original study describing the model development (one model)[170], or in the updating study (two models, corresponding to the original model and updated model)[189].

2.3.7 Risk of Bias Assessment

Risk of bias assessments for all model developments and external validations are presented in Table 2.3, alongside measures of discrimination and details of model presentation and reproducibility. Further details on variable selection and handling of missing data are presented in the [Supplementary Material](#). Responses to the risk of bias signalling questions are also provided in the [Supplementary Material](#).

All 12 model developments were judged at an overall high risk of bias.

The analysis domain was assessed at high risk of bias across all model developments. One model obtained a sufficient sample size (event to predictor parameter ratio > 10), and adequately reported model performance, including calibration[170]. All models (excluding model updating) were developed using a univariable selection stage and did not adjust model predictions to account for optimism due to overfitting. In four model developments, the presented risk scores were not reproducible from the regression model coefficients(full calculations are presented in the [Supplementary Material](#))[167].

Five models (two studies) were assessed as having a low risk of bias in the predictors domain[166, 167]. Both studies provided evidence that the model predictors were defined consistently for all patients and were assessed without knowledge of outcome data. The remaining seven models (six studies) were considered at high risk of bias, with one model including predictors that were likely measured after the time of intended model implementation[186].

The outcome domain was assessed at low risk of bias for all model developments, except for two models where bias risk was unclear[170, 186].

The participants domain was assessed at low risk of bias for four models (one study)[167], with the remaining models considered high risk due to using retrospectively collected data.

All 19 external validations were also judged at high risk of bias, although assessment across the domains was limited by a lack of reporting. Briefly, sources of bias were similar to those identified for model developments, including small sample sizes, the use of retrospectively collected data, and the absence of calibration measure reporting. Please refer to the [Supplementary Material](#) for further details.

2.4 Discussion

Across a range of diseases, the number of prediction model studies has surged over the last two decades, driven by an increasing focus on personalised medicine, the need to provide evidence for guideline development, and a growing number of tools available for model development[103, 163, 190]. VL is no exception, with a total of 12 prognostic models identified, of which nine were published since 2013. All models were developed in Brazil or East Africa and predict either in-hospital or registry-reported mortality.

When using a prognostic model to predict mortality, for example, in a hospital ward or outpatient clinic, the clinician should be confident that after inputting model predictors (for example, age, haemoglobin, clinical signs and symptoms), they receive a trusted estimate of the probability (risk) of the outcome occurring. Empowered with this information, the patient can then be counselled, and important treatment decisions agreed on. Having confidence in the model output is fundamental, since inaccurate risk predictions can lead to suboptimal decision-making, inequitable care, and at times, patient harm.

2.4.1 Prognostic Model Assessment

Three important considerations should be taken into account when assessing whether a prognostic model's estimated risks are reliable for a target patient. These include (1) applicability: whether the model has been developed or evaluated in a setting and population similar to that of the target patient; (2) performance, including both discrimination and calibration, in the populations and settings chosen for evaluation; and (3) risk of bias: whether the studies performing model development or evaluation are subject to systematic errors that may distort risk estimates and performance measures.

Applicability

First, we consider model applicability. Has the model been developed, and/or evaluated in patients similar to the patient I'm interested in? Here, the onus is on the model user to compare their target patient and setting to those in which the model was developed and/or evaluated (Table 2.2, [Supplementary Material](#)). In VL, important mismatches can result from differences in (i) HIV status, (ii) patient age, (iii) geographical setting (both locally within a country, and between endemic regions), (iv) treatments used (where reported), (v) temporal differences, and (vi) treatment setting (e.g. inpatient vs outpatient). For example, estimating the mortality risk of a patient with HIV co-infection using a model developed from HIV-negative patients is likely to underestimate the true risk. Similarly, using a model developed in Brazil to estimate mortality risk in patients from India, may well overestimate the true risk, given the overall lower mortality rate in South Asia[3]. One may also question the contemporary risk estimates of models developed using data from 10-20 years ago. For example, is it fair to assume that models developed using hospital or registry data in Brazil in the 2000s and early 2010s are still accurate, given significant changes in treatment, and an evolving disease epidemiology[191]? Questions regarding model applicability, such as these, often have no clear consensus answer, although are important to consider.

Performance

Second, even if a model were applicable to my patient population, how can I be sure it performs well? As we summarise here (Table 2.3 and [Supplementary Material](#)), model discrimination, presented as the c-statistic, is universally reported across all identified models, both when evaluated in the same patients used for model development (apparent performance) and when evaluated in new patients (external validation). Furthermore, given the reported c-statistics are frequently over 70–80%, even when evaluated in new data, we are reassured that in the appropriate population, most models do a fairly good job at assigning higher risk to those who progress to death compared to those who do not. Crucially, while a highly discriminative model can reliably rank patients by risk, discrimination alone provides no information on the accuracy of the predicted probabilities. A model that estimates a mortality risk of 20%, where the true risk is 2%, may still have 100% discriminative performance, despite dramatically overestimating the true risk.

Instead, we should be reviewing a model's calibration, ideally presented as a plot of observed vs. estimated risks, to assess this fundamental, yet frequently overlooked aspect of model performance[192]. However, with estimated and observed risks

only reproducible for one model risk score[170], we find that the VL prognostic model landscape bucks the broader trend: calibration is neglected[162, 172, 193]. In contrast, measures of discrimination and classification (sensitivity, specificity), are preferentially reported, despite their limited clinical utility[176].

Risk of Bias Assessment

Finally, we consider risk of bias, alongside the closely related issue of model reporting, both of which directly influence the interpretability, reproducibility, and clinical use of prognostic models.

All model development and external validation studies were judged to be at high risk of bias according to PROBAST. Importantly, this classification does not imply that the identified models are inherently flawed or clinically uninformative. Rather, PROBAST highlights aspects of study design, analysis, or reporting that can challenge interpretation of estimated risks and performance measures[176]. Below, we briefly describe several recurrent issues and direct readers to accessible guidance on best practice in prognostic modelling.

A frequent concern was model overfitting, which occurs when models capture random variation specific to the development dataset rather than true underlying risk patterns. This is more likely when many predictors are considered relative to the number of outcome events, and can lead to exaggerated performance estimates and inflated risk predictions when models are applied to new patients[194, 195]. While rules of thumb often suggest a minimum of 10–20 events per predictor parameter, most identified models fell well below this threshold. Formal sample size calculations are now available for prediction model development, which can also be used to assess the number of predictors that can be reliably supported for a given sample size[46].

Internal validation provides a means of adjusting performance estimates and risk estimates for the effects of overfitting. However, only two models applied internal validation methods, either using cross-validation[166] or a split-sample approach[170]. Although historically common, data splitting is increasingly discouraged, particularly in smaller datasets, as it reduces the effective sample size available for model development. Resampling methods, such as bootstrapping, allow full use of the available data while providing more reliable estimates of model performance[161].

Another recurring source of bias was predictor selection based on univariable analyses. This approach can result in unstable models and misleading predictor inclusion, as variables are selected based on isolated statistical associations rather than their joint contribution to risk prediction. Alternative strategies include

pre-specifying predictors based on clinical relevance, or applying penalisation or dimension-reduction techniques to limit model complexity[176, 196].

We also identified reporting-related concerns that directly affected reproducibility of derived risk scores, with one study (four models) presenting scores that did not correspond to reported regression coefficients. While these discrepancies may reflect reporting errors, they undermine confidence in score implementation and external use.

Other concerns related to model reporting and reproducibility were commonplace, limiting both model appraisal and application to the individual patient. Notably, several studies did not report absolute risk estimates corresponding to the presented risk scores[171, 186, 187, 189]. When models are presented primarily as binary classifiers (e.g. high vs low risk), much of their potential clinical utility is lost, as clinicians are unable to interpret or communicate individualised risk estimates.

Similarly, incomplete presentation of the full model equation—including the intercept term—restricted reproducibility and external evaluation. Without this information, users are limited to applying simplified risk scores, which may differ in performance from the underlying regression model[176]. As only two studies reported the full model equation[170, 189], independent evaluation of the remaining models would require direct contact with the original authors.

Additional sources of bias—such as categorisation of continuous predictors, uncertain or suboptimal handling of missing data, reliance on retrospective data sources, and model evaluation in datasets with few outcomes—were also common. Importantly, the issues identified in this review are not unique to VL, but mirror challenges repeatedly highlighted in systematic reviews of prognostic models across a wide range of disease areas[103, 162, 176, 197, 198]. For readers interested in addressing these issues, we signpost accessible and authoritative guidance on best practices in model development[196], evaluation[199, 200], sample size calculation[201, 202], model reporting[103], model presentation, (including risk score derivation)[185], and updated guidance on risk of bias assessment[203].

2.4.2 Predictors of Mortality

Where outcome timing was reported, death frequently occurred within days of hospitalisation, indicating a short prediction horizon. Two of the included prediction model studies reported time to death: one-third of patients died within 48 hours of admission in South Sudan[188], and the average time to death was just over five days in a study reporting from Teresina (Piauí, Brazil)[171]. In this context,

predictors retained across models predominantly reflect advanced disease and imminent physiological decompensation.

Severe VL is increasingly understood as a progressive inflammatory syndrome — described as ‘leishmanial sepsis’ and characterized by cytokine storm, disseminated intravascular coagulation, secondary bacterial sepsis, and evolving multiple organ dysfunction[51]. It is therefore unsurprising that frequently identified predictors of mortality—including jaundice, bleeding, dyspnoea, oedema, and bacterial co-infection (cf. Figure 2.2) represent clinical markers of established organ failure and broadly align with factors identified in systematic reviews of prognostic factors for VL mortality in East Africa[204] and Brazil[205].

Conversely, factors plausibly important earlier in the disease course, such as symptom duration, nutritional status, or relapse history, were infrequently selected. While these variables may influence susceptibility or delayed care-seeking, they appear to have limited discriminatory value once patients reach hospital with severe disease.

The short prediction horizon also explains the relatively high discrimination reported across studies, with c-statistics frequently approaching or exceeding 0.85. Under these conditions, discrimination is driven by late-stage clinical features rather than early prognostic signals. As a result, existing models primarily identify patients at high risk of near-term mortality, rather than supporting earlier risk stratification or anticipation of clinical deterioration.

2.4.3 Implications for Future Research

Several important evidence gaps were identified in this review. Most notably, no prediction model studies were identified outside East Africa and Brazil. This is striking given that South Asia has, until recently, accounted for the lion’s share of the global VL burden. Mortality rates in South Asia are, however, relatively low compared with Brazil and East Africa, meaning that relatively large sample sizes would be required to develop mortality models without substantial risk of overfitting. In this setting, routinely collected programme data and national surveillance systems may offer a pragmatic opportunity to support adequately powered model development, provided data quality and outcome ascertainment are sufficient.

A further major evidence gap is the absence of models predicting relapse or post-kala-azar dermal leishmaniasis. These outcomes are of particular importance in elimination settings, since both represent persistent infection reservoirs that sustain transmission[85, 206]. The lack of current prognostic tools in this area limits our ability to target follow-up, secondary prophylaxis, or intensified surveillance.

Recently the WHO released a target product profile for a test-of-cure following treatment, such that patients at high risk of subsequent relapse can be identified early[88]. A prognostic model for relapse could serve as a surrogate for such an in vitro test. Leveraging IPD meta-analysis allows harmonisation of data across heterogeneous studies, increases statistical power, and maximises the value of existing datasets—particularly in disease areas where high-quality data are scarce.

In Brazil, existing models — including those informing current national guidelines[60] — are based on data collected 10–20 years ago. Since that time, the epidemiology of VL has evolved, and treatment practices have shifted substantially, for example, with higher rates of VL/HIV co-infection, an increasing age at diagnosis, and better access to liposomal amphotericin B[191]. These changes raise questions about the continued validity of older models and highlight the need for model evaluation with recent patient data and, where necessary, model updating. In addition, future studies should carefully consider whether it is appropriate to combine patients with and without HIV within a single model, given their distinct clinical trajectories, immunological profiles, and risk factors for adverse outcomes[106].

Finally, emerging applications of artificial intelligence and machine learning warrant cautious consideration in the VL prediction modelling landscape. Recent studies have demonstrated the feasibility of applying machine learning methods to VL datasets[207, 208]. However, as emphasised in PROBAST+AI and TRI-POD+AI guidance, such approaches are not immune to bias, overfitting, or poor transparency[103, 203]. Without rigorous reporting, external validation, and explicit consideration of clinical use-cases, machine learning models risk offering limited real-world utility. Future research should therefore prioritise methodological robustness, interpretability, and clinical relevance over algorithmic complexity.

2.4.4 Limitations

The principal limitations of this review relate to its scope. We excluded unpublished manuscripts, including conference abstracts and educational theses, due to concerns about variable methodological quality and access; however, this may have led to the omission of emerging or ongoing work.

In line with our prespecified inclusion criteria, we also excluded studies reporting diagnostic models[209, 210], models that were not developed using a multivariable approach[144, 152, 211], prognostic factor studies[13, 114, 212, 213], and registered but unpublished prognostic model studies[214]. While these exclusions were necessary to maintain a focussed research question, they limit our ability to provide a comprehensive overview of the full VL prediction research landscape.

2.4.5 Conclusion

We present the first systematic review that identifies, summarises, and critically appraises prognostic models for all VL clinical outcomes. Using established methodological guidance, we provide a comprehensive, objective, and transparent synthesis of model characteristics, performance, applicability, and risk of bias. Our findings highlight substantial gaps in the current evidence base.

All identified models predict mortality, were developed exclusively in Brazil or East Africa, and are almost all based on data collected over a decade ago. No models were developed in South Asia or the Mediterranean region, nor addressed relapse or PKDL — outcomes that are central to elimination efforts. These represent important evidence gaps where future research efforts should be prioritised.

Clinicians, researchers, and policymakers can refer to this review to assess the strengths and limitations of existing VL prognostic models in this highly neglected and often fatal disease.

We direct interested readers to expert guidance to support transparent reporting and reduce common sources of bias in the development and evaluation of prediction models.

3

Methodology

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3.1 Introduction

As outlined in Chapter 1, VL relapse is consequential not only for individual patients but also poses a threat to sustained elimination efforts. Accordingly, the development of a non-invasive tool to predict relapse — functioning as a ‘test-of-cure’ following initial treatment — has been identified by the WHO as a research priority[86, 88, 206]. A prognostic model represents a potential solution: by quantifying the relationship between patient characteristics and subsequent relapse

events, relapse risk in future patients can be estimated and clinical decision-making informed. However, as demonstrated in Chapter 2, no prognostic models for VL relapse have been published to date.

To address this evidence gap, four prognostic models are developed using IPD from the IDDO VL data platform: two for patients from the ISC and two for patients from East Africa. Within each region, one model includes parasite grade at baseline (pre-treatment) assessment, and one model excludes parasite grade, reflecting differences in data availability and clinical practice. All models use routinely collected information available at the time of initial cure assessment to predict six-month relapse among VL patients without HIV co-infection.

The development and evaluation of clinical prediction models is supported by an extensive and growing methodological literature. Over the past decade, reporting guidelines for prediction model studies have been established[103, 161, 179, 215], alongside an increasing number of reviews and recommendations that define best practice[196, 199, 200, 202, 216]. In particular, the application of meta-analysis techniques to IPD from multiple studies presents exciting new opportunities for prediction model development and evaluation[215, 217]. Notable opportunities include increased sample sizes leading to greater statistical power, and the ability to explore heterogeneity in predictor effects and model performance across different settings. Additionally, IPD can be used to standardise inclusion criteria and outcome/predictor definitions across included studies. However, as we lay out in this chapter, the use of IPD in prediction model research also introduces challenges; specifically (i) the need for statistical models that account for clustering of participants within studies and (ii) the presence of missing data, which can be sporadically missing within studies, or entirely missing from one or more studies[215].

The aim of this chapter is to describe and justify the methodology used for model development and evaluation — from data acquisition to final model presentation. Guidelines and methodological texts are cited accordingly, and checklists provided for current reporting guidelines on prediction model studies (TRIPOD+AI and TRIPOD-Cluster)[103, 215]. Additional material is presented in Appendix B and in the [Supplementary Material](#).¹ A protocol is available on the [Open Science Framework](#).²

This chapter’s structure closely mirrors the methodological workflow as outlined in Figure 3.1, with sections on data harmonisation, model development, and internal validation. In keeping with best practice, and similar to the approach

¹Available at <https://github.com/jpwil/dphil>.

²Created Nov 8, 2024, available at <https://osf.io/z4bdn>.

adopted in Chapter 2, the research question is presented in Box 3.1 using the PICOTS (population, index model, comparator model, outcome, timing, and setting) framework[177, 217]. Further elaboration of the eligibility criteria, and standardised definitions of predictors and the outcome are considered in the following section.

All analyses were performed using R version 4.4.1[218], with R packages cited in the relevant sections below. R scripts used for model development and evaluation are provided in the [Supplementary Material](#).

Box 3.1: Definition of the research question: a PICOTS approach

Population HIV-negative patients that are prospectively recruited into a clinical trial with a diagnosis of visceral leishmaniasis, confirmed either serologically or parasitologically. No restrictions are placed on age, sex or treatment regimen.

Index models For each setting, two prognostic models are developed; one including baseline parasite grade from a tissue aspirate, and one without. The models predict the *future* occurrence of relapse, using patient information collected at treatment baseline. The intended time of model use is following a successful assessment of treatment response.

Comparator model As established in Chapter 2, no published relapse models are available for comparison or updating.

Outcome Relapse is defined as the recurrence of signs and symptoms of VL requiring rescue treatment, and following demonstration of an initial treatment response.

Timing Relapse occurring within 6-months of test-of-cure (typically occurring at the time of treatment completion, or within 30 days of starting treatment).

Setting Participants from either the Indian subcontinent or East Africa.

3.2 Data Harmonisation

Here, data harmonisation refers to the process of data acquisition, curation, and any subsequent data manipulation required to produce a single analysis dataset ready for model development.

In the interest of full disclosure, data acquisition was completed by IDDO colleagues prior to the commencement of this DPhil project. The first stage of data curation — conversion of the contributed datasets to the Clinical Data Interchange Standards Consortium (CDISC) Study Data Tabulation Model (SDTM) standard — was performed by the IDDO data engineering team with support from the IDDO science team (myself included). Subsequent methodological steps were led by myself.

3.2.1 Data Acquisition

A systematic review of the scientific literature was first performed in 2016, with the aim of comprehensively cataloguing all existing VL clinical trials (PROSPERO: [CRD42021284622](#))[219]. 145 trials were initially identified (1980–2016, n = 26,986 patients), with further trials added during periodic updates according to an open protocol[220]. Between 2018–2022, corresponding authors of the identified VL clinical trials were invited to share their IPD with the IDDO VL data platform, in line with the General Data Protection Regulation (GDPR)-compliant IDDO data sharing governance[221, 222].

3.2.2 Data Curation

Conversion of the contributed datasets to an analysis-ready dataset of all eligible IPD occurred in two key stages.

Stage 1: CDISC SDTM Curation

To facilitate reusability and interoperability, contributed datasets were standardised to a common storage format: the CDISC-compliant SDTM standard[223], adapted by IDDO for VL[224]. During this process, contributed datasets underwent *pseudonymisation*,³ prior to being available for data sharing requests. Briefly, SDTM format datasets comprise a number of standardised domains (tables) containing related information (e.g. patient demographics, laboratory results, treatment administration, clinical signs and symptoms). Each domain contains a set of standard variables (table columns, e.g. STUDYID, USUBJID, VISITDY) alongside VL-specific variables defined by IDDO (e.g. parasite grade, spleen size). Further details of the curation process are available in the [IDDO SDTM Implementation Guide](#).⁴

³ ‘...processing of personal data in such a manner that the personal data can no longer be attributed to a specific data subject without the use of additional information’ <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/data-sharing/anonymisation/pseudonymisation/> (accessed 15 Dec 2025).

⁴Available at <https://www.iddo.org/tools-and-resources/data-tools>, free registration required.

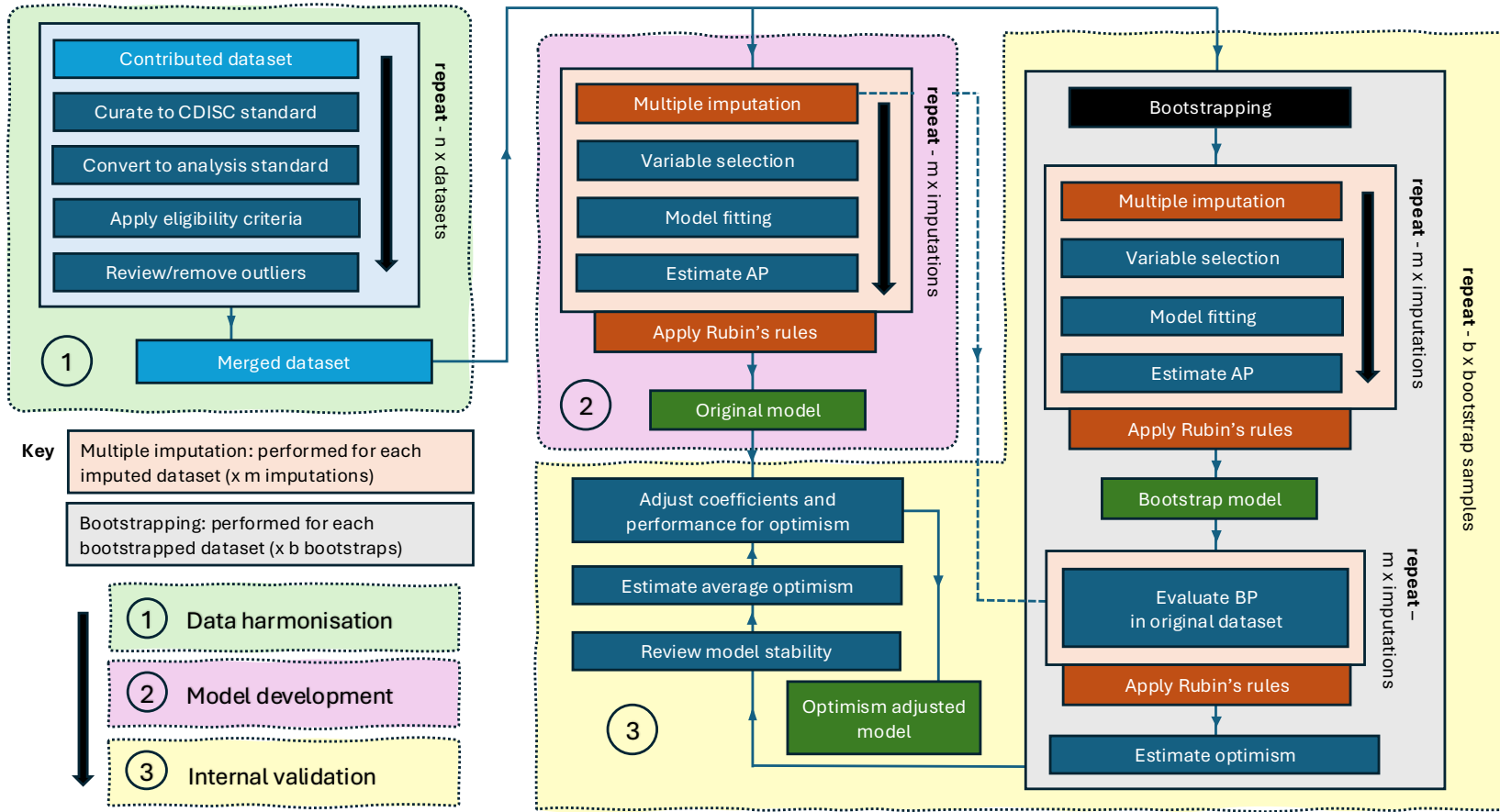


Figure 3.1: Schema of methodological workflow. ① Data harmonisation is performed for each contributed dataset including: initial curation (by the IDDO data engineering team) to the CDISC SDTM format, application of inclusion and exclusion criteria, and removal of outliers. Curated and cleaned datasets are converted into an analysis (wide) format and merged prior to ② model development. Multiple imputation is used to create ($m = 30$) imputed (complete) datasets. Variable selection, model fitting, and apparent performance evaluation is performed on all imputed datasets. Estimates are pooled using Rubin's Rules. Bootstrapping is used to perform ③ internal validation, allowing (i) review of model stability and (ii) optimism adjustment of performance measures and original model coefficients. All model development steps, including multiple imputation, are repeated for each of the ($b = 500$) bootstrap datasets. The resulting bootstrap models ($b = 500$) are evaluated both in the corresponding bootstrap (imputed) datasets and the original (imputed) datasets, and pooled using Rubin's Rules. The mean of the differences of the pooled performance measures (in bootstrap vs. original dataset) are used to shrink the original model coefficients and apparent performance measures, resulting in the final optimism-adjusted model. AP: apparent performance; b: number of bootstraps; BP: bootstrap model performance; CDISC: Clinical Data Interchange Standards Consortium; IDDO: infectious diseases data observatory; m: number of imputations; n: number of contributed studies. SDTM: Study Data Tabulation Model.

Stage 2: Analysis-Ready Dataset Curation

Subsequently, SDTM format datasets were converted to a single analysis-ready dataset, primed for model development. This stage consisted of multiple steps, refined iteratively over several months and in close consultation with the IDDO data engineering team:

- Identification and removal of spurious data points (e.g. outliers, discussed below)
- Application of study and participant eligibility criteria (Section 3.2.3)
- Creation of a standardised outcome variable according to a pre-defined definition (Section 3.2.4)
- Conversion of the datasets from a long to wide format, consisting of one row per participant
- Merging of all datasets into a single analysis dataset

Data wrangling during the second curation stage was performed with the *tidyverse* suite of R packages[225]. Identification and removal of spurious data points was performed through subgroup tabulations and visual inspection of histograms and scatter plots. Where two incongruous data points were identified, for example, incompatible height and weight values, a third variable, such as BMI or age, would be used to identify the spurious value. Data points considered to be outliers were converted to missing values. A complete record of all data cleaning steps, including outlier identification and removal, was maintained and documented in commented R scripts.

In Section 1.3 of Chapter 1, relapse was defined broadly as ‘the reappearance of VL signs and symptoms following an initial treatment response’, and ‘typically confirmed by direct visualisation of the parasite on a tissue aspirate smear’. Despite appearing a clear definition, on closer inspection it can be appreciated that even subtle variations in eligibility criteria, study design, and the definition of efficacy endpoints can, at times unexpectedly, impact the proportion of patients experiencing relapse as a study outcome.

3.2.3 Population at Risk

Clear specification of the population at risk is fundamental to understanding the model’s real-world applicability[103, 217]. Particular attention is given to the definition of initial cure, since (i) relapse can only occur following an initial

Box 3.2: Eligibility criteria

- **Study-level inclusion criteria**
 - Studies conducted in either the ISC (India, Nepal, Bangladesh) or East Africa (Ethiopia, Sudan, South Sudan, Kenya, Uganda)
 - Prospective design, defined as participants having provided informed consent
 - Participants recruited with a diagnosis of VL as defined by a combination of clinical symptoms and either parasitological or serological confirmation
 - Studies that report, as a minimum, the treatment regimen including at least the drug name(s), dose and duration
 - Recruited a minimum of 6 patients
 - Included a minimum of 6 months of prospective follow-up from treatment initiation
 - Reported VL relapse events during the 6-month follow-up period
- **Participant-level exclusion criteria**
 - Participants with HIV co-infection or from a setting with high HIV co-infection prevalence and without a negative HIV test
 - Participants who were confirmed pregnant at the time of treatment initiation
 - Participants with symptomatic treatment failure requiring rescue treatment, identified either before or at initial cure assessment

treatment response, and (ii) heterogeneity in study-level cure definitions can be partly addressed through IPD-based standardisation.

Inclusion and exclusion criteria were applied at the study and participant levels and are presented in Box 3.2. Criteria are chosen according to (i) the eligibility criteria applied in the original systematic review from which identified study authors were invited to contribute their IPD[219], (ii) the range of studies available in the IDDO VL data platform, and (iii) the resulting impact and applicability of models developed.

Study-Level Criteria

Study-level inclusion criteria were applied to ensure that contributed studies were sufficiently comparable in terms of epidemiological context, study design, and outcome ascertainment to permit meaningful harmonisation and pooled analysis.

Studies were limited to those conducted in the ISC and East Africa, reflecting both the public health relevance of relapse prediction in regions with ongoing VL

elimination programmes, and the availability of IPD the IDDO VL data platform. On review of the IDDO inventory[226], only two studies were conducted outside these regions — one in Greece conducted in the 1990s [227], and one in Brazil in the 2010s [228]. These were excluded to preserve geographical and epidemiological coherence.

Only prospectively conducted studies were included. Prospective designs allow for systematic and active follow-up, predefined outcome definitions, and contemporaneous outcome recording, all of which are important for the reliable identification of initial cure and subsequent relapse. However, reliance on IPD from clinical trial settings limits model applicability to real-world patients — those who are managed outside trial settings, and may not meet the often-stringent eligibility criteria. These limitations are discussed further in Chapter 6.

A minimum study size of at least 6 patients was imposed to exclude very small cohorts with unstable relapse estimates. Finally, studies were required to report relapse events during follow-up, either explicitly or in a form that allowed relapse to be inferred from the available IPD.

Participant-Level Criteria

With respect to clinical presentation, treatment response, and outcomes, patients with VL/HIV co-infection constitute an important but distinct clinical population. Accordingly, patients with and without VL/HIV co-infection were *not* combined within a single prediction model, given the substantial uncertainty in extrapolating relapse associations derived from HIV-negative patients to those with VL/HIV co-infection. Since the majority of contributing studies excluded patients with VL/HIV co-infection, insufficient IPD were available to develop a separate model for this group. Patients without evidence of HIV testing were excluded if they were otherwise considered to be from a high risk area. For the East Africa models, patients without HIV testing were only included if they were recruited from Sudanese sites.

As with VL/HIV co-infection, very few contributing studies included pregnant participants, reflecting their systematic exclusion from VL trials and precluding the development of a separate prediction model.

Initial Cure

Understanding study-specific definitions of initial cure is important, as all studies require the patient to demonstrate a treatment response, measured with a test-of-cure, in order to be at subsequent risk of relapse. Consequently, patients *not* achieving initial cure — described as initial treatment failure — should be excluded.

A direct consequence of excluding these patients is that model-derived risk estimates are only applicable to patients demonstrating initial cure.

Initial cure definitions based solely on clinical improvement, as is common in routine practice, are likely to classify some patients as cured despite persistently positive tissue aspirates, were these assessed. These patients form a subgroup at increased risk of relapse and would instead be classified as initial treatment failures under more stringent, parasitology-based test-of-cure criteria, thereby being excluded from subsequent follow-up. Consequently, all else being equal, studies applying stricter definitions of initial cure will observe a lower subsequent relapse risk.

Recently, Dahal and colleagues performed a systematic review of the design, conduct, analysis, and reporting of VL therapeutic efficacy studies, published between 2000 and 2021. Of the 89 studies identified, 71 (79.8%) included parasitological assessment, with or without demonstration of clinical improvement, as part of the test-of-cure, while 13 (14.6%) required clinical improvement only. The remaining studies did not provide a definition. Timing also varied considerably, with the 68 (76.4%) of studies performing the test-of-cure between 15–30 days following treatment completion[90]. Similar patterns are observed in the contributed studies, as reported in the [Supplementary Material](#), and discussed further in subsequent results chapters. Importantly, criteria for ‘clinical improvement’ are often not specified. Further complicating interpretation, many studies describe a subgroup of ‘slow responders’, who remain in the study despite a positive tissue aspirate in the test-of-cure. These patients may undergo repeat assessment at subsequent time points (e.g. 2–4 weeks later), with or without treatment extension, and may or may not ultimately be classified as having achieved initial cure.

Such variation in the initial cure definition can challenge standardisation efforts, leading to differences in observed relapse rates stemming from differences in the population at risk. These differences, however, can often be mitigated through interrogation of the IPD. Box 3.3 provides a working definition for initial cure, which is applied during data harmonisation.

3.2.4 Outcome

Relapse itself, where described at the study-level, is also subject to substantial variation with respect to its (i) definition — including the severity of symptoms required to trigger a repeat aspirate and the tissue type chosen for aspirate, and (ii) timing — including whether patients were actively screened at set time points with clinical examination $\pm$ routine aspirates, or whether dependent on patients attending voluntarily based on recurrent symptoms and discharge advice. In line

Box 3.3: IPD-based working definitions

Initial cure Where initial cure (or initial treatment failure) is *not* directly recorded in the IPD as an efficacy outcome, or where it is recorded but the study definition considers ‘slow responders’ as initial treatment failures, it can be inferred from (i) improvement of signs and symptoms between baseline and test-of-cure, and (ii) not requiring rescue treatment during initial treatment. Importantly, reflecting both routine clinical practice and a number of study definitions, detection of parasites at test-of-cure should not preclude the subsequent development of relapse, so long as points (i) and (ii) are met.

Relapse Where relapse is not directly recorded in the IPD as an efficacy outcome, the event can be inferred from two or more of: (i) the need for rescue treatment within 6 months of initial cure assessment (test-of-cure), (ii) the presence of a positive tissue aspirate, and (iii) in addition to a recurrence of compatible signs and symptoms.

with findings by Dahal et al, a significant proportion of contributing studies do not directly define relapse as a study outcome[90]. Instead, for most studies, a relapse event can be inferred from patients achieving initial cure who subsequently do not meet the definition of ‘definite cure’, which itself is typically defined as patients requiring rescue treatment.

Similar to the approach described for identifying patients that achieve initial cure, access to IPD allows relapse events to be inferred from other variables, including: definite cure status, tissue aspirates, timing of rescue treatment initiation, and patient signs and symptoms. Box 3.3 provides an IPD-based working definition of relapse.

Relapse is recorded and modelled as a binary outcome variable (occurred vs. not occurred). Unfortunately, modelling relapse as a time-to-event variable is not feasible, since *timing* information is (i) inconsistently presented across the contributed IPD, and (ii) where presented, is often limited to fixed, predetermined study visits (e.g. 3 months, 6 months).

3.3 Model Development

3.3.1 Sample Size

A common ‘rule of thumb’ is that at least 10–20 outcome events per predictor parameter (EPP) are needed to prevent model overfitting[229, 230]. However, both the validity of this threshold, and the broader premise that a single rule applies universally, have been increasingly debated in the prediction modelling literature[231].

A common ‘rule of thumb’ is that at least 10–20 outcome events per predictor parameter (EPP) are needed to prevent model overfitting[229, 230]. However, both the validity of this threshold, and the broader premise that a single rule applies universally, have been increasingly debated in the prediction modelling literature[231].

Responding to these concerns, Riley et al. (2018) proposed a principled sample size calculation framework for prediction model development[232], implemented here using the `pmsampsize` R package[233]. For binary or time-to-event outcomes, this approach defines minimum sample size requirements based on three criteria: (i) limited overfitting, operationalised as a global shrinkage factor ≥ 0.9 ; (ii) a small absolute difference (≤ 0.05) between apparent and adjusted Nagelkerke’s R^2 ; and (iii) precise estimation (within ± 0.05) of the average outcome risk. In the present study, where sample size and relapse rate are fixed, this framework was instead used to calculate the *maximum number of predictor parameters* supported by the data. In the absence of previously published relapse prediction models, a Nagelkerke R^2 of 0.15 was assumed, as recommended by the authors[232]. The maximum number of supported predictor parameters based on variations of criterion (i) are summarised in Table 3.1 for the ISC and EA datasets across their respective sample sizes and relapse rates.

Table 3.1: Maximum number of predictor parameters supported by the available IPD in the ISC and East Africa datasets. Calculations performed for a range of minimum permitted shrinkage factors (criterion 1) as per Riley et al’s methodology[232]. The remaining two criteria were unchanged from Riley et al’s recommendations (see text). For all calculations, the minimum permitted shrinkage factor was the criterion limiting the number supported predictor parameters. Fixed sample sizes — ISC: 4,599 patients, 228 relapses; East Africa: 2,051 patients, 99 relapses. Assuming Nagelkerke’s $R^2 = 0.15$.

Shrinkage factor	Maximum supported predictor parameters	
	ISC models	EA models
0.95	12	5
0.90 ¹	25	11
0.85	40	17
0.80	58	25

¹ Threshold suggested by Riley et al[232].

ISC Model

With a total of 228 relapses identified in 4,599 participants (5.0% event rate), the maximum number of predictor parameters satisfying Riley et al’s three criteria,

with a global shrinkage factor of at least 0.90, is 25 (corresponding to 8.86 EPP). Relaxing the permitted shrinkage factor from 0.90 to 0.85 allows for up to 40 predictor parameters (corresponding to 5.57 EPP).

East Africa Model

With a total of 99 relapses identified in 2,051 participants (4.8% event rate), the maximum number of predictor parameters satisfying Riley et al.'s three criteria, with a global shrinkage factor of at least 0.90, is 11 (corresponding to 8.81 EPP). Relaxing the permitted overall shrinkage from 0.90 to 0.85 allows for 17 predictor parameters, corresponding to 5.82 EPP.

3.3.2 Candidate Predictors

All model variables, including candidate predictors, are listed in Table 3.2.

For the ISC models, a total of 17 candidate predictor parameters are included (16 when excluding parasite grade), corresponding to a study-specific random intercept term and 12 (11) participant-level candidate predictors, $EPP = 13.41$ (14.25). Assuming a Nagelkerke's R^2 of 0.15, this EPP should comfortably satisfy sample size considerations.

For the East Africa models, a total of 14 candidate predictor parameters are included (13 when excluding parasite grade), corresponding to the same variables included in the ISC models minus treatment group, due to convergence issues discussed below, $EPP = 7.07$ (7.62). Assuming a Nagelkerke's R^2 of 0.15, this EPP should satisfy sample size calculations when the minimum permitted global shrinkage factor is reduced from 0.90 to 0.85.

Full specification of all candidate predictors are presented in Table 3.2.

The following points were considered when selecting candidate predictors:

- To avoid excessive missing data, predictors must be available for at least 50% of participants.
- Predictors should be routinely measured, or at least available for measurement, in the majority of treatment centres in endemic areas.
- Since the model is applied at the time of initial cure, predictors should be available at or *prior* to this time point (typically measured within a month of starting treatment).
- Predictors should be *preferentially* included if (i) they have previously been shown to predict relapse, or (ii) other compelling reasons exist to include — such as expert opinion and arguments supporting a causal association between the predictor and relapse. Discussions in Section 1.3 provide further insight.

Variable	Specification (categories)	dof	log	ISC		EA	
				PG	$\overline{\text{PG}}$	PG	$\overline{\text{PG}}$
Relapse (outcome)	Categorical (2)	-	-	✓	✓	✓	✓
Study	Random intercept	1	-	✓	✓	✓	✓
Treatment	Categorical (3)	2	-	✓	✓	✗	✗
Sex	Categorical (2)	1	-	✓	✓	✓	✓
Malnutrition	Categorical (3)	2	-	✓	✓	✓	✓
Anaemia severity	Categorical (2)	1	-	✓	✓	✓	✓
Age	Continuous ¹	3	No	✓	✓	✓	✓
Fever duration	Continuous	1	Yes	✓	✓	✓	✓
WBCs	Continuous	1	Yes	✓	✓	✓	✓
Spleen size	Continuous ²	1	No	✓	✓	✓	✓
Platelets	Continuous	1	Yes	✓	✓	✓	✓
ALT	Continuous	1	Yes	✓	✓	✓	✓
Creatinine	Continuous	1	Yes	✓	✓	✓	✓
Parasite grade ³	Continuous	1	No	✓	✗	✓	✗

¹ Age is modelled as a cubic polynomial.

² A small additive constant (+1) is added to spleen size to allow inclusion of patients with non-palpable spleens.

³ Parasite grade is recorded on a semi-quantitative logarithmic scale, ranging from 1+ to 6+.

Table 3.2: Variables used in the development of the Indian subcontinent (ISC) and East Africa (EA) models. Models are fitted either with parasite grade (PG) or without parasite grade ($\overline{\text{PG}}$). -: not applicable, ✓: included in model, ✗: not included in model, dof: degrees of freedom, log: whether modelled on a natural logarithmic scale, ALT: alanine aminotransferase, WBC: white blood cell.

- Sample size considerations explored in Section 3.3.1 should guide the maximum number of predictor parameters to prevent model overspecification and overfitting.
- To facilitate model convergence, excessive collinearity should be avoided.

When considering the final point, it is worth highlighting that model convergence issues can occur due to collinearity not only between predictors, but also between predictors and random-effects (clustering) structures. Given the significant heterogeneity in study design and outcome definitions, and further reasons addressed further in Section 3.3.5, each contributing study is modelled as a random-intercept term. Consequently, caution must be exercised with categorical predictors that are uniquely, or near-uniquely, identified at the study level.

Treatment

Treatment regimen — already established as an important predictor relapse — represents a categorical predictor affected by collinearity at the study level. For example, including a treatment category of *14 days of paromomycin* in the ISC model would lead to convergence failure when included in a model with study as a random-intercept term. As can be appreciated from Figure 4.2, only one contributing study (Sundar 2009) includes patients receiving this treatment regimen. Consequently, a model including both would not be able to distinguish between relapse risk related to the study or the treatment regimen.⁵

Considering the distribution of treatment regimens across studies within the ISC, treatment will be categorised into three categories that occur in at least three studies (see Table 4.1 and Figure 4.2):

- Single dose liposomal amphotericin B (10 mg/kg)
- 28 days miltefosine (standard dose)
- Other

As a consequence of creating a ‘catch-all’ treatment group: *Other*, some of the relationship between treatment and relapse not already accounted for at the study level will be lost. This is an important limitation and addressed further in the discussion (Chapter 6).

Unfortunately, convergence issues preclude the inclusion of treatment as a categorical predictor in the East Africa models. This is not surprising, given fewer participants, fewer studies, and higher treatment-study collinearity when compared to the ISC models (see Figure 5.2). Convergence issues persisted despite exploring different treatment groupings, including separate groups for SSG and SSG/PM combination therapy. Therefore, in the East Africa models, the impact of treatment on relapse risk is incorporated into the study level random intercept.

Malnutrition

Malnutrition is a well-established determinant of progression from asymptomatic infection to clinical VL and of adverse outcomes following initial treatment, including treatment failure and mortality[1, 204]. Evidence linking malnutrition to relapse, however, remains sparse. This may partly reflect the absence of a unified framework for defining malnutrition across age groups, leading to frequent omission

⁵Or more technically, this induces near-non-identifiability between the fixed treatment effect and the study-level random intercept, resulting in an ill-conditioned information matrix and failure of numerical optimization.

Table 3.3: Malnutrition severity definitions by age group. BMI: body-mass-index, WFH: weight-for-height, yrs: years, [$\cdot$]: including range limit, ($\cdot$): not including range limit

Age group	Metric	Severe	Moderate	Mild/normal
[0, 5) yrs	WFH z-score	$(-\infty, -3]$	$(-3, -2]$	$(-2, \infty)$
[5, 19) yrs	BMI z-score	$(-\infty, -3]$	$(-3, -2]$	$(-2, \infty)$
[19, ∞) yrs	BMI (kg/m^2)	(0, 16.0)	[16.0, 18.5)	[18.5, ∞)

or inconsistent classification in prognostic factor and prediction model studies. Anthropometric assessment of malnutrition is inherently age-specific, and no formal guidance exists on how to combine measures across the life course.

We therefore adopt a pragmatic three-level severity scale using age-appropriate metrics (Table 3.3). In children under five years, malnutrition is classified using weight-for-height (WFH) z-scores, consistent with WHO guidelines on the definition of acute malnutrition[234]. For individuals aged five to under nineteen years, BMI-for-age z-scores (BMI z-scores) are used with identical cut-points, supported by the WHO 2007 growth reference, which was explicitly constructed to ensure continuity with the under-five standards at age five[235]. In adults aged 19 years and over, malnutrition severity is defined using established BMI thresholds. Cole et al. showed that adult BMI cut-offs of 16 and 17 kg/m^2 at 18 years correspond approximately to BMI-for-age z-scores of -3 and -2 in children and adolescents, providing a statistical basis for approximate continuity of severity definitions across the adolescent-adult boundary[236]. While imperfect, this approach preserves broadly comparable degrees of nutritional deficit across age groups.

For modelling purposes, individuals with mild ($18.5 \leq \text{BMI} < 25 \text{ kg}/\text{m}^2$) or obesity ($\text{BMI} \geq 25 \text{ kg}/\text{m}^2$), or equivalent z-scores ($z > -1$), were combined into a single *mild/normal* category due to small numbers.

The approach adopted here is consistent with methods used in previous studies of VL outcomes that also use age-specific anthropometric indicators to define pragmatic malnutrition severity groupings[96, 112, 125, 133]. z-scores were calculated using the `anthro` and `anthroplus` R packages for children < 5 years and ≥ 5 years, respectively[237, 238].

Anaemia

Anaemia was grouped into two categories: *severe* and *non-severe*, using haemoglobin cut-offs stratified by age and sex thresholds as defined by 2024 WHO guidelines[239]. Additional subdivision of the non-severe anaemia group was limited by the small number of participants in the mild and normal categories.

Age

To account for an anticipated non-linear relationship between age and relapse, age was modelled as a third-degree polynomial term (including linear, squared, and cubic components).

Parasite Grade

Baseline parasite grade, when available, was assessed from splenic, bone marrow, or lymph node aspirates. When reported, the logarithmic counting method of Chulay and Bryceson (1983) was either described or directly cited Additional file 1[52].

Logarithmic Transformations

A number of continuous predictors — including fever duration, spleen length, and all blood tests — were log-transformed to reduce skewness and thus stabilise variance and better approximate normality. For spleen size, +1 was added prior to log-transformation to accommodate zero values and avoid undefined logarithmic results. While recognising that this transformation implicitly assumes that non-palpable spleens can be modelled on the same continuous scale with palpable spleens, the approach was considered a pragmatic solution to avoid model overspecification.

All models assume that the adjusted log(odds) of relapse varies *linearly* with each continuous predictor when transformed as per Table 3.2. These assumptions are evaluated prior to model fitting through visual assessment of the univariable associations (as described below in Section 3.3.3, and presented in Figure C.17 for the ISC and Figure D.17 for East Africa) and, following model fitting, through inspection of the calibration plots.

3.3.3 Descriptive Analysis

All candidate predictors (marginal and study-specific) are summarised in both tabular and graphical forms.

The correlation between continuous predictors is illustrated with faceted scatter plots using the `ggpairs()` function from the `GGally` R package[240]. The correlation between continuous and categorical predictors is presented with faceted box-and-whisker plots.

Unadjusted relationships between the candidate predictors and the outcome (relapse) are presented in graphical form with 95% confidence intervals. Since this stage of the analysis is exploratory and descriptive, p-values are not presented to

avoid multiple hypothesis testing and inflation of type I error, with emphasis instead placed on the shape, magnitude, and uncertainty of observed associations[241].

For continuous predictors (excluding parasite grade), relapse is modelled using a generalised additive model (GAM) with a binomial error distribution and logit link, fitted using the `gam()` function in the R package `mgcv`[242]. The effect of the continuous predictor is represented by a smooth function estimated using penalised thin-plate regression splines. Model fitting is performed by penalised maximum likelihood, with the degree of smoothness selected automatically via restricted maximum likelihood (REML). Both relapse % and log(odds) of relapse are presented, allowing for both direct inspection of the relapse risk and visual assessment of the linearity assumption between the log(odds) of relapse and transformed continuous predictors.

For categorical predictors and parasite grade, relapse is presented using bar charts with 95% confidence intervals calculated using the Wilson method[243].

3.3.4 Missing Data

Data can be missing entirely from a study (systematic missingness), or affecting only certain patients within a study (sporadic missingness). Three principal sources of missing data were identified across both the ISC and East Africa harmonised datasets:

- planned non-capture at the study or site level, often resulting in systematic missingness,
- unplanned incomplete capture of predictors at the study level, resulting in sporadic missingness, and
- incomplete data retrieval, likely resulting in both systematic and sporadic missingness.

Patterns and extent of missingness in candidate predictors were explored and summarised using both tabular and graphical approaches. Study-stratified density plots were used to visualise missingness patterns stratified by study and are presented in the results chapters (Figures 4.7 and 5.7 presented in the Results chapters).

Multiple imputation with chained equations (MICE)⁶ was used to generate multiple imputed datasets[244, 245]. MICE was chosen for its flexibility in accommodating different variable types (continuous, binary, categorical, and count), its suitability for complex data structures including multilevel data, and the availability

⁶also known as fully conditional specification

Table 3.4: Multiple imputation with chained equations (MICE) methods used for univariable imputation of variables with missing data. ALT: alanine aminotransferase; BMI: body mass index; Crt: creatinine; Hb: haemoglobin; ISC: Indian subcontinent; n/a: not applicable; Plt: platelets; WBC: white blood cell count; WFH: weight-for-height.

Method	Variables	Description	R package
2l.lmer	age, weight, height, fever duration, all bloods (Hb, ALT, WBC, Plt, Crt), spleen size	Two-level normal model using <code>lme4::lme4()</code>	<code>mice</code> [246] <code>micemd</code> [249]
2l.zip	parasite grade (East Africa only)	Two-level zero-inflated Poisson model using Bayesian estimation	<code>countimp</code> [250]
2l.poisson	parasite grade (ISC only)	Two-level Poisson model using Bayesian estimation	<code>countimp</code> [250]
Passive imputation	BMI, BMI-for-age z-score, WFH z-score, age ² , age ³	Calculated during imputation from age, weight and height.	<code>mice</code> [246]
n/a	outcome, treatment (ISC only), sex, study (random intercept term)	No imputation performed (no missing data)	n/a

of well-established software implementations in R, notably the `mice` package and its extensions[246, 247].

Imputation was performed under the missing at random (MAR) assumption, whereby the probability of missingness depends only on observed data[248]. Congeniality between the imputation and analysis models was promoted by (1) including in the imputation model all candidate predictors, the outcome, and variables used to derive predictors (including haemoglobin, height, and weight), and (2) accounting for between-study heterogeneity using a multilevel imputation framework with study included as a random intercept[244]. The imputation methods used are summarised in Table 3.4.

The imputation predictor matrix was initially specified to be saturated and include all variables listed in Table 3.4 as well as variables without missing data (outcome, treatment, and sex), and derived predictors (age polynomial terms, BMI,

and anthropometric z-scores). Study was included as a random intercept. Derived predictors were passively imputed from age, weight, and height at the end of each iteration, and, to avoid circularity in the imputation model, were excluded as independent variables when imputing age, weight, and height.

Specification of the imputation model was an iterative process. Candidate imputation models were evaluated with respect to (1) convergence of the chained equations and (2) the extent to which the distributions of imputed values aligned with the corresponding observed data distributions. Model specification and selection were informed by visual inspection of diagnostic plots, including trace plots of the mean and standard deviation of imputed values across iterations, density plots comparing observed and imputed distributions, and scatter plots comparing observed and imputed data for key anthropometric relationships (weight versus height, weight versus age, and height versus age). Diagnostic plots are included in the [Supplementary Material](#). On visual inspection of the trace plots, twenty iterations per imputation were considered more sufficient to reach stable convergence.

Parasite grade was modelled as count data. One unit was subtracted from the original grading scale (1+ to 6+) to permit zero values. The final choice of model was guided by matching the distribution of parasite grades from the imputation model with the observed distribution in the non-missing data. In the ISC dataset, a Poisson model provided the best fit to the observed data. In contrast, for the East Africa dataset, a zero-inflated Poisson model better captured the excess of 1+ grades.

For each model development dataset, 30 imputations were generated with 20 iterations per imputation. This represented a balance between computational feasibility and the need to stabilise parameter estimates. As a general rule, it has been stated that the number of imputations should at least match the percentage of incomplete cases[244]. For datasets with many variables, however, van Buuren suggests that this requirement can be relaxed such that the minimum number of imputations approximates the average missing data rate[246]. The average missing data rates across candidate predictors were 14.3% for the ISC dataset and 14.9% for the East Africa dataset, whilst the proportion of cases with any missing data was over 80%. Since 80 iterations was considered too computationally intensive, a compromise of 30 imputations was chosen.

Unless specified otherwise, subsequent analyses were performed separately within each imputed dataset, with results pooled across imputations using Rubin's Rules[248].

Grouping of anaemia severity and malnutrition status was performed after imputation, as described in Section 3.3.2.

3.3.5 Model Specification

Relapse was modelled using a multivariable generalised linear mixed-effects model (GLMM) with a logit link function. Predictors were transformed as previously described. Following the aforementioned transformations, continuous predictors were centred and scaled by their means and standard deviations, respectively, to improve model stability. Anticipated between-study heterogeneity was accounted for by including study as a random intercept term[251]. While introducing significant methodological complexity, a number of benefits gained by accounting for between-study heterogeneity[215]:

- Ignoring clustering can result in relapse probability estimates that are biased towards the marginal population estimate[252].
- Variation in model performance measures can be compared and contrasted across the included studies, allowing insights into sources of heterogeneity[253].
- Allows for improved generalisability of the model to new settings and populations[215, 253].

GLMMs were fitted using the `glmer()` function from the `lme4` R package[254]. This approach estimates fixed effects while accounting for random effects by maximising the marginal likelihood, which integrates over the random-effects distribution. For GLMMs, this integration cannot be solved exactly and is approximated using the Laplace approximation for binomial outcomes. Model parameters were estimated using maximum likelihood. To enhance model stability and improve convergence, optimisation was performed using the `nloptwrap` optimizer with an increased number of iterations and strict convergence criteria (maximum evaluations = 200,000; absolute tolerances for the objective function and parameters = 1×10^{-8}).

The saturated ISC model (with parasite grade and treatment groups included) is presented in Box 3.4. For the East Africa models, the treatment group predictors are omitted due to collinearity at the study level, as previously discussed. For both ISC and East Africa, models are also fitted without parasite grade, given its frequent absence from routine clinical practice.

Box 3.4: Saturated model specification

$$\log \left(\frac{\Pr(Y_{ij} = 1)}{\Pr(Y_{ij} = 0)} \right) = \beta_0 + \beta_{\text{sex}} \cdot \text{sex}_{ij} + \beta_{\text{age}} \cdot \text{age}_{ij} + \beta_{\text{age}^2} \cdot \text{age}_{ij}^2 + \beta_{\text{age}^3} \cdot \text{age}_{ij}^3 + \\ \beta_{\text{treat-sda}} \cdot \text{treat-sda}_{ij} + \beta_{\text{treat-oth}} \cdot \text{treat-oth}_{ij} + \beta_{\text{mal-mod}} \cdot \text{mal-mod}_{ij} + \\ \beta_{\text{mal-sev}} \cdot \text{mal-sev}_{ij} + \beta_{\text{anaemia-sev}} \cdot \text{anaemia-sev}_{ij} + \beta_{\text{fever-dur}} \cdot \text{fever-dur}_{ij} + \\ \beta_{\text{spleen-cm}} \cdot \text{spleen}_{ij} + \beta_{\text{para}} \cdot \text{para}_{ij} + \beta_{\text{wbc}} \cdot \text{wbc}_{ij} + \beta_{\text{plat}} \cdot \text{plat}_{ij} + \\ \beta_{\text{alt}} \cdot \text{alt}_{ij} + \beta_{\text{creat}} \cdot \text{creat}_{ij} + \mu_j.$$

Where: i represents the i th participant in study j , Y is the relapse outcome (1 = relapse, 0 = no relapse), and

$$Y_{ij} \sim \text{Bernoulli}(E[Y_{ij}]) \\ \mu_j \sim \mathcal{N}(0, \tau^2).$$

β_0 is the fixed model intercept, and β_k are the fixed effect coefficients for each candidate predictor k , corresponding to sex (1 = male, 0 = female), age, treatment group (reference = standard dose miltefosine, **treat-sda**: 10 mg/kg single-dose liposomal amphotericin B, **treat-oth**: other), malnutrition severity (reference = mild/normal, **mal-mod**: moderate, **mal-sev**: severe), anaemia severity (1 = severe, 0 = non-severe), fever duration, spleen size, **para**: parasite grade, **wbc**: white blood cell count, **plat**: platelet count, **alt**: alanine transaminase, and **creat**: creatinine. The random intercept μ_j captures between-study heterogeneity, with variance τ^2 . All continuous candidate predictors, except age, spleen size, and parasite grade, are log-transformed, centred by their mean, and scaled by their standard deviation. Age is centred and scaled, and modelled as a third-degree polynomial term. Spleen size is log-transformed after adding +1.

3.3.6 Variable Selection

The final predictor set was determined using backwards variable selection with cutoff $p < 0.10$. Univariable selection was avoided to prevent exclusion of predictors that may demonstrate significance only in the presence of other predictors, and to reduce the risk of overfitting[176].

A variety of methods have been explored for performing variable selection with multiple imputed datasets[255, 256]. In accordance with recommendations by Austin et al. and Wood et al., Rubin's Rules were used to combine p-values for predictor significance at each selection stage across the imputed datasets. At each stage, the full model is fitted in each of the 30 imputed datasets and the estimated regression coefficients and their standard errors are pooled using Rubin's Rules. The candidate predictor with the highest pooled p-value above the 0.10 threshold was removed,

and the process repeated until all remaining predictors had pooled p-values below 0.10[248, 255, 256]. For categorical predictors with over two groups (malnutrition, treatment), predictor significance was assessed with the D1 multivariate Wald test as implemented in the `mice` R package[246, 257]. For age, lower-order polynomial terms were retained in the model, whilst higher-order terms remained.

Variable selection was performed in a bespoke R script, available in the [Supplementary Material](#).

3.3.7 Model Performance

In accordance with the TRIPOD-Cluster reporting guidelines, measures of calibration and discrimination are presented for the prediction models overall and for each study (cluster) separately[215]. Wynants et al. address a number of decisions that must be made when evaluating prediction model performance in clustered data[252]. As described by Wynants et al, within-study (conditional) performance measures are presented reflecting model performance when evaluated at the study level[252, 258]. Assessment of model performance heterogeneity across studies is presented with forest plots using a combination of fixed and random effects meta-analyses[177, 258].

Discrimination

Model discrimination was assessed using the *c* (concordance)-statistic. For models with binary outcomes, the *c*-statistic is equivalent to the AUC, and is the probability that a randomly selected case (patient who experiences a relapse event) has a higher predicted relapse probability than a randomly selected non-case (patient who does not experience a relapse event).

Study-specific *c*-statistics were estimated for each imputed dataset using the R package: `pROC`[259]. Standard errors and 95% confidence intervals were calculated using the bootstrap method with 2000 resamples.

As suggested by Burgess et al., study-specific *c*-statistics were first pooled across imputed datasets using Rubin’s Rules *prior* to estimating the summary (overall) *c*-statistic using meta-analysis[260]. As recommended by van Klaveren et al.[258] and Debray et al.[177], overall *c*-statistics are estimated using a random-effects meta-analysis approach. Between-study variance (τ^2) is estimated using restricted maximum likelihood (REML), and variance of the pooled *c*-statistic is estimated with the Hartung-Knapp-Sidik-Jonkman method[261].

For comparison, a fixed-effects meta-analysis approach to estimating the overall *c*-statistic is also presented, weighted by the number of event/non-event pairs in

each study. This estimate is equivalent to the c-statistic obtained from comparing pairs of events and non-events within the same study only[252, 262].

Given the often small number of relapses in individual studies, estimates of study-specific c-statistic uncertainty (variance, confidence intervals) are likely to be imprecise and should be interpreted cautiously. To reduce potential bias introduced from small sample sizes, only studies with >5 events are included in the meta-analyses[263, 264].

Calibration

Model calibration assesses agreement between predicted and observed relapse probability, and was evaluated with calibration plots, calibration slope, and calibration intercept (calibration-in-the-large, CITL). Calibration slope and intercept are estimated conditional on the study, and represent within-study performance measures[252].

Calibration intercept describes whether a model’s average predicted risk differs from the overall observed event rate. A positive intercept indicates that the model systematically underestimates relapse risk, whilst a negative intercept indicates systematic overestimation. The calibration intercept is estimated by fitting a logistic GLMM with relapse as the outcome, study as a random intercept, and the linear predictor from the original model as an offset term. The estimated fixed intercept then corresponds to the within-study calibration intercept, with an expected value of zero when assessed in the development dataset (apparent performance)[252].

Calibration slope assesses whether a model’s predicted risks are systematically too extreme or too conservative. A slope of 1 indicates perfect calibration and is the expected value when evaluating apparent performance in the development dataset. Values less than 1 indicate overfitting (overly extreme predictions), and values greater than 1 indicate underfitting (overly moderate predictions). It was estimated by fitting a further logistic GLMM with relapse as the outcome, study as a random intercept, and the linear predictor from the original model included as both a fixed and random effect (i.e. a random slope term)[252]. In this formulation, the fixed effect of the linear predictor represents the within-study calibration slope. To mitigate model convergence issues arising from the inclusion of studies with small sample sizes, we adopted a Bayesian approach. Parameters were estimated using the Hamiltonian Monte Carlo algorithm as implemented in Stan, accessed via the R package `brms`[265, 266]. Posterior sampling was conducted using four chains with 2,500 iterations per chain (including warm-up), and the

target acceptance rate was increased (`adapt_delta` = 0.95) to reduce divergent transitions. Default weak priors were used.

Heterogeneity in calibration slope and intercept across studies is summarised using a the same random effects approach to that described for the c-statistic. Study-specific estimates were initially calculated within each imputed dataset prior to pooling across imputations using Rubin’s Rules[260]. Summary measures were then estimated using random-effects meta-analysis with REML estimation of between-study variance and Hartung-Knapp-Sidik-Jonkman confidence intervals[261].

Calibration plots allow direct visual assessment of the relationship between observed (y -axis) and predicted (x -axis) relapse probabilities. Firstly, all imputed datasets were stacked into a single dataset. Predicted relapse probabilities were then calibrated to each study using logistic recalibration of the intercept and divided into deciles. Observed probabilities were calculated for each decile and 95% confidence intervals estimated using Wilson’s method[243]. The smoothed relationship between observed and predicted relapse probabilities was estimated using a generalised additive model, as previously described. To account for the stacking of 30 imputed datasets, all standard errors were inflated by $\sqrt{30}$.

Following the same methodology for standard calibration plots, calibration was also illustrated for selected predictor subgroups by plotting both observed and predicted probabilities on the y -axis against predictor subgroups on the x -axis.

3.4 Internal Validation

Internal validation is performed using bootstrap resampling with 500 bootstrap samples, as recommended by Collins et al[199]. Bootstrapping evaluates all aspects of the model building process to quantify and adjust for overfitting and optimism in model performance[199]. Bootstrapping is performed at the overall patient, which as described by Bouwmeester et al., provides accurate optimism estimates in clustered data[267].

For each of the 500 bootstrap samples of the original dataset, all aforementioned model development steps, including multiple imputation and variable selection, were undertaken (see Figure 3.1). For each of the resulting 500 bootstrap models, pooled performance measures were evaluated across (i) the imputed bootstrap datasets used to derive the bootstrap model and (ii) the imputed original datasets used to derive the original model. The mean pooled performance differences between (i) and (ii) were then subtracted from the original model apparent performance measures to obtain the optimism-adjusted performance measures. Uniform shrinkage

of the parameter estimates was performed by multiplying the fitted regression coefficients by the optimism-adjusted calibration slope and re-estimating the model intercept with logistic recalibration.

Given the computational intensity of the model development process (~4 hours per model development), bootstrap validation was implemented using a high performance computing cluster based in the Nuffield Department of Medicine at the University of Oxford.⁷

3.5 Summary

In summary, this chapter has described the methods used to develop and internally validate four prognostic models for six-month VL relapse (HIV-negative patients) using IPD from the IDDO VL platform: two models for the ISC (with and without baseline parasite grade) and two analogous models for East Africa.

Contributed datasets were standardised to the CDISC-SDTM format, converted to a single analysis-ready dataset, and filtered using pre-specified study- and participant-level eligibility criteria that ensure the population at risk comprises patients who achieved an initial cure. Candidate predictors were selected based on clinical relevance and data availability, and were transformed as appropriate (e.g. logarithmic or polynomial transformations). Predictor specification was informed by considerations of collinearity, including grouping of treatment variables to reduce study-level dependence and support model convergence. Sample-size considerations followed the framework of Riley et al. to guide the maximum allowable degrees of freedom.

Modelling used multivariable GLMMs with study as a random intercept, fitted on multiply imputed datasets (MICE, 30 imputations, multilevel imputation with study random intercept and diagnostics to check plausibility). Backwards selection (pooled p-values, cutoff $p < 0.10$) determined the final predictor set, with coefficients and uncertainty combined with Rubin's rules at each variable selection stage. Internal validation applied full bootstrap resampling ($b = 500$), repeating multiple imputation and selection within each bootstrap to quantify optimism. The final models were uniformly shrunk by the optimism-adjusted calibration slope.

⁷<https://www.medsci.ox.ac.uk/for-staff/resources/bmrc>

4

Results: Indian Subcontinent Models

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4.1 Descriptive Analysis

On application of the eligibility criteria to the IDDO VL data platform, a total of 19 studies and 4,599 patients were selected for inclusion[64, 67, 71, 94, 268–282]. At the participant selection stage, 27 participants (0.6%, 27/4860) were excluded due to having a positive HIV test, and 201 (4.1%, 201/4860) were excluded due to not achieving initial cure. A flow diagram is presented in Figure 4.1.

4.1.1 Study Characteristics

Key characteristics of the 19 included studies are presented in Table 4.1.

A median of 144 patients were recruited in each study, ranging from 6 to 928 patients (IQR: 87 to 330 patients). Patients were recruited between 2000 and 2017 from sites in India (15 studies, 4,171 [90.7%] patients)[64, 67, 71, 94, 268, 269, 271,

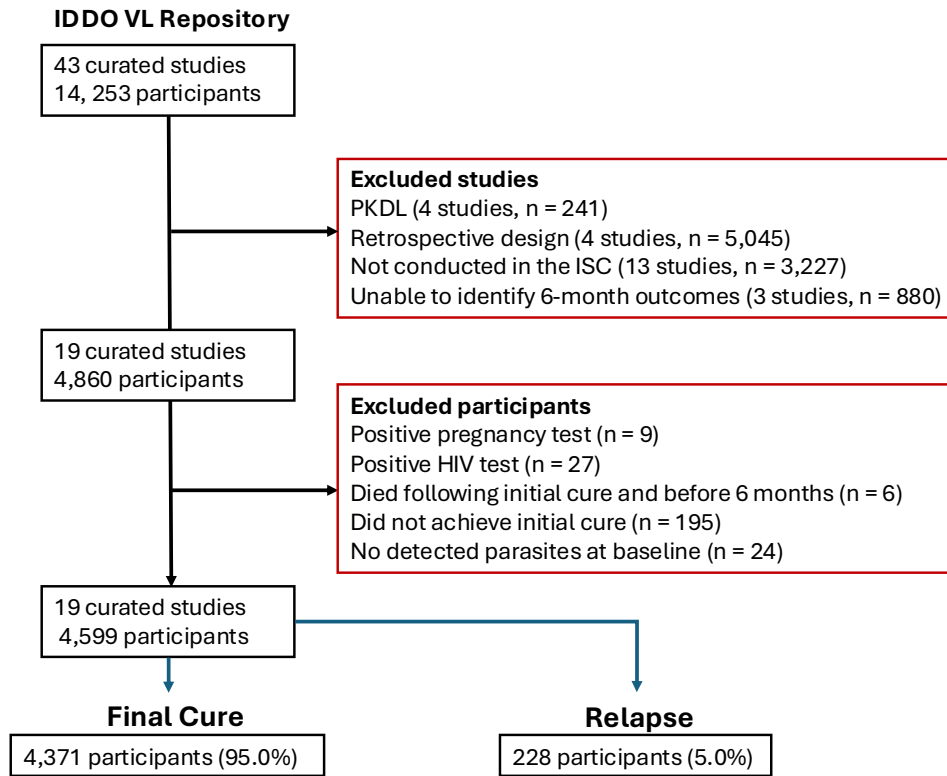


Figure 4.1: Flow diagram showing the studies and patients excluded from Indian subcontinent model development. HIV: human immunodeficiency virus; IDDO: Infectious Diseases Data Observatory; ISC: Indian subcontinent; PKDL: post kala-azar dermal leishmaniasis; VL: visceral leishmaniasis.

[272, 276–282] and Nepal (4 studies, 428 [9.3%] patients)[270, 273–275]. All Indian studies recruited patients from the state of Bihar in northeastern India, with the majority of studies led by investigators from either a government referral hospital in Patna (Rajendra Memorial Research Institute of Medical Sciences: RMRIMS) (4 studies, 1,125 [24.5%] patients)[64, 269, 271, 272], or a private research facility in Muzaffarpur (Kala-azar Medical Research Centre: KAMRC) (9 studies, 2,652 [57.7%] patients)[67, 94, 268, 276–280, 282]. All Nepalese studies were led by investigators based at the B.P. Koirala Institute of Health Sciences, a health sciences university in the city of Dharan, Koshi Province, eastern Nepal.

Table 4.1: Key characteristics of included studies, ordered by lead author and year of publication/protocol. Where information not presented in the publication, information is extracted from the study protocol. Total patient number, number of relapses, and treatment arms reflect information provided in the contributed datasets. -: not reported; ABLE: Amphotericin B lipid emulsion (Bharat Serum and Vaccines Ltd.); alt.: alternative; ABD: Amphotericin B deoxycholate; BD: twice daily (bis die); BMGF: Bill and Melinda Gates Foundation; BPKIHS: B.P. Koirala Institute of Health Sciences; D: Day(s); EC: European Commission; Govt.: Government; ICMR: Indian Council of Medical Research; IM: Intramuscular; KAMRC: Kala-azar Medical Research Center; LAMB: Liposomal amphotericin B (Gilead formulation unless otherwise specified); MF: Miltefosine; mg/kg: milligrams per kilogram; NCVBDC: National Center for Vector Borne Diseases Control; OD: once daily (omni die); P: Publication; PATH: Program for Appropriate Technology in Health; PO: per os (oral) PM: Paromomycin; Pr: Protocol; Ref: Reference; RMRIMS: Rajendra Memorial Research Institute of Medical Sciences; SSG: Sodium stibogluconate; TDR: UNICEF/UNDP/World Bank/WHO Special Programme for Research and Training in Tropical Diseases; UN: United Nations; WHO: World Health Organization

Study ¹	Title	Journal	Sponsor/funding	Location(s)	Study design	Study arm(s)	Age (years)	Study period	n (model)	Relapses (%)
Bhattacharya 2007[64]	Phase 4 trial of miltefosine for the treatment of Indian visceral leishmaniasis	J Infect Dis	ICMR	13 locations in Bihar, India (outpatient setting)	Open label; phase 4; safety/efficacy	(1) MF, PO, 28 D, dosed according to age and weight ^{2,3}	2–65	2002–2004	352 ⁴	22 (6.3)
Chakraborty 2008[268]	Human placental extract offers protection against experimental visceral leishmaniasis: a pilot study for a phase-I clinical trial	Am J Trop Med Hyg	Indian Council of Scientific and Industrial Research; Albert David Ltd.	KAMRC, Muzaffarpur, India	“Pre-phase 1”; pilot/preliminary	(1) Human placental extract, 2.06 mg, IM, single-dose; (2) ABD, 1 mg/kg, IV, alt. days for 30 D ⁵	≥ 5	2003–2005	6	0 (0)
Das 2009[269]	A controlled, randomized nonblinded clinical trial to assess the efficacy of amphotericin B deoxycholate as compared to pentamidine for the treatment of antimony unresponsive visceral leishmaniasis cases in Bihar, India	Ther Clin Risk Manag	-	RMRIMS, Patna, India	Randomised; open label; efficacy	(1) ABD, 1 mg/kg, IV, alt. days for 30 D; (2) pentamidine, 4 mg/kg, IM, alt. days for 30 D	6–60	2002	73	5 (6.8)
Koirala 2003[270]	Phase IV trial of miltefosine in the treatment of visceral leishmaniasis	Protocol only	-	BPKIHS, Dharan, Nepal	Open label; phase 4; safety/efficacy	(1) MF, PO, 28 D, dosed according to age and weight ^{2,3}	2–65	2003–2004	116	12 (10.3)
Pandey 2016[271]	Pharmacovigilance of miltefosine in treatment of visceral leishmaniasis in endemic areas of Bihar	Am J Trop Med Hyg	NCVBDC, Govt. of India; World Bank	4 locations in Bihar, India	Open label; safety/efficacy	(1) MF, PO, 28 D, dosed according to age and weight ^{2,6}	6–70	2012–2015	600	45 (7.5)
Pandey 2017[272]	Efficacy and safety of liposomal amphotericin B for visceral leishmaniasis in children and adolescents at a tertiary care center in Bihar, India	Am J Trop Med Hyg	-	RMRIMS, Patna, India	Open label; safety/efficacy	(1) LAMB, 10 mg/kg, IV, single-dose	<15	2014–2016	100	2 (2.0)

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Table 4.1: continued

Study ¹	Title	Journal	Sponsor/funding	Location(s)	Study design	Study arm(s)	Age (years)	Study period	n (model)	Relapses (%)
Rijal 2003[273]	Treatment of visceral leishmaniasis in south-eastern Nepal: decreasing efficacy of sodium stibogluconate and need for a policy to limit further decline	Trans R Soc Trop Med Hyg	WHO, Geneva University Hospital; Novartis Foundation	BPKIHS, Dharan, Nepal	Non-randomised; efficacy	SSG, 20 mg/kg, IV/IM ⁷ OD, 30 D, extended to 40 D if positive aspirate at 30 D. Either (1) in hospital or (2) first 5–7 days in hospital;	All	2000–2001	102	1 (1.0)
Rijal 2010(A)[274]	Efficacy and safety of liposomal amphotericin B in Nepalese patients with visceral leishmaniasis	Protocol only	TDR	BPKIHS, Dharan, Nepal	Phase 2/3; safety/efficacy	(1) LAMB, 3 mg/kg, IV, OD, D1-5 (21 mg/kg total dose)	12–65	2010–2011	32	1 (3.1)
Rijal 2010(B)[275]	Clinical risk factors for therapeutic failure in kala-azar patients treated with pentavalent antimonials in Nepal	Trans R Soc Trop Med Hyg	EC (5th Framework Programme)	BPKIHS, Dharan, Nepal	Prospective cohort	(1) SSG, 20 mg/kg, IV/IM ⁷ OD, 30 D	-	2001–2003	178	1 (0.6)
Sundar 2007[71, 283]	Injectable paromomycin for visceral leishmaniasis in India	N Engl J Med	PATH (including UN, BMGF, TDR)	4 locations in Bihar, India	Randomised; open label; phase 3; non-inferiority; safety/efficacy	(1) PM, 15 mg/kg, IM, OD, 21 D; (2) ABD, 1 mg/kg, IV, alt. days for 30 D	5–55	2003–2004	250 ⁴	4 (1.6)
Sundar 2008(A)[276, 284]	New treatment approach in Indian visceral leishmaniasis: single-dose liposomal amphotericin B followed by short-course oral miltefosine	Clin Infect Dis	Banaras Hindu University	KAMRC, Muzaffarpur, India	Partially randomised; open label; phase 2; non-comparative; sequential; triangular	LAMB, 5 mg/kg, IV, single-dose, either (1) alone, or (2,3,4) in combination with MF, 50 mg, PO, BD, D2–9, D2–11, or D2–15; and (5) combination of LAMB, 3.75 mg/kg, IV, single-dose, and MF, 50 mg, PO, BD, D2–15	≥ 12	2006–2007	225	7 (3.1)
Sundar 2008(B)[277]	Safety of a pre-formulated amphotericin B lipid emulsion for the treatment of Indian kala-azar	Trop Med Int Health	Bharat Serum and Vaccines Ltd.	KAMRC, Muzaffarpur, India	Non-randomised; non-comparative; open label; phase 2; safety/efficacy	ABLE, IV, OD, 3 D at (1) 5 mg/kg (2) 4 mg/kg, or (3) 3 mg/kg	12–65	2004–2005	45	4 (8.9)
Sundar 2009[278, 285]	Short-course paromomycin treatment of visceral leishmaniasis in India: 14-day vs 21-day treatment	Clin Infect Dis	Banaras Hindu University	KAMRC, Muzaffarpur, India	Randomised; open label; phase 3; safety/efficacy	PM, 15 mg/kg, IM, OD, either (1) 14 D or (2) 21 D	5–55	2007–2008	307	26 (8.5)
Sundar 2010[279, 286]	Single-dose liposomal amphotericin B for visceral leishmaniasis in India	N Engl J Med	Banaras Hindu University	KAMRC, Muzaffarpur, India	Randomised; open label; phase 3; safety/efficacy	(1) ABD, 1 mg/kg, IV, alt. days for 30 D; (2) LAMB, 10 mg/kg, IV, single-dose	2–65	2008–2009	412	14 (3.4)

continued on next page

Table 4.1: continued

Study ¹	Title	Journal	Sponsor/funding	Location(s)	Study design	Study arm(s)	Age (years)	Study period	n (model)	Relapses (%)
Sundar 2011[280, 287]	Ambisome plus miltefosine for Indian patients with kala-azar	Trans R Soc Trop Med Hyg	Banaras Hindu University	KAMRC, Muzaffarpur; RMRIMS, Patna, India	Open label; phase 2; safety/efficacy	(1) LAMB, 5 mg/kg, IV, single-dose, with MF, 2.5 mg/kg/day, PO, D2-D15 ⁸	2–65	2007–2009	128	5 (3.9)
Sundar 2012[67]	Efficacy of miltefosine in the treatment of visceral leishmaniasis in India after a decade of use	Clin Infect Dis	EC (Kaladrug-R); Sitaram Memorial Trust	KAMRC, Muzaffarpur, India	Open label; safety/efficacy	(1) MF, PO, 28 D, dosed according to age and weight ^{2, 3}	6–70	2009–2010	571	34 (6.0)
Sundar 2014[281, 288]	Efficacy and safety of amphotericin B emulsion versus liposomal formulation in Indian patients with visceral leishmaniasis: a randomised, open-label study	PLoS Negl Trop Dis	Bharat Serums and Vaccines Ltd; Department of Science and Technology, Govt. of India	4 locations in Bihar, India.	Randomised; open label; phase 3; safety/efficacy	(1) LAMB, 15 mg/kg, IV, single-dose; (2) ABLE, 15 mg/kg, IV, single-dose	5–65	2009–2011	144 ⁴	9 (6.3)
Sundar 2015[282, 289]	Single-dose indigenous liposomal amphotericin B in the treatment of Indian visceral leishmaniasis: A phase 2 study	Am J Trop Med Hyg	Lifecare Innovations; Department of Science and Technology, Govt. of India	KAMRC, Muzaffarpur, India	Non-randomised; non-comparative; open label; phase 2; safety/efficacy	LAMB (Lifecare Innovations), IV, single-dose (1) 10 mg/kg or (2) 15 mg/kg	12–60	2012–2013	30	3 (10.0)
Sundar 2019[94, 290]	Effectiveness of single-dose liposomal amphotericin B in visceral leishmaniasis in Bihar	Am J Trop Med Hyg	Banaras Hindu University	KAMRC, Muzaffarpur, India	Observational; efficacy	(1) LAMB, 10 mg/kg, IV, single-dose	All	2013–2017	928	33 (3.6)

¹ Study name is composed of the lead author and year of publication or most recent protocol.

² Miltefosine dosing (i) ≥ 12 years and ≥ 25 kg: 50 mg BD; (ii) ≥ 12 years and <25 kg: 50 mg OD; (iii) <12 years and <25 kg: 2.5 mg/kg/day.

³ Dosing in <12 years and <25 kg: in divided doses

⁴ Contributed number of participants from KAMRC site only.

⁵ ABD arm not presented in publication.

⁶ Dosing in <12 years and <25 kg: not further described.

⁷ Dosing not further described.

⁸ SSG route of administration (IV vs IM) not specified.

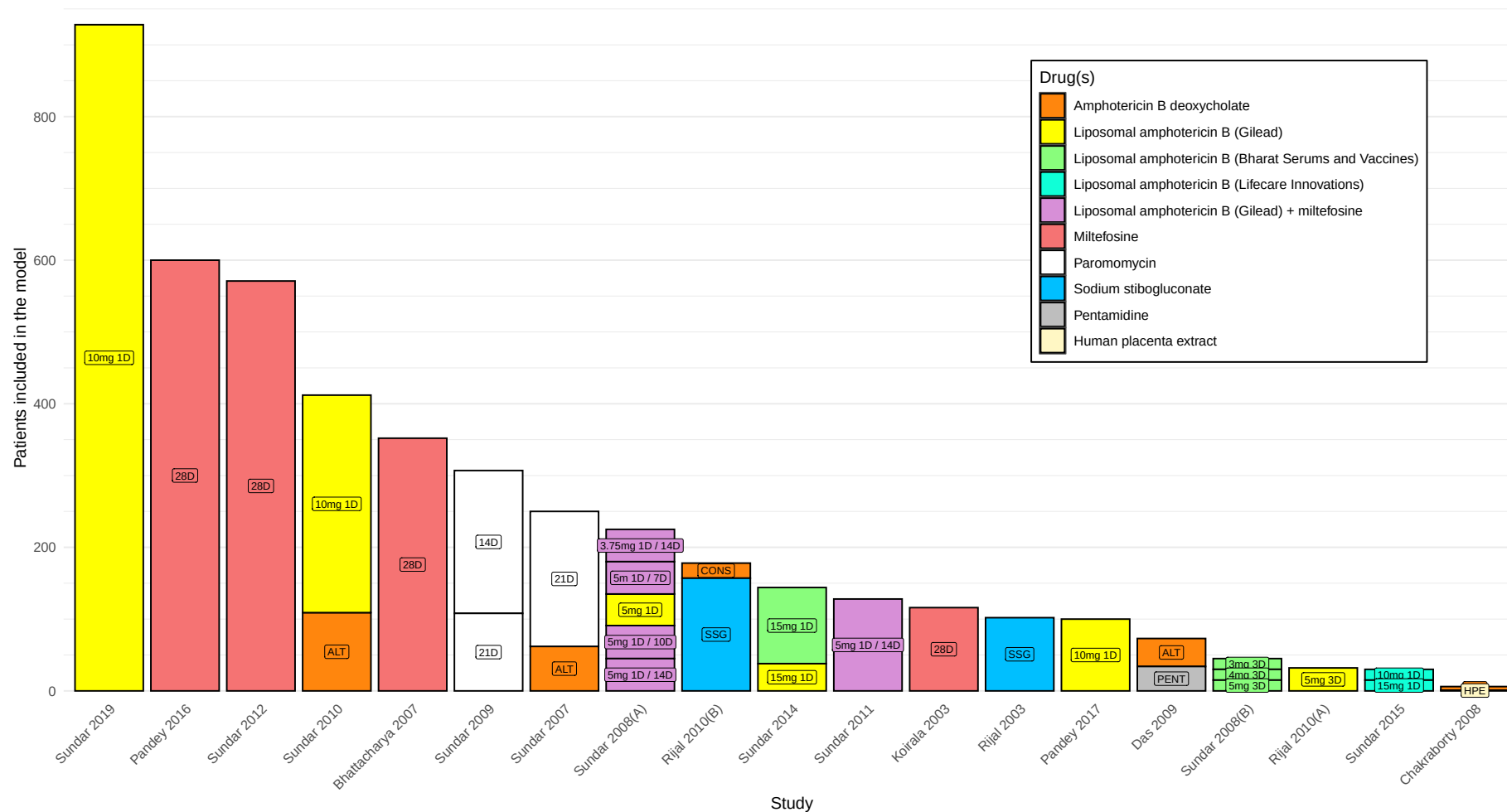


Figure 4.2: Bar chart showing the distribution of treatment regimens across contributing studies from the Indian subcontinent. Important distinguishing dosing information are provided in the overlaid labels, as space allows. Full treatment details are presented in Table 4.1. 6 patients are included from Chakraborty 2008: 5 receiving alternate day amphotericin B deoxycholate and 1 receiving human placenta extract. ABD: amphotericin B deoxycholate; ALT: alternate days; CONS: consecutive days; D: days; HPE: human placenta extract; MF: miltefosine; PENT: pentamidine; mg: milligrams/kilogram; SSG: sodium stibogluconate.

Study Design and Treatment Arms

Study designs are summarised in Table 4.1 as described in the publication or protocol. Four studies (428 [9.3%] patients) were described as Phase 2 trials[276, 277, 280, 282], four studies (1,113 [24.2%] patients) as Phase 3 trials[71, 278, 279, 281], two studies (468 [10.2%] patients) as Phase 4 trials[64, 270], and one study (32 [0.7%] patients) as a Phase 2/3 trial[274].

Ten studies (1,625 [35.3%] patients) allocated patients to more than one treatment arm[71, 268, 269, 275–279, 281, 282], of which six studies (1,411 [30.7%] patients) were randomised[71, 269, 278, 279, 281] or partially randomised[276].

Treatment arms are summarised by study in Figure 4.2. The three largest studies were all monotherapy trials that treated with either 10 mg/kg single-dose liposomal amphotericin B (1 study, 928 [20.2%] patients)[94], or 28-day miltefosine (2 studies, 1,171 [25.5%] patients).

Figure 4.2 presents the different treatment regimens across all studies.

Study Eligibility Criteria

All studies recruited both male and female patients, and all but three studies restricted inclusion by age. Four studies only recruited patients aged 12 years and above (314 [6.8%] patients)[274, 276, 277, 282]. Where specified, the *lower* age limit was otherwise between 2 and 6 years. One study (100 [2.2%] patients) recruited children only (< 15 years)[272]. Otherwise, the *upper* age limit ranged between 60 and 70 years.

In the 16 studies that included baseline haemoglobin level in their eligibility criteria, the median lower threshold was 50 g/L (range 35 to 60 g/L).¹ Almost all studies excluded patients with serious illness or the presence of significant co-existing diseases. No study explicitly excluded patients based on malnutrition severity.

Twelve studies (2,021 [43.9%] patients) explicitly reported performing HIV testing on all recruited patients[64, 66, 71, 268, 270, 274, 276–279, 281, 282], and patients with confirmed VL/HIV co-infection were excluded from all but two studies (280 [6.1%] patients)[273, 275]. Patients with VL/HIV co-infection from these two studies were excluded at the IPD level, where reported.

Inclusion criteria based on prior VL treatment were reported in 12 studies (2,629 [57.2%] patients)[71, 94, 269, 273–275, 277–282], with timing since last VL treatment ranging from 10 days[277] to lifelong[94]. One study (73 [1.6%] patients) only recruited patients who had failed treatment with SSG[269].

¹with one study reporting boosting haemoglobin with blood transfusions prior to recruitment[269].

Specific wording of the inclusion and exclusion criteria, and exclusion thresholds for platelets, white blood cells, clotting, renal, and liver function, are provided in the [Supplementary Material](#).

Diagnostic Criteria

The majority of studies specified clinical criteria for inclusion, with six studies (1,984 [43.1%] patients) requiring ≥ 2 weeks' fever and splenomegaly[94, 271, 273–275, 281]. The remaining studies either did not specify the clinical criteria (two studies, 36 [0.8%] patients)[268, 282], or referred more generally to the presence of typical signs and symptoms (11 studies, 2,579 [56.1%] patients).

VL was confirmed by tissue aspirate in all but two studies, where the rK39 RDT was instead used as the primary diagnostic method (1,528 [33.2%] patients)[94, 271]. Three studies (276 [6.0%] patients)[272, 274, 281], screened patients with the rK39 RDT prior to confirmation with a tissue aspirate.

Relapse

Nine studies directly defined relapse in their publications or protocols (2,131 [46.3%] patients)[64, 71, 94, 269, 270, 272–275]. In the remaining studies, relapse events could be inferred indirectly from definitions of initial cure and treatment failure/success. Confirmation with tissue aspirates was performed in 13 studies, and reported, at least partially, in the IPD of 12 studies (see [Supplementary Material](#) for study-specific details).

In the three studies that reported relapse beyond 6 months[94, 275, 276], IPD interrogation allowed identification of the subset of patients where relapse occurred within 6 months of initial cure assessment. Active follow-up strategies were used to identify relapse events in all but one study (928 [20.2%] patients)[94].

Initial Cure

Initial cure was assessed 28–30 days after treatment initiation in 14 studies (3,740 [81.3%] patients)[64, 67, 94, 268–273, 275, 279–282]. In the remaining five studies, initial cure was assessed either at two different time points, depending on the treatment arm (two studies, 532 [11.6%] patients)[71, 278], or between days 16 and 19 (three studies, 302 [6.6%] patients)[274, 276, 277]. Clinical criteria for initial cure were described in all but two studies[64, 268], although often loosely defined (see [Supplementary Material](#) for study-specific details).

Tissue aspirate formed part of the initial cure assessment in all but one study (928 [20.2%] of patients without tissue aspirate)[94]. In nine studies (1,378 [30.0%]

patients)[64, 71, 270, 273, 274, 277, 278, 281, 282], patients with an initial 1+ parasite grade underwent a repeat aspirate 10 to 60 days later. If the repeat aspirate were negative, the patient would often be considered a ‘slow-responder’, allowing progression to either definite cure or relapse.

4.1.2 Patient Characteristics

Overall (marginal) distributions of categorical and continuous variables are tabulated in Tables 4.2 and 4.3, respectively. These distributions are also displayed graphically in Figure 4.4 for bar charts of categorical variables, and Figures 4.6 and 4.5 for histograms of laboratory and non-laboratory variables, respectively. Study-specific distributions are presented for age, sex, and relapse in Figure 4.3. In the Appendix, study-specific distributions of all categorical and continuous variables are presented in Figures C.1 and C.2–C.14, respectively.

Patient numbers and proportions presented in this section exclude missing data. See Section 4.1.4 for further information on missing data.

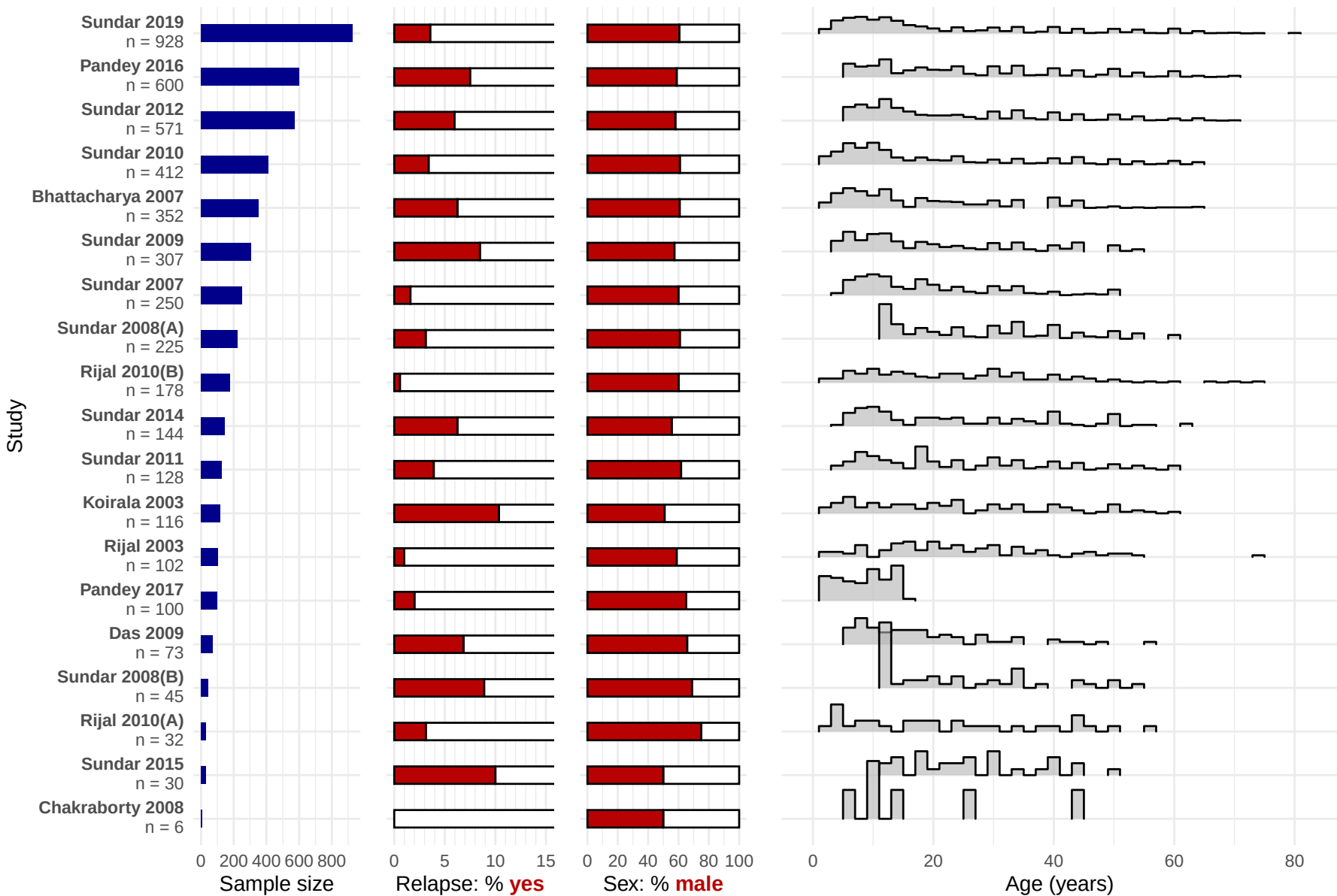


Figure 4.3: Graphical summary of the Indian subcontinent study-specific sample sizes and distributions of relapse status, sex, and age.

Categorical Variables

Across all studies, 2,745 (59.7%) of patients were male, ranging from 50.0%[\[268, 282\]](#) to 75.0%[\[274\]](#) at the study level.

Relapse within 6 months was identified in 228 (5.0%) patients, varying from 0%[\[268\]](#) to 10.3%[\[270\]](#) at the study level.

The majority of patients (52.4%) had mild/normal malnutrition according to the definition provided in Section [3.3.2](#). Moderate and severe malnutrition affected 28.7% and 18.9% of patients overall, although these proportions varied significantly across studies and were affected by considerable missing data (see Appendix Figure [C.1](#)).

Approximately half of all patients (2,051 [\[49.8%\]](#) patients) had severe anaemia at the time of recruitment, as defined by the 2024 WHO guidelines[\[239\]](#).

At the patient level, the most common treatment regimen was 28 days of oral miltefosine (1,639 [\[35.6%\]](#) patients), followed by 10 mg/kg single-dose liposomal amphotericin B (1,331 [\[28.9%\]](#) patients). The remaining treatment regimens (1,629 [\[35.4%\]](#) patients), consisted of a broad range of experimental and non-experimental regimens, including amphotericin B deoxycholate, other (non-Gilead) lipid formulations of amphotericin B, paromomycin, SSG, pentamidine, and human placenta extract (Figure [4.2](#)).

The most common baseline parasite grade on tissue aspirate was 1+ (1,201 [\[40.7%\]](#) patients), while the median grade was 2+ (764 [\[25.9%\]](#) patients at this grade; IQR: 1 to 3). The proportion of patients decreased with increasing parasite grade, with 54 (1.8%) patients with a parasite grade of 5+. Where reported, 92.0% (2,717 patients) of tissue aspirates were obtained from the spleen. The remaining aspirates were from bone marrow and accounted for almost all aspirates performed in three studies from Nepal[\[270, 273, 274\]](#) (Appendix Figure [C.1](#)).

Continuous Variables

The median age was 18 years, ranging from 1 to 80 years (IQR: 10 to 32 years). Overall the distribution was right-skewed, with study-specific distributions reflecting age-specific inclusion criteria. 138 (3.0%) of patients were under 5 years, and 47 (1.0%) were 65 years or over.

In adults (≥ 19 years) the median BMI was 18.2 kg/m² (IQR: 16.4 to 20.8 kg/m²) (1,313 patients). In children (≥ 5 and < 19 years), the median BMI-for-age z-score was -1.68 (IQR: -2.67 to -0.72) (1,338 patients), and in younger children (< 5 years) the median weight-for-height z-score was -1.80 (IQR: -2.97 to -0.99) (27 patients).

Variable	Overall (%) n = 4,599	Final cure (%) n = 4,371	Relapse (%) n = 228
Sex			
Female	1,854 (40.3)	1,771 (40.5)	83 (36.4)
Male	2,745 (59.7)	2,600 (59.5)	145 (63.6)
Malnutrition			
Normal/mild	1,403 (30.5)	1,333 (30.5)	70 (30.7)
Moderate	769 (16.7)	732 (16.7)	37 (16.2)
Severe	506 (11.0)	485 (11.1)	21 (9.2)
(Missing)	1,921 (41.8)	1,821 (41.7)	100 (43.9)
Anaemia			
Non-severe	2,089 (45.4)	1,973 (45.1)	116 (50.9)
Severe	2,071 (45.0)	1,985 (45.4)	86 (37.7)
(Missing)	439 (9.5)	413 (9.4)	26 (11.4)
Treatment			
Miltefosine ¹	1,639 (35.6)	1,526 (34.9)	113 (49.6)
Other	1,629 (35.4)	1,562 (35.7)	67 (29.4)
LAMB ²	1,331 (28.9)	1,283 (29.4)	48 (21.1)
Parasite grade			
1+	1,201 (26.1)	1,152 (26.4)	49 (21.5)
2+	764 (16.6)	733 (16.8)	31 (13.6)
3+	610 (13.3)	567 (13.0)	43 (18.9)
4+	323 (7.0)	301 (6.9)	22 (9.6)
5+	54 (1.2)	53 (1.2)	1 (0.4)
(Missing)	1,647 (35.8)	1,565 (35.8)	82 (36.0)
Aspirate source³			
Bone	235 (8.0)	221 (7.9)	14 (9.5)
Spleen	2,717 (92.0)	2,585 (92.1)	132 (90.4)

¹ 28 days of linear-dosed miltefosine at standard dosing.

² Single dose liposomal amphotericin B (Gilead) 10mg/kg.

³ Denominator for % in aspirate source: number of patients with documented parasite grade (overall: 2,952; final cure: 2,806; relapse: 146; no missing data).

Table 4.2: Summary of categorical candidate predictors and parasite source across contributed studies from the Indian subcontinent. Missing data are described where present. SDA: Single dose liposomal amphotericin B 10mg/kg.

The median spleen size was 4 cm (IQR: 2 to 7 cm), and ranged from 0 to 22 cm. Eight studies reported patients without splenomegaly at baseline[64, 67, 71, 94, 272, 276, 278, 279], corresponding to 218 (3.7%) patients with recorded spleen sizes.

The median duration of fever prior to patient recruitment was 30 days (IQR: 20 to 60 days). The distribution was markedly right-skewed, ranging from 1 to 730 days.

For distributions of laboratory results (haemoglobin, white blood cells, platelets, creatinine, and alanine aminotransferase), please refer to Table 4.3 and Figure 4.6.

Variable	Overall (n = 4,599)		Final cure (n = 4,371)		Relapse (n = 228)	
	Median (IQR)	Missing ¹ (%)	Median (IQR)	Missing (%)	Median (IQR)	Missing ¹ (%)
Age (years)	18 (10 – 32)	7 (0.2)	18 (10 – 33)	6 (0.1)	14 (8 – 32)	1 (0.4)
Height (cm)	150.0 (125.0 – 161.5)	1,914 (41.6)	150.0 (126.0 – 161.0)	1,815 (41.5)	149.4 (124.0 – 164.6)	99 (43.4)
Weight (kg)	37 (21 – 46)	418 (9.1)	36.8 (21 – 46)	389 (8.9)	35 (19 – 49)	29 (12.7)
BMI (kg/m ²) ²	18.22 (16.37 – 20.81)	841 (39.0)	18.17 (16.33 – 20.70)	804 (39.1)	19.77 (17.24 – 24.07)	37 (38.1)
BMI z-score ³	-1.68 (-2.67 – -0.72)	962 (41.8)	-1.67 (-2.65 – -0.71)	903 (41.5)	-1.99 (-2.81 – -1.11)	59 (47.6)
WFH z-score ⁴	-1.80 (-2.97 – -0.99)	111 (80.4)	-1.53 (-2.51 – -0.97)	108 (81.8)	-2.73 (-3.13 – -2.27)	3 (50.0)
Spleen size (cm)	4 (2 – 7)	642 (14.0)	4 (2 – 7)	595 (13.6)	3 (2 – 6)	47 (20.6)
Fever duration (days)	30 (20 – 60)	1,779 (38.7)	30 (20 – 60)	1,667 (38.1)	20 (15 – 30)	112 (49.1)
Parasite grade	2 (1 – 3)	1,647 (35.8)	2 (1 – 3)	1,565 (35.8)	2 (1 – 3)	82 (36.0)
WBC ($\times 10^9$ /L)	3.4 (2.5 – 4.5)	435 (9.5)	3.4 (2.4 – 4.5)	409 (9.4)	3.4 (2.7 – 4.5)	26 (11.4)
Platelets ($\times 10^9$ /L)	112 (77 – 155)	434 (9.4)	112 (77 – 155)	408 (9.3)	119.5 (83 – 160)	26 (11.4)
Haemoglobin (g/L)	79 (67 – 93)	433 (9.4)	79 (67 – 92)	407 (9.3)	82.5 (69 – 96)	26 (11.4)
ALT (IU/L)	31 (20 – 52)	449 (9.8)	31.2 (20 – 52)	421 (9.6)	30 (19 – 52)	28 (12.3)
Creatinine (μ mol/L)	63.7 (51.3 – 79.6)	646 (14.0)	63.7 (51.3 – 79.6)	598 (13.7)	63.7 (51.3 – 76.0)	48 (21.1)

¹ Denominator for missing %: total number of patients in respective group (overall, relapse or final cure). For measures of malnutrition (BMI, BMI-for-age z-score, and weight-for-height z-score), see below.

² Denominator for missing %: number of patients aged ≥ 19 years, n = 2,154 (relapse: 97, final cure: 2,057).

³ Denominator for missing %: number of patients aged 5–18 year inclusive, n = 2,300 (relapse: 124, final cure: 2,176).

⁴ Denominator for missing %: number of patients aged < 5 years, n = 138 (relapse: 6, final cure: 132).

Table 4.3: Summary of continuous candidate predictors across contributed studies from the Indian subcontinent. Including additional variables used for the derivation of malnutrition status (height, weight, BMI, BMI-for-age z-score, weight-for-height z-score). ALT: alanine aminotransferase; BMI: body mass index; cm: centimetres; IQR: inter-quartile range, IU: international units; kg: kilograms; L: litres; m: metres; WBC: white blood cells; WFH: weight-for-height; g: grams; μ mol: micromoles.

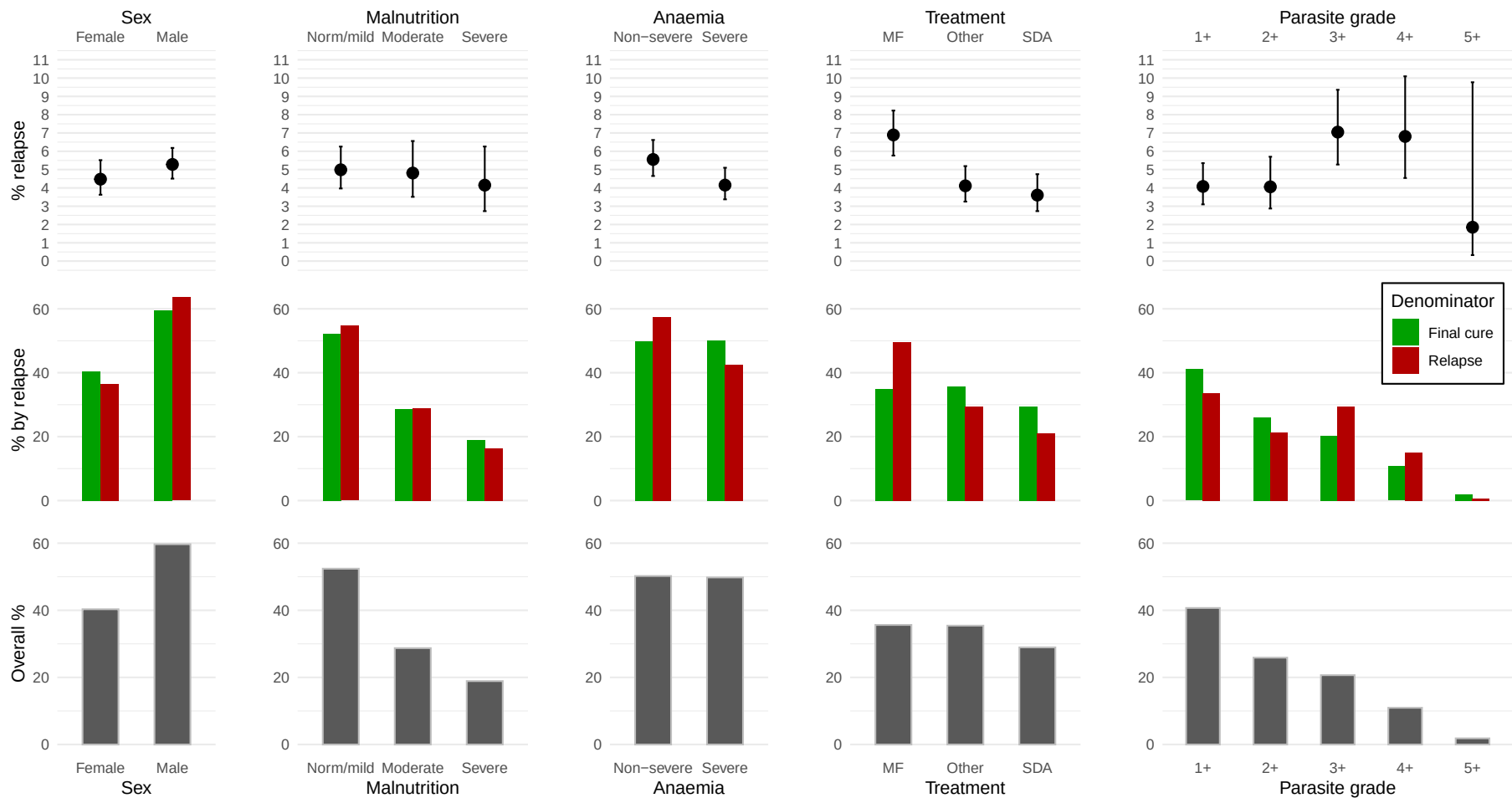


Figure 4.4: Marginal distributions and predictor-outcomes relationships for categorical candidate predictors. Excluding missing data. 95% binomial confidence intervals calculated using the Wilson method. MF: miltefosine; Norm: normal; SDA: single-dose liposomal amphotericin B.

4.1.3 Univariable Associations

Unadjusted relationships between variables (including all candidate predictors, excluding missing data) and relapse risk are presented in tabular form (Tables 4.2, 4.3) and visually alongside their distributions (Figure 4.4 for categorical variables, and Figures 4.5 and 4.6 for continuous non-laboratory and laboratory variables, respectively). The relationships are presented on the log-odds (logit) scale for continuous candidate predictors in Appendix Figure C.17.

Across continuous candidate predictors, GAM smooths suggested often non-linear relationships with relapse risk, with wider uncertainty at the extremes of the predictor distributions where data were sparse. Apparent trends were most evident for age, duration of fever and spleen size. Age showed a shallow U-shaped pattern, reaching a minimum relapse risk at approximately 20 years, while duration of fever showed a marked downward trend, with longer fever durations associated with lower relapse risk. With spleen size, a notable downward trend in risk was seen for spleen sizes over 2 cm, perhaps better appreciated on the logit scale in Appendix Figure C.17. For the laboratory predictors, weak upward trending monotonic patterns were observed for white blood cells and platelets, with higher values associated with increased relapse risk.

Trends in non-candidate predictors, including weight, height, and haemoglobin, were also apparent. A marked upward trend was seen with haemoglobin and relapse risk, which was also evidence in the categorical associations, with severe anaemia associated with a lower relapse risk compared to non-severe anaemia.

Treatment with miltefosine was associated with a higher unadjusted relapse risk when compared with 10 mg/kg single-dose liposomal amphotericin B or ‘Other’ treatment. Among patients who relapse, 113/228 (49.6%) were treated with miltefosine, compared to 1526/4371 (34.9%) of those who achieved final cure.

Patients with parasite grades of 3+ and 4+ were associated with higher relapse risk compared to patients with grades 1+ and 2+. Extrapolation of any trend to patients with 5+ parasite grade was limited by small numbers.

A correlation matrix showing associations between continuous variables is presented in Appendix Figure C.15, and between continuous and categorical variables in Appendix Figure C.16. These correlations are considered further in the Discussion section.

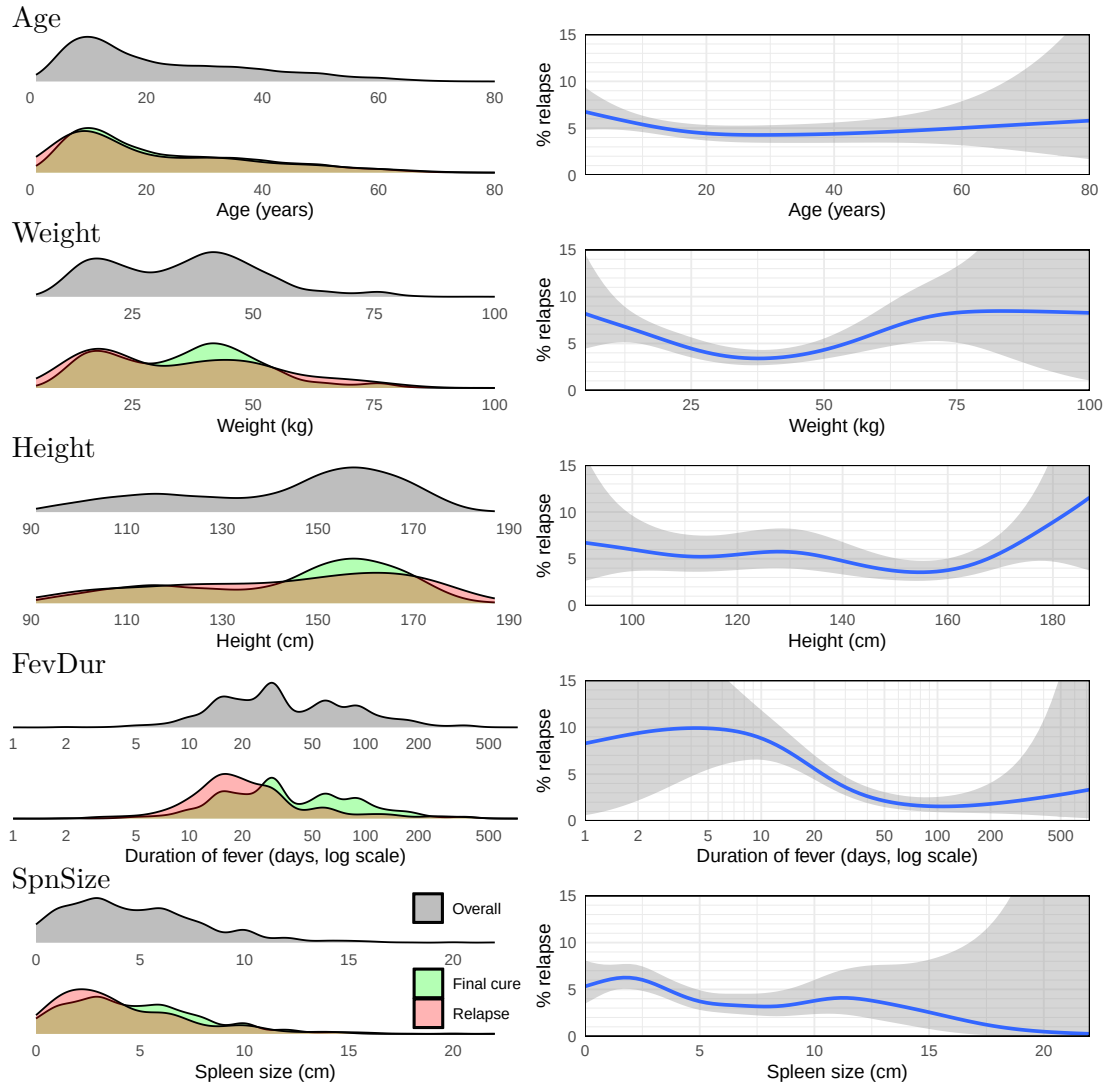


Figure 4.5: Marginal distributions and predictor-outcome relationships for continuous non-laboratory candidate predictors. FevDur: duration of fever; SpnSize: spleen size. For each candidate predictor, left upper panel shows the overall density across studies and the left lower panel shows overlapping densities normalised by relapse status. The right panel shows a univariable generalised additive model spline fit, with 95% confidence interval, of relapse.

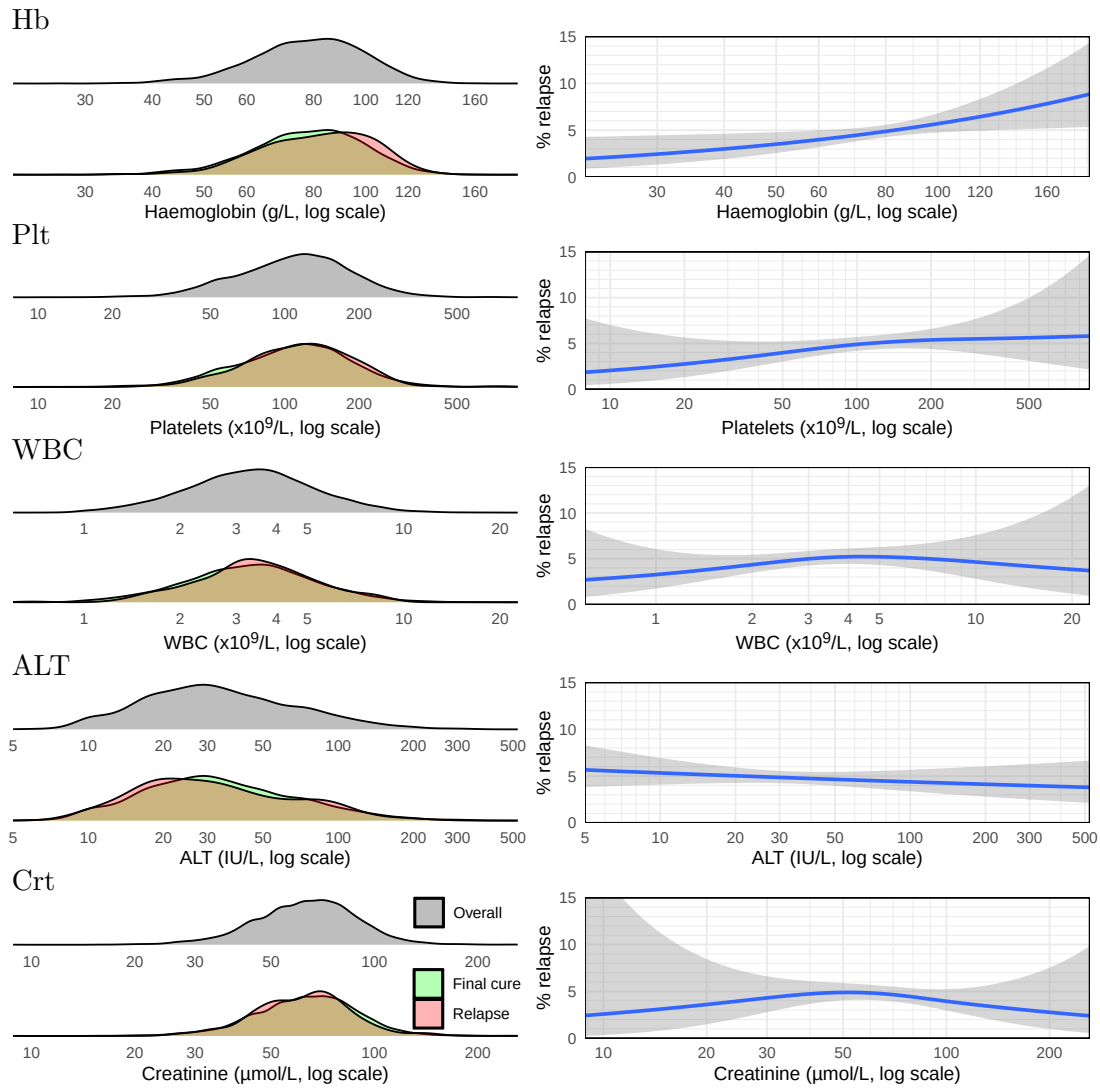


Figure 4.6: Marginal distributions and predictor-outcome relationships for continuous laboratory candidate predictors. All predictors presented on log scale. Hb: haemoglobin; Plt: platelet; WBC: white blood cells; ALT: alanine aminotransferase; Crt: creatinine. For each candidate predictor, left upper panel shows the overall density across studies and the left lower panel shows overlapping densities normalised by relapse status. The right panel shows a univariable generalised additive model spline fit, with 95% confidence interval, of relapse.

4.1.4 Missing Data

Missing data were frequent among the candidate predictors, with 3,697 (80.4%) of patients missing at least one candidate predictor data point. Just under half of all patients (2,177 [47.3%] patients) were missing exactly one data point, and 1,520 (33.1%) patients were missing two or more data points. No missing data were present in the outcome (relapse) or random-effect (study) variables.

The candidate predictor with the most missingness was malnutrition, which

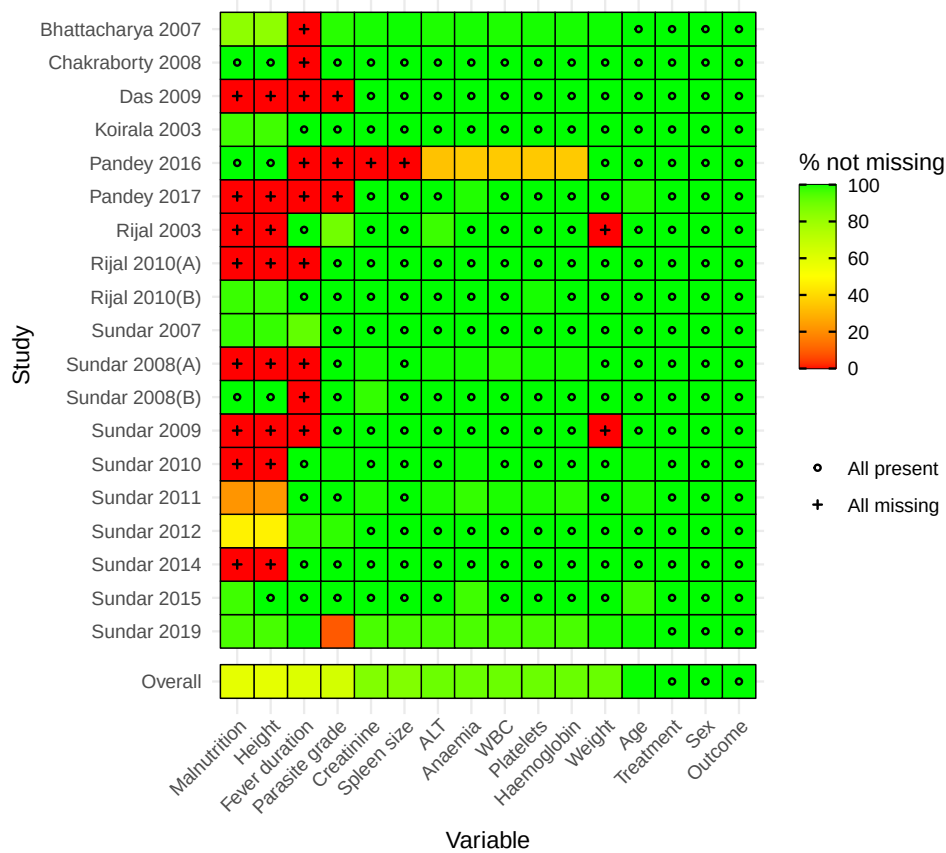


Figure 4.7: Density plot illustrating the amount of missing data overall and across contributing studies from the Indian subcontinent. Study ordered by lead author and year of publication (or protocol). Variables ordered by amount of missingness. ALT: alanine aminotransferase; WBC: white blood cells.

affected 1,921 (41.8%) patients due to a lack of height information. After malnutrition, fever duration was missing in 1,779 (38.7%) of patients, and parasite grade was missing in 1,647 (35.8%) patients. Remaining predictors had under 15% missingness. Missingness patterns, ordered by missingness in the candidate predictors, are presented in Figure 4.7 at the study level and overall.

Multiple Imputation

On review of the multiple imputation diagnostic plots (as described in Section 3.3.4 and available for review in the [Supplementary Material](#)), no clear or consistent patterns indicating violations of the missing-at-random assumption were apparent. Convergence was obtained well before the 20 iterations for the majority of imputed predictors, and the distributional assumptions of the imputation model appeared robust on inspection of the diagnostic density and scatter plots.

4.2 Model Results

Two models were fitted to the ISC IPD — one model including parasite grade as a candidate predictor, and one model excluding parasite grade.

4.2.1 Model Specification and Coefficient Estimates

Forest plots of the final model predictors are presented in Figure 4.8. Full specification of the final models, including intercept terms, p-values, and predictor transformations, are presented in Appendix Tables C.1 and C.2.

For both models, final predictors included: age (both linear and squared terms), duration of fever, treatment regimen, and presence of severe anaemia. Parasite grade was also included when considered as a candidate predictor. As can be appreciated from Figure 4.8, the adjusted odds ratios were similar across the two models. The similarity between models can also be appreciated in Figure 4.9, which shows relapse probability predictions with the model intercept recalibrated to the Sundar 2019 dataset[94].

In both models, U-shaped relationships between age and relapse risk were shown, reflecting a significant positive age squared coefficient. Inverse relationships between fever duration and relapse risk were also demonstrated. In the model including parasite grade, the relapse odds decreased by 36.0% (95% CI: 23.1–46.7%) for each doubling of fever duration, with similar findings with the model excluding parasite grade. When included, a unit increase in parasite grade (e.g. from 1+ to 2+) was associated with a 30.6% increase in relapse odds (95% CI: 12.1–52.2%). Relapse odds in the 10 mg/kg single-dose liposomal amphotericin B, or ‘Other’ treatment groups were significantly lower (approximately half) than the odds in the standard dose miltefosine regimen, although with marked uncertainty. The presence of severe anaemia was also found to be associated with decreased odds of relapse by approximately one third (32.0%, 95% CI: 7.0–50.3%) in the model including parasite grade. Appendix Tables C.1 and C.2 provide the complete predictor coefficients and standard errors for both models.

Pooled estimates of intraclass correlation coefficients (ICC)² were 3.5% for the model including parasite grade, and 4.4% for the model excluding parasite grade.

²ICC quantifies the proportion of total variance in the latent propensity for the outcome attributable to between-study differences.

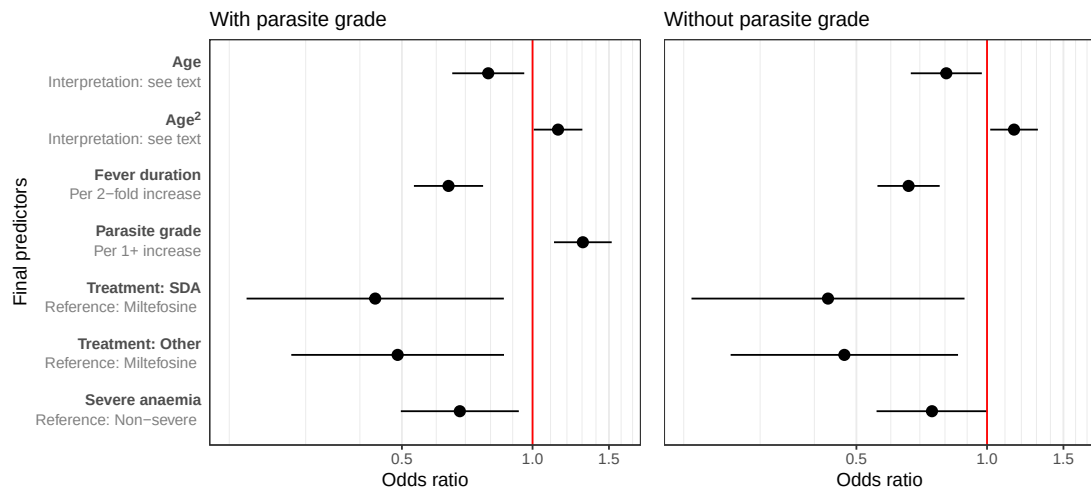


Figure 4.8: Forest plots of adjusted odds ratios with 95% confidence intervals for final model predictors. For age, the odds ratio represents a combination of linear and squared effects, with age centred by the mean and scaled by the standard deviation. Please refer to Figure 4.9 for the adjusted relapse probabilities after model intercept recalibration to the observed relapse rate in the Sundar 2019 dataset[94]. SDA: single-dose liposomal amphotericin B 10 mg/kg.

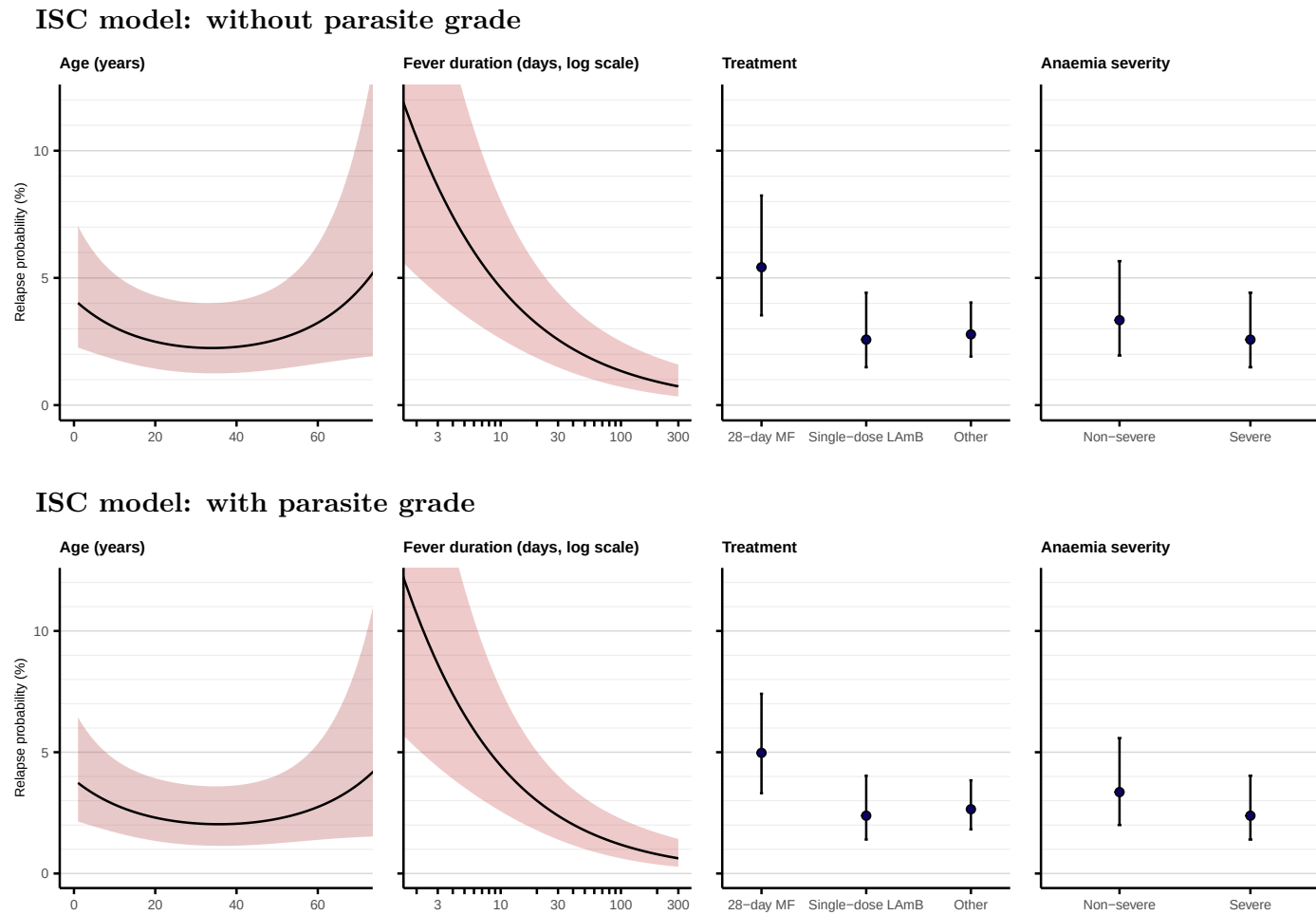


Figure 4.9: Adjusted associations between final predictors and predicted relapse probability, as estimated from the final ISC prognostic models. Probabilities were calculated from optimism-adjusted models and following logistic recalibration (intercept-term only) to data contributed from Sundar 2019[94]. Where not varying in the plot, predictions are standardised to a representative reference participant: median age (18 years), median fever duration (30 days), treated with 10 mg/kg single-dose liposomal amphotericin B, with severe anaemia, and — for the model including parasite grade — a median parasite count of 2+.

4.2.2 Model Performance and Internal Validation

Apparent and optimism-adjusted c-statistics and calibration slopes are presented in Table 4.4. When assessed in studies with >5 relapse events, model discrimination was estimated at 0.70 (95% CI: 0.66–0.74) in the model including parasite grade, and 0.69 (95% CI: 0.64–0.74) in the model excluding parasite grade. These estimates reduced to 0.68 and 0.67, respectively, after adjusting for optimism. No evidence of significant between-study heterogeneity in discrimination was identified in either model, with forest plots presented in Figure 4.10 (model including parasite grade) and Appendix Figure C.18 (excluding parasite grade).

	Estimate (95% CI)	Average optimism	Optimism-adjusted performance
Model: with PG			
C-statistic	0.70 (0.66–0.74)	0.014	0.68
Calibration slope	1.01 (0.69–1.32)	0.093	0.91
Model: without PG			
C-statistic	0.69 (0.64–0.74)	0.016	0.67
Calibration slope	1.02 (0.68–1.35)	0.098	0.92

Table 4.4: Apparent and optimism-adjusted performance measures. Abbreviations: CI: confidence interval; PG: parasite grade

Optimism in calibration slopes were estimated at 0.093 and 0.098 in the models including and excluding parasite grade, respectively. Resulting uniform shrinkage factors were 0.91 and 0.92, respectively. Multiplying together these shrinkage factors with the estimated model coefficients will result in the optimism-adjusted coefficients (Appendix Tables C.1 and C.2). No significant evidence of between-study heterogeneity in calibration slope was identified, with forest plots presented in Figure 4.11 (model including parasite grade) and Appendix Figure C.19 (model excluding parasite grade).

Calibration intercepts (calibration-in-the-large) were found to vary significantly between studies for both models, as can be appreciated from the forest plots in Figure 4.11 for the model including parasite grade, and Appendix Figure C.19 for the model excluding parasite grade (test for heterogeneity; $p = 0.01$ and $p = 0.002$ for models including and excluding parasite grade).

Visual inspection of the calibration plots comparing predicted and observed relapse probabilities showed minimal deviation from perfect calibration (Figure 4.12). Calibration plots showing predicted and observed relapse probabilities for fever duration (Figure 4.13 for model including parasite grade), and other predictors

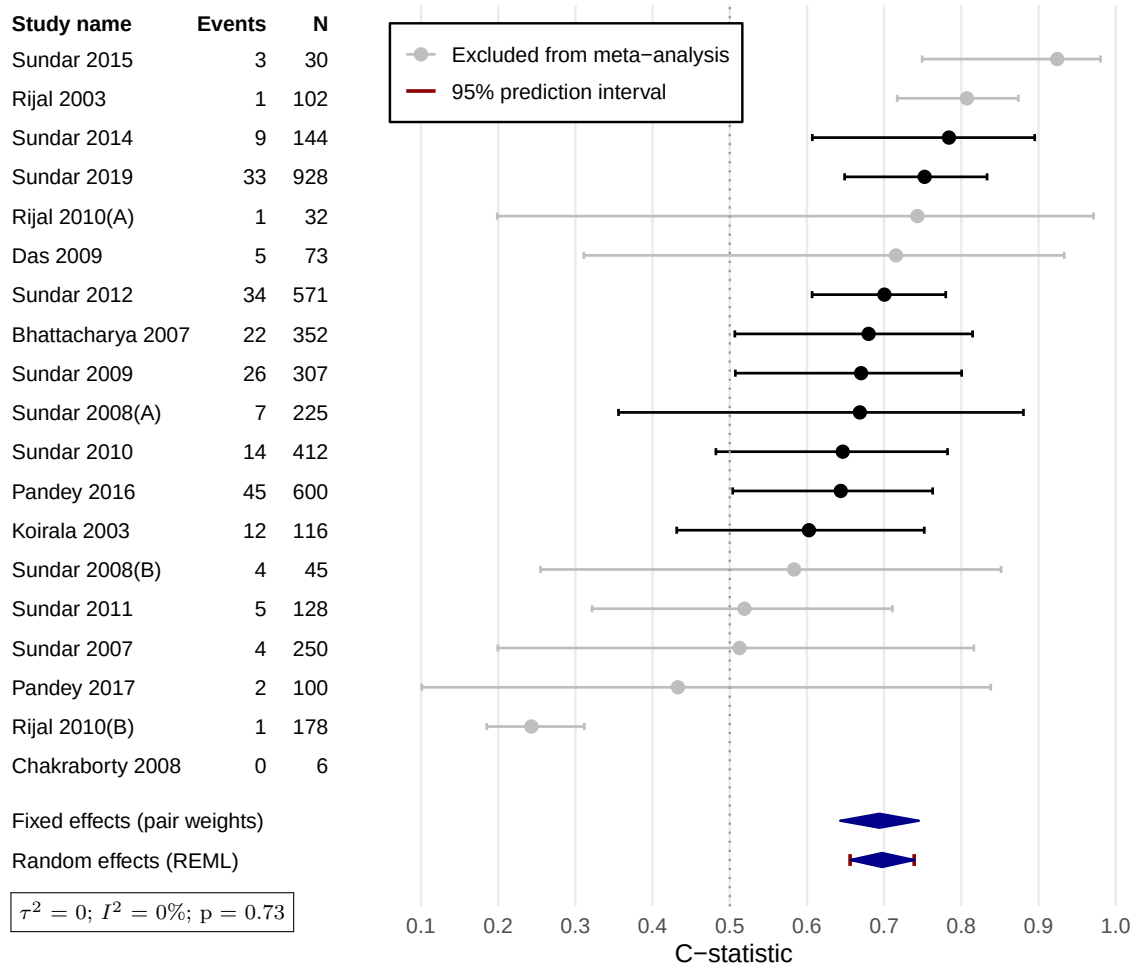


Figure 4.10: Forest plot showing individual and pooled study c-statistics, for the model **including** parasite grade. For the model excluding parasite grade refer to Appendix C.18. Pooled c-statistics are presented from both fixed-effects and random-effects meta-analysis models, after excluding studies with ≤ 5 relapse events. Blue diamonds: pooled summary estimates with 95% confidence intervals. For Chakraborty 2008, no relapse events occurred and c-statistic is therefore undefined.

(Appendix Figures C.20 to C.24 for model including parasite grade and C.25 to C.28 excluding parasite grade) and also showed good overall agreement, with predicted relapse probabilities lying within the 95% confidence intervals of the observed probabilities.

The distribution of the final predictors selected across the 2×500 bootstrap models are presented in Appendix Table C.3. Reassuringly, the most frequently selected predictors correspond to the final predictors selected in both final models. Of note, fever duration was selected across all bootstraps for both models. When considered as a candidate predictor, parasite grade was selected in 472 (94.4%) of bootstrap models. A degree of instability was apparent, with predictors such as

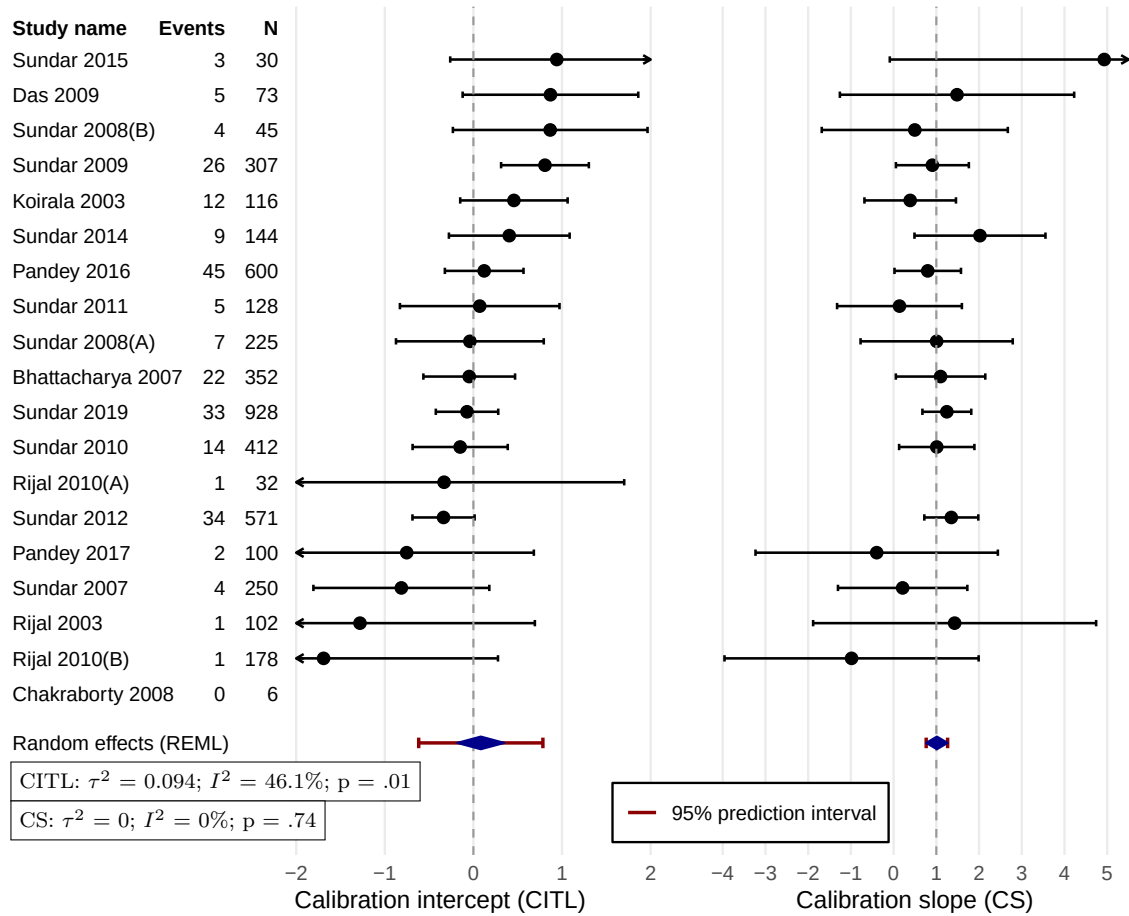


Figure 4.11: Forest plots showing individual and pooled study calibration measures for the model **including** parasite grade. Left: calibration intercept (calibration-in-the-large, CITL); Right: calibration slope (CS). Blue diamonds: summary estimates with 95% confidence intervals. Calibration measures not presented for Chakraborty 2008 due to no relapse events. Calibration slope not presented for Rijal 2010(A) due to only one relapse event and few total participants leading to failure of model convergence.

spleen size, sex, alanine aminotransferase, and the cubic age term being selected in 100–250 bootstraps.

4.3 Summary

In summary, two prognostic models predicting relapse were fitted to the ISC IPD — one including parasite grade as a candidate predictor and one excluding it. Both final models retained the same core predictors: age (modelled using linear and squared terms, yielding a U-shaped association), duration of fever (inverse association), treatment regimen (lower relapse risk with LAMB and ‘Other’ regimens compared with miltefosine), and severe anaemia (lower relapse risk with increased

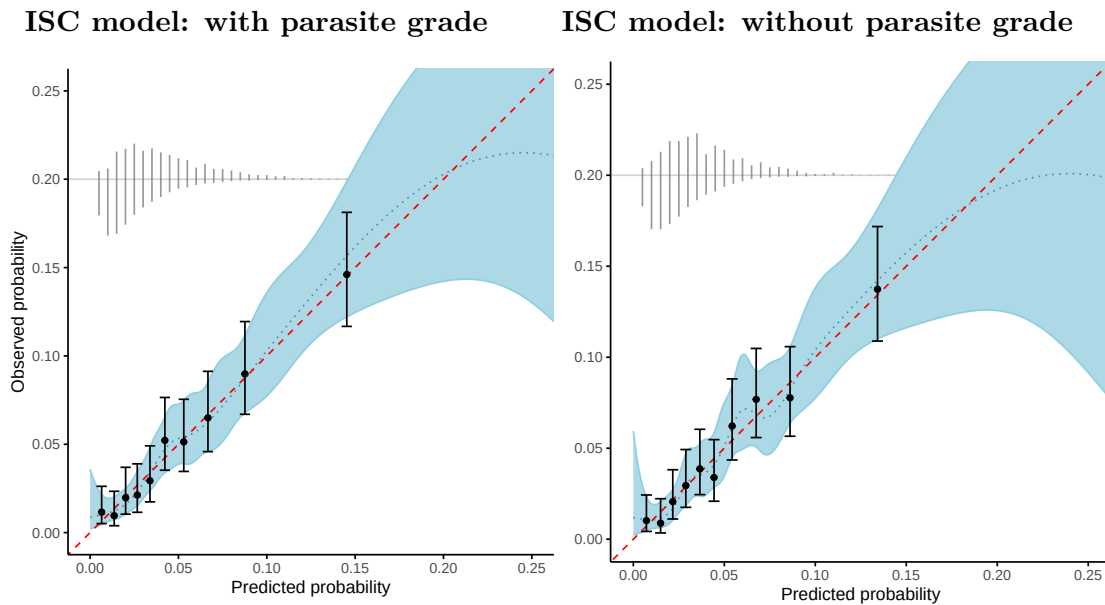


Figure 4.12: Calibration plots showing observed versus predicted probabilities for deciles of predicted probability. Red dashed line represents perfect calibration. Observed probabilities are presented with 95% confidence intervals (black error bars). A generalised additive model is fitted to show the smoothed mean observed probability (blue dotted line) with 95% confidence intervals (blue ribbon). Histograms, normalised by outcome, are overlaid to illustrate the distribution of relapses and cures across the expected probabilities.

severity). When parasite grade was included, it was retained and associated with a higher risk of relapse.

Discriminative performance was modest. The pooled apparent c-statistics were 0.70 (95% CI 0.66–0.74) for the model including parasite grade and 0.69 (95% CI 0.64–0.74) for the model excluding it, reducing after optimism correction to 0.68 and 0.67, respectively. Overall agreement between observed and predicted relapse risk was good across predictors, although significant between-study variation in calibration intercepts (calibration-in-the-large) was observed for both models.

The following section presents an analogous summary of the East Africa models using the same structure, enabling direct comparison with the ISC model results.

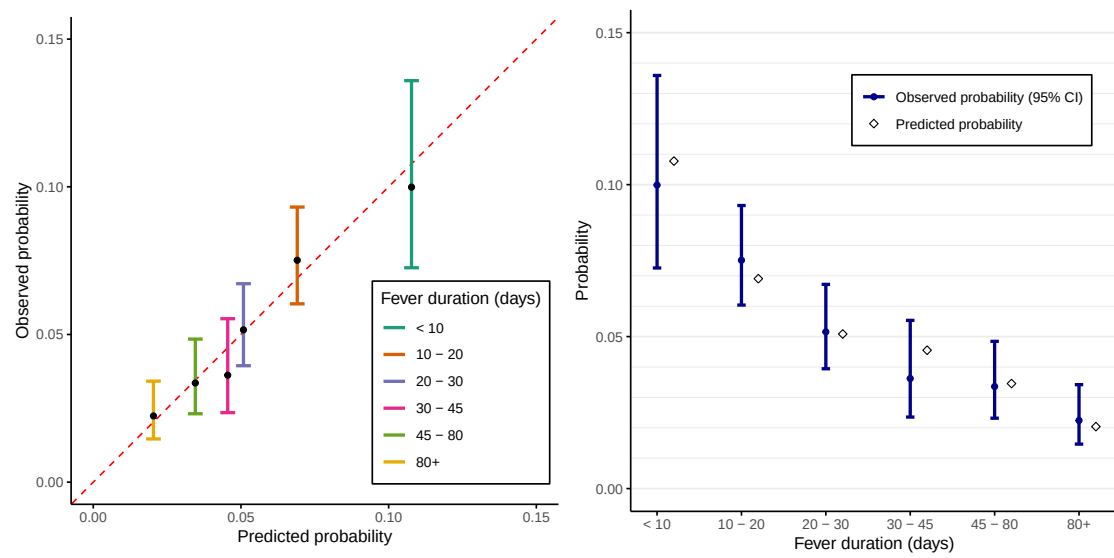


Figure 4.13: Calibration plots for different fever durations (model including parasite grade).

5

Results: East Africa Models

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5.1 Descriptive Analysis

A total of 2,095 patients from nine East African studies were identified after applying the eligibility criteria to the IDDO VL data platform[72, 121–123, 126, 291–294]. At the participant selection stage, 370 participants (13.5%, 370/2749) were excluded due to either having a positive HIV test, or lacking a HIV test but recruited from a site outside of Sudan (considered high-risk). 323 (11.7%, 323/2749) participants were excluded due to not achieving initial cure. A flow diagram is presented in Figure 5.1.

5.1.1 Study Characteristics

Important study characteristics are presented in Table 5.1.

A median of 138 patients per contributed study were included in model development, ranging from 29 to 638 patients (IQR: 92 to 289 patients). Five studies

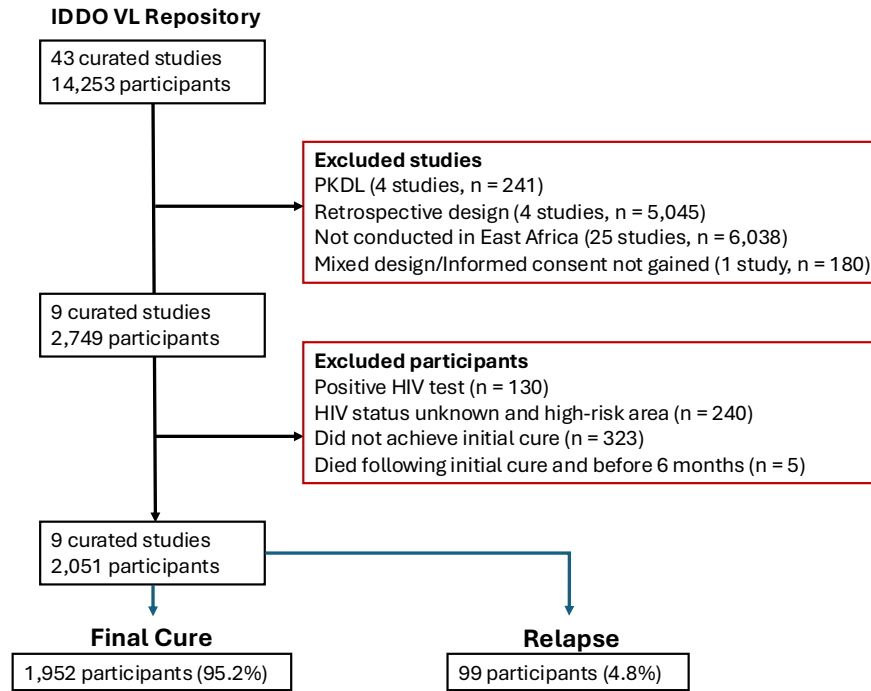


Figure 5.1: Flow diagram showing the studies and participants excluded from East Africa model development, following application of the eligibility criteria. HIV: human immunodeficiency virus; IDDO: Infectious Diseases Data Observatory; EA: East Africa; PKDL: post kala-azar dermal leishmaniasis; VL: visceral leishmaniasis.

recruited from sites across multiple countries (1,186 [57.8%] patients)[72, 121, 123, 126, 294], and the remaining four studies recruited from site(s) in single countries, including Sudan (two studies, 505 [24.6%] patients)[122, 293] and Ethiopia (two studies, 360 [17.6%] patients)[291, 292]. A full breakdown of patients by country is included in Section 5.1.2 below.

The majority of studies were sponsored by DNDi and conducted across the Leishmaniasis East Africa Platform (LEAP) (six studies, 1,226 [59.8%] patients, conducted between 2004 and 2016). These included LEAP 0104 — a large pivotal study comparing short course SSG & paromomycin to SSG or paromomycin monotherapy (three contributed studies, 967 [47.1%] patients)[72, 122, 123]. LEAP 0106 was a smaller study with the aim of identifying the minimum safe and effective dose of liposomal amphotericin B (LAMB) (one study, 92 [4.5%] patients)[121]. LEAP 0208 compared miltefosine with two different combination therapies (LAMB + miltefosine, LAMB + SSG) (138 [6.7%] patients)[294], and LEAP 0714 was a smaller allometric dosing study of miltefosine in paediatric patients (29 [1.4%] patients)[126].

The remaining studies were led by MSF and conducted between 1998 and 2005, including two studies comparing generic and branded SSG (577 [28.1%] patients)[291, 293], and a further study comparing SSG and miltefosine (248 [12.1%] patients)[292].

Table 5.1: Key characteristics of included studies from East Africa, ordered by lead author and year of publication/protocol. -: not reported; BD: twice daily (bis die); D: day(s); IM: intramuscular; IV: intravenous; LAMB: liposomal amphotericin B (Gilead formulation); MF: miltefosine; mg/kg: milligrams per kilogram; OD: once daily (omni die); PO: per os (oral); PM: paromomycin; SSG: sodium stibogluconate;

Study ¹	Title	Journal	Sponsor/ funding	Location(s)	Study design	Study arm(s) ⁴	Age (years)	Study period	n (model)	Relapses (%)
Hailu 2010[123]	LEAP 0104. Geographical variation in the response of visceral leishmaniasis to paromomycin in East Africa: a multicentre, open-label, randomized trial	PLoS Negl Trop Dis	DNDi/MSF	5 centres: Ethiopia (Gondar University Hospital, Arba Minch Hospital); Sudan (MSF Treatment Centre, Um el Khjer; Ministry of Health Hospital, Kassab); Kenya (CCR, KEMRI, Nairobi)	Multicentre, open-label, randomised trial	(1) SSG, 20 mg/kg, IV or IM, OD, 30 D; (2) PM, 15 mg/kg, IM, OD, 21 D; (3) Combination of PM and SSG, same dose, frequency and route, 17 D	4–60	2004–2008	289	21 (7.3)
Khalil 2014[121]	LEAP 0106. Safety and efficacy of single-dose versus multiple doses of AmBisome for treatment of visceral leishmaniasis in eastern Africa: a randomised trial	PLoS Negl Trop Dis	DNDi (multiple funders)	3 centres: Ethiopia (Gondar University Hospital, Arba Minch Hospital); Sudan (Ministry of Health Hospital, Kassab)	Multicentre, open-label, non-inferiority, randomised trial with adaptive design	LAMB, IV, either single-dose at (1) 7.5 mg/kg; (2) 10 mg/kg; (3) 12.5 mg/kg; (4) 15 mg/kg, or multiple dose at (5) 3 mg/kg, OD, D1-5, 14, 21 (total 21 mg/kg)	≥ 4	2009–2011	92	11 (12.0)
Mbui 2019[126]	LEAP 0714. Pharmacokinetics, Safety, and Efficacy of an Allometric Miltefosine Regimen for the Treatment of Visceral Leishmaniasis in Eastern African Children: An Open-label, Phase 2 Clinical Trial	Clin Infect Dis	DNDi (multiple funders)	2 clinical sites: Kachelibra, West Pokot County, Kenya; Amudat, Karamoja sub-region, Uganda	Open-label, phase 2, clinical trial	(1) MF, allometric dosing according to sex, height, and weight, BD, PO, 28 D ²	4–12	2015–2016	29	2 (6.9)
Musa 2010[122]	LEAP 0104. Paromomycin for the treatment of visceral leishmaniasis in Sudan: a randomized, open-label, dose-finding study	PLoS Negl Trop Dis	DNDi (multiple funders)	Ministry of Health Hospital, Kassab, Sudan	Open-label, phase 2, randomised, dose-finding study	(1) PM, 15 mg/kg, IM, OD, 28 D; (2) PM, 20 mg/kg, IM, OD, 21 D	4–60	2005–2006	40	8 (20.0)
Musa 2012[72]	LEAP 0104. Sodium stibogluconate (SSG) & paromomycin combination compared to SSG for visceral leishmaniasis in East Africa: a randomised controlled trial	PLoS Negl Trop Dis	DNDi/MSF (multiple funders)	6 centres: 5 centres described in Khalil 2014 (above) and Amudat Hospital, Uganda.	Multicentre, open-label, parallel-arm, randomised trial	(1) SSG, 20 mg/kg, IV or IM, OD, 30 D; (2) PM, 20 mg/kg, IM, OD, 21 D; (3) combination of SSG, 20 mg/kg, IV or IM, OD, 17 D and PM, 15 mg/kg, IM, OD, 17 D	4–60	2004–2010	638	30 (4.7)
Ritmeijer 2001[291]	Ethiopian visceral leishmaniasis: generic and proprietary sodium stibogluconate are equivalent; HIV co-infected patients have a poor outcome	Trans R Soc Trop Med Hyg	MSF	Temporary MSF treatment centre, Densha, Ethiopia	Open-label, pseudo-randomised controlled trial	(1) SSG (generic); ³ 20 mg/kg, IM, 30 D; (2) SSG (Pentostam, GlaxoWellcome), 20 mg/kg, IM, 30 D	all	1998–1999	112	1 (0.9)

continued on next page

Table 5.1: continued

Study ¹	Title	Journal	Sponsor/ funding	Location(s)	Study design	Study arm(s) ⁴	Age (years)	Study period	n (model)	Relapses (%)
Ritmeijer 2006[292]	A comparison of miltefosine and sodium stibogluconate for treatment of visceral leishmaniasis in an Ethiopian population with high prevalence of HIV infection	Clin Infect Dis	MSF	2 centres in Ethiopia: Humera Hospital, Mycadra Health Center	Open-label, randomised controlled trial	(1) SSG, 20 mg/kg/day, IM, OD, 30 D (extended if HIV positive); (2) MF, 100 mg/day, PO, OD, 28 D	≥ 15	2003–2005	248	6 (2.4)
Veeken 2000[293]	A randomized comparison of branded sodium stibogluconate and generic sodium stibogluconate for the treatment of visceral leishmaniasis under field conditions in Sudan	Trop Med Int Health	MSF	2 MSF treatment centres in Gedaref State, Sudan: Um Kuraa, Kassab	Open-label, pseudo-randomised controlled trial	(1) SSG (generic); ³ 20 mg/kg, IM, 30 D; (2) SSG (Pentostam, GlaxoWellcome), 20 mg/kg, IM, 30 D	all	1998–1999	465	4 (0.9)
Wasunna 2016[294]	LEAP 0208. Efficacy and Safety of AmBisome in Combination with Sodium Stibogluconate or Miltefosine and Miltefosine Monotherapy for African Visceral Leishmaniasis: Phase 2 Randomized Trial.	PLoS Negl Trop Dis	DNDi (multiple funders)	3 centres: Kenya (Kimalel Health Centre); Sudan (Dooka Hospital and Ministry of Health Hospital, Kassab)	Phase 2, open-label, non-comparative randomised trial (adaptive-sequential design)	(1) Combination of LAMB, 10 mg/kg, IV, single-dose, D1 and SSG, 20 mg/kg, IM, OD, D2-11; (2) Combination of LAMB, 10 mg/kg, IV, single-dose, D1, and MF, 2.5 mg/kg, PO, OD, D2-11; (3) MF ⁵ 2.5 mg/kg, PO, OD, 28 D	7–60	2010–2012	138	16 (11.6)

¹ Study name is composed of the lead author and year of publication.

² Refer to the supplementary material of the publication for allometric dosing table[126].

³ SSG tested and dispensed by International Dispensary Association, The Netherlands; manufactured by Albert David Ltd, India.

⁴ For both arms in Veeken 2010, Ritmeijer 2001, and Ritmeijer 2006, if positive initial test-of-cure, treatment would continue with SSG, including if previously taking MF, until two subsequent consecutive tests of cure, performed weekly, were negative.

⁵ Actual dosing ranged from 2.0–3.33 mg/kg/day after rounding to nearest 10 mg tablets; full regimen described in publication[294]

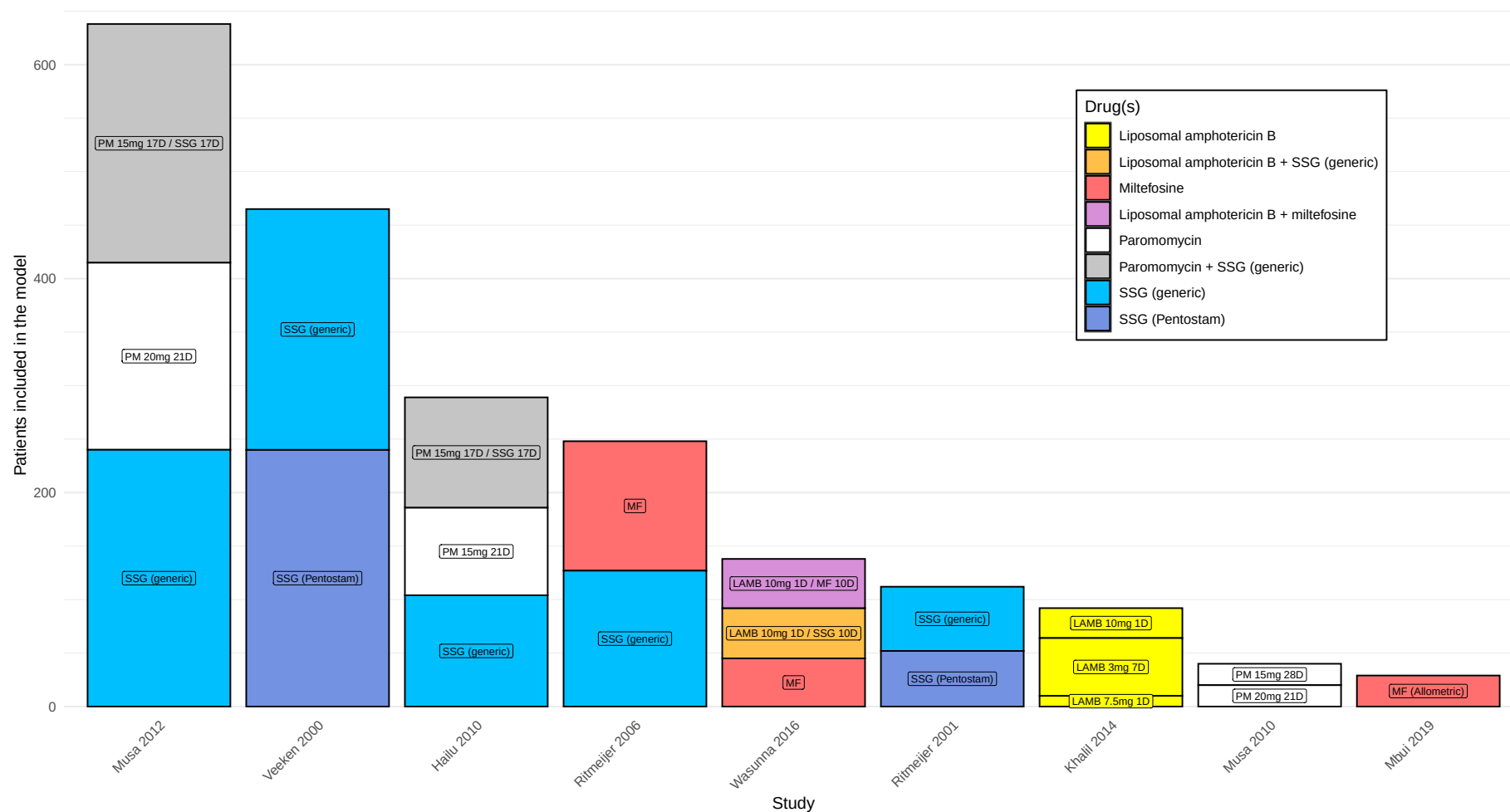


Figure 5.2: Bar chart showing the distribution of treatment regimens across contributing studies from East Africa. Drugs are colour-coded (see legend). Important distinguishing dosing information provided in the overlaid labels. Full treatment details presented in Table 5.1. D: days; LAMB: liposomal amphotericin B (Gilead); PM: paromomycin; MF: miltefosine; mg: milligrams/kilogram; SSG: sodium stibogluconate.

Study Design and Treatment Arms

All studies bar one allocated patients to more than one treatment arm (eight studies, 2,022 [98.6%] patients)[72, 121–123, 291–294]. Six studies randomly allocated patients to treatment arms (1,445 [70.5%] patients)[72, 121–123, 292, 294] and two studies were pseudo-randomised¹ (577 [28.1%] patients)[291, 293]. Three studies were described by the authors as Phase 2 trials (207 [10.1%] patients)[122, 126, 294].

Treatment arms are summarised in Figure 5.2. Treatment allocations at the patient level are described further in Section 5.1.2.

Study Eligibility Criteria

All but two studies described age limits in their inclusion criteria[72, 121–123, 126, 292, 294]. Two studies specified lower age limits only (≥ 15 years in one study; 248 [12.1%] patients, and ≥ 4 years in a further study; 92 [4.5%] patients). Three studies limited inclusion to patients between 4 and 60 years (967 [47.1%] patients)[72, 122, 123], and one study limited inclusion to patients between 7 and 60 years (138 [6.7%] patients)[294]. One study recruited children between 4 and 12 years (29 [1.4%] patients)[126].

Six studies (all DNDi, 1,226 [59.8%] patients) reported exclusion criteria based on platelet count ($\geq 40 \times 10^9/L$) and haemoglobin (≥ 40 g/L in one study, 92 [4.4%] patients[121], and ≥ 50 g/L in five studies, 1,134 [55.2%] patients[72, 122, 123, 126, 294]). Four studies described excluding patients with ‘severe’ VL (1,048 [51.1%] patients)[72, 121, 123, 126], and four studies excluded patients based on malnutrition severity (three studies excluded patients with severe protein and/or caloric malnutrition; 467 [22.8%] patients[122, 123, 294], and one study excluded children with weight-for-height or BMI-for-age z-scores < -3 ; 29 [1.4%] patients[126]). Seven studies (1,474 [71.9%] patients) excluded patients with severe concomitant illness and/or co-infection[72, 121–123, 126, 292, 294].

Four studies (299 [14.6%] patients) required a negative HIV test as a prerequisite for inclusion[121, 122, 126, 294]. The remaining studies either did not report HIV testing or HIV-related exclusion criteria (one study, 465 [22.7%] patients)[293], or permitted the inclusion of patients with VL/HIV co-infection (with variable testing strategies, 4 studies, 1,019 [49.7%] patients)[72, 123, 291, 292].

Of the eight studies that included adults (2,022 [98.6%] patients), four studies (1059/2022 [52.4%] patients) excluded women who were pregnant or lactating[72, 121–123], one study excluded women of childbearing age (138/2022 [6.8%] patients)[294], one study excluded all female patients regardless of age (248/2022

¹Based on whether the direct agglutination test (DAT) batch number was odd or even.

[12.3%] patients)[292],² and two studies did not report excluding women based on pregnancy or lactating status(577/2051 [28.1%] patients)[291, 292].

Patients with prior VL treatment were excluded in seven studies; either within 6 months in five studies (1,088 [53.0%] patients)[72, 121–123, 126], or at any prior time point in two studies (577 [28.1%] patients)[291, 293].

Full study-specific eligibility criteria can be found in the [Supplementary Material](#).

Diagnostic Criteria

A positive tissue aspirate was required for VL confirmation in all six DNDi studies (1,226 [59.8%] patients), with tissue type including bone marrow, spleen, or lymph node, depending on practice at the recruiting site. All studies required clinical evidence of VL, although the exact working varied across studies (presented in [Supplementary Material](#)). The three earlier MSF studies (825 [40.2%] patients)[291–293] confirmed VL with a DAT (titre $\geq$ 1:6,400), with borderline titres (1:800–1:3,200) requiring a confirmatory tissue aspirate. One study (112 [5.5%] patients) reported some patients receiving a VL diagnosis on clinical grounds due to occasional disruption to the supply of DAT kits[291].

Relapse

Relapse was directly defined in two studies (288 [14.0%] patients)[122, 292], and could be inferred in the remaining studies through definitions of definite cure, initial cure, and treatment failure. All relapse cases were confirmed with a tissue aspirate, with aspirate results available in the contributed IPD. All studies adopted an active follow-up strategy following test-of-cure. However, loss-to-follow-up rates varied across studies, with particularly high rates seen in the earlier pragmatic MSF studies[291–293]. Four studies (1,059 [51.6%] patients)[72, 121–123] performed routine 6 month aspirates.

Initial Cure

Definitions of initial cure varied considerably across studies, and are presented in the [Supplementary Material](#). In the six DNDi studies, test-of-cure occurred at variable time points within 31 days of treatment initiation, depending on the treatment arm (range 18–31 days)[72, 121–123, 126, 294]. In the three MSF studies, treatment was extended with SSG until two consecutive tests-of-cure were negative[291–293]. Routine aspirates were performed in all studies, although two studies explicitly

²Due to concern re: teratogenicity of miltefosine.

stated that aspirates were only performed in the presence of palpable splenomegaly or lymphadenopathy (360 [17.6%] patients)[291, 292].

In five studies (1,197 [58.4%] patients), those showing clinical improvement despite a *positive* initial test-of-cure aspirate, regardless of parasite grade, were considered to have failed the outcome of initial cure[72, 121–123, 294]. However, these patients remained in the study and could subsequently achieve definite cure at 6 months if clinical symptoms did not recur and a subsequent (e.g. 3 or 6 month) aspirate were negative. These patients, sometimes referred to as ‘slow-responders’, were included in model development despite not meeting the study definition of initial cure.

5.1.2 Patient Characteristics

Overall (marginal) distributions of categorical and continuous variables are tabulated in Tables 5.2 and 5.3, respectively. These distributions are also displayed graphically in Figure 5.4 for bar charts of categorical variables, and Figures 5.5 and 5.6 for histograms of non-laboratory and laboratory variables, respectively. Study-specific distributions are presented for age, sex, and relapse in Figure 5.3. In the Appendix, study-specific distributions of all categorical and continuous variables are presented in Figures D.1 and D.2–D.14, respectively.

Patient numbers and proportions presented in this section exclude missing data. See Section 5.1.4 for further information on missing data.

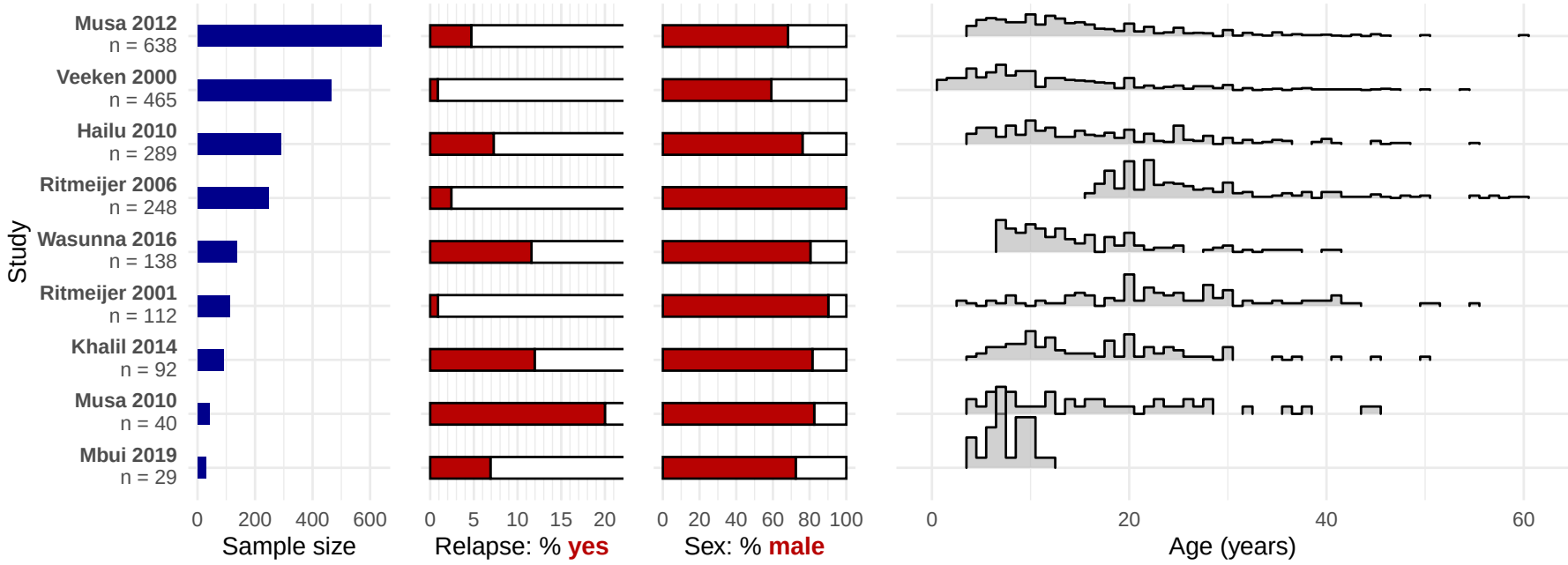


Figure 5.3: Graphical summary of East Africa study-specific sample sizes and distributions of relapse status, sex, and age.

Categorical Variables

Almost three quarters of patients (1,519 [74.1%]) were male, ranging from 59.1%[293] to 100%[292] at the study level.

Relapse within 6 months was reported in 99 (4.8%) patients overall, and varied considerably at the study level from 0.9%[291, 293] to 20%[122].

As defined in Section 3.3.2, almost a quarter of all patients were severely malnourished (509 [24.9%] patients). Moderate malnutrition affected 800 (39.1%) patients, and 735 (36.0%) had mild/normal malnutrition. Higher rates of severe malnutrition were seen in the two early MSF studies conducted in the late 1990s (33.1%)[291, 293] (presented in Appendix Figure D.1).

Severe anaemia affected 999 (48.8%) of patients overall.

A variety of treatment regimens were investigated across the studies. At the patient level, the single most common drug used was SSG (either generic or branded) — used as monotherapy in 1,048 (51.1%) patients, or as part of combination therapy with paromomycin or LAMB in 326 (15.9%) and 47 (2.2%) patients, respectively. The second most frequently used drug was paromomycin (received by 623 [30.3%] of patients overall) followed by LAMB (185 [9.1%] patients) and miltefosine (195 [9.5%] patients). The full range of treatment regimens, including dosing, are presented in Figure 5.2.

Parasite grade ranged from 1+ to 6+, with 1+ being the most common (511 [36.8%] patients who underwent tissue aspirate), followed by 2+ (237 [17.1%] patients), and decreasing to 77 (5.5%) patients with 6+. Tissue aspirate source was not reported in 702 (50.5%) of patients overall. Where reported, 393 (29.1%) were from the spleen, 163 (12.1%) were from lymph nodes, and 131 (9.7%) from bone marrow.

Approximately half of all patients were recruited from sites in Sudan (1,104 [53.8%] patients), followed by Ethiopia (695 [27.8%] patients), Kenya (227 [11.1%]), and Uganda (25 [1.2%] patients).

Continuous Variables

The median age was 14 years, ranging from 1 to 60 years (IQR: 9 to 22 years). Age distributions by study are presented in Figure 5.3, showing right-skewed distributions reflecting the study-specific age inclusion criteria. 129 (6.3%) of patients were under 5 years at the time of recruitment.

In adults the median BMI was 17.6 kg/m² (IQR: 16.3 to 18.7 kg/m²). In children (≥ 5 and < 19 years), the median BMI-for-age z-score was -2.32 (IQR: -3.10 to -1.50), and in younger children (< 5 years) the median weight-for-height z-score was -2.36 (IQR: -3.12 to -1.58) (data available for 129 patients).

Variable	Overall (%) n = 2,051	Final cure (%) n = 1,952	Relapse (%) n = 99
Sex			
Female	532 (25.9)	503 (25.8)	29 (29.3)
Male	1,519 (74.1)	1,449 (74.2)	70 (70.7)
Malnutrition			
Normal/mild	735 (35.8)	710 (36.4)	25 (25.3)
Moderate	800 (39.0)	760 (38.9)	40 (40.4)
Severe	509 (24.8)	476 (24.4)	33 (33.3)
(Missing)	7 (0.3)	6 (0.3)	1 (1.0)
Anaemia			
Non-severe	1,049 (51.1)	1,015 (52.0)	34 (34.3)
Severe	999 (48.7)	934 (47.8)	65 (65.7)
(Missing)	3 (0.1)	3 (0.2)	0 (0.0)
Parasite grade			
1+	511 (24.9)	488 (25.0)	23 (23.2)
2+	237 (11.6)	219 (11.2)	18 (18.2)
3+	192 (9.4)	181 (9.3)	11 (11.1)
4+	179 (8.7)	169 (8.7)	10 (10.1)
5+	193 (9.4)	180 (9.2)	13 (13.1)
6+	77 (3.8)	60 (3.1)	17 (17.2)
(Missing)	662 (32.3)	655 (33.6)	7 (7.1)
Aspirate source¹			
Bone	131 (9.4)	110 (8.6)	21 (22.8)
Spleen	393 (28.3)	369 (28.9)	24 (26.1)
Lymph node	163 (11.7)	151 (11.8)	12 (13.0)
(Missing)	702 (50.5)	649 (50.7)	35 (38.0)

¹ Denominator for % in aspirate source: number of patients with documented parasite grade (overall: 1,389; final cure: 1,279; relapse: 92).

Table 5.2: Summary of categorical candidate predictors and parasite source across contributed studies from East Africa. Missing data are described where present.

The median spleen size was 8 cm (IQR: 5 to 11 cm), and ranged from 0 to 30 cm. Where measured, 138 (6.9%) patients had non-palpable spleens.

The median duration of fever prior to recruitment was 40 days (IQR: 25 to 91 days), and ranged from 3 to 761 days.

Distributions of laboratory results are presented in Table 5.3 and Figure 5.5.

A correlation matrix showing associations between continuous variables is presented in Appendix Figure D.15, and between continuous and categorical variables in Appendix Figure D.16. These correlations are considered further in the Discussion section.

Variable	Overall (n = 2,051)		Final cure (n = 1,952)		Relapse (n = 99)	
	Median (IQR)	Missing ¹ (%)	Median (IQR)	Missing (%)	Median (IQR)	Missing ¹ (%)
Age (years)	14 (9 – 22)	1 (0.0)	15 (9 – 22)	1 (0.1)	12 (9 – 20)	0 (0.0)
Height (cm)	153 (126 – 168)	9 (0.4)	154 (126 – 168)	8 (0.4)	143 (127 – 164)	1 (1.0)
Weight (kg)	35 (21 – 49)	1 (0.0)	35 (21 – 49)	1 (0.1)	28 (20.6 – 44.5)	0 (0.0)
BMI (kg/m ²) ²	17.56 (16.29 – 18.69)	4 (0.5)	17.54 (16.29 – 18.71)	3 (0.4)	17.63 (16.65 – 18.22)	1 (3.2)
BMI z-score ³	-2.32 (-3.10 – -1.50)	2 (0.2)	-2.30 (-3.09 – -1.47)	2 (0.2)	-2.58 (-3.43 – -1.89)	0 (0.0)
WFH z-score ⁴	-2.36 (-3.12 – -1.58)	0 (0.0)	-2.32 (-3.01 – -1.55)	0 (0.0)	-3.42 (-3.89 – -3.24)	0 (0.0)
Spleen size (cm)	8 (5 – 11)	69 (3.4)	8 (5 – 11)	69 (3.5)	7 (4 – 10)	0 (0.0)
Fever duration (days)	40.4 (25.0 – 91.3)	349 (17.0)	40.4 (25.0 – 91.3)	320 (16.4)	30.4 (20.3 – 60.9)	29 (29.3)
Parasite grade	2 (1 – 4)	662 (32.3)	2 (1 – 4)	655 (33.6)	3 (2 – 5)	7 (7.1)
WBC ($\times 10^9$ /L)	2.5 (1.8 – 3.5)	887 (43.2)	2.5 (1.8 – 3.5)	871 (44.6)	2.7 (1.8 – 3.6)	16 (16.2)
Platelets ($\times 10^9$ /L)	106 (73 – 157)	890 (43.4)	105 (73 – 155)	874 (44.8)	110 (74.5 – 171)	16 (16.2)
Haemoglobin (g/L)	79 (67 – 91)	2 (0.1)	80 (67 – 92)	2 (0.1)	72 (62 – 83.5)	0 (0.0)
ALT (IU/L)	21 (14 – 31)	920 (44.9)	21 (14 – 31)	901 (46.2)	19 (14 – 30)	19 (19.2)
Creatinine (μ mol/L)	61.9 (44.2 – 88.4)	826 (40.3)	61.9 (44.2 – 88.4)	815 (41.8)	61.9 (44.2 – 88.4)	11 (11.1)

¹ Denominator for missing %: total number of patients in respective group (overall, relapse or final cure). For measures of malnutrition (BMI, BMI-for-age z-score, and weight-for-height z-score), see below.

² Denominator for missing %: number of patients aged ≥ 19 years, n = 756 (relapse: 30, final cure: 726).

³ Denominator for missing %: number of patients aged 5–18 year inclusive, n = 1,165 (relapse: 65, final cure: 1,100).

⁴ Denominator for missing %: number of patients aged < 5 years, n = 129 (relapse: 4, final cure: 125).

Table 5.3: Summary of continuous candidate predictors across contributed studies from East Africa. Including additional variables used for the derivation of malnutrition status (height, weight, BMI, BMI-for-age z-score, weight-for-height z-score). ALT: alanine aminotransferase; BMI: body mass index; cm: centimetres; IQR: inter-quartile range, IU: international units; kg: kilograms; L: litres; m: metres; WBC: white blood cells; WFH: weight-for-height; g: grams; μ mol: micromoles.

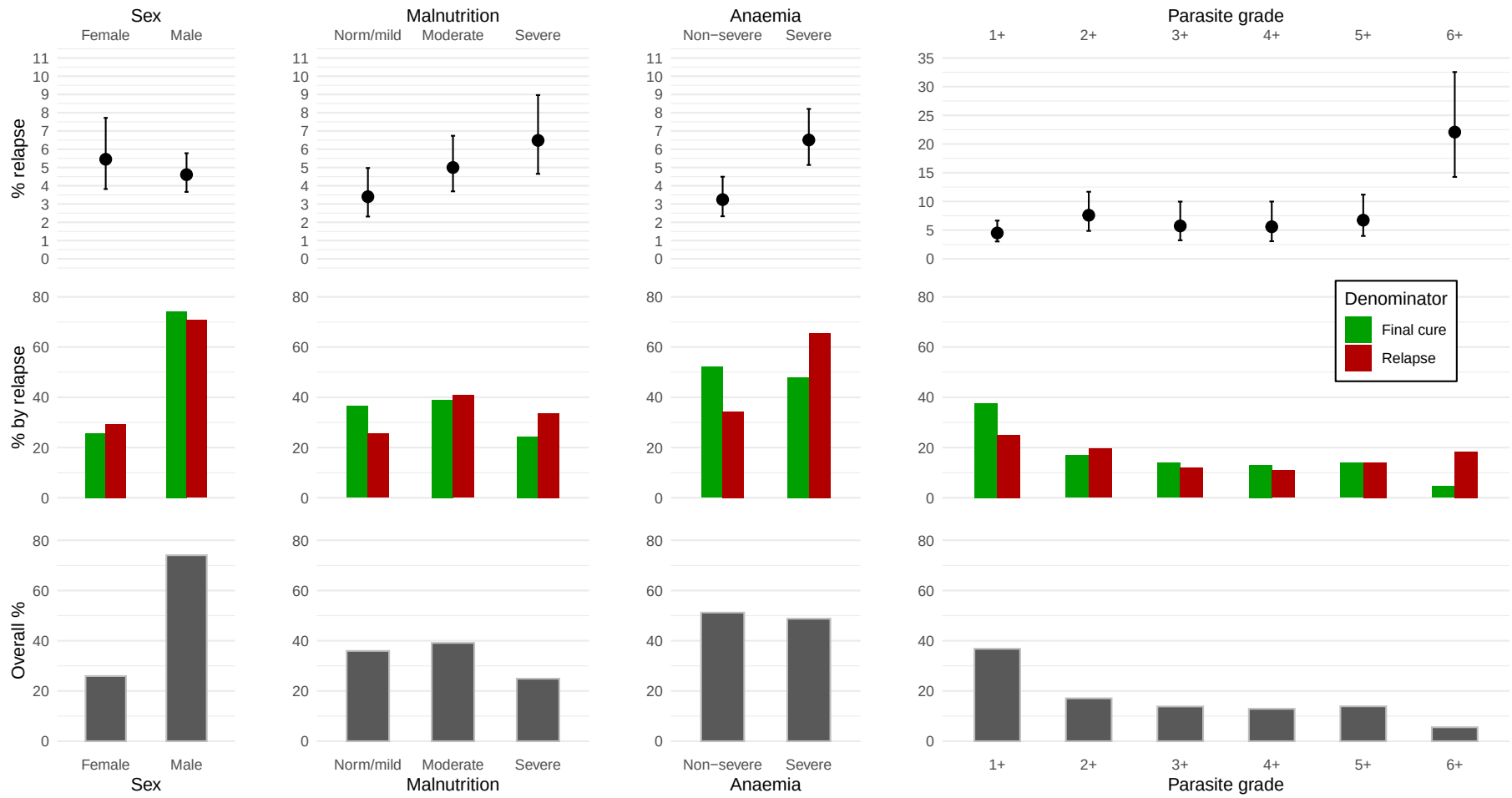


Figure 5.4: Marginal distributions and predictor-outcomes relationships for categorical candidate predictors. Excluding missing data. 95% binomial confidence intervals calculated using the Wilson method. Note: for parasite grade relapse %, y -axis is rescaled to accommodate increased risk in 6+ group. Norm: normal.

5.1.3 Univariable Associations

Unadjusted relationships between variables (including all candidate predictors) and relapse risk are presented both in tabular form (Tables 5.2, 5.3) and visually alongside their distributions (Figure 5.4 for categorical variables, and Figures 5.5 and 5.6 for non-laboratory and laboratory variables, respectively). Predictor-outcome relationships are presented on the log-odds (logit) scale for continuous candidate predictors in Appendix Figure D.17.

On review of the associations between the categorical predictors and relapse risk (upper facet row, Figure 5.4), clear trends were found for malnutrition, anaemia, and parasite grade. Increased relapse risk was seen with higher malnutrition severity, ranging from approximately 3.5% for mild/normal to 6.5% for severe categories. Perhaps more marked, with non-overlying confidence intervals, was the increased relapse risk seen in the severe anaemia group compared to the non-severe anaemia group, from approximately 3% to 6.5%. For relapse cases, 65/99 (65.7%) had severe anaemia, compared to 476/1946 (24.5%) of final cure cases (excluding missing data).

With parasite grade, a substantial increase in relapse risk was seen for patients with a 6+ grade compared to < 6+ grades, increasing from approximately 5–7.5% to over 20%. Among relapse cases, 17/92 (18.5%) of patients had a 6+ grade aspirate, compared to 60/1297 (4.6%) of the final cure cases (after excluding missing data).

Clear trends were observed between all non-laboratory continuous variables and relapse risk (Figure 5.5). For age, relapse risk followed a non-linear trend, peaking at approximately 10 years and decreasing to a nadir in the early 20s. Fluctuations in risk with height and weight were also noted, mirroring the relationship between age and relapse. Less marked downward trends between relapse risk and spleen size and fever duration were observed.

Among the laboratory continuous variables (Figure 5.6), a pronounced inverse relationship between relapse risk and haemoglobin was demonstrated, concordant with the aforementioned increased risk seen in patients with severe vs. non-severe anaemia. Appreciable upwards trends between relapse risk and both WBC count and platelet count were also seen, although with uncertain significance.

A correlation matrix showing associations between continuous variables is presented in Appendix Figure D.15, and between continuous and categorical variables in Appendix Figure D.16. These correlations are considered further in the Discussion section.

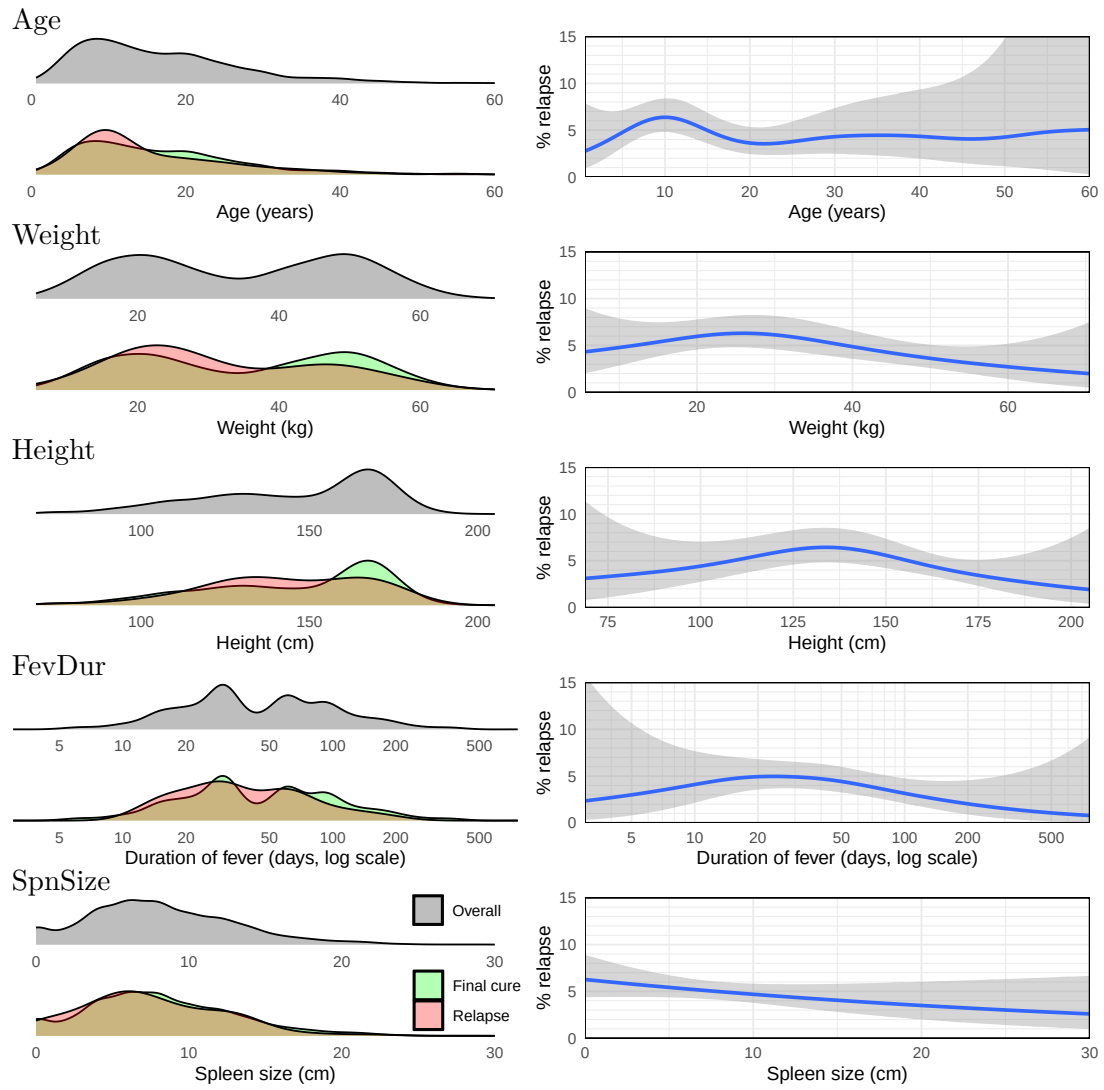


Figure 5.5: Marginal distributions and predictor-outcome relationships for continuous non-laboratory candidate predictors. FevDur: duration of fever; SpnSize: spleen size. For each candidate predictor, left upper panel shows the overall density across studies and the left lower panel shows overlapping densities normalised by relapse status. The right panel shows a univariable generalised additive model spline fit, with 95% confidence interval, of relapse.

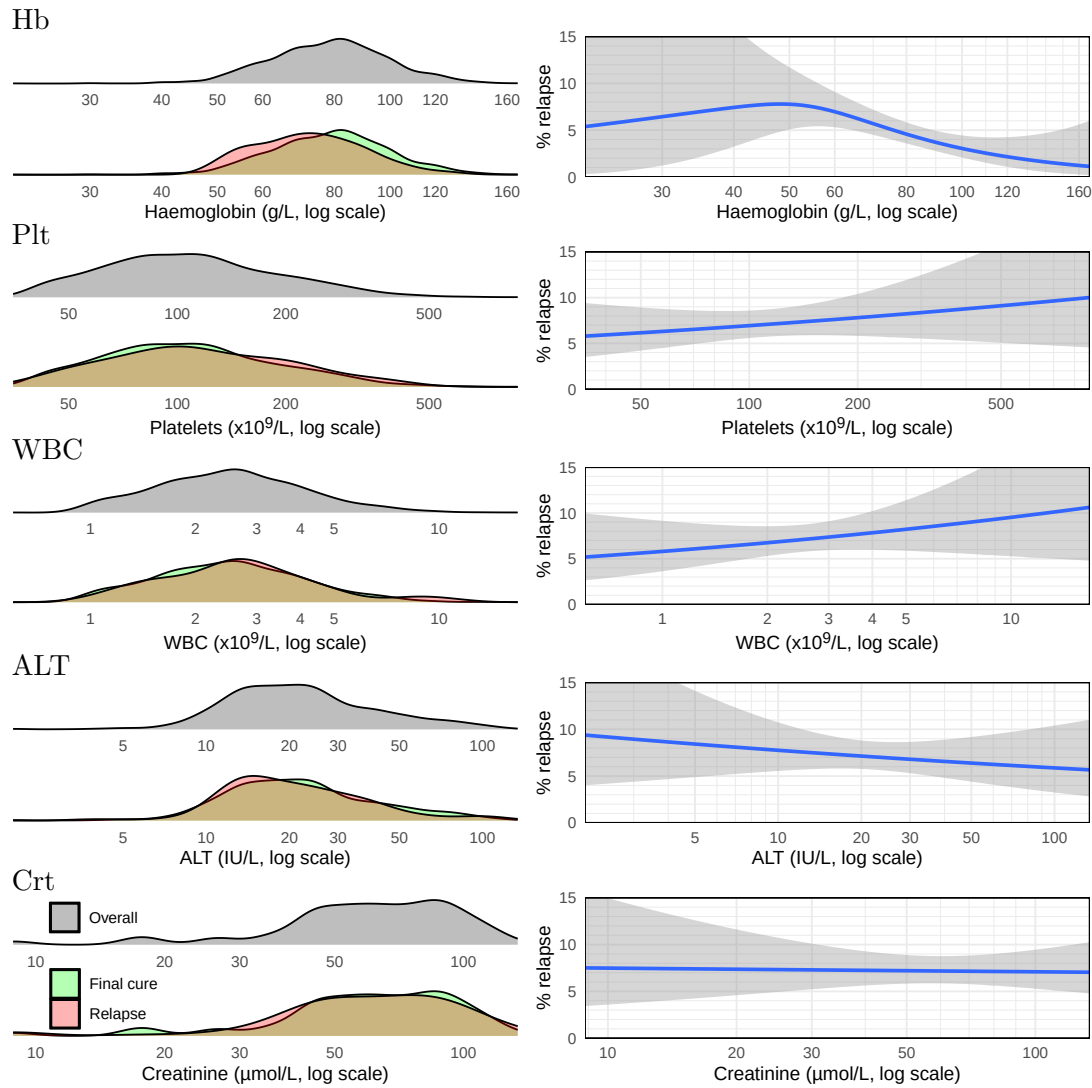


Figure 5.6: Marginal distributions and predictor–outcome relationships for continuous laboratory candidate predictors. All predictors presented on log scale. Hb: haemoglobin; Plt: platelet count; WBC: white blood cell count; ALT: alanine aminotransferase; Crt: creatinine. For each candidate predictor, left upper panel shows the overall density across studies and the left lower panel shows overlapping densities normalised by relapse status. The right panel shows a univariable generalised additive model spline fit, with 95% confidence interval, of relapse.

5.1.4 Missing Data

Missing data were common, affecting one or more candidate predictors in 1,168 (56.9%) patients. 215 (10.5%) patients were missing exactly one candidate predictor. As appreciated from Figure 5.7, the majority of missingness originated from the earlier MSF studies, where ALT, platelet count, WBC count, creatinine, and parasite grade were not routinely collected, resulting in almost one-third of all patients (634 [30.9%] patients) missing 5 candidate predictors.

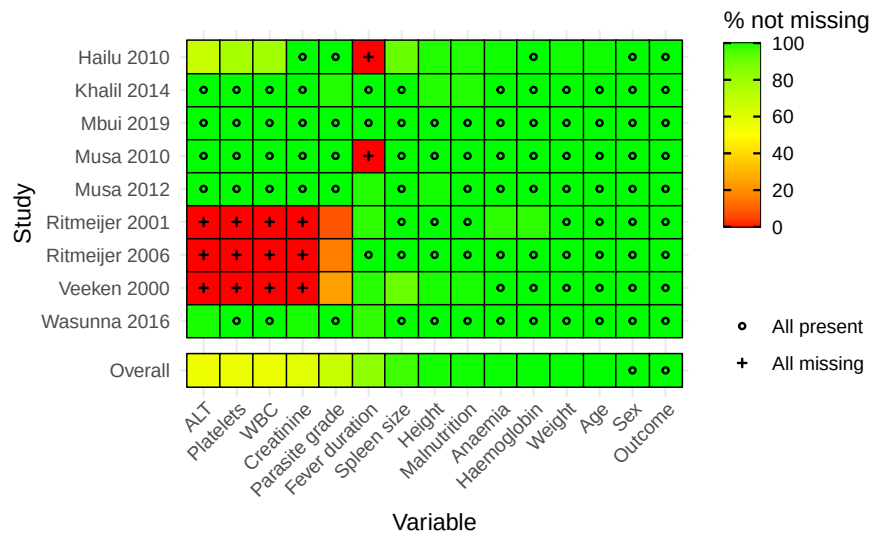


Figure 5.7: Density plot illustrating the amount of missing data overall and across contributing studies from East Africa. Study ordered by lead author and publication year. Variables ordered by amount of missingness. ALT: alanine aminotransferase; WBC: white blood cells.

Across all studies, ALT, platelet count, WBC count, and creatinine were missing in 40–45% of all patients. Parasite grade was missing for approximately one-third of the total cohort (662 [32.3%] of patients).

Multiple Imputation

Diagnostic plots for the 30 imputed datasets are available in the [Supplementary Material](#). On review of the trace plots, there was good evidence for convergence of the imputation model. Density and scatter plots revealed overall good correlation between the imputed and original variables.

5.2 Model Results

As described in the Methodology, two separate prediction models were developed from the East Africa IPD; one including parasite grade as a candidate predictor, and one excluding parasite grade.

5.2.1 Model Specification and Coefficient Estimates

Final (retained) predictors across both models are presented in Figure 5.8. Full specification of the final models, including intercept terms, p-values, and predictor transformations, are presented in Appendix Tables D.1 and D.2.

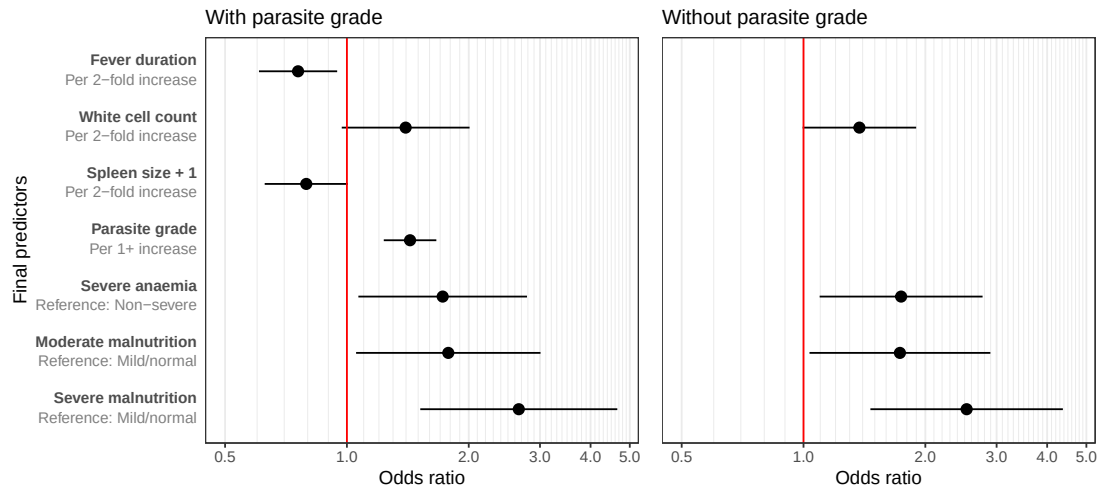


Figure 5.8: Forest plot of adjusted odds ratios with 95% confidence intervals for final model predictors. Odds ratios are displayed on a logarithmic scale.

Shared final predictors across both models included WBC count, anaemia, and malnutrition. The effect size (odds ratios) between these predictors and relapse are approximately the same across both models, and are described here for the model with parasite grade. For WBC count, the relapse odds increases by approximately 40% per doubling of the WBC count (OR: 1.40, 95% CI: 0.97–2.01). Increasing relapse odds are also seen in patients with severe vs. non-severe anaemia (OR: 1.72, 95% CI: 1.07–2.79), and in patients with moderate and severe malnutrition (OR: 2.66, 95% CI: 1.52–4.66 comparing severe vs. mild/normal malnutrition). For the similar estimates in the model without parasite grade please refer to Appendix Table D.2 and aforementioned figures.

In the model that considered parasite grade, three additional predictors were retained: parasite grade, spleen size, and fever duration. Odds of relapse were predicted to increase by over 40% per 1+ increase (OR: 1.43, 95% CI: 1.23–1.66). As observed in the unadjusted predictor-relapse relationships, inverse relationships were seen with spleen size and fever duration. Relapse risk decreased by approximately 25% per doubling of fever duration (OR: 0.75, 95% CI: 0.61–0.95) and by approximately 20% per doubling of [spleen size + 1], for example, from 2 to 5 cm, or 4 to 9 cm (OR: 0.79, 95% CI: 0.63–1.01).

As presented in Appendix Tables D.1 and D.2, considerable between-study heterogeneity was present, with pooled ICC estimates of over 20% in both models.

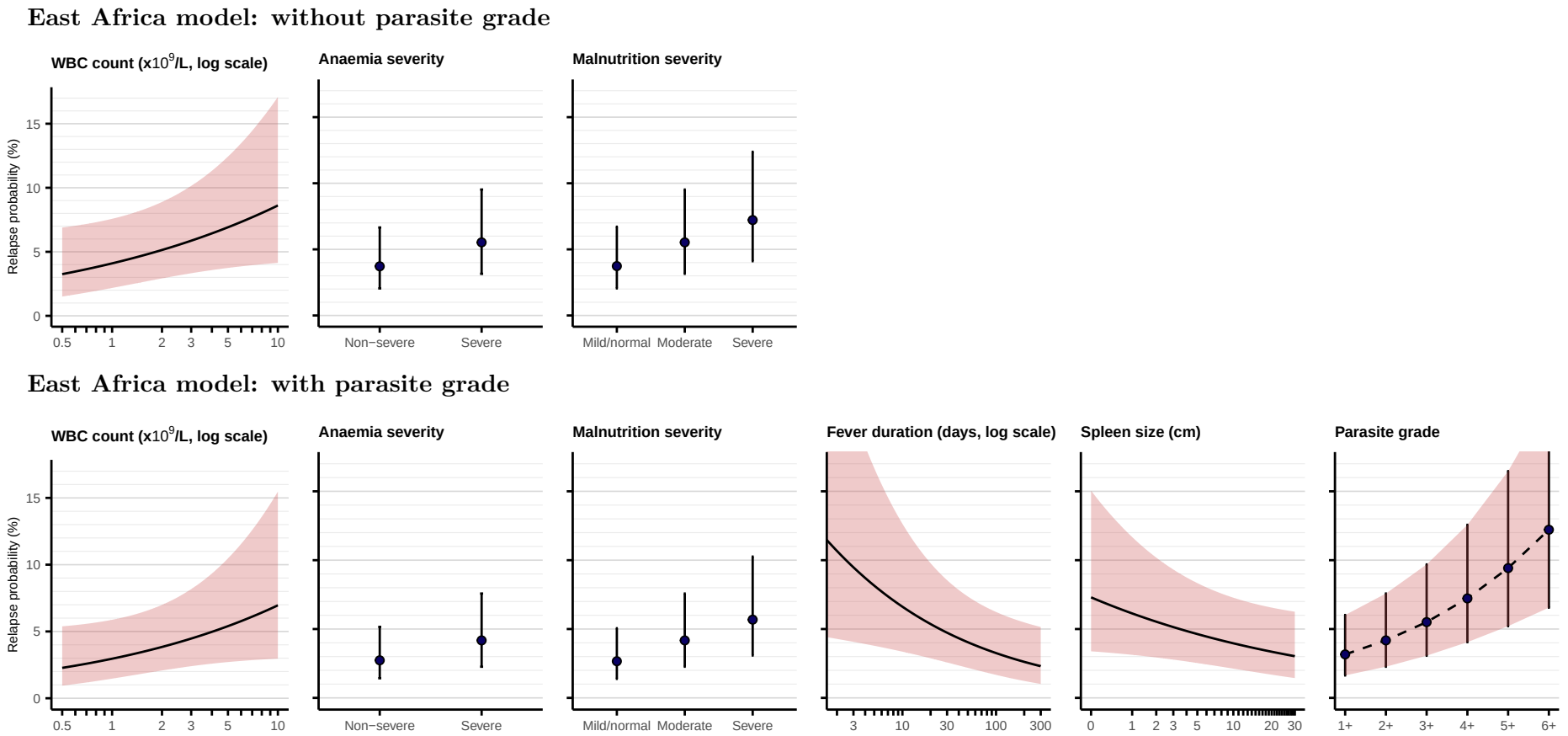


Figure 5.9: Adjusted associations between final predictors and predicted relapse probability, as estimated from the final East Africa prognostic models. Probabilities were calculated from optimism-adjusted models and following logistic recalibration (intercept-term only) to data contributed from Musa 2012[72]. Where not varying in the plot, predictions are standardised to a representative reference participant: median age (13 years), median fever duration (45 days), median white blood cell count ($2.5 \times 10^9/\text{L}$), with severe anaemia, moderate malnutrition, and — for the model including parasite grade — a median parasite count of 2+.

5.2.2 Model Performance and Internal Validation

Apparent and optimism-adjusted performance measures are presented in Table 5.4. After excluding studies with ≤ 5 events, the overall c-statistics were estimated at 0.72 (95% CI: 0.63–0.82) and 0.66 (95% CI: 0.56–0.76) in the models including and excluding parasite grade, respectively. After adjusting for optimism, these estimates shrank to 0.68 and 0.61, respectively. Both models showed some evidence of between-study heterogeneity in discrimination ($I^2 = 46.6\%$, $p = 0.11$ for the model including parasite grade, and $I^2 = 53.1\%$, $p = 0.056$ for the model excluding parasite grade), as illustrated in Figure 5.10 and Appendix Figure D.18.

	Performance estimate (95% CI)	Average optimism	Optimism-adjusted performance
Model: with PG			
C-statistic	0.72 (0.63–0.82)	0.046	0.68
Calibration slope	1.03 (0.64–1.42)	0.224	0.81
Model: without PG			
C-statistic	0.66 (0.56–0.76)	0.050	0.61
Calibration slope	1.08 (0.47–1.68)	0.329	0.75

Table 5.4: Apparent and optimism-adjusted performance measures. Abbreviations: CI: confidence interval; PG: parasite grade

Optimism in calibration slope was moderate to high, estimated at 0.22 and 0.33 for the models including and excluding parasite grade, respectively. Uniform shrinkage factors were 0.81 and 0.75, respectively (Appendix Tables D.1 and D.2). There was no evidence for significant between-study heterogeneity in calibration slope.

Calibration intercepts were shown to vary significantly across the contributing studies for both models — with $I^2 = 89.0\%$ for the model including parasite grade, and $I^2 = 90.7\%$ for the model excluding parasite grade (both $p < 0.0001$ for test of heterogeneity).

Forest plots showing the heterogeneity of calibration performance measures across the contributing studies are presented in Figure 5.11 (model including parasite grade) and Appendix Figure D.19 (model excluding parasite grade).

Calibration plots demonstrated concordance between predicted and observed relapse risk across both models (see Figure 5.12). Concordance is also demonstrated when comparing predicted and observed probabilities for malnutrition severity (Figure 5.13 for the model including parasite grade and Appendix Figure D.27 for the model excluding parasite grade). Calibration plots for all final predictors are presented in the Appendix Figures D.20 to D.28.

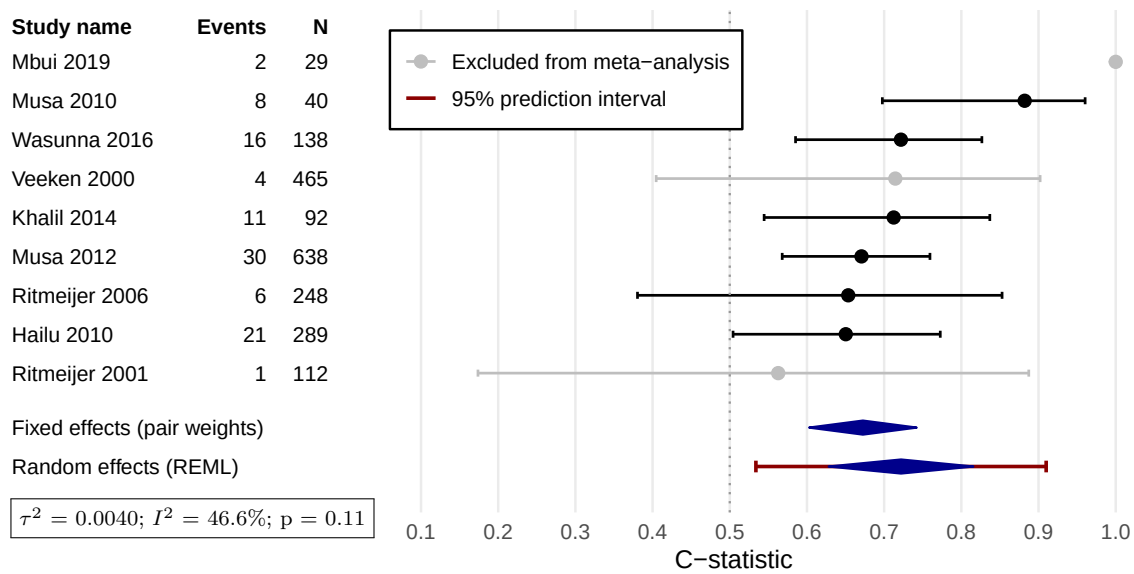


Figure 5.10: Forest plot showing individual and pooled study c-statistics, for the model **including** parasite grade. For the model excluding parasite grade refer to Appendix D.18. Pooled c-statistics are presented from both fixed-effects and random-effects meta-analysis models, after excluding studies with ≤ 5 relapse events. Study-specific confidence intervals with few events should be interpreted with caution (as discussed in the Methodology 3.3.7). Blue diamonds: pooled summary estimates with 95% confidence intervals. For Mbui 2019, all bootstrapped c-statistic estimates were 1 exactly and therefore no variance is presented.

The distribution of final predictor selection across the 2×500 bootstrap models are presented in Appendix Table D.3. When included as a candidate predictor, parasite grade consistently selected for inclusion across almost all bootstrap models (499 [99.8%] bootstrap models). In both models, the selection of malnutrition was also consistently seen (477 [95.4%] and 461 [92.2%] bootstrap models when including and excluding parasite grade, respectively). WBC count was the least stable predictor included in both models (254 [50.8%] and 188 [37.6%] bootstrap models when including and excluding parasite grade, respectively).

5.3 Summary

In summary, two prognostic models predicting relapse were fitted to the East Africa IPD — one including parasite grade as a candidate predictor and one excluding it. Both final models retained the same three core predictors: WBC count (higher relapse odds with increasing count), severe anaemia (higher relapse odds compared with non-severe anaemia), and malnutrition (higher relapse odds with increasing severity). When parasite grade was included, it was retained and strongly associated

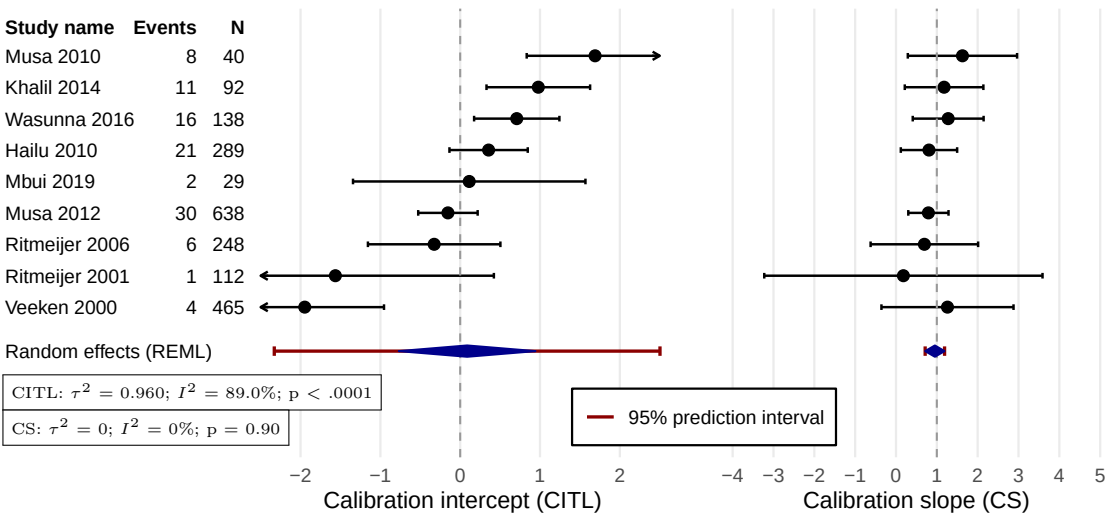


Figure 5.11: Forest plots showing individual and pooled study calibration measures for the model **including** parasite grade. Left: calibration intercept (calibration-in-the-large, CITL); Right: calibration slope (CS). Blue diamonds: summary estimates with 95% confidence intervals. Calibration slope for Mbui 2019 not shown due to small sample size causing failure of model convergence.

with higher relapse risk, and two additional clinical predictors were also selected: spleen size and duration of fever, both showing inverse associations with relapse risk.

Discriminative performance was modest overall and better when parasite grade was available. The pooled apparent c-statistics were 0.72 (95% CI 0.63–0.82) for the model including parasite grade and 0.66 (95% CI 0.56–0.76) for the model excluding it, reducing after optimism correction to 0.68 and 0.61, respectively. Calibration was broadly concordant between observed and predicted relapse risk across predictors, although substantial between-study variation in calibration intercepts (calibration-in-the-large) was observed for both models, alongside moderate-to-high optimism in calibration slope.

In the next chapter, the East Africa and ISC models will be explored through a more critical lens. Reflections will be made on both the clinical plausibility of the identified predictor-outcome relationships and on the practical considerations of model implementation in routine conditions, including how they can be used to improve clinical decision making in the context of elimination.

East Africa model: with parasite grade East Africa model: without parasite grade

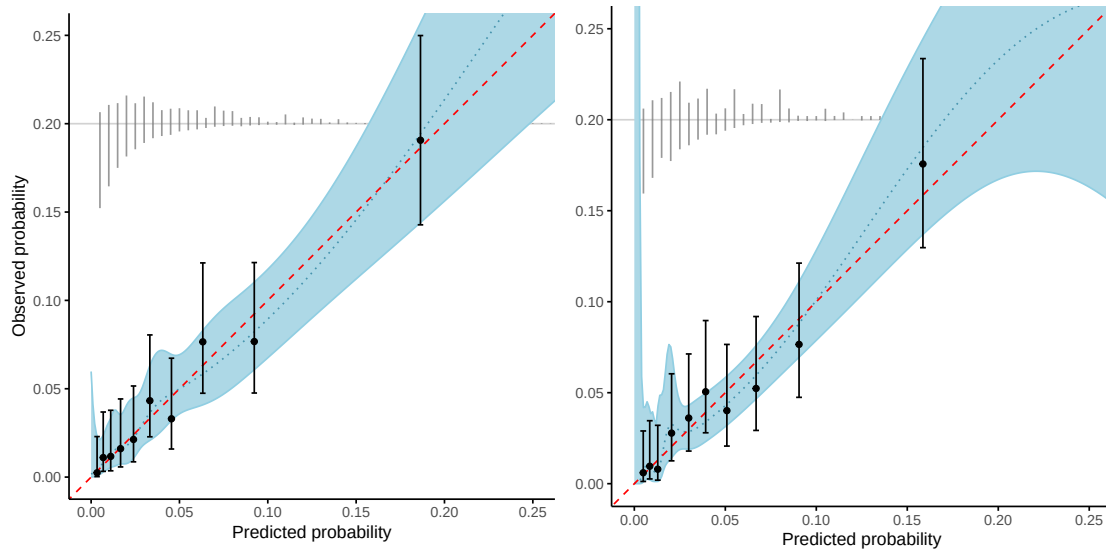


Figure 5.12: Calibration plots showing observed versus predicted probabilities for deciles of predicted probability. Red dashed line represents perfect calibration. Observed probabilities are presented with 95% confidence intervals (black error bars). A generalised additive model is fitted to show the smoothed mean observed probability (blue dotted line) with 95% confidence intervals (blue ribbon). Histograms, normalised by outcome, are overlaid to illustrate the distribution of relapses and cures across the expected probabilities.

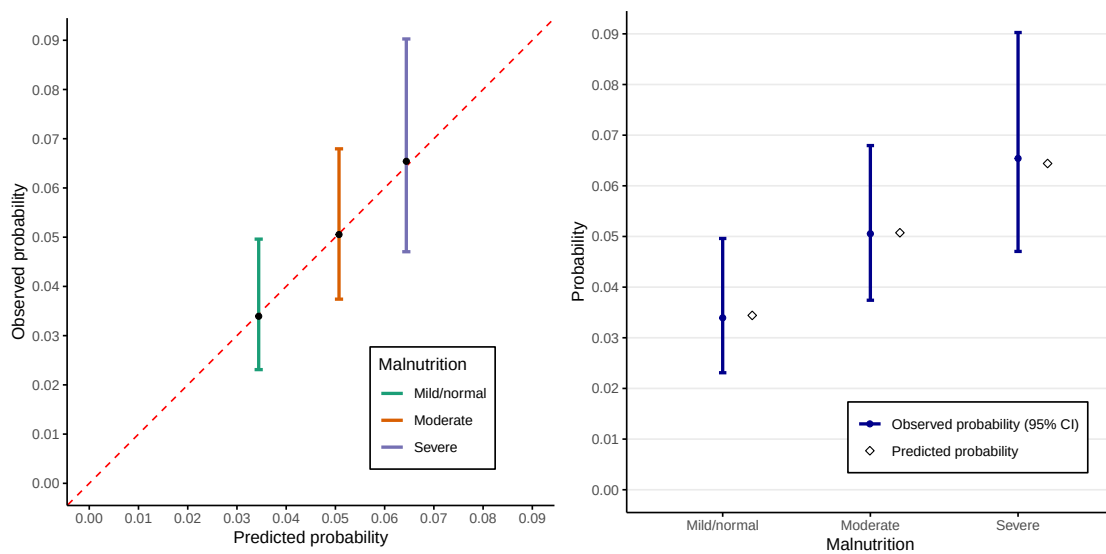


Figure 5.13: Calibration plots for malnutrition severity (model including parasite grade).

6

Discussion

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6.1 Overview

Drawing on prospectively collected data from 6,650 participants across 19 prospective studies, the models presented in this thesis constitute the first attempt to predict visceral leishmaniasis relapse at the individual patient level.

Despite considerable heterogeneity across the contributed studies and substantial missing data — necessitating the use of novel statistical approaches — several important predictors of relapse were identified. Fever duration and baseline parasite grade showed effects shared across both East Africa and the ISC, supporting their role as core prognostic factors. Other predictors demonstrated region-specific effects: the association between anaemia and relapse differed in direction between regions, WBC count and malnutrition were predictive only in East Africa, and age only in the ISC. Models incorporating parasite grade achieved superior discrimination, especially in East Africa. However, in the ISC, excluding parasite grade still allowed for meaningful model performance.

6.1.1 Chapter Structure

This final chapter provides a structured reflection on the presented models. Insights from preceding chapters are drawn upon to provide context and assess the implications of the presented prediction models.

First, the mechanistic plausibility of the identified predictor-relapse relationships is explored — ‘*what do the models teach us about the nature of VL relapse?*’. Specifically, a causal framework is developed that links the model predictors with VL relapse. This approach supports user confidence in the models, and provides a tool for discovery that can generate new hypotheses and insights into the underlying disease process.

Second, key strengths and limitations are explored relating to the underlying data, study design, and assumptions made during the modelling process. Issues related to model performance, interpretability, and generalisability are discussed. The potential impact of the developed models on clinical practice is considered.

Third, directions for future research are explored. These include how data-driven approaches can be leveraged to shine new light onto both the nature and determinants of VL relapse, and how such efforts can be translated into policy.

6.2 Plausibility of Relapse Predictors

It merits emphasis that the identified relapse-predictor relationships, as illustrated in Figures 4.8 and 5.8, simply reflect statistical associations arising under the stated modelling assumptions. Predictive accuracy does not imply biological plausibility. However, offering a mechanistic framework can add substantial value. Specifically:

User confidence. Biological plausibility improves the face validity of the models, increasing user trust in their predictions.

Model applicability. By clarifying which associations are likely to remain stable, a mechanistic understanding informs the extent to which a model can be applied across different populations, settings, and time periods.

Tool for discovery. Model findings, when interpreted through a mechanistic lens, can generate hypotheses and contribute to a broader understanding of the underlying disease process.

Given these benefits, this section aims to build on the disease pathophysiology set out in Chapter 1, and provide additional context to the model findings.

6.2.1 Parasite Dynamics

Supporting a more mechanistic interpretation, Figure 6.1 schematically illustrates the temporal dynamics of parasite load before, during, and after treatment. The figure situates relapse within the broader natural history of infection and treatment response, contextualised as four interconnected stages: asymptomatic stage → symptomatic stage → treatment stage → post-treatment stage.

In Chapter 1, studies identifying factors predictive of relapse were systematically reviewed and presented in Table 1.2. By considering the natural history of disease, and how relapse predictors relate to the underlying pathophysiological processes occurring at each stage, we can better understand the mechanistic plausibility of the observed predictor-relapse relationships.

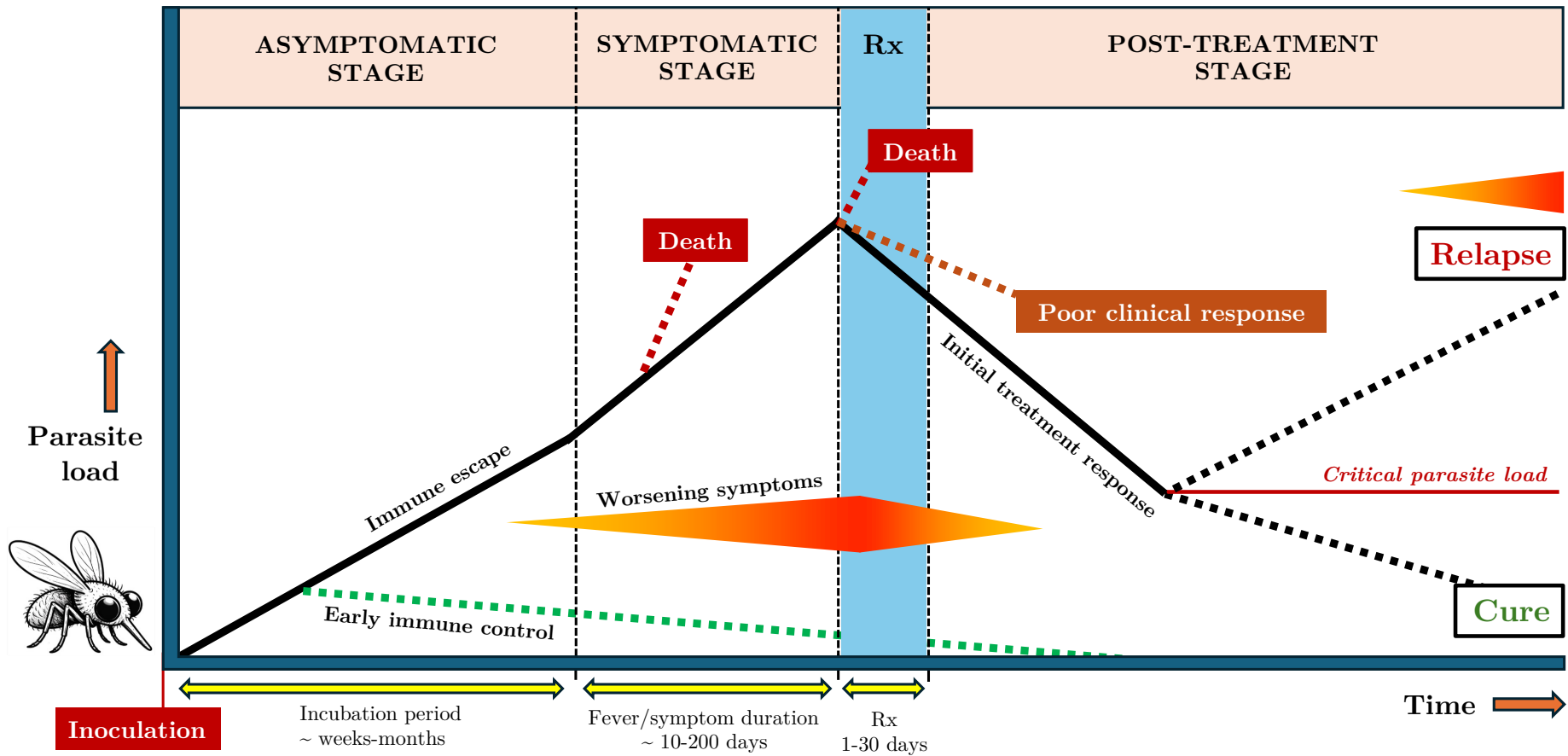


Figure 6.1: Schematic of the evolution of parasite load during VL infection. Events occurring during the post-treatment stage are critical in determining subsequent relapse. It is hypothesised that for each individual patient, a post-treatment ‘critical parasite load’ defines the threshold below which the host immune system controls parasite replication resulting in lasting cure. The threshold itself depends on (i) the pre-treatment parasite load, (ii) the degree of parasite depletion by treatment, (iii) host immunity, and (iv) the intrinsic parasite virulence. Rx: treatment, PKDL: post-kala-azar dermal leishmaniasis.

From Asymptomatic Infection to Symptomatic Disease

In endemic areas, cross-sectional surveys demonstrate that approximately 10–30% of individuals harbour asymptomatic infection[295]. However, only a minority of these individuals progress to symptomatic disease, underscoring that clinical illness represents the exception rather than the norm in infected populations[296]. Primary disease progression reflects a complex interplay of host genetic susceptibility, immune competence, and parasite-related factors, although these relationships remain poorly understood[297].

Relapse shares a key conceptual feature with primary disease progression — namely, a failure to control parasite expansion and the transition from an asymptomatic to symptomatic state. Therefore, studies examining determinants of primary disease progression can also provide insights into the mechanisms underlying relapse. High-quality prospective incidence studies are particularly informative in this regard, and provide important context for upcoming discussions[296, 298, 299].

Progression of Symptomatic Disease

As symptoms evolve, immune control is progressively undermined by parasite immune evasion, facilitating intracellular persistence and dissemination within the mononuclear phagocytic system. Host immune responses become increasingly skewed toward an immunoregulatory profile associated with dysfunctional cell-mediated immunity. This shift drives a vicious cycle of immunosuppression, heightened susceptibility to co-infections, and progressive end-organ damage[51, 297].

Historical descriptions from the pre-treatment era reveal marked variability in the natural history of disease. Nineteenth-century outbreaks in Assam and Bengal were characterized by devastating epidemic cycles, with rapid disease progression and mortality exceeding 90–95%[14, 300]. In contrast, in settings with endemic disease, a more chronic, relapsing-remitting course is described that could persist for years[300].

Understanding how clinical and laboratory factors evolve during symptomatic disease can help explain a number of relapse predictors, including fever duration, splenomegaly, anaemia, WBC count, and parasite grade. Drawing on literature that describes the natural progression and inter-relatedness of these factors can offer a framework for interpreting their mechanistic links to relapse.

Treatment Stage

Treatment decisions — including the choice of drug, dose, and duration — are central to determining whether, and to what extent, a patient improves clinically and biochemically. As discussed in Chapter 1, with effective treatment, quantitative molecular markers of blood parasite load decrease from the order of $\sim 10^3$ – 10^5 parasites/ml at treatment initiation to $\sim 10^0$ – 10^1 parasites/ml by day 28, although with substantial variation between treatment regimens and individuals. Symptomatic improvement and subsequent relapse have been shown to be correlated with end-of-treatment parasite burden[55, 143, 144, 301].

Do Predictors of Initial Treatment Failure Also Predict Relapse?

Relapse, by definition, is contingent on an initial clinical, and often parasitological response to treatment, and reflects failure to achieve sustained parasite clearance. This raises the question of whether risk factors for initial treatment failure are also risk factors for relapse. If so, predictors identified in studies of treatment failure could be extrapolated to better understand relapse risk. However, answering this question requires careful consideration of both how initial treatment failure is defined and the underlying pathophysiological processes it captures.

In analyses based on routine care data, where restrictive inclusion criteria are not applied, the most common cause of initial treatment failure is often death during treatment. This is illustrated by Yeshaw et al. (2020)[302], who reported that among 568 adult patients admitted to the University of Gondar Hospital, Ethiopia, between 2013 and 2017, 65 (11.1%) patients died during treatment, whereas only 9 (1.5%) were classified as treatment failures due to poor clinical response or persistence of parasites at the end of treatment. Notably, 60% of deaths occurred within the first 10 days of treatment. Similarly, in Brazil, Maia-Elkhoury et al. (2019)[303] showed that among 1,589 visceral leishmaniasis deaths reported nationally between 2007 and 2014, the median time from notification to death was 9 days (IQR: 4–17 days). Comparable patterns have been reported in other settings[171, 188, 304].

In these contexts, early mortality — the most common cause of initial treatment failure — occurs due to advanced disease at presentation with severe organ dysfunction. Consequently, and as described in Chapter 2, risk factors for mortality reflect the degree of organ dysfunction prior to treatment. These patients never reach a post-treatment state in which relapse can occur. Extrapolating predictors of initial treatment failure from these settings to relapse is therefore mechanistically inappropriate, given the distinct underlying pathophysiologies.

By contrast, in low-mortality settings without significant drug resistance, or in studies that exclude patients with severe disease, initial treatment failure is a rare event. When initial treatment failure occurs, it often reflects a positive test-of-cure, despite evidence of clinical improvement[67, 73]. In this context, initial treatment failure better reflects inadequate parasite suppression, which is already established as an important predictor of relapse[55, 143, 144]. Therefore, in such settings, observed treatment failure may function more as a surrogate marker of future relapse risk than as an indicator of true clinical non-response. However, depending on study outcome definitions, initial treatment failure may also include patients who discontinue treatment due to drug toxicity, or those who abandon therapy for non-biological reasons, further contributing to outcome heterogeneity.

On reflection, initial treatment failure is a heterogeneous construct that varies substantially across studies and settings, frequently encompassing multiple underlying processes, including death due to advanced disease, treatment discontinuation due to adverse events, and inadequate parasite suppression. The relative contribution of these components depends on the patient population, clinical setting, and study design. While initial treatment failure defined by parasitological non-response in low-mortality settings may serve as a proxy for relapse risk (inadequate parasite suppression), in high-mortality settings, or when treatment also captures discontinuation due to adverse events, extrapolating risk factors from initial treatment failure to relapse is likely misleading at best.

Post-treatment Stage

Expert consensus holds that VL treatment rarely results in sterile cure, as evidenced by animal models[305], the persistence of serological markers following clinical cure[306], the frequent detection of viable parasites at test-of-cure despite no subsequent relapse[144], case reports of disease reactivation many years after initial exposure[307, 308], and the recent detection of quiescent parasites within bone marrow niches[309]. Instead, treatment is thought to suppress parasite burden to a level that allows for reconstitution of the host's cell-mediated immune response and subsequent parasite control[305, 310]. In this context, relapse reflects a failure to achieve sustained parasite suppression, despite an initial treatment response. Events that occur following treatment are therefore critical in determining whether relapse occurs.

Verrest et al. (2024)[55] hypothesise that post-treatment parasite dynamics are dependent on four key factors: (i) the pre-treatment parasite burden, (ii) the degree of parasite depletion by the treatment, (iii) the intrinsic parasite virulence, and,

crucially, (iv) the host’s immune system. It can be further hypothesised that, during the post-treatment stage, for each individual there exists a ‘critical parasite load’ (Figure 6.1). The threshold itself would be expected to vary between individuals, reflecting both differences in host immunity and in parasite virulence. For parasite loads below this threshold, the host immune system controls parasite replication resulting in lasting cure. For parasite loads above this threshold, parasite replication outpaces immune control, leading to relapse (Figure 6.1).

6.2.2 Causal Model for Relapse

Building on Verrest et al.’s hypothesis, a causal (mechanistic) model can be assembled. In this putative model, three key drivers of relapse, all shaped by events prior to and during treatment, determine whether relapse occurs: (i) post-treatment parasite grade, (ii) host immunity, and (iii) parasite fitness. Post-treatment parasite grade itself can be considered a direct consequence of both pre-treatment parasite grade and the treatment regimen. If presented as a directed acyclic graph (DAG), these drivers constitute the final common pathway (mediators) for relapse. In Figure 6.2, text boxes (‘nodes’) represent relapse determinants, and lines (‘vertices’) indicate the direction of causality between nodes. For example, pre-treatment host immunity directly affects the initial rate of symptom progression.

Caution is advised, however, when interpreting such a model. As with DAGs for any disease process, they represent a broad approximation of true biological events and often fail to capture the nuanced dynamic interactions between outcome determinants as the disease evolves. Rarely do disease experts agree on the ‘correct’ DAG[311]. Limitations aside, the proposed DAG at least provides an explicit, albeit speculative, framework on the causes of relapse, motivating further debate and hypothesis generation.

6.2.3 Parasite Grade

In both the ISC and East Africa, higher pre-treatment parasite grade was strongly associated with increased relapse risk. For each unit increase in parasite grade (i.e. from 1+ to 2+), the adjusted odds of relapse increased by 31% in the ISC (aOR 1.31, 95% CI 1.12–1.52) and 43% in East Africa (aOR 1.43, 95% CI 1.23–1.66).

This association is supported by evidence from diverse patient populations. Among 146 patients with VL/HIV co-infection in Ethiopia, of whom 44 (30%) relapse, Abongomera et al. (2017) reported that a pre-treatment parasite grade of 6+ was strongly associated with relapse (HR 6.6, 95% CI 2.6–16.6)[118]. In 177 immunocompetent patients enrolled in East African DNDi trials (27 [15%]

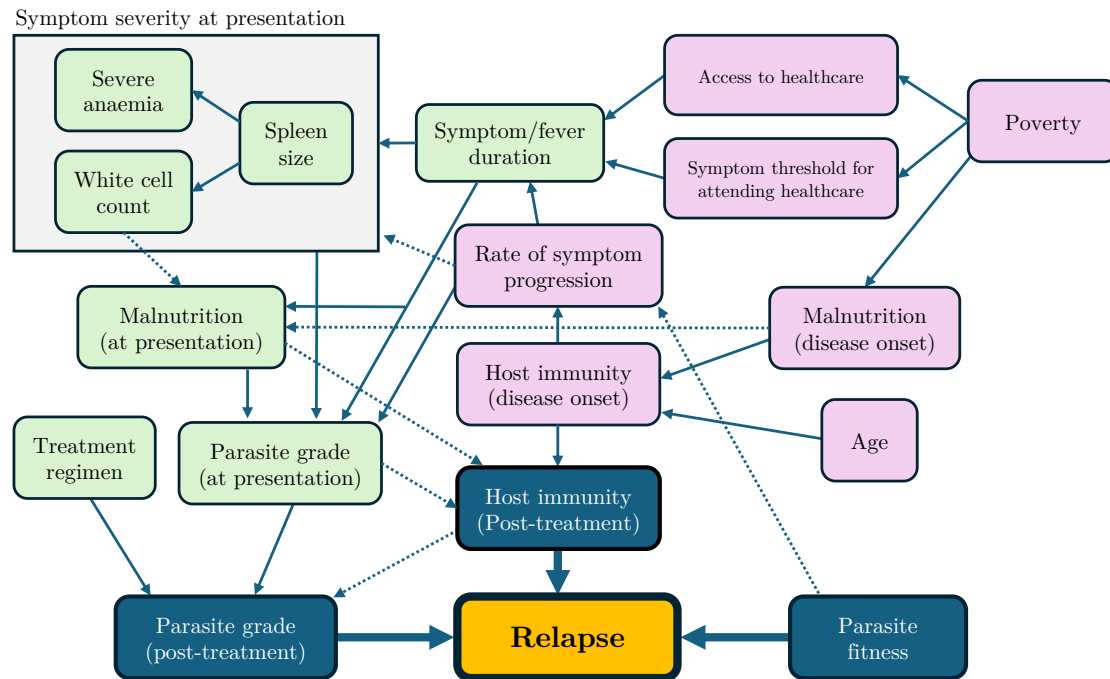


Figure 6.2: Directed acyclic graph (DAG) describing putative effectors of visceral leishmaniasis relapse. Factors are colour coded:

Factors measurable at the time of treatment initiation

Factors measurable at time of disease onset, or shortly thereafter

Factors on the final common pathway to relapse

Relapse

relapse rate), Verrest et al. (2021) demonstrated that pre-treatment parasitaemia, measured by blood qPCR, was significantly lower in those who achieved definitive cure compared to those who relapsed ($p = 0.03$)[144]. Similar findings were reported in India by Sudarshan et al. (2014)[312]. Parasite load assessed by blood qPCR has been shown to be strongly correlated with parasite grade measured on tissue aspirates, thereby supporting inferences made from blood qPCR to tissue microscopy[143, 144, 313, 314].

Of note, pre-treatment parasite grade was not considered in any of the studies included in the literature review of relapse predictors in HIV-negative patients (Section 1.3.1, Table 1.2). This reflects routine clinical practice, where non-invasive serology is preferred as the initial diagnostic test. Spleen size, however, was significantly associated with relapse in five of the seven identified studies where considered[94, 96, 110, 112, 114]. Given the strong correlation between pre-treatment spleen size and parasite grade, demonstrated both in this thesis (Appendix D, Figures C.16 and D.16) and elsewhere[315], it can be inferred that if pre-treatment parasite grade *were* considered as a relapse predictor, a positive association would have been shown.

Indirect evidence on the link between parasite grade and relapse can be drawn from prospective studies examining the incidence of asymptomatic infection and evolution to symptomatic disease. Sudarshan et al. (2014) followed an Indian cohort of 511 asymptomatic individuals who were qPCR positive, of whom 10 progressed to symptomatic disease within a year. Those with higher qPCR levels were at increased risk of disease progression[312]. Similar studies show that asymptomatic individuals with higher antibody titres (DAT, rK39) are also more likely to progress to symptomatic disease[149, 296], although data on the relationship between antibody titres and tissue aspirate parasite grade are lacking.

Drawing on infectious diseases beyond VL provides further support for a plausible association between pre-treatment parasite grade and relapse. In pulmonary tuberculosis, higher pre-treatment bacillary loads — reflected by sputum smear grade, culture time-to-positivity, or cavitary disease — are associated with delayed culture conversion and increased risk of relapse[316, 317]. Furthermore, in uncomplicated malaria, higher pre-treatment parasite density is associated with an increased recrudescence rate following artemisinin-based combination therapy[318, 319]. Similar findings have been observed across a range of other infectious diseases, including HIV[320], leprosy[321], and even *Clostridioides difficile* colitis[322].

Mechanistically, pre-treatment parasite grade can be associated with relapse through three interrelated pathways. First, higher pre-treatment parasite grades increase the probability that residual parasites persist through treatment, thereby raising the post-treatment parasite burden relative to the individual patient’s putative ‘critical parasite load’ required for immune control (Figure 6.1). In the proposed DAG, post-treatment parasite grade represents a mediator on the causal pathway between pre-treatment parasite grade and relapse (Figure 6.2). Second, higher parasite grades are both a cause of, and caused by an impaired host immunity. Animal models demonstrate that the degree of immunosuppression is associated with parasite density[323, 324], and in humans, markedly higher parasite grades are observed in patients with VL/HIV co-infection[1, 3, 152]. Lastly, higher parasite grades may be associated with more virulent parasite strains, which are more likely to evade immune control and cause relapse[131].

Taken together, the consistent regional effects observed in this thesis, the supporting evidence from prior studies, and a strong biological rationale, all reinforce pre-treatment parasite grade as a core prognostic determinant of VL relapse.

6.2.4 Fever Duration

Duration of fever, a close surrogate for the duration of symptomatic disease, was also identified as a significant relapse predictor in both regions. In both ISC models, a doubling of fever duration was associated with an approximate one third decrease in relapse odds (Figure 4.8). A less marked association between fever duration and relapse risk was identified in the East Africa model with parasite grade, where a doubling of fever duration was associated with a more modest 20% decrease in relapse odds, although with an upper 95% confidence interval bordering the null effect (Figure 5.8).

In the East Africa model without parasite grade, fever duration was not selected as a final predictor. This is likely a manifestation of the ‘suppressor’ effect of parasite grade on fever duration, resulting from a high correlation between fever duration and parasite grade (Figure D.16), and discrepant correlations between the predictors (parasite grade, fever duration) and relapse (Figures 5.4 and 5.5)[325]. A similar effect is seen with spleen size, discussed in Section 6.2.7.

In the ISC, a strong *unadjusted* inverse association between fever duration and relapse risk was also seen, supporting the final ISC model findings (Figure 4.5). In the combined East Africa dataset, a similar but less pronounced inverse association was observed in the unadjusted analysis (Figure 5.5).

As previously referenced in Chapter 1, two large studies in the ISC have demonstrated significant associations between fever (or symptom) duration and relapse in patients without VL/HIV co-infection.

In 2014, Burza et al. presented a retrospective cohort study of 8,588 HIV-negative patients from Bihar, India, treated in an MSF programme between 2007 and 2012 with a cumulative dose of 20 mg/kg liposomal amphotericin B over 4–10 days[96]. Of the 8,537 (99.4%) patients discharged with initial cures, 119 (1.4%) re-attended with parasitologically confirmed relapse (median time of 10.1 months). After adjusting for age, sex, and change in spleen size, the authors found a statistically significant inverse association between relapse and fever duration: compared with patients reporting < 4 weeks of symptoms, those with 4–8 weeks had 38% lower odds of relapse, and those with > 8 weeks had 55% lower odds. This result was unexpected, as prolonged symptoms would ordinarily be associated with greater illness severity and poorer outcomes. The authors proposed that in the Indian context, earlier presentation to care may itself reflect more severe disease or poor immune status.

In 2020, Goyal et al. reported the long-term outcomes of a large non-randomised multicentre phase IV trial (2012–2017), comparing relapse and PKDL outcomes in immunocompetent Indian VL patients treated with three liposomal amphotericin

B-based regimens[108, 326]. With a median follow-up time of approximately 40 months and 1,750 patients who successfully completed treatment, a disease duration of under 8 weeks was found to be significantly associated with increased relapse hazard, both in the univariable analysis and after adjusting for sex, age (< 12 or ≥ 12 yrs), and treatment regimen (aHR 3.6, 95% CI: 1.4–9.1). Similar findings were found when analysing 1-year outcomes[97]. The authors did not offer any explanation for this finding.

In the one East Africa study that considered symptom duration as a relapse predictor, no association was found. However, with only 17 relapses, this may reflect type 2 error[113]. In a retrospective study from Georgia, a *positive* association was found between symptom duration and relapse risk[114]. Extrapolating these findings to the ISC or East African context, however, would be considered inappropriate by most disease experts.

Mechanistically, the inverse association between symptom duration and relapse is likely explained by confounding through host immunity. In a recent expert disease review, Pareyn et al. (2025) describe that in patients with immunosuppression, or at the extremes of age, the disease can be present more acutely[1]. Disease progression and outcomes are also considered more severe in patients with no prior disease exposure, as exemplified by the high death rates seen in patients affected by forced migration due to famine and civil war in East Africa[13, 41], and historical descriptions of epidemic ‘waves’ crossing Bengal in the early to mid-19th century[14, 15]. If disease severity were the principal driver of healthcare attendance, then the association between symptom duration and relapse is clear: patients with weaker host immunity develop symptoms more rapidly, and therefore present earlier to healthcare facilities. While host immunity during initial symptom progression is not the same as post-treatment host immunity — a final mediator of relapse in the proposed causal model (Figure 6.2) — they are certainly closely correlated.

This proposed link between symptom duration and relapse becomes muddled when additional determinants of healthcare-seeking behaviour are considered alongside disease severity. The interval between symptom onset and presentation is influenced not only by clinical progression, but also by structural and contextual barriers to care. These include geographic distance to treatment facilities, direct and indirect patient costs, political instability, diagnostic delays arising from limited disease familiarity among patients and healthcare professionals, restricted access to diagnostic testing, and reliance on traditional healers. Such barriers are likely more prominent in parts of East Africa, where health system constraints and political instability may weaken the relationship between symptom severity and

healthcare attendance[86]. By contrast, in the ISC, sustained elimination efforts have increased disease awareness, strengthened health system capacity, and reduced financial barriers to care. In this context, variation in time to presentation is likely to better reflect underlying symptom severity rather than structural barriers. This may explain why symptom duration is a stronger predictor of relapse in the ISC compared to East Africa.

6.2.5 Treatment Regimen

In both ISC models, treatment with miltefosine is associated with an approximately 2 to 2.5 times higher relapse odds when compared to single-dose liposomal amphotericin B (10 mg/kg), or other treatments combined (see Figure 4.8, and Appendix Tables C.1 and C.2). This association is also seen in the unadjusted associations (Figure 4.4).

The finding that miltefosine is associated with higher relapse rates compared to single-dose liposomal amphotericin B is consistent with observations made in the ISC during the late 2000s and early 2010s. In Bihar, India, Sundar et al. showed that 6-month relapse rates had approximately doubled, from 3.0% observed in the pivotal RCT conducted between 1999 and 2000[63], to 6.8% in a subsequent trial conducted between 2009 and 2010[67]. High relapse rates were also demonstrated in subsequent pharmacovigilance studies in India[271], Bangladesh[327], and Nepal[66]. While increased relapse rates were largely blamed on the emergence of resistance during widespread use in the early elimination campaign, alongside poor stewardship practices[328], work by the Kaladrug-R project suggested that the true explanation may be more nuanced. Compared to non-relapsing strains, relapsing strains were not shown to demonstrate increased resistance based on *in vitro* promastigote cultures, but were instead found to be more infective[131]. Regardless of the underlying mechanism, concerns about increased relapse rates with miltefosine, together with strong emerging trial evidence demonstrating the superior efficacy of single-dose liposomal amphotericin B, ultimately led to its adoption in place of miltefosine in the elimination programme[75].

Interpretation of the decreased relapse rate in the miltefosine versus ‘Other’ treatment category is challenging because the ‘Other’ group combines heterogeneous treatment regimens (presented in Figure 4.2). This represents an important limitation in the prediction model development, discussed further in Section 6.3.2.

6.2.6 Anaemia Severity

The presence of severe anaemia is found to be a significant predictor of relapse in both the ISC and East Africa, although with opposing directions of association. In the ISC, the presence of severe anaemia is predictive of an approximate 30% *reduction* in relapse odds, contrasting with an approximate 70% *increase* in relapse odds in East Africa. Effects are similar across models with and without parasite grade, and while remaining statistically significant at the 5% level, both confidence intervals skirt the null effect. Similar associations are seen in the unadjusted associations, demonstrated graphically in Figures 4.6 (ISC) and 5.6 (East Africa).

Severe anaemia is highly prevalent in patients with VL, and most often found to be normocytic and normochromic[47, 48]. Even in the relatively well patients used in model development, almost 50% were found to be severely anaemic according to the age- and sex-adjusted 2024 WHO definition[239] (Tables 4.2 and 5.2).

Ferrokinetic studies performed in the 1970s and 80s show that erythrocytes in VL-infected patients have reduced survival times due to increased splenic haemolysis, with relatively preserved bone marrow haematopoiesis[329, 330]. The finding of increased intracellular iron stores and reduced circulating iron suggests that an anaemia of chronic disease also contributes to the overall picture[331]. These findings support observations that spleen size and duration of illness correlate with the anaemia severity, demonstrated both in this thesis (Appendix Figures C.15, C.16 [ISC] and D.15, D.16 [East Africa]) and elsewhere[47, 48, 332].

In the literature review of relapse determinants presented in Chapter 1, one study from the ISC[110], and one from East Africa[113] demonstrated positive unadjusted associations between anaemia severity and increased relapse risk, although interpretation is challenged by small sample sizes and possible confounding by age and spleen size. In the proposed DAG (Figure 6.2), anaemia affects relapse through its association with symptom severity (including spleen size, resulting in splenic sequestration), symptom duration (linked to anaemia of chronic disease), rate of symptom progression, and malnutrition (linked to iron deficiency anaemia).

In the ISC, it is plausible that anaemia severity is primarily a surrogate of symptom duration (i.e. reflecting an anaemia of chronic disease); with longer symptom durations representing milder, more indolent disease caused by improved host immunity (as described in Section 6.2.4), and therefore associated with a lower relapse risk.

Conversely, it could be argued that in East Africa, where patients are more malnourished (24.9% with severe malnutrition vs. 18.9% in the ISC, excluding missing data, cf. Tables 4.2, 5.2), anaemia severity is driven more by pre-existing

anaemia due to malnutrition, or perhaps better correlated with disease severity at presentation due to the aforementioned increased barriers to care. In either case, both associations would align the severity of anaemia with a less competent host immunity and/or a higher parasite burden.

Overall, these contrasting findings suggest that severe anaemia is unlikely to act as a direct biological determinant of relapse, but rather functions as a context-dependent surrogate marker for underlying disease severity and host immunity. The direction of association observed in each region likely reflects differences in the causal pathways underlying anaemia rather than a fundamentally different biological effect of anaemia itself.

6.2.7 Spleen Size

Despite spleen size being significantly correlated with both parasite grade and illness duration, it was only identified as a significant predictor of relapse in the East Africa model that included parasite grade. With borderline significance ($p = 0.056$), the odds of relapse were found to *decrease* by approximately 40% for each doubling of the spleen size. Interestingly, the direction of this relationship is reproduced in the unadjusted analyses in both regions (Figures 5.5 and 4.5).

As presented in Section 1.3.3, spleen size was the second most frequently identified relapse predictor in the literature review. However, interpretation, and comparison with the prediction model findings, is complicated by differences in reported measures and differences in the covariates adjusted for in the multivariable models. In the ISC, for example, significant adjusted associations were identified between relapse and (i) smaller changes of spleen size during treatment[96], (ii) larger spleen sizes at discharge[110], and (iii) larger spleen sizes at admission[94]. In East Africa, Gorski et al. (2010) showed that in a multivariable model containing treatment, admission, and discharge spleen sizes, the odds of relapse were higher in patients with larger admission and discharge spleens.

The apparent discordance between earlier reports and the findings presented here likely reflects differences in study populations, clinical settings, model construction, and the covariates adjusted for. Drawing on a similar inference used to explain the relationship between anaemia severity and relapse, and given the strong correlation between symptom duration and splenic enlargement, it can be hypothesised that patients with smaller spleens present after a shorter, more rapidly progressive disease course. In this context, a smaller spleen at admission may indicate weaker host immunity, shorter illness duration, and consequently, a greater risk of subsequent relapse. Although the East African dataset includes fewer patients, the median

spleen size at admission is twice that observed in the ISC cohort (8 cm vs. 4 cm). This striking contrast parallels the higher parasite grades and longer symptom duration also seen in the East Africa dataset. The broader distribution of spleen sizes in East Africa likely contributes to its ability to discriminate which patients progress to relapse, and therefore its inclusion in the final model.

6.2.8 Age

Age, including both linear and squared terms, was retained as a significant predictor in both ISC models. Plotting predicted relapse probability (y-axis) against age (x-axis) resulted in a ‘U’-shaped association (Figure 4.9), with a similar appearance seen in the unadjusted association (Figure 4.5). Conversely, no age terms were retained in the final models developed from the East Africa dataset.

In the literature review of relapse determinants in immunocompetent patients, of the seven identified studies from the ISC, all but one demonstrated a significant relationship between relapse risk and age, with younger age always associated with increased risk[66, 94, 96, 97, 108, 110]. Of these studies, two grouped age into more than two strata: using routinely collected data from MSF treatment clinics, Burza et al. (2014, India)[96] and Lucero et al. (2015, Bangladesh)[110] demonstrated that patients under 5 years and over 40–45 years were at increased risk of 12-month relapse. Authors have attributed the higher risk in younger patients to a less mature host immunity[66, 96, 108], and less effective treatment efficacy due to altered pharmacokinetics and pharmacodynamics[66, 108].

Increased relapse risk in older patients is also likely related to host immunity, given a higher prevalence of comorbidities and age-related immunosenescence[333]. The conjecture that the age-relapse relationship is driven by differences in host immunity is supported indirectly by drawing on observations on the speed of clinical deterioration. Similar to patients with VL/HIV co-infection, patients at the extremes of age are considered more likely to present with an acute disease[1, 2]. Furthermore, analysis of VL surveillance data collected in Bihar and Jharkhand states, India, show that younger patients (< 15 years) more often present after a shorter period of symptoms, when compared to older age groups[334]. While host immunity is not the only determinant of symptom duration prior to treatment, the association between the extremes of age with a shorter pre-treatment symptomatic period is consistent with a reduced capacity to contain parasite replication, leading to more rapid clinical progression and earlier care-seeking. Taken together, these observations support the hypothesis that impaired or dysregulated host immunity

at the extremes of age contributes both to acute presentation and to an increased risk of relapse following treatment.

The absence of age as a relapse predictor in the East Africa models is also consistent with a large observational study of patients attending MSF clinics in South Sudan. Gorski et al. (2010) compared the ages of 166 patients with confirmed relapse with almost 8,000 patients without a record of relapse, and found no significant difference[112]. One possible explanation is that patients in this region often present with more advanced disease, such that severity-related factors — including malnutrition, anaemia, and parasite burden — dominate relapse risk, thereby attenuating more subtle host determinants such as age. Differences in treatment regimens may also contribute to this effect.

6.2.9 White Blood Cell Count

In both East African models, doubling the pre-treatment WBC count was associated with an approximately 40% increase in relapse odds, with borderline significance (Figure 5.8). This direction of effect was consistent in the unadjusted analysis (Figure 5.6).

This appears to be the first report of an association between WBC count and relapse, likely reflecting limited prior investigation. Only one of 11 studies in the literature review evaluated WBC as a candidate predictor[114].

The direction of effect is perhaps counterintuitive, since lower WBC counts are typically associated with a more prolonged and severe disease[1, 3]. Consistent with this, both datasets in this thesis show that lower WBC counts correlate with larger spleens, higher pre-treatment parasite grades, and lower haemoglobin and platelet counts (Figures C.15, C.16, D.15, D.16). These relationships likely reflect established mechanisms of cytopenia in VL — splenic sequestration, bone marrow infiltration, and, for haemoglobin, anaemia of chronic disease[47, 335].

Given the strong negative correlation between spleen size and WBC count, the previously described causal arguments linking smaller spleen sizes to relapse can also be applied to higher WBC counts (Section 6.2.7). It is plausible that more moderate (higher) WBC counts, like smaller spleen sizes, act as a surrogate for a more impaired host immunity, reflecting an accelerated evolution of symptoms.

6.2.10 Malnutrition

In contrast to the ISC, a strong adjusted association between malnutrition severity and relapse risk is observed in East Africa. Using the age-adjusted malnutrition definitions described in Table 3.3, compared to patients with mild or normal nutritional status, the odds of relapse increased by approximately 70–80% in those with moderate malnutrition and by more than 150% (i.e. over 2.5-fold) in those with severe malnutrition (Figure 5.8). A similar trend is evident in the unadjusted analysis in East Africa (Figure 5.4), but not in the ISC (Figure 4.4).

Among the seven studies identified in the literature review where the relationship between malnutrition and relapse was assessed, only one study reported a significant association. In a large prospective trial including over 1,000 patients, Sundar et al. (2019) found that individuals weighing under 30 kg had approximately double the adjusted odds of relapse, concluding that malnutrition was a risk factor[94]. However, as this analysis did not adjust for age, residual confounding cannot be excluded. The remaining studies used heterogeneous definitions of malnutrition and were predominantly conducted in the ISC, limiting their comparability and generalisability to the East African setting.

The observed association between malnutrition and relapse is biologically plausible and likely mediated through its impact on host immunity. Within a causal framework, similar to parasite grade, malnutrition can act both as a driver and a consequence of an impaired host immunity[336]. Patients with VL are much more likely to be malnourished compared to healthy controls[1, 3]. However, evidence linking malnutrition to progression from asymptomatic infection to clinical disease remains limited, primarily due to the need for large prospective cohort studies[104, 337]. In a landmark study of individuals with asymptomatic infection, Bern et al. (2007) demonstrated that higher red meat consumption was protective against progression to symptomatic VL, and that symptomatic individuals had lower levels of zinc and retinol[299]. Together, these findings support the interpretation that malnutrition acts as a proxy for impaired host immunity.

A possible explanation for why malnutrition is significantly associated with relapse in East Africa, but not the ISC, is the greater severity of malnutrition observed in East African patients (24.9% vs. 18.9% with severe malnutrition in East Africa and the ISC, respectively). In the East African context, malnutrition may capture a threshold of immune compromise sufficient to impair parasite clearance, resulting in a detectable increase in relapse risk.

6.3 Strengths and Limitations

The models developed in this thesis result from a decade-long collaborative effort to collate and harmonise datasets from multiple clinical trials. Predictors of relapse have been identified using robust statistical methods, accounting for both substantial missing data and heterogeneity across the contributed datasets. The resulting models are the first of their kind in the field of neglected tropical diseases, and represent a significant advance in our understanding of relapse determinants in patients treated for VL. However, model findings must be interpreted in light of several limitations, including the representativeness of the development datasets to the target population, the assumptions made during model development, and the need for external validation with independent datasets.

6.3.1 Sample Size and Internal Validity

Like many neglected tropical diseases, VL studies are few in number and often small in sample size; a consequence of a disproportionate lack of research funding relative to the disease burden. This is particularly true for studies of VL relapse, which require both follow-up for at least 6 months after treatment completion and large patient numbers to capture sufficient relapse events. Such studies are logistically challenging and expensive to conduct, especially in resource-limited settings[62]. As identified in prediction models for VL mortality[158], and in models across other diseases[161–164], the majority are underpowered to support the number of candidate predictors. This leads to overfitting, where the model captures random noise rather than true relationships, resulting in poor performance when evaluated in new patient populations[161]. By combining studies using IPD meta-analysis, the effective sample size, and therefore statistical power, can be increased[215].

Indeed, one of the principal strengths of this thesis was the use of IPD meta-analysis to develop relapse prediction models using data from a total of 4,599 patients in the ISC (including 228 relapses) and 2,095 patients in East Africa (including 99 relapses). These represent the largest datasets used to investigate relapse predictors in VL.

Using recently developed sample size calculation methods for prediction model development[338], it was estimated that the ISC dataset would support the development of a model with up to 25 predictors whilst maintaining a recommended uniform shrinkage of ≥ 0.9 (Table 3.1). With the actual number of predictor parameters chosen at 17 (16) for the ISC model with (without) parasite grade (Table 3.2), it was reassuring to find that after performing internal validation,

the calculated uniform shrinkage was 0.91 (0.92) (Table 4.4). Since both uniform shrinkage factors were above the 0.9 threshold, this suggests that overfitting is unlikely a major concern in the ISC models.

When applying similar calculations to the East Africa dataset, it was estimated that the effective sample size would support a model with up to 11 predictor parameters to maintain a uniform shrinkage factor of ≥ 0.9 , or up to 17 parameters when relaxing the uniform shrinkage factor to ≥ 0.85 (Table 3.1). With the actual number of predictor parameters chosen at 15 (14) for the East Africa model with (without) parasite grade (Table 3.2), the calculated uniform shrinkage factors were 0.81 (0.75) (Table 5.4). These shrinkage estimates were lower than expected, and suggest that overfitting is a larger concern in the East Africa models, particularly for the model without parasite grade. Reasons for the discrepancy between the expected and measured uniform shrinkage factors likely reflect increased model instability introduced by combining multiple imputation with a hierarchical model structure — a combination that is not clearly supported by the sample size methodology[338]. The robust methodology used for internal validation, which involved bootstrapping the entire model development process, including imputation and model selection, also likely contributed to the lower shrinkage factors observed in the East Africa models, as it more accurately captures the true model optimism[161].

Model overfitting was mitigated through internal validation, allowing for calculation of optimism-adjusted coefficients and performance measures. On balance, the benefits of including a comparable set of predictors across both ISC and East Africa models were considered to outweigh the risks of overfitting, particularly given the exploratory nature of this thesis and the interest in comparing the predictor-relapse across regions. This being said, the potential for overfitting in the East Africa models highlights the importance of external validation with independent datasets.

6.3.2 Predictor Selection

Across both the ISC and East Africa models, predictor selection was shaped not only by prior clinical knowledge of relapse risk, but also by substantial practical constraints arising from data structure, heterogeneity, and variable availability. While a broad range of potential predictors was initially considered, decisions regarding inclusion were ultimately guided by considerations of identifiability, consistency of measurement, and real-world applicability of the final models. The following sections outline key limitations and trade-offs encountered in selecting predictors, focusing in particular on treatment, clinical variables, and measurements obtained at initial cure assessment.

Treatment

An important limitation to recognise is that treatment was only partially accounted for in the ISC models, and not included at all in the East Africa models. This is despite the fact that treatment is a well-established determinant of relapse risk, as described in Chapter 1 (Section 1.3.3).

As briefly discussed in Chapter 3 (Section 3.3.2), and illustrated in Figures 4.2 and 5.2 for the ISC and East Africa studies respectively, treatments were highly heterogeneous. Many regimens — when considering (i) choice of agent(s), (ii) dosing, and (iii) duration — were only represented by a single study. Consequently, if a model were to include both treatment as a grouped predictor *and* study as a random intercept term, model convergence would fail due to the high multicollinearity between treatment and study.¹ As an imperfect compromise, for the ISC models, the treatments of single-dose liposomal amphotericin B (10 mg/kg) and miltefosine (standard dose) were retained as separate categories, given that they were shared across sufficient studies to permit identifiability of both study and treatment. The remaining ‘*Other*’ category included a heterogeneous mix of different regimens, resulting in a coefficient that is difficult to interpret. Unfortunately, despite attempts, this compromise was not possible in the East Africa models, given the even greater heterogeneity in treatment regimens and the smaller number of studies.

Alternative modelling approaches were considered to permit inclusion of a broader range of treatment regimens.

First, a non-hierarchical model was considered, where study was not included as a random intercept term. This approach would remove the collinearity between treatment and study, but would also fail to account for the between-study clustering of the predictor-relapse relationships, which can occur due to differences in study design, setting, and patient populations. The consequences of not using a hierarchical model are considered in Section 6.3.6.

Second, treatment groups could be defined purely based on the pharmacological agents used, without considering dosing or duration. This approach would allow for the inclusion of more treatment groups, but would also falsely assume that all regimens containing a particular agent have the same effect on relapse risk, which is, of course, incorrect.

¹Alternatively described as an approximately non-identifiable model. As an example, consider the extreme case, where a model includes two studies, with each study using a different treatment. Including both study and treatment as predictors would lead to a failure of model convergence, given that the effects of study and treatment cannot be uniquely identified.

Clinical Signs and Symptoms

Among the clinical variables included as candidate predictors, only spleen size and fever duration were considered in the model building process. Other signs and symptoms were explored, but not considered as candidate predictors despite their potential utility in predicting relapse. These included the magnitude of fever, the presence of weight loss, cough, dyspnoea, bleeding, ascites, and oedema.

Their exclusion was, in part, due to the lack of consistent measurement and reporting across the contributed datasets. For example, ‘presence of cough’ was frequently present in the harmonised datasets, but it would often be unclear how the variable was defined (i.e. whether the cough was observed by a clinician vs. reported by the patient voluntarily vs. reported by the patient on direct questioning), when the variable was measured (at admission vs. during treatment), the time period or time point the cough occurred (prior to recruitment, or at the time of initial patient assessment), and whether missing values reflected a true absence of the symptom, or simply a lack of enquiry.

A further reason for the exclusion of many clinical variables was their lack of discriminatory power in predicting relapse. The majority of contributed studies excluded patients with severe illness (with full eligibility criteria presented in the [Supplementary Material](#)). As a result, only a small proportion of patients in a minority of the contributed datasets had any markers of severe infection, such as jaundice, co-infection, bleeding, ascites, or oedema. The relatively stringent eligibility criteria raise broader questions about the applicability of the developed models, addressed further in Section 6.3.5.

Variables Measured at Initial Cure Assessment

Many of the contributed datasets provided patient information measured at the time(s) of initial cure assessment, including repeat blood tests, and patient signs and symptoms such as spleen size. In an ideal world, these variables would have been considered in a model predicting relapse, since (i) the model is intended for use at the time of initial cure assessment (once initial treatment failure has been ruled out), and (ii) more generally, using patient information measured closer to the timing of the outcome event often results in superior model performance (as seen with models predicting VL mortality in Chapter 2).

When considering the inclusion of these variables, however, a number of challenges were identified.

First, the timing of the initial cure assessment varied significantly across studies, ranging from 18 to 31 days following treatment initiation in East Africa, and from

16 to 30 days in the ISC (full details in the [Supplementary Material](#)). Furthermore, in the majority of studies, in the event of a failed initial test-of-cure (e.g. a positive aspirate), a repeat test-of-cure would be performed at subsequent time points, perhaps accompanied by an extended treatment course. This variability in timing would have introduced significant heterogeneity in the model building process and challenged the interpretation of the resulting model coefficients. A possible ‘work-around’ would have been to instead model the *rate of change* in these variables from treatment initiation to initial cure assessment; for example, change in spleen size per day, or change in haemoglobin per day. However, these calculated ‘rate’ variables would likely be highly correlated with the initial value at admission. For example, a patient with a larger spleen at admission would be expected to have a faster rate of spleen reduction during treatment. Also, the assumption of a constant change over time — allowing comparisons regardless of the timing of the initial cure assessment — would likely be violated, since the rate of change in these variables is likely to be greater during the early stages of treatment, and then slow down as patients approach initial cure assessment.

Second, during the exploratory stage of data harmonisation, a much higher proportion of missing data was observed for variables measured at the time of initial cure assessment. Accounting for this missing data using multiple imputation techniques would have introduced additional complexity, and likely instability, to the model building process.

Last, but certainly not least, including variables measured at approximately one month after treatment initiation would directly impact a model’s real-world utility. This is especially true in the ISC, where patients are treated with single-dose liposomal amphotericin B and discharged from care shortly after. Despite recommendations that patients return for assessment at 1 and 6 months, in routine practice, this often does not reflect real-world practice[1, 68]. In East Africa, in contrast, since current first-line treatment mandates 17 days of injections, an end-of-treatment assessment could further inform a relapse prediction model. However, whether information provided at 17 days would align with information provided at 30 days, as is more often reported by the contributing studies, is uncertain.

6.3.3 Performance Considerations

After optimism adjustment, both ISC relapse models provide moderate to good discrimination (c-statistics of 0.68 for the model with parasite grade, 0.67 for the model without parasite grade), and are well-calibrated on internal validation. Consequently, for patients with non-severe disease and without significant immunosuppression,

clinicians and policy-makers should be reassured that the presented models provide an effective tool for risk-stratification and clinical decision-making.

In contrast, in East Africa, despite both models being well-calibrated, only the model *with* parasite grade can reasonably discriminate between those who relapse vs. those who do not (c-statistic 0.68). In settings where parasite grade is *not* routinely available (i.e. where diagnosis is made with the rK39 rapid test and/or the direct agglutination test), the model *without* parasite grade would need to be used, with an optimism-adjusted c-statistic of 0.61. While still meaningful in certain settings, most experts would consider a c-statistic of < 0.65 to reflect poor discrimination[339]. Compared to the ISC, the added benefit in model discrimination from including parasite grade in the East Africa models likely reflects the broader range of parasite grades in the East Africa datasets (cf. Figures 4.4 and 5.4). Increased variation in parasite grade may reflect both higher rates of immunosuppression (e.g. malnutrition is overall more severe in the East Africa datasets), and increased delays in diagnosis (symptom duration is overall high in the East Africa datasets).

6.3.4 Model Assumptions

“Since all models are wrong the scientist must be alert to what is importantly wrong. It is inappropriate to be concerned about mice when there are tigers abroad.” — George Box (1976)[340]

According to the well-known aphorism, all models are wrong.² Models presented in this thesis are, of course, no exception. Model assumptions are implicit in the model equations (specifications) and represent a necessary simplification of a complex and often unknown interplay of factors that determine relapse risk. Here, key assumptions made during model development are reviewed, and their validity considered. Hopefully, the reader will conclude that while many mice are unavoidably present, the existence of tigers lurking in the shadows is unlikely.

In keeping with expert guidance on model development, continuous predictors were not categorised into discrete groups[103]. Instead, a linear relationship between the log-odds of relapse and each continuous predictor, with appropriate transformations where deemed necessary, was assumed. The choice of transformation was guided by both prior knowledge (for example, modelling age as a polynomial function to capture increased risk at the extremes of age), and through visual inspection of the univariable (unadjusted) relationships between relapse and the candidate predictors (presented in Sections 4.1.3 and 5.1.3 for relapse probabilities, and Figures C.17 and D.17 for the log-odds of relapse). Univariable associations

²https://en.wikipedia.org/wiki/All_models_are_wrong

were shown to be approximately linear, especially when accounting for the margin of error around the line of best fit.

It could be argued that more flexible modelling approaches, such as the use of splines, would have allowed for more complex relationships to be captured between the continuous predictors and relapse risk. Despite the challenges faced in developing models that combine splines with multiple imputation (for example, uncertainty in how to best identify the spline knot locations), it would not be an unprecedented approach[215, 341]. However, given the limited number of relapse events — thereby limiting the number of candidate predictor parameters to prevent overfitting — a decision was made to avoid the use of splines, favouring model parsimony over model complexity.

Calibration plots provide the strongest evidence supporting the overall choice of model specification. Deciles of predicted vs. observed relapse probabilities are presented in Figures 4.12 and 5.12 for the ISC and East Africa models respectively. Comparisons of predicted and observed relapse probabilities are also presented for select subgroups of all final predictors in the Appendices (Figures C.20 to C.28 for the ISC, and Figures D.20 to D.28 for East Africa). Overall, when considering the margins of error, the calibration plots show good agreement between predicted and observed relapse probabilities.

An implicit assumption in the final models is that the predictor-relapse relationships are the same across all predictor subgroups. In the interest of model parsimony and to avoid overfitting, interaction terms were not considered during predictor selection. This assumption is, to some extent, tested by (1) reviewing the heterogeneity in calibration slope in the models when evaluated across the individual contributing studies, and (2) assessing the calibration plots in two important age subgroups; (i) in patients 18 years and older, and (ii) under 18 years. Overall, the calibration slopes appear reasonably consistent across studies, with no evidence for significant heterogeneity ($I^2 = 0$ for all models across both regions, as displayed in Figures 4.11 and 5.11) for the models with parasite grade. Evaluation of the calibration plots in the two age subgroups is limited by the reduced number of relapse events, especially in the East Africa models. When considering the large confidence intervals in the probability estimates, the age-specific calibration plots do not show any obvious deviations from the line of perfect calibration (presented in the [Supplementary Material](#)).

6.3.5 Model Applicability

If a model demonstrates good performance and is constructed using reasonable assumptions, its predictions should be applicable to the patient populations and settings used for model development and/or updating[203]. This raises an important question: “How similar are the patients used in model development to the intended target population?”

In patients with VL/HIV co-infection, or other forms of severe immunosuppression, treatment failure or relapse is considered the norm rather than the exception[118]. As all models presented here were developed in immunocompetent patients, extrapolation to immunosuppressed populations would likely underestimate true relapse risk and result in poor calibration. These models should therefore *not* be used in this high-risk group.

Patients with severe disease were largely excluded from the contributing clinical trials. Model performance in this subgroup is therefore uncertain. On the one hand, severe disease may be associated with a higher risk of relapse. These patients are likely to be more immunosuppressed as a consequence of disease severity, and therefore more likely to relapse. Furthermore, severe disease itself can occur as a causal consequence of shared risk factors for relapse, including immunosuppression (malnutrition, immune-naivety), or infection with more virulent parasite strains[131]. On the other hand, severe disease may also be associated with a lower relapse risk. In many settings, patients with severe disease are managed more intensively, including extended treatment regimens and the preferential use of liposomal amphotericin B. They may also undergo post-treatment tissue aspiration to confirm cure. Given these competing considerations, the use of the presented models in patients with severe disease should be avoided.

A further important limitation that merits reflection relates to the definitions of initial cure across the contributed studies. Specifically, how these definitions contrast with real-world initial cure assessments and can result in the systematic underestimation of true 6-month relapse risk. As discussed in the Methodology (Section 3.2.3), a range of methods were used to assess initial cure in the contributing studies, often requiring a repeat clinical and biochemical assessment with tissue aspirate performed approximately 1 month after treatment initiation (full descriptions in the [Supplementary Material](#)). In the trial setting, patients failing this assessment would either have an extended treatment course with a follow-up aspirate, or would simply be classified as ‘initial treatment failure’ and excluded from subsequent trial outcomes. In routine practice, however, the definition of ‘initial cure’ is poorly defined, resulting in a systematically higher relapse risk due to

the inclusion of patients who would have failed a formal initial cure assessment under trial conditions. For example, in the ISC, where most patients are treated with single-dose liposomal amphotericin B and discharged the same day, in the absence of routine active follow-up, initial cure assessment within 1 month only occurs in patients who return due to persistent or recurrent symptoms. In East Africa, in contrast, current first-line treatment mandates 17 days of injections. This allows for a clinical assessment at the end of treatment, but without the routine tissue aspirate that would normally occur in the trial setting. The extent to which the differences in initial cure assessments between the development datasets and real-world practice affects model applicability is uncertain, and likely varies considerably across settings.

Additional concerns arise when patients are treated with novel or non-standard regimens not represented in the development data. For example, updated WHO guidelines are expected to introduce paromomycin and miltefosine as treatment options in East Africa[73].

In summary, the models are most applicable to patients with non-severe disease, without significant immunosuppression, and treated with regimens similar to those used in model development. External validation using independent datasets, ideally with prospective real-world follow-up, is required to further assess model applicability and performance.

6.3.6 Modelling Between-Study Heterogeneity

Substantial between-study heterogeneity in relapse risk was anticipated in the final models, given differences in eligibility criteria, treatment regimens, definitions of relapse and initial cure, patient populations, settings, and other study design factors. Further details on these differences are presented in the [Supplementary Material](#) and described in the study characteristics sections of the results chapters (Sections 4.1.1 [ISC] and 5.1.1 [East Africa]).

To account for this heterogeneity, study was modelled as a random intercept term. As described in Chapter 3 (Section 3.3.5), the principal benefits of adopting a hierarchical model include (i) improved generalisability of the model to new settings and populations, and (ii) avoidance of probability estimates that would otherwise be biased towards the marginal population estimate[162, 215, 253].

Despite the added statistical complexities introduced by modelling study as a random intercept, the approach is validated when assessing the variation in calibration intercept (calibration-in-the-large) between studies using a two-stage random-effects meta-analysis. In the ISC model including parasite grade, approximately half of the variation in calibration intercepts is explained by between-study heterogeneity

($I^2 = 46.1\%$, $p = 0.01$, Figure 4.11). In the East Africa model including parasite grade, even more between-study heterogeneity is demonstrated, with almost 90% of the variation in calibration intercepts explained by between-study heterogeneity ($I^2 = 89.0\%$, $p < 0.0001$, Figure 5.11). Similar findings were observed in the models without parasite grade (Figures C.19 and D.19).

6.4 Model Implementation

It is well established that the majority of published prediction models are never used in routine clinical practice[342]. In part, this reflects a saturated market,³ but also stems from limited user confidence due to a lack of external validation, the absence of transparent reporting, the use of variables that are not readily available in routine practice, and crucially, a lack of stakeholder engagement[161, 342].

The models presented in this thesis answer an important clinical question and are the only tools for VL relapse risk stratification. However, several barriers must be overcome before their adoption in routine clinical practice. Some of these barriers, including choice of intercept, method of implementation, and potential use-cases, are discussed here. Engagement of policymakers, external validation, and updating with new data are all vital if these models are to achieve meaningful clinical impact and be sustainably integrated into routine care pathways.

6.4.1 Choice of Intercept

As a consequence of adopting a random-effects modelling approach to account for between-study heterogeneity in relapse risk, the user must select a suitable model intercept that best represents their target patient population. Overall model calibration (calibration-in-the-large) will be directly affected by the intercept choice. Suggested intercepts are provided in the full model specifications, including the average (marginal) intercepts across all contributed studies and intercepts following model recalibration to the largest datasets that use current first-line treatments, as provided by Sundar et al. (2019) for the ISC, and Musa et al. (2012) for East Africa[72, 94]. Full model specifications are presented in the Appendices (Tables C.1 and C.2 for the ISC models, and Tables D.1 and D.2 for the East Africa models). As discussed in Section 6.5, the optimal intercept for a particular setting should be selected based on data describing local relapse rates.

³The number of new prediction models being published continues to double every 5 years[343].

6.4.2 Implementation Method

Models presented in this thesis can be implemented either as risk (sum) scores or as electronic calculators accessible through a mobile phone application or web interface[103, 185]. Risk scores integrate more naturally into routine bedside practice but require model simplification, with consequent loss of predictive accuracy. In contrast, electronic calculators preserve the full model specification and their predictive potential. With smartphone use now near-ubiquitous and barriers to application development and deployment continuing to fall, integrating all four models into a unified application is likely the most practical and accessible approach[344, 345].

6.4.3 Potential Applications

Since the models were developed from patients who achieved an initial cure (i.e. excluding patients with initial treatment failure), the intended time of risk stratification must therefore be the time of initial cure assessment. However, in the absence of routine follow-up at 1 month — as is typically the case in the ISC — risk stratification must instead be performed at the time of treatment, with the caveat that the predictions only become valid once the patient achieves initial cure (i.e. remains symptom-free at approximately one month after treatment initiation). In East Africa, risk stratification could be performed after completion of the 17-day treatment course of PM/SSG.

A major benefit of all clinical prediction models is that, where the variables required for model application are already captured, calculating an outcome (relapse) risk costs only a few seconds of someone’s time. Subsequent actions would then depend on the resources available. As a low-expenditure strategy, model-derived relapse risk could simply form part of patient counselling on discharge. Patients could be informed if they are in a high-risk group, promoting higher vigilance for new symptoms and facilitating earlier presentation for re-treatment. More intensive management strategies could range from establishing routine follow-up of high-risk patients (e.g. at 1, 3 and 6 months after treatment), to targeting high-risk patients with tissue aspirates to confirm initial cure (i.e. one month after treatment), or even by offering more intensive treatment regimens to reduce the relapse risk altogether.

Further engagement with clinicians, programme staff, and policymakers is vital to determine how exactly the models can be best integrated with day-to-day practice. Statistical performance alone is not sufficient: routine use must also fit clinical need, workflow, and resource availability.

6.5 Implications for Future Research

It is hoped that the models presented here will provide a catalyst for further research into the determinants of VL relapse. Such research could focus on the external validation and/or updating of VL relapse prediction tools, or provide further novel insights into the mechanistic link between VL relapse and host, parasite, and treatment factors.

6.5.1 Prediction Modelling

The reported model performance measures are only valid when applied to participants of the contributed studies: patients without severe disease, without HIV, and treated under trial conditions. To build user confidence in the models' real-world applicability, future research should focus on model evaluation (external validation) using data collected from routine practice, with updating where necessary[200].

Routine programmatic data collection can make this possible. Electronic surveillance systems, such as the Kala-azar Management Information System (KAMIS) in India[346] and the Brazilian national notifiable diseases information system (Sistema de Informação de Agravos de Notificação — SINAN)[347] offer important opportunities for external validation and updating. However, in order to evaluate the full models, sufficient predictor information must be captured at the time of treatment initiation and it must also be possible to confidently link relapse events with the index presentation. Whilst on paper this may appear straightforward, in practice this is often not the case. Missing identifiers, patient movement, and fragmented records can preclude the linking of consecutive VL episodes. The experience of MSF in South Sudan demonstrates how challenging this can be outside of research settings[133].

Once a model is externally validated, with proven accuracy in the target setting, the next question is 'does the model lead to better decisions and improved patient outcomes?' Model-based clinical decisions are typically based on an absolute risk threshold. For example, 'all patients with an estimated relapse risk of over 10% will be invited to a follow-up appointment at 3 months', or 'all patients with a relapse risk over 20% will be offered a tissue aspirate one month after treatment completion.' Selection of the optimal threshold should, ideally, be guided by an assessment of the direct risks and benefits to the patient, and costs. Analysis of 'net benefit', and how this varies with the cutoff threshold, is advocated in the literature, but rarely performed[348].

Any model deployment, for example, as a component of the ISC or East Africa elimination campaign, should be accompanied by research that evaluates both

the model’s predictive accuracy (external validation), and the model’s impact on decision-making.

6.5.2 Mechanistic Understanding

As explored in Chapter 1, Section 1.3.3, a growing body of evidence supports the link between relapse and measures of host immunity and/or parasite burden. The importance of these biomarkers is strongly echoed in the model-supported framework presented in Section 6.2.2. In the proposed causal model (Figure 6.2), mechanistic links between the majority of the identified predictors and relapse can be plausibly traced back to markers of host immunity — a framework that challenges a more intuition-based approach, where markers of disease severity would be the strongest predictors of relapse. New evidence presented in this thesis is not only compatible with our current understanding of relapse determinants, but also offers fresh insight into how readily-available host factors (e.g. symptom duration, age, anaemia, malnutrition) can act as surrogates of host immunity, and therefore also relapse. This is especially important given the challenges faced in measuring novel markers of immunity (e.g. IgG subclass titres, T-cell subsets, IFN γ and PD1 levels), or parasite burden (e.g. molecular methods) outside of research or high-resource settings.

Reflecting on the above, researchers should not focus on biomarkers in isolation when building models predicting VL outcomes, regardless of whether the models are intended as predictive or exploratory, or whether in immunocompetent or immunosuppressed patients. Future research should ideally incorporate clinical correlates of host immunity *in addition* to novel biomarkers when optimising their statistical models. Routinely measured clinical correlates — including symptom duration, spleen size, severity of malnutrition and anaemia — can provide important extra information with strong predictive potential.

6.6 Conclusion

As elaborated in Chapter 1, relapse represents a growing and often underappreciated adverse event in patients treated for VL. Of particular note, relapse strains are more likely to harbour resistance mutations and demonstrate higher infectivity, thereby threatening WHO-supported elimination efforts underway in both the ISC and East Africa. The magnitude of the threat is understated, since most well-designed prospective studies exclude patients at higher relapse risk due to stringent eligibility and test-of-cure criteria, and also because relapses occurring beyond 6 months after treatment are not recorded.

Despite the importance of relapse both at the individual and public health levels, a systematic review of VL prediction models — presented in Chapter 2 and published early 2026[158] — reveals the absence of any VL relapse prediction model. To address this important evidence gap, whilst also supporting a WHO-led call for research into a non-invasive test-of-cure[88], clinical trial data from the IDDO VL data platform were leveraged to develop the first VL relapse prediction models.

In Chapter 3, a robust methodology is presented for the development and internal validation of four relapse models across the ISC and East Africa. Statistical techniques are described that combine hierarchical logistic regression modelling — accounting for heterogeneity across the contributed studies — with multiple imputation techniques to account for data missing both sporadically within studies, or systematically across studies. Many statistical methods were drawn from recently published methodological literature, for example, on sample size determination and the reporting of aggregate performance measures for hierarchical models derived from multiple imputations.

Chapters 4 and 5 present the model findings in full, identifying a small number of biologically plausible predictors of relapse across two distinct endemic regions (ISC and East Africa, respectively). In particular, parasite grade and fever duration emerged as core prognostic factors across both regions. Other predictors, including anaemia, spleen size, age, WBC count, and malnutrition, were more context-dependent, highlighting important differences in the clinical and epidemiological pathways that shape relapse risk in the ISC and East Africa. Together, these findings are consistent with the mechanistic model developed in this Chapter, in which readily available clinical variables serve as surrogates of underlying host immunity and disease severity.

In summary, this thesis presents a data-driven and clinically focused exploration of both the causes and nature of VL relapse. Through the application of robust IPD meta-analysis techniques to data from over 6,500 participants across 19 prospective studies, predictive models are developed to forecast VL relapse at the individual patient level. Although the models would benefit from external validation before routine deployment, they already offer an important foundation for clinical risk stratification, patient counselling, and future programme planning. More broadly, this work reinforces relapse as a clinically meaningful and programmatically important outcome. It demonstrates that even in low-resource settings, routinely measured host factors can be leveraged to generate meaningful predictions of VL relapse risk at the individual patient level.

Appendices



Background Literature Search

A literature search was performed with the aim of writing a narrative review on the VL relapse. Specifically to address the following questions:

- What is the burden of relapse?
- When does relapse occur?
- What is the cause (mechanism) of relapse?
- What are the determinants of relapse?

All searches performed on August 11th 2025 on PubMed, Embase and Web of Science (Boxes A.1–A.3). Search performed from database inception. After deduplication with Covidence[[183](#)] 1,891 articles were identified. Following title and abstract review, 55 articles were identified for inclusion in the narrative review.

Inclusion criteria:

Inclusion criteria: (1) Publications looking at associations between VL relapse occurrence and host/parasite characteristics during, before, or shortly after treatment of index case. (2) Systematic/literature reviews looking at relapse determinants. (3) All ages, geographical locations.

Exclusion criteria: (1) Conference abstracts, protocols, case reports. (2) Articles not published in English. (3) Non-human studies.

Box A.1: PubMed search terms (932 hits)

```
("Leishmaniasis, Visceral"[Mesh] OR "visceral leishmaniasis" OR  
"leishmaniasis, visceral" OR "kala azar" OR "kala-azar") AND  
("Recurrence"[Mesh] OR relapse* OR recurrent OR recurrence OR  
recrudescence OR "treatment failure")
```

Box A.2: Embase search terms (1303 hits)

1. visceral leishmaniasis.mp. or exp visceral leishmaniasis/
2. kala-azar.mp.
3. 1 or 2
4. exp relapse/ or relapse*.mp.
5. recurrence*.mp. or exp recurrent disease/
6. recurrent.mp.
7. recrudescence.mp.
8. 4 or 5 or 6 or 7
9. treatment failure.mp. or exp treatment failure/
10. 8 or 9
11. 3 and 10

Box A.3: Web of Science (1097 hits)

1. TS=((Leishmaniasis and Visceral) or (Kala azar) or (Kala azar))
2. TS=((relapse\$) or (recurren\$) or (recrudescence) or (treatment failure\$))
3. #2 AND #1

B

Appendix — Model methodology

C

Appendix — Results: Indian Subcontinent Model

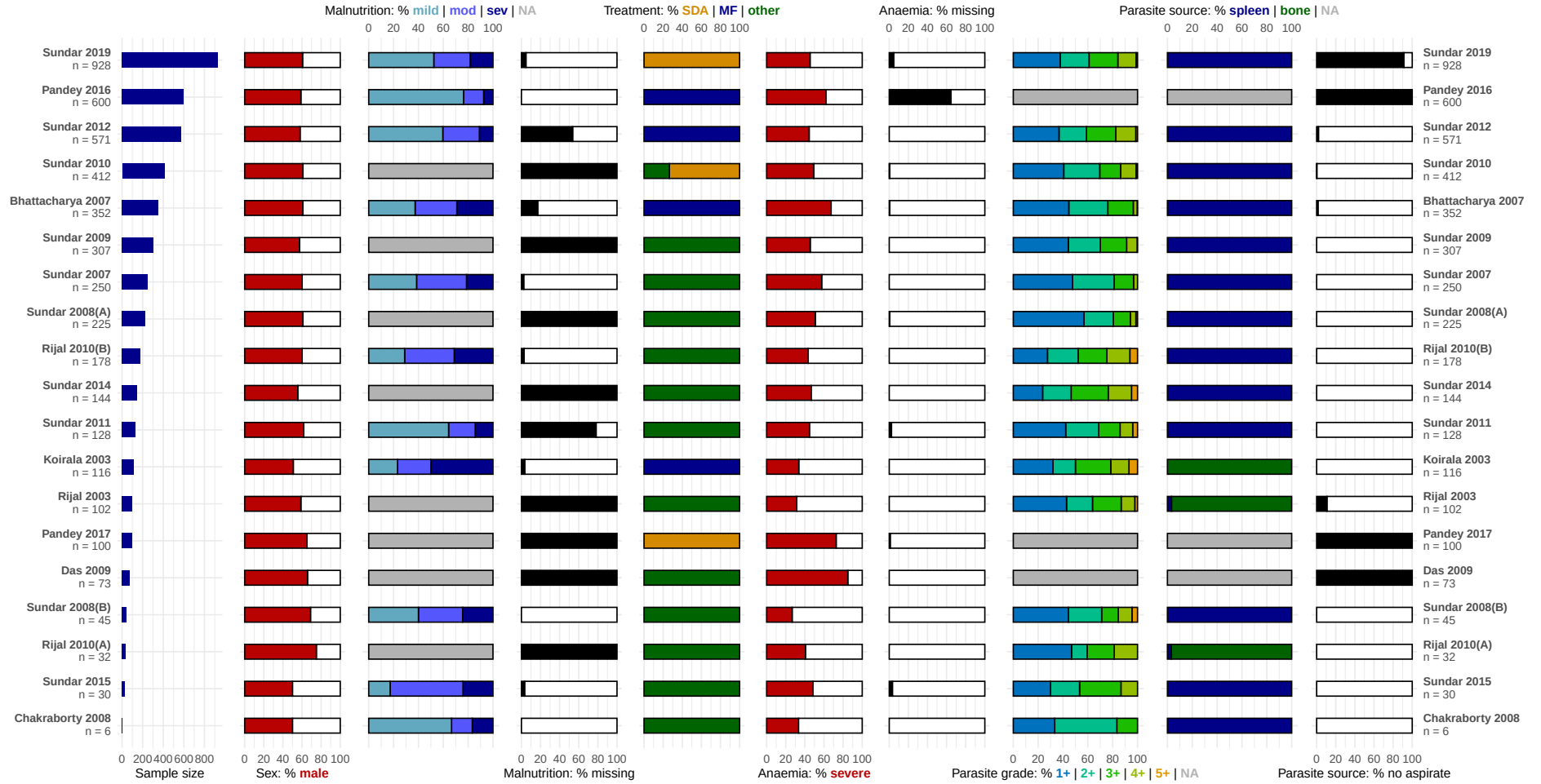


Figure C.1: Distribution of categorical predictors by study. Missing data excluded from stacked bar charts for malnutrition, parasite grade. Where aspirates were performed, the source was never missing. NA: unable to show distribution as all study-specific data are missing. MF: miltefosine; SDA: single-dose liposomal amphotericin B

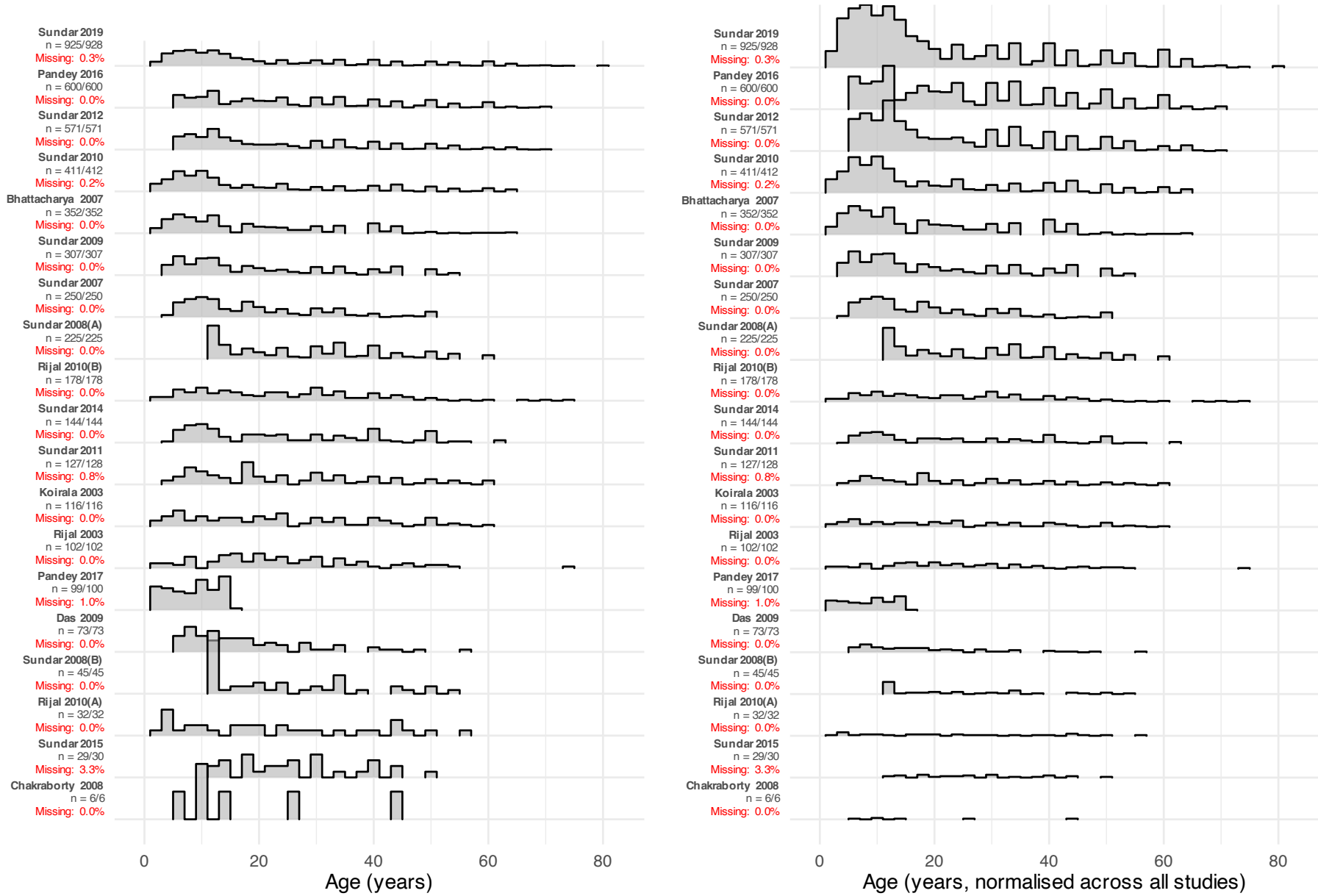


Figure C.2: Distribution of age across studies from the Indian subcontinent. Missing age data described by study.

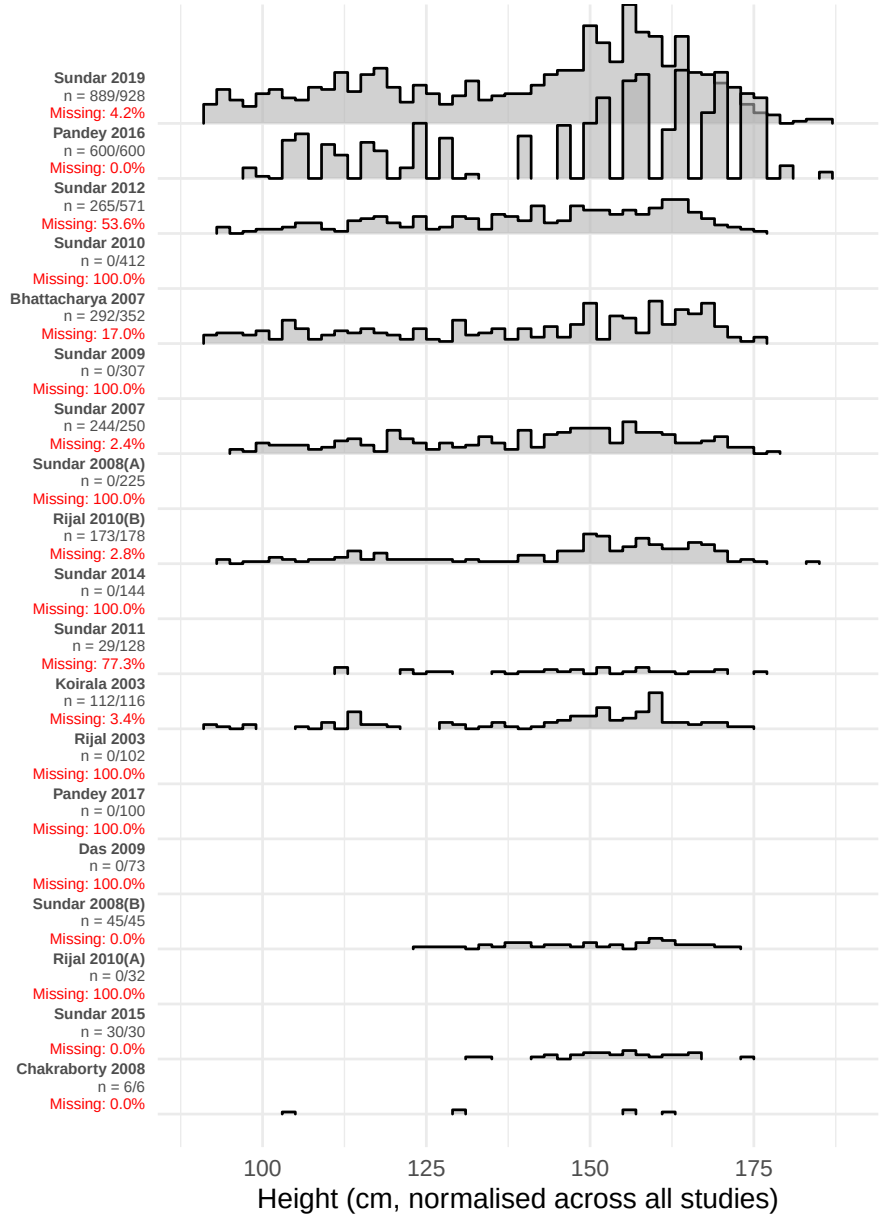
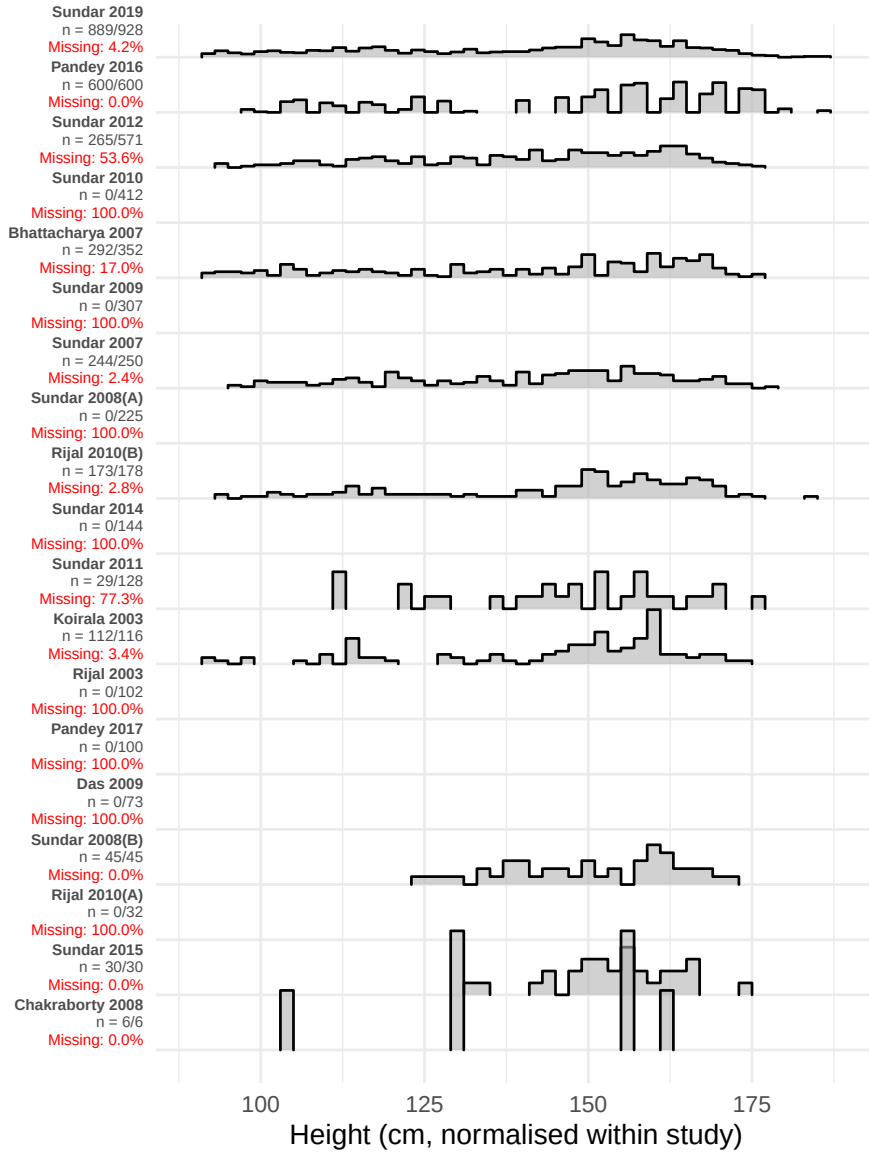


Figure C.3: Distribution of height across studies from the Indian subcontinent. Missing data described by study.

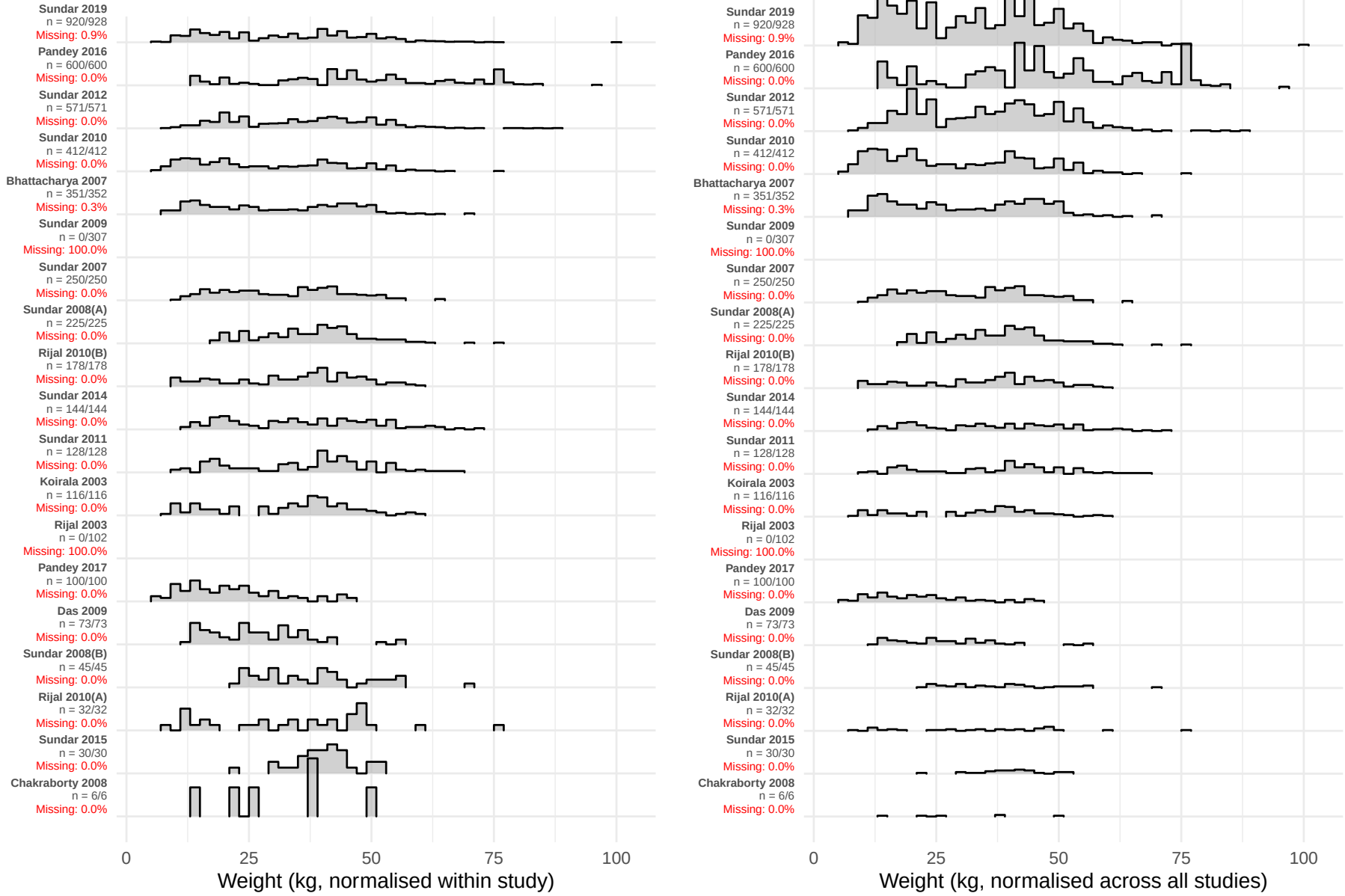


Figure C.4: Distribution of weight across studies from the Indian subcontinent. Missing data described by study.

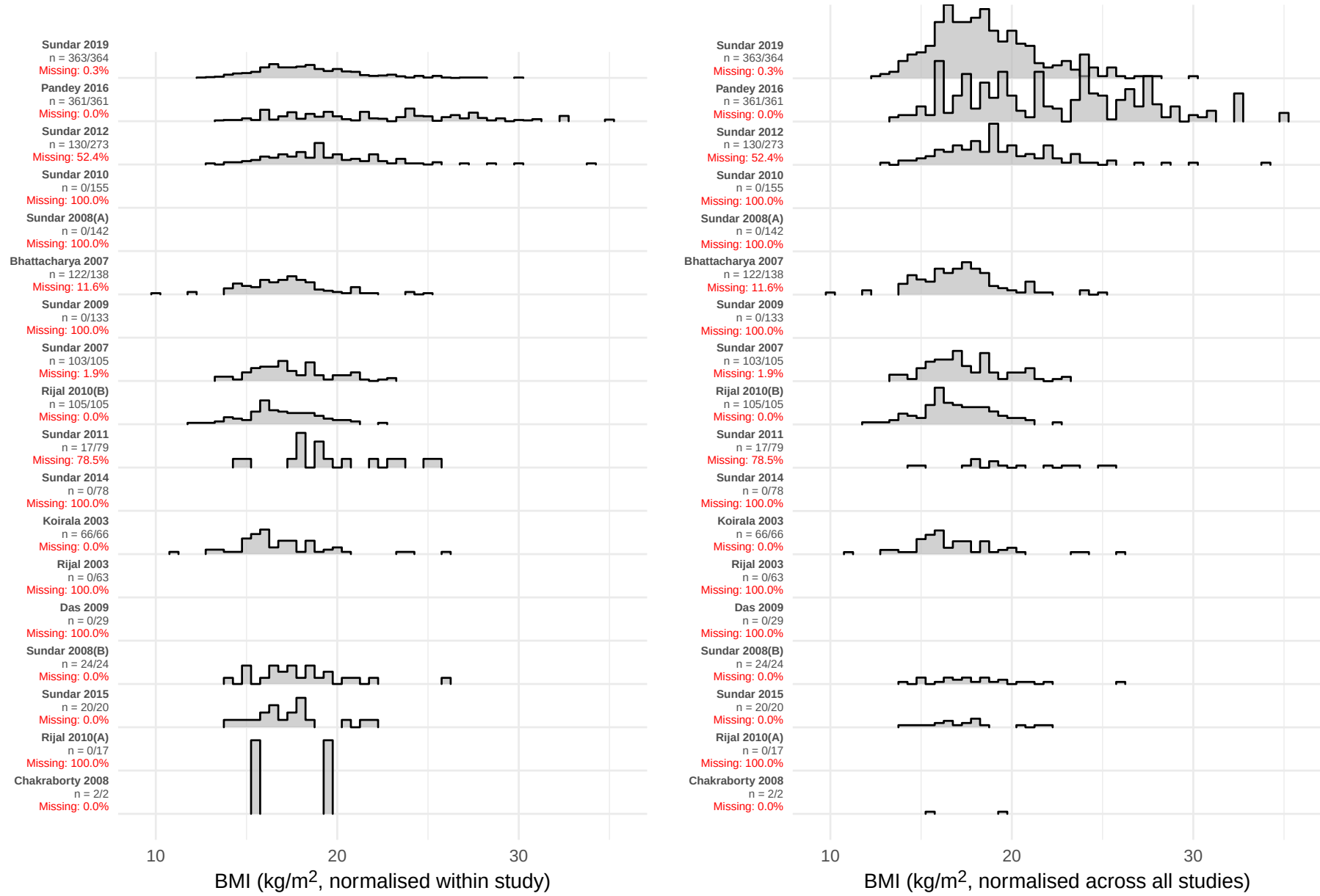


Figure C.5: Distribution of BMI across studies from the Indian subcontinent. Including only participants aged 19 and over (Pandey 2017 excluded). Missing data described by study.

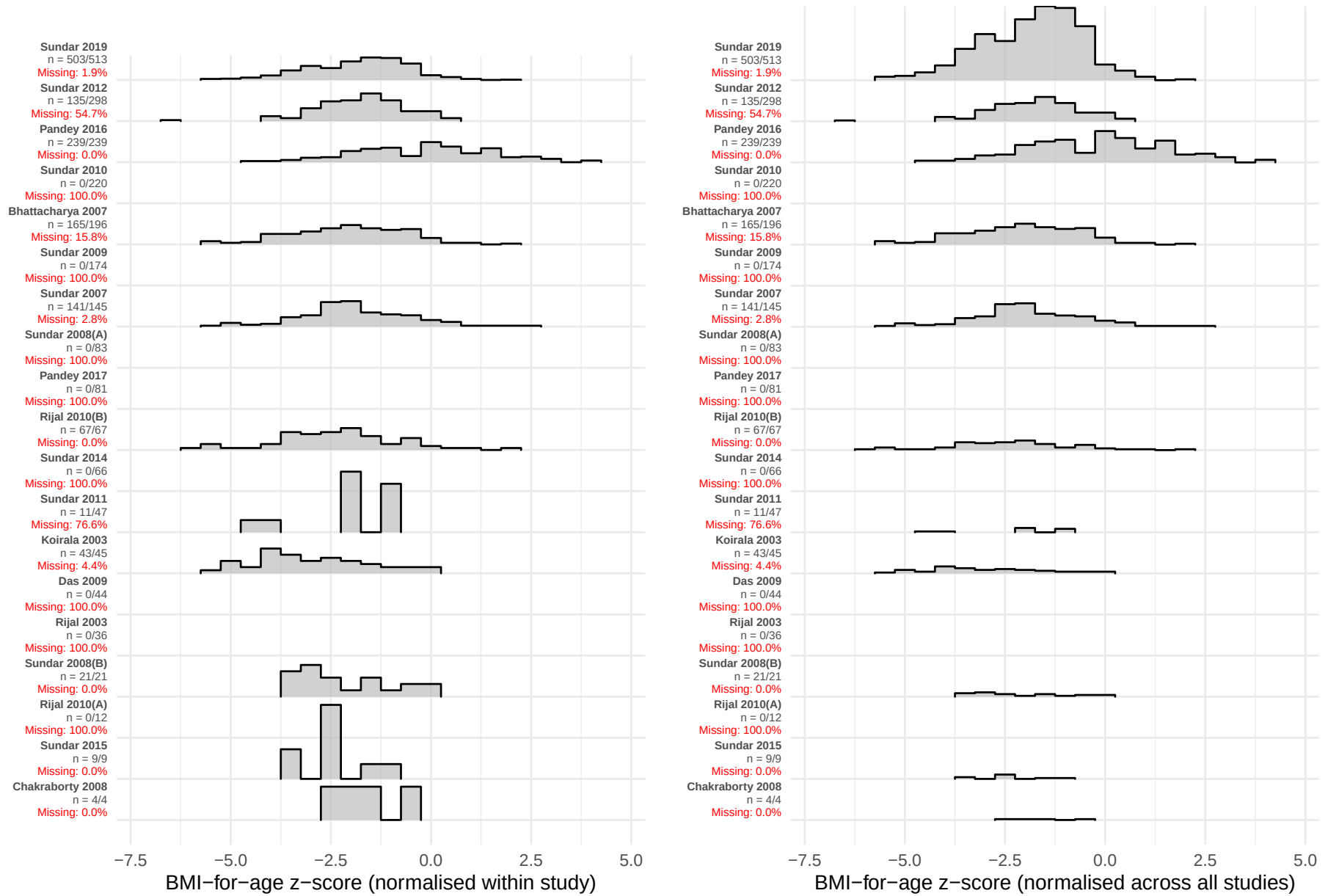


Figure C.6: Distribution of BMI-for-age z score across studies from the Indian subcontinent. Including only participants aged from 5–18, inclusive. Missing data described by study.

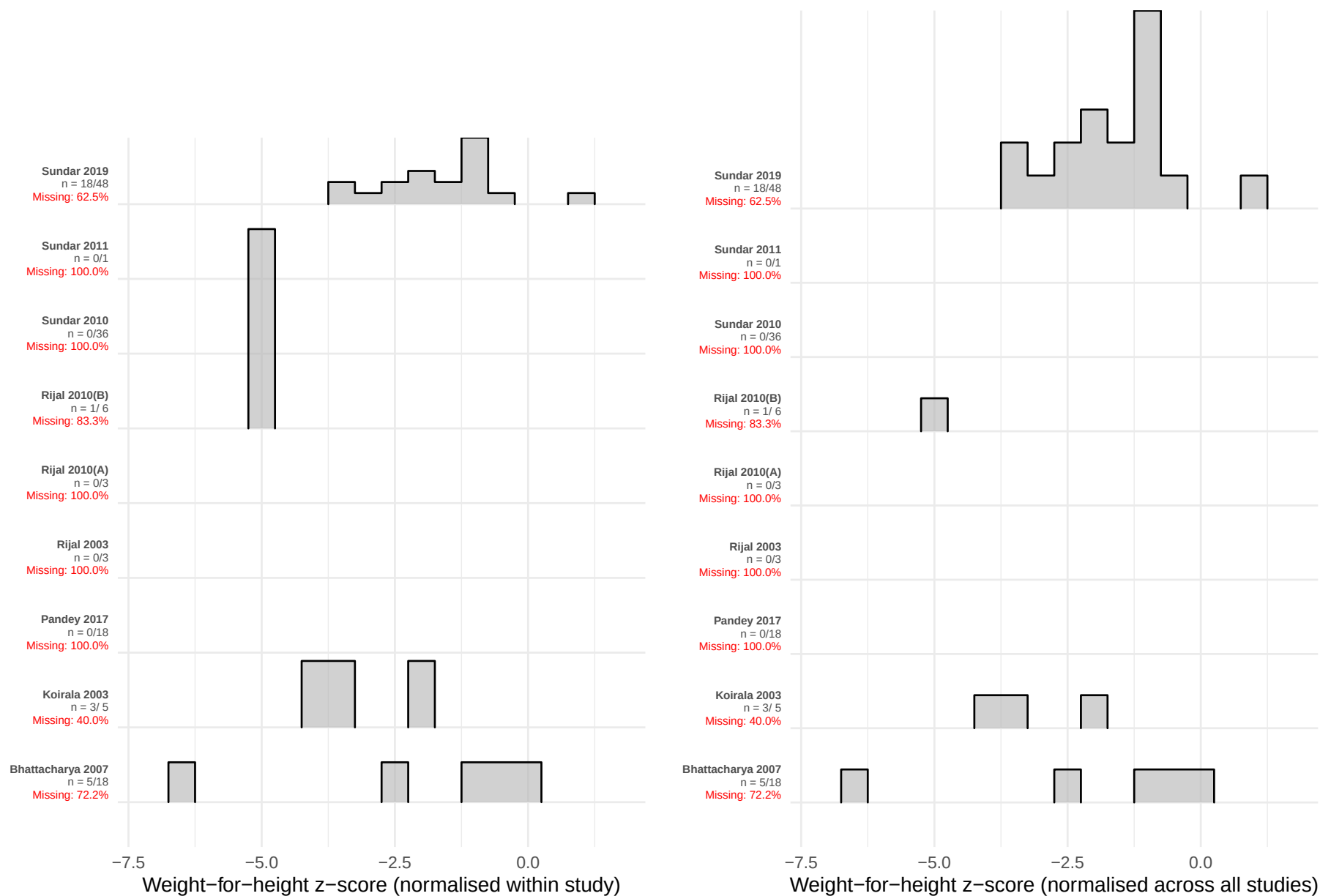
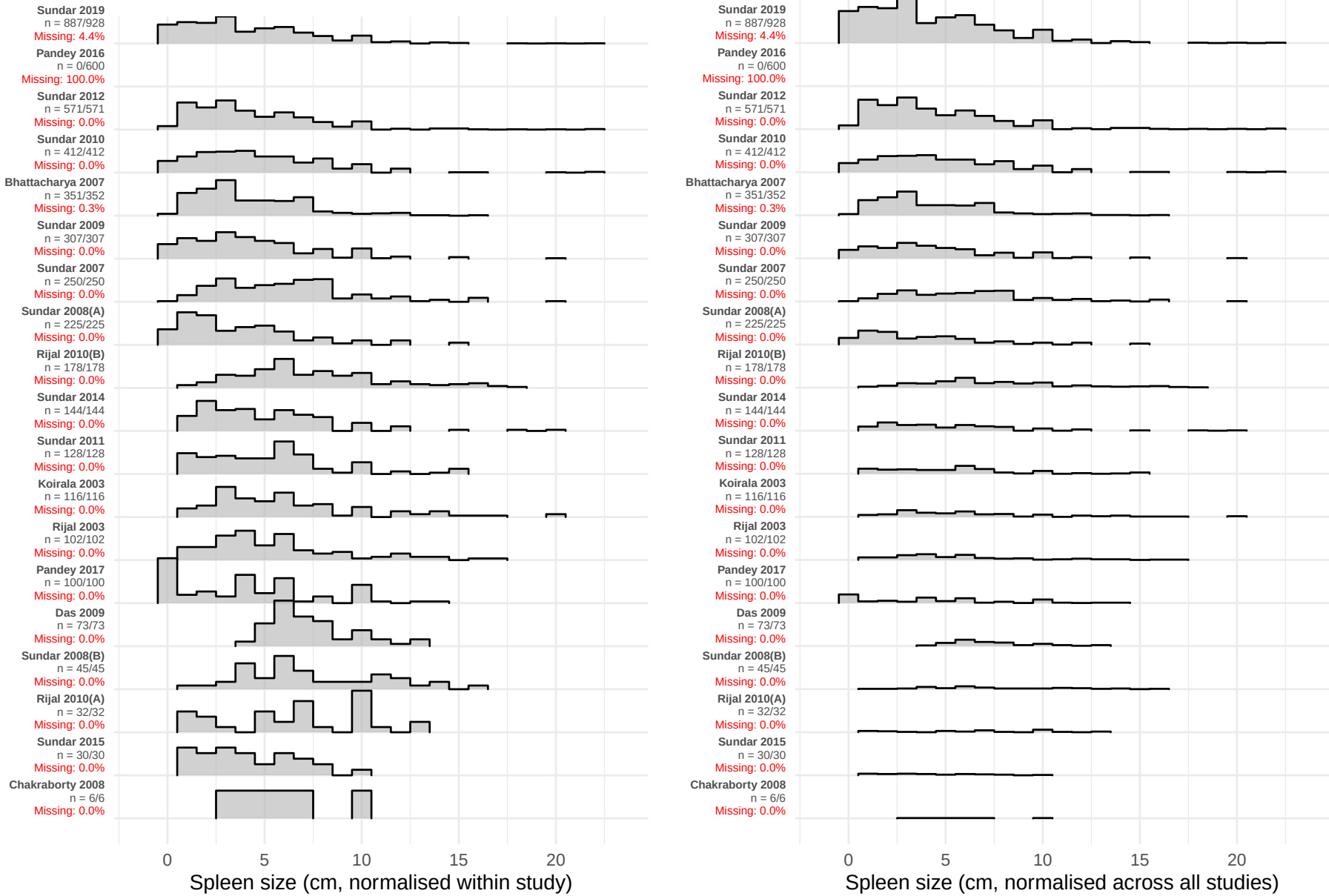


Figure C.7: Distribution of weight-for-height z score across studies from the Indian subcontinent. Including only participants aged under 5. Missing data described by study.



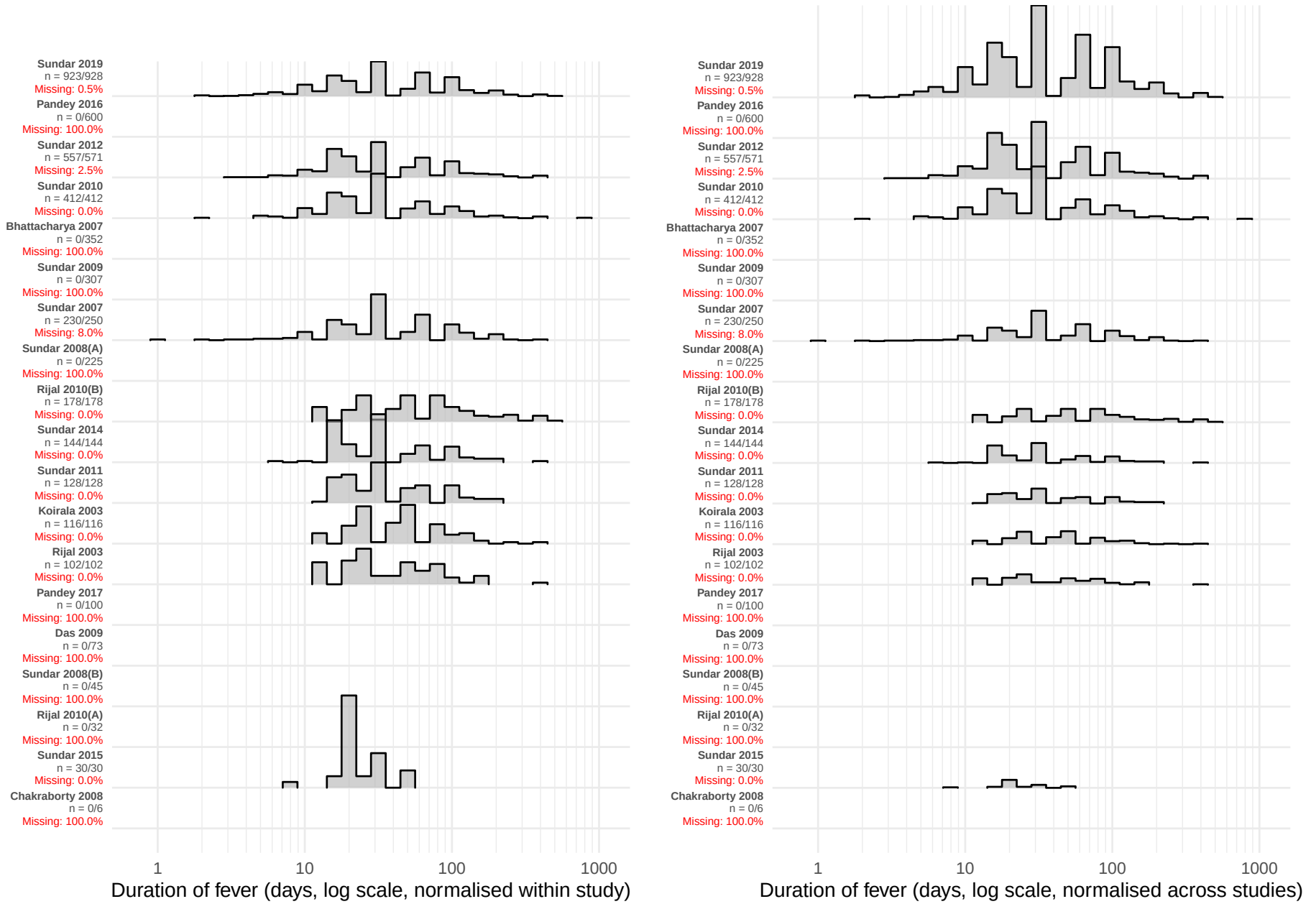


Figure C.9: Distribution of fever duration (log scale) across studies from the Indian subcontinent. Missing data described by study.

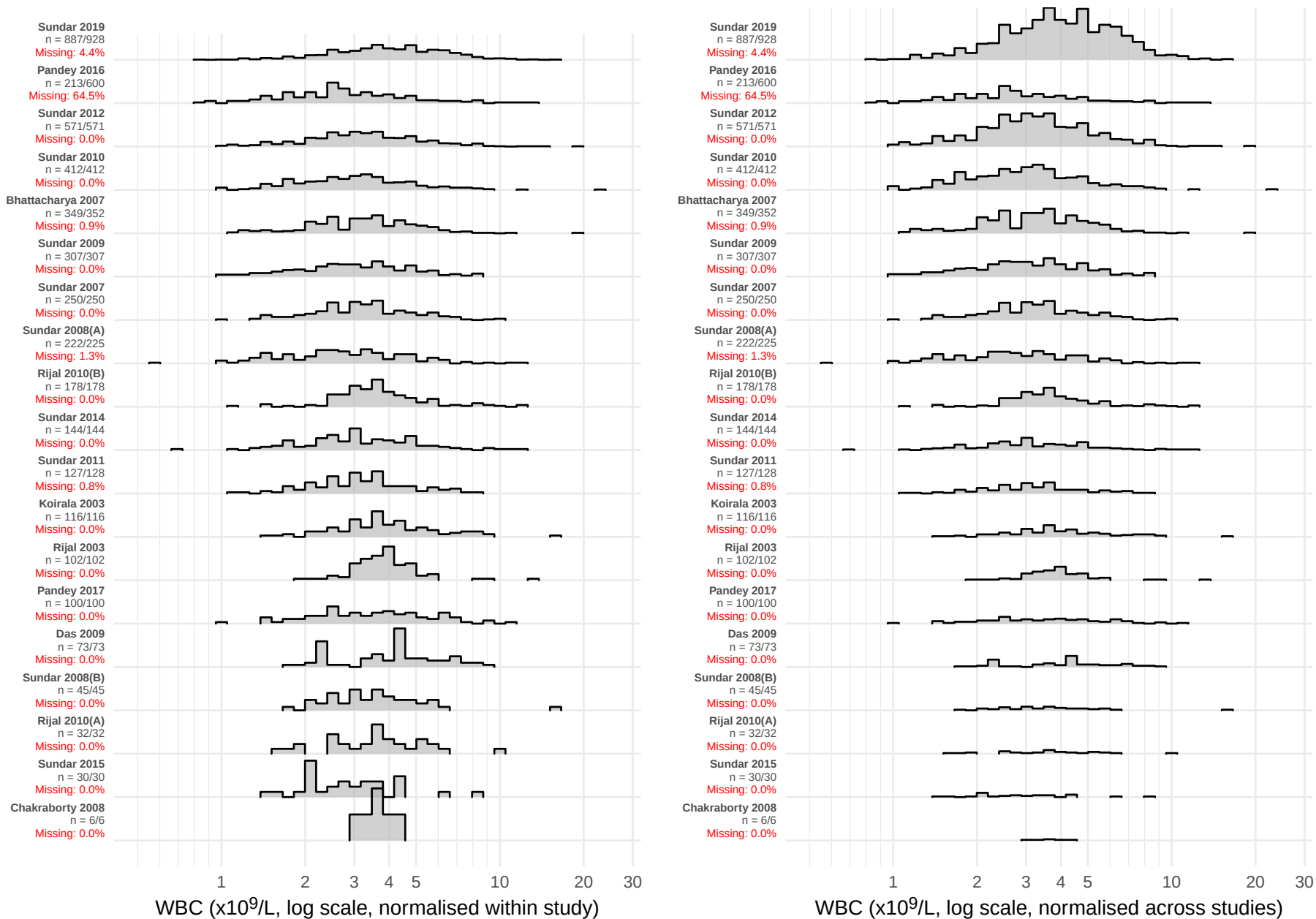


Figure C.10: Distribution of white blood cell count (log scale) across studies from the Indian subcontinent. Missing data described by study.

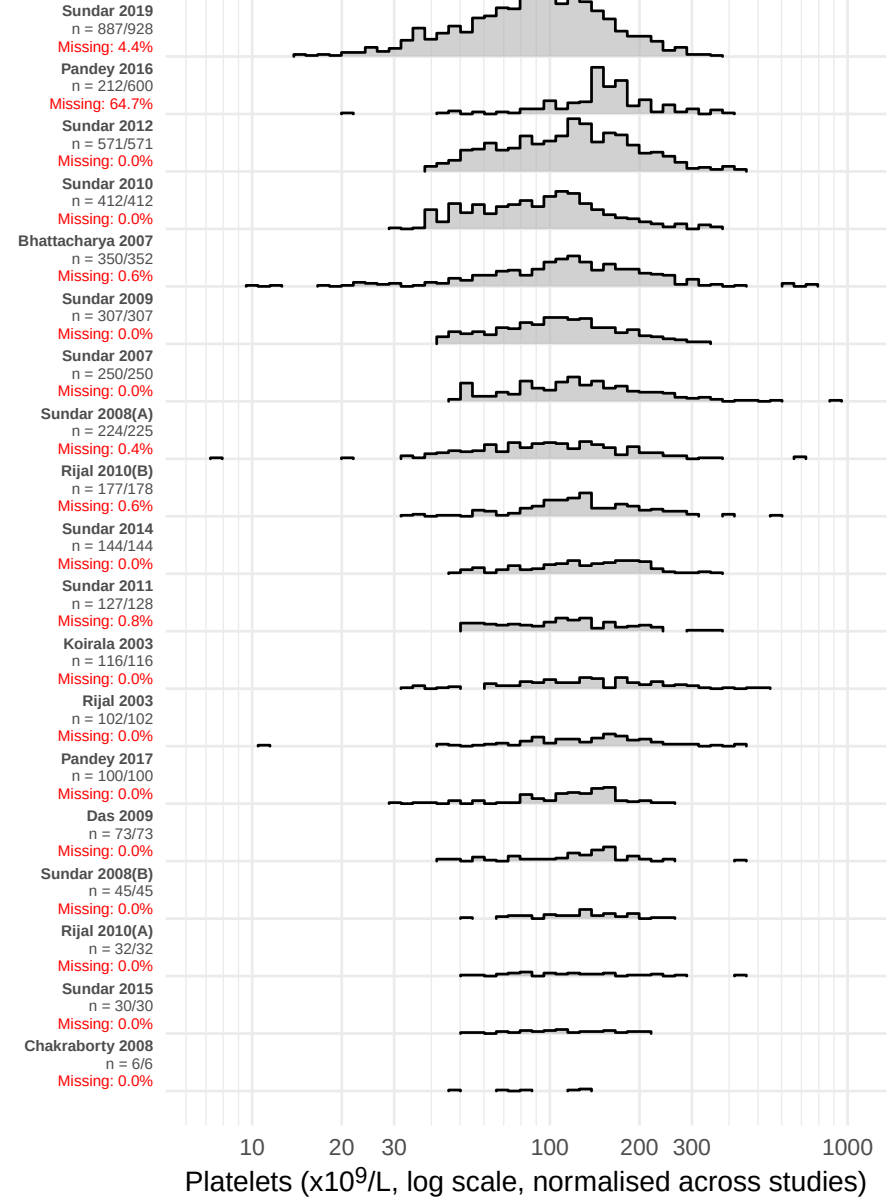
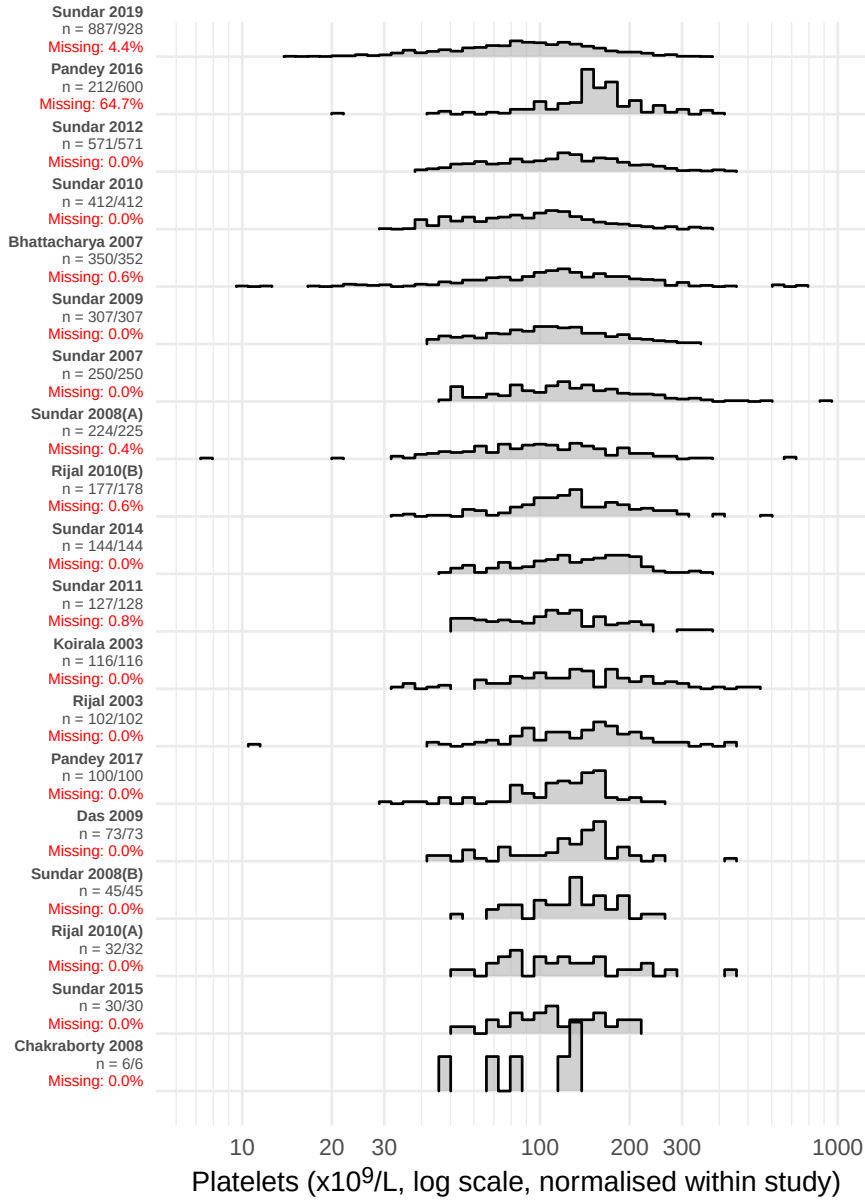


Figure C.11: Distribution of platelet count (log scale) across studies from the Indian subcontinent. Missing data described by study.

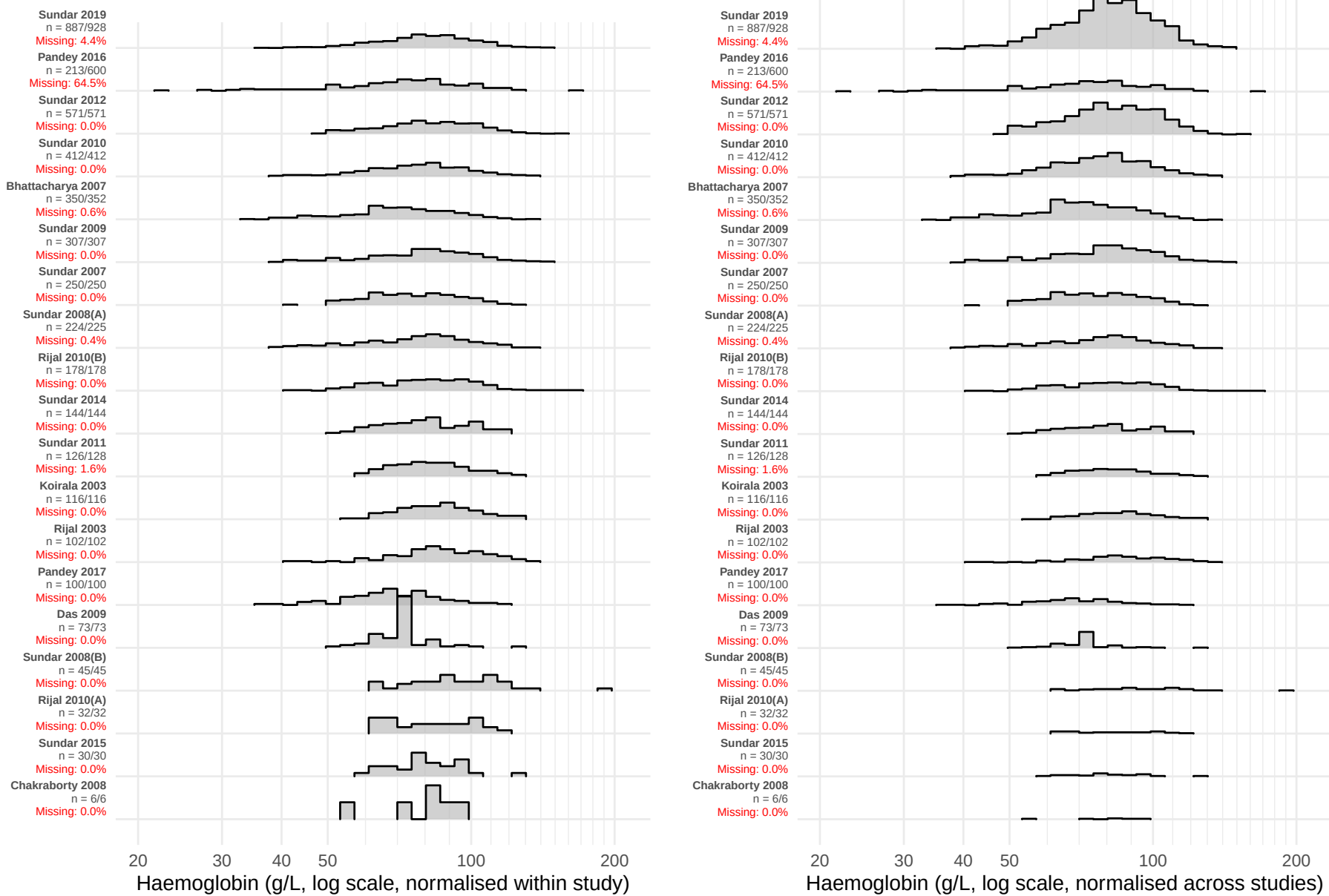


Figure C.12: Distribution of haemoglobin (log scale) across studies from the Indian subcontinent. Missing data described by study.

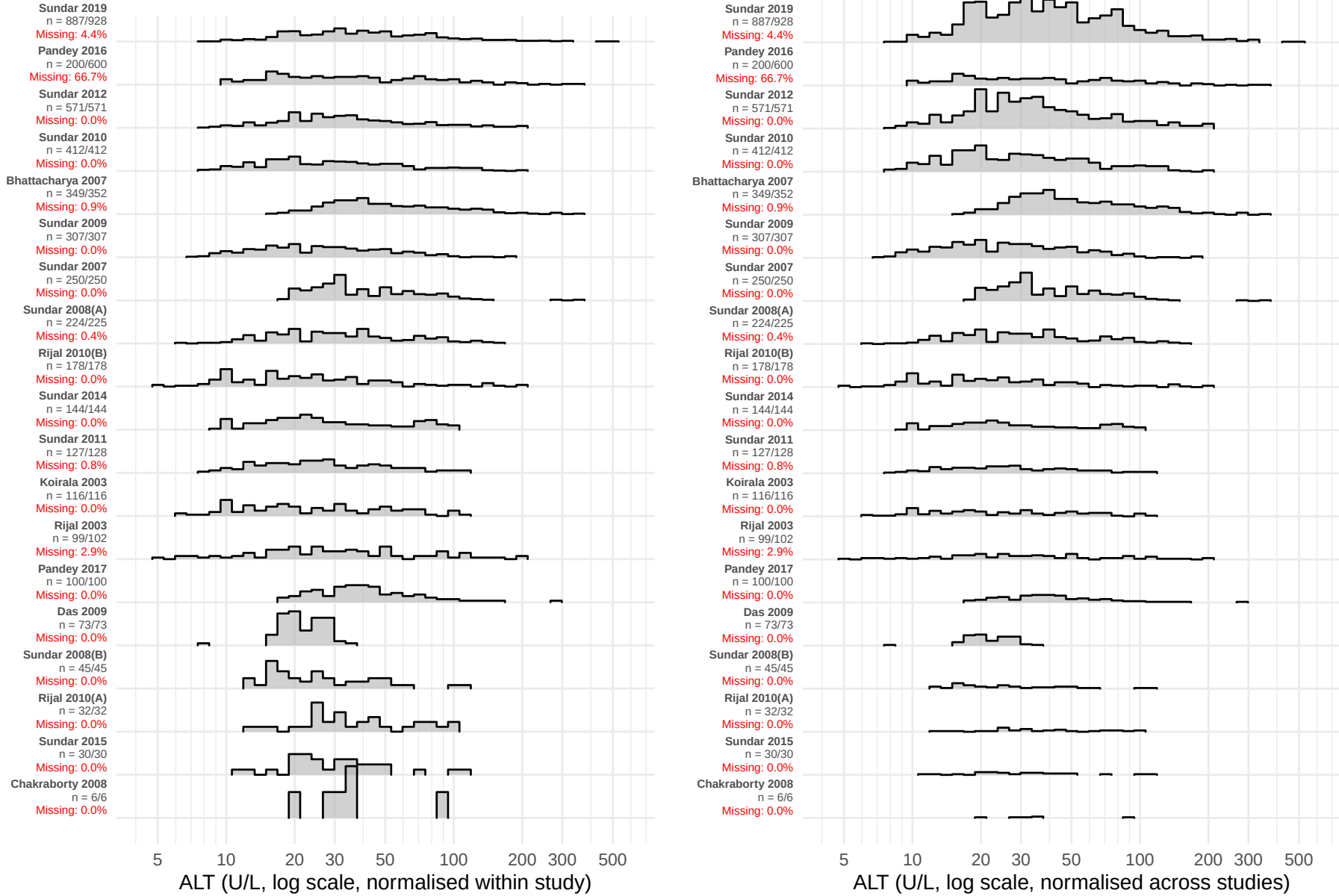


Figure C.13: Distribution of alanine transaminase (ALT, log scale) across studies from the Indian subcontinent. Missing data described by study.

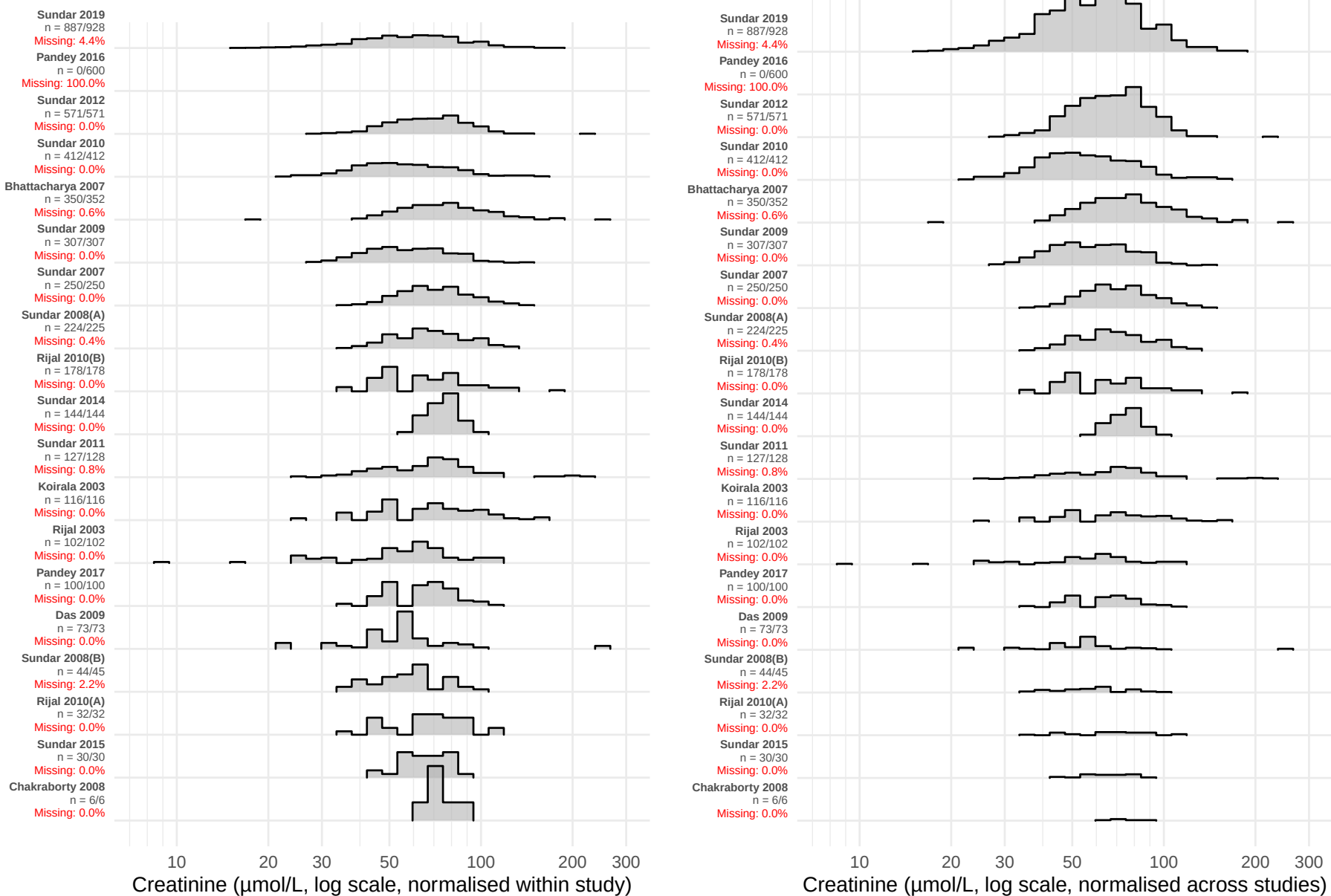


Figure C.14: Distribution of creatinine (log scale) across studies from the Indian subcontinent. Missing data described by study.

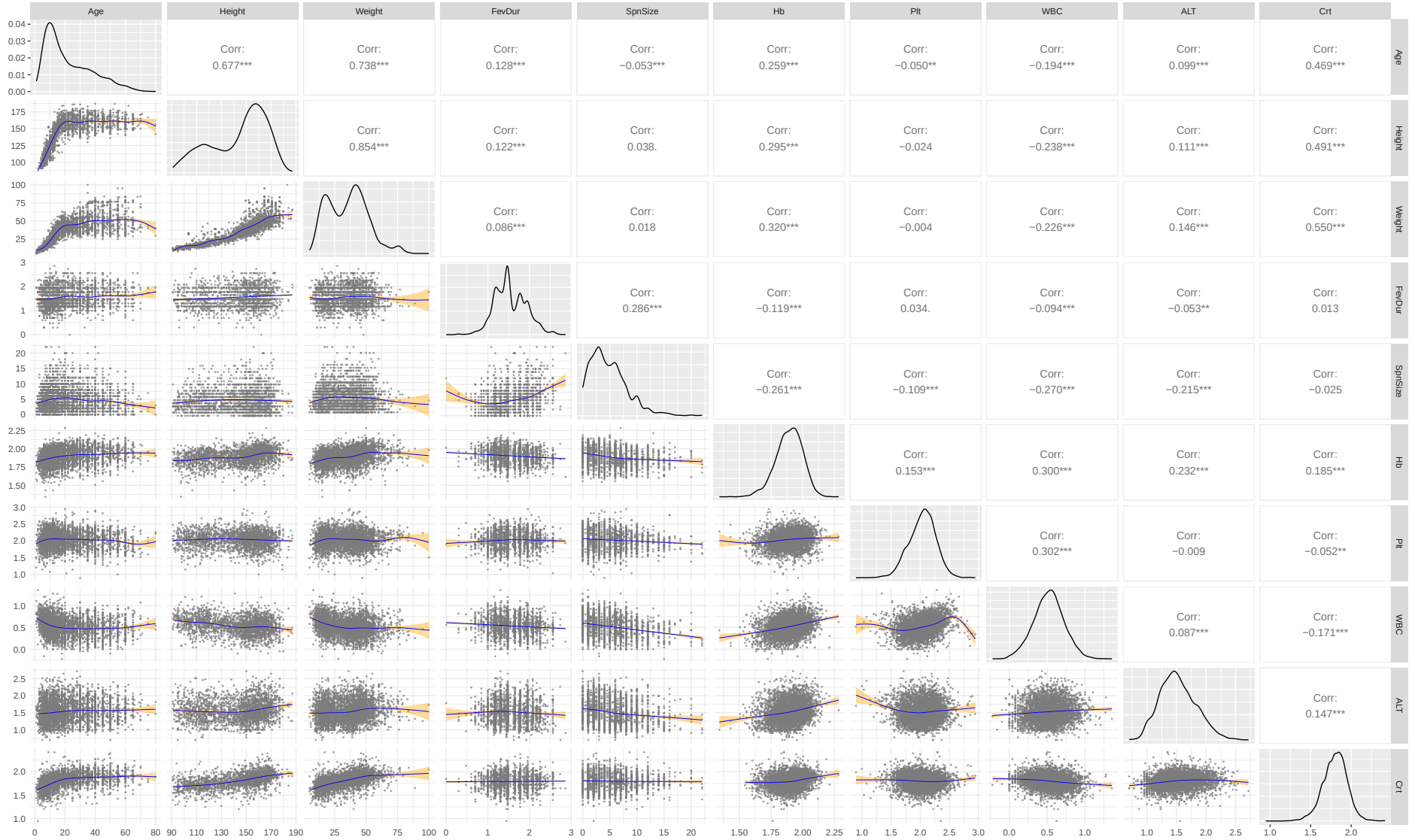


Figure C.15: Correlation between continuous variables. For scatter plots, a univariable generalised additive model is fitted (blue line) with 95% confidence interval ribbon filled (orange area). Pearson correlation coefficients are presented, ‘***’ $p < 0.001$; ‘**’ $p < 0.01$; ‘*’ $p < 0.05$; ‘.’ $p < 0.10$. Age: years; Height: cm; Weight: kg; FevDur: duration of fever, $\log_{10}(\text{days})$; SpnSize: spleen size (cm); Hb: haemoglobin, $\log_{10}(\text{g/L})$; Plt: platelets, $\log_{10}(\times 10^9/\text{L})$; WBC: white blood cells $\log_{10}(\times 10^9/\text{L})$; ALT: alanine aminotransferase, $\log_{10}(\text{U/L})$; Cr: creatinine $\log_{10}(\mu\text{mol/L})$.

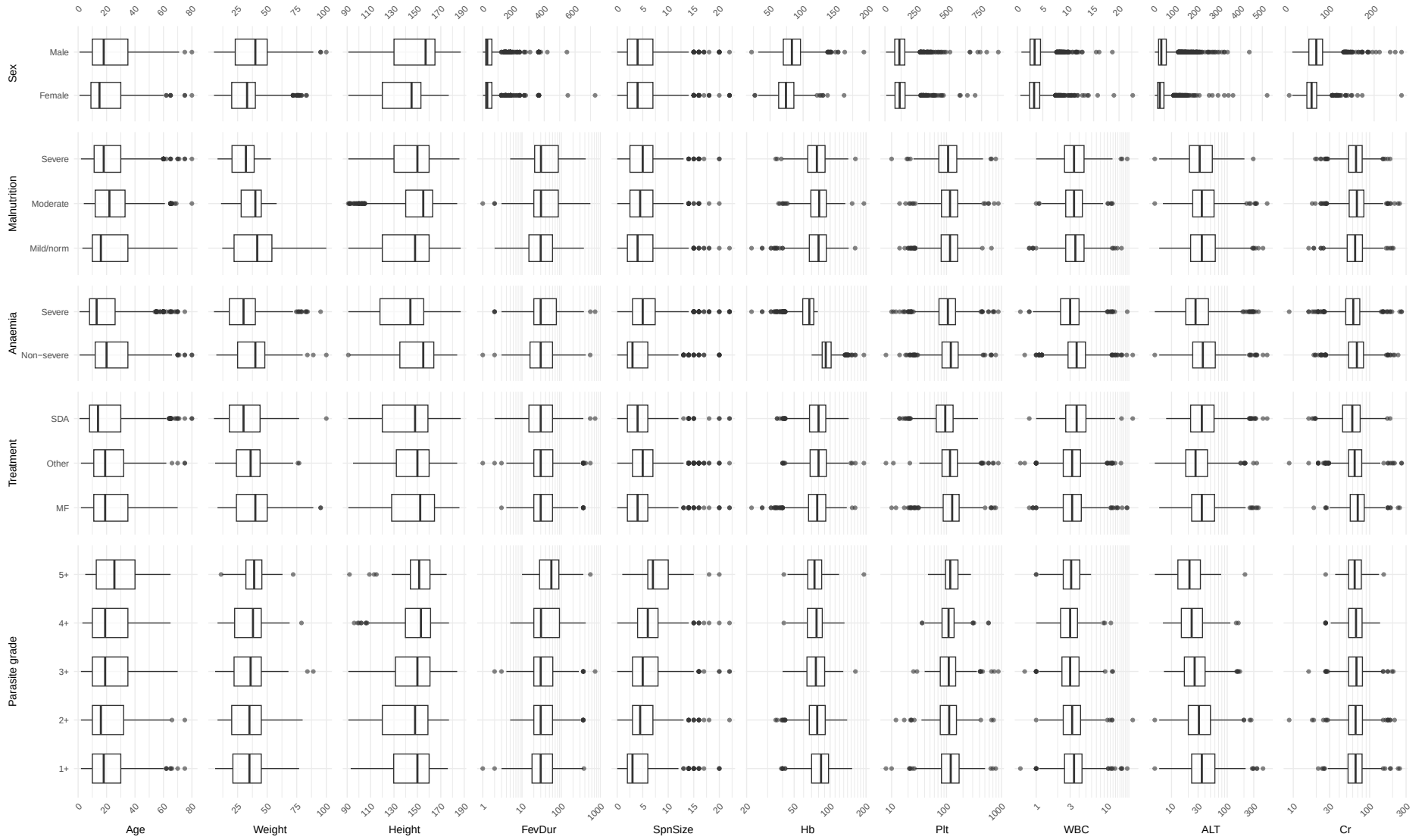


Figure C.16: Correlation between continuous and categorical variables. FevDur and laboratory tests axes transformed to log10 scale. Age: years; Height: cm; Weight: kg; FevDur: duration of fever, days; SpnSize: spleen size, cm; Hb: haemoglobin, g/L; Plt: platelets, $\times 10^9/L$; WBC: white blood cells $\times 10^9/L$; ALT: alanine aminotransferase, U/L; Cr: creatinine $\mu\text{mol/L}$.

term	estimate ¹	std.error	t statistic	df (effective)	p value
Fixed intercept	-2.9166	0.2592	-11.25	1123.60	6.54E-28
Parasite grade ²	0.2669	0.0781	3.42	174.73	7.88E-04
Age (linear) ³	-0.2352	0.0975	-2.41	3222.98	1.59E-02
Age (squared) ³	0.1352	0.0658	2.05	4178.35	4.00E-02
Anaemia: Severe ⁴	-0.3860	0.1597	-2.42	875.81	1.59E-02
Fever duration ⁵	-0.5594	0.1175	-4.76	77.95	8.68E-06
Treatment: SDA ⁶	-0.8349	0.3483	-2.40	3059.66	1.66E-02
Treatment: Other ⁶	-0.7159	0.2880	-2.49	2905.85	1.30E-02
Unadjusted intercept (Sundar 2019): -2.9888					
Uniform shrinkage factor: 0.9132					
Adjusted intercepts after shrinkage: Average: -3.1608; Sundar 2019: -3.2514					
Pooled τ^2 : 0.1209. Pooled ICC: 0.0354					

¹ Logit scale before uniform shrinkage.

² Continuous variable, from 1+ to 6+ as described by Chulay and Bryceson[52]. Transformed $x \mapsto x - 1$.

³ Units: years. Standardised by mean (22.301633) and standard deviation (15.4159649).

⁴ Reference group: non-severe anaemia.

⁵ Units: days. Converted to natural log scale and then standardised by mean (3.550509) and standard deviation (0.8693318).

⁶ Reference group: standard dose miltefosine.

Table C.1: Model coefficients pooled across 30 imputed datasets using Rubin's Rules. Model **including** parasite grade. df: degrees of freedom; SDA: single-dose liposomal amphotericin B; std.error: standard error

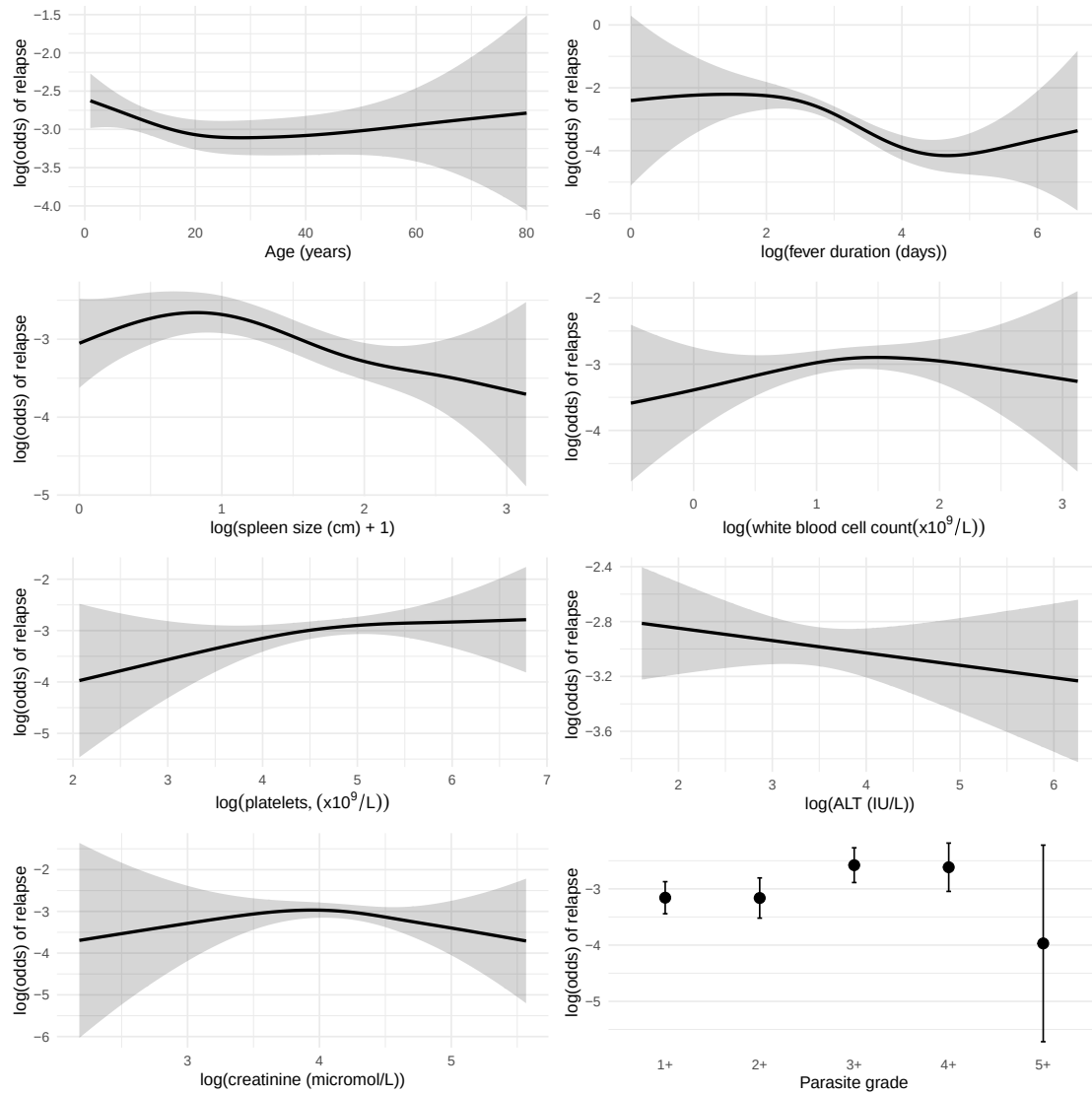


Figure C.17: Associations between transformed continuous predictors and log(odds) of relapse. For each predictor (excluding parasite grade), a univariable generalised additive model spline fit is shown, with 95% confidence intervals. For parasite grade, 95% confidence intervals are calculated for each grade using the Wilson method.

term	estimate ¹	std.error	t statistic	df	p.value
Fixed intercept	-2.6042	0.2547	-10.22	2137.17	5.51E-24
Age (linear) ²	-0.2161	0.0962	-2.25	3968.52	2.47E-02
Age (squared) ²	0.1434	0.0645	2.22	4413.49	2.62E-02
Anaemia: Severe ³	-0.2921	0.1501	-1.95	2350.48	5.17E-02
Fever duration ³	-0.5221	0.1057	-4.94	107.97	2.86E-06
Treatment: SDA ⁴	-0.8438	0.3696	-2.28	3813.56	2.25E-02
Treatment: Other ⁴	-0.7571	0.3078	-2.46	2506.34	1.40E-02
Unadjusted intercept (Sundar 2019): -2.7030					
Uniform shrinkage factor: 0.9174					
Adjusted intercepts after shrinkage: Average: -2.8425; Sundar 2019: -2.9543					
Pooled τ^2 : 0.1507. Pooled ICC: 0.0437					

¹ Logit scale before uniform shrinkage.

² Units: years. Standardised by mean (22.301633) and standard deviation (15.4159649).

³ Reference group: non-severe anaemia.

⁴ Units: days. Converted to natural log scale and then standardised by mean (3.550509) and standard deviation (0.8693318).

⁵ Reference group: standard dose miltefosine.

Table C.2: Model coefficients pooled across 30 imputed datasets using Rubin’s Rules. Model **excluding** parasite grade. df: degrees of freedom; SDA: single-dose liposomal amphotericin B; std.error: standard error

Predictors	Model with PG	Model without PG
Fever duration	500 ¹	500 ¹
Parasite grade	472 ¹	n/a
Age (linear)	418 ¹	415 ¹
Age (squared)	355 ¹	392 ¹
Treatment	333 ¹	337 ¹
Anaemia	314 ¹	263 ¹
Spleen size	244	120
Sex	209	217
Age (cubic)	114	150
ALT	86	184
Platelets	78	78
WBC	51	55
Malnutrition	33	26
Creatinine	30	30

¹ Selected in final model.

Table C.3: Frequency of predictor selection during 500 bootstrap model developments. Predictors sorted by selection frequency in the model with parasite grade. ALT: alanine aminotransferase; n/a: not applicable; PG: parasite grade; WBC: white blood cell count

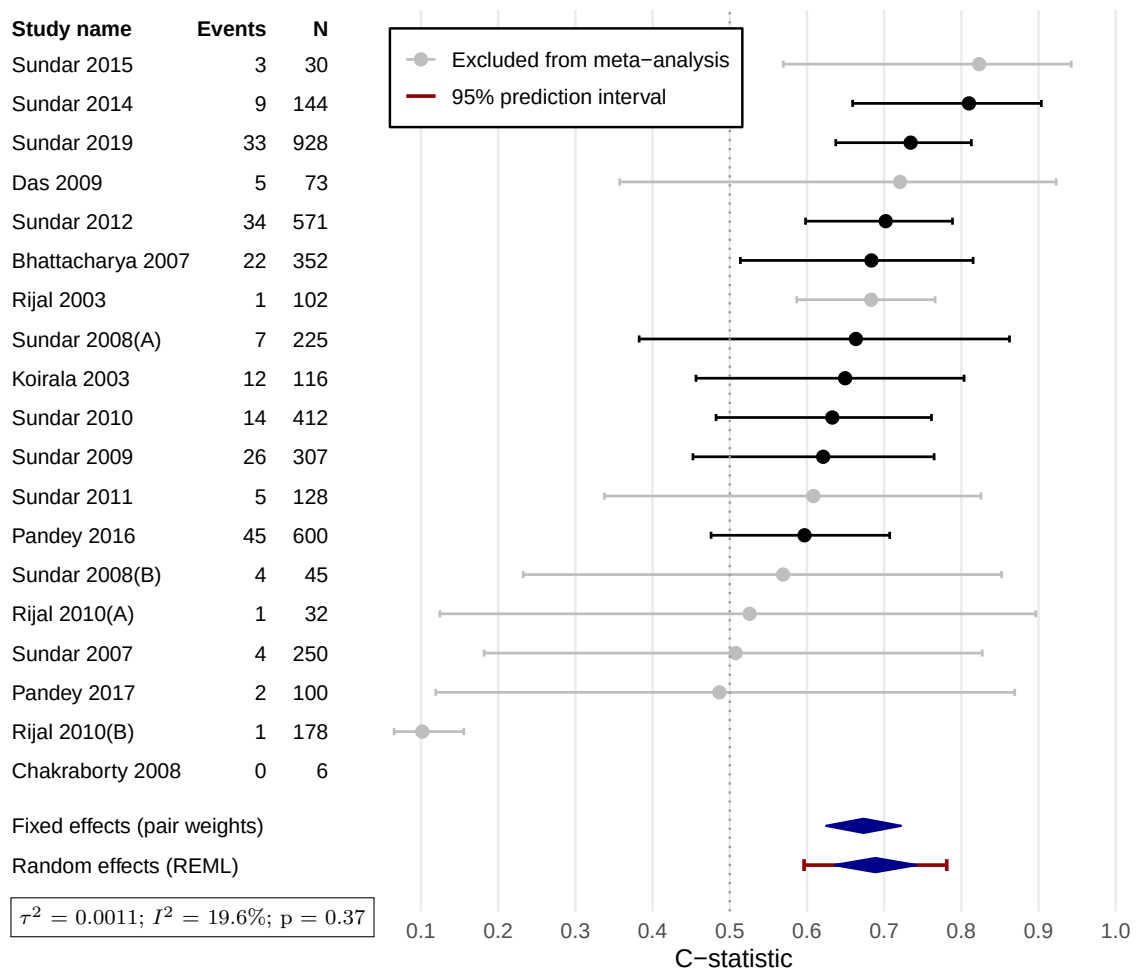


Figure C.18: Forest plot showing individual and pooled study c-statistics for the model **excluding** parasite grade. Pooled c-statistics are presented from both fixed-effects and random-effects meta-analysis models. Blue diamonds: pooled summary estimates with 95% confidence intervals. For Chakraborty 2008, no relapse events occurred and the c-statistic is therefore undefined. Study-specific confidence intervals should be interpreted with caution due to small sample sizes and relapse events in some studies (see Methodology Section 3.3.7).

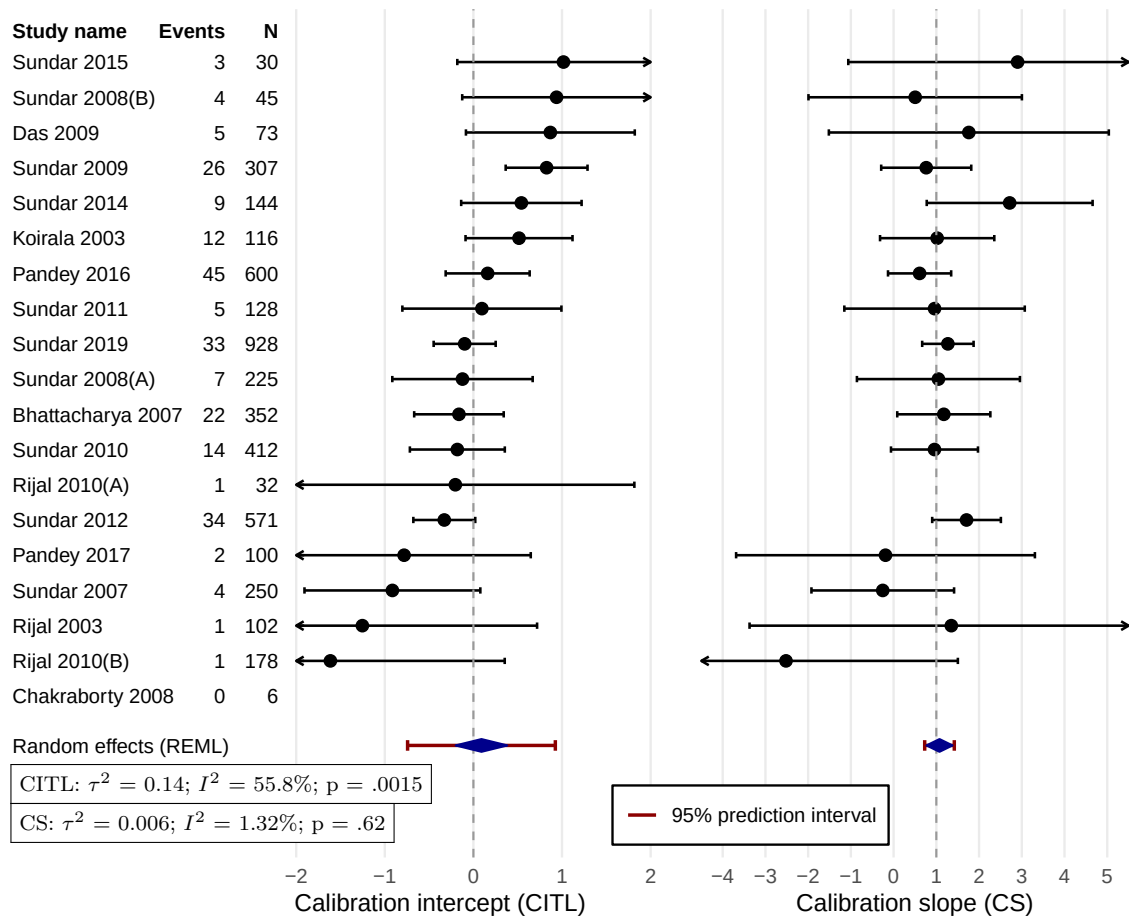


Figure C.19: Forest plots showing individual and pooled study calibration measures for the model **excluding** parasite grade. Left: calibration intercept (calibration-in-the-large, CITL); Right: calibration slope (CS). Blue diamonds: summary estimates with 95% confidence intervals. Calibration measures not presented for Chakraborty 2008 due to no relapse events. Calibration slope not presented for Rijal 2010(A) due to only one relapse event and few total participants leading to failure of model convergence.

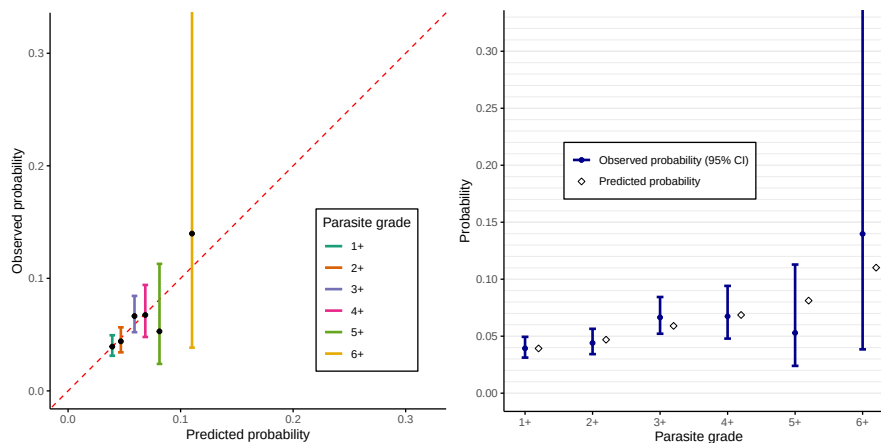


Figure C.20: Calibration plots for parasite grades — model **including** parasite grade.

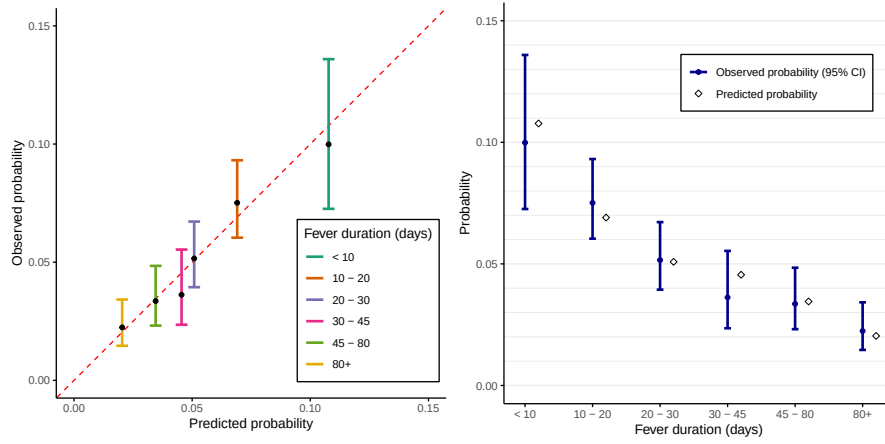


Figure C.21: Calibration plots for fever duration — model **including** parasite grade.

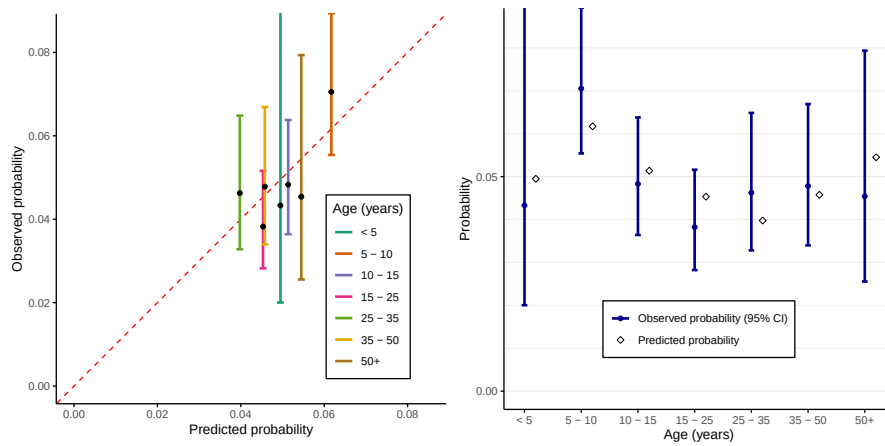


Figure C.22: Calibration plots for age — model **including** parasite grade.

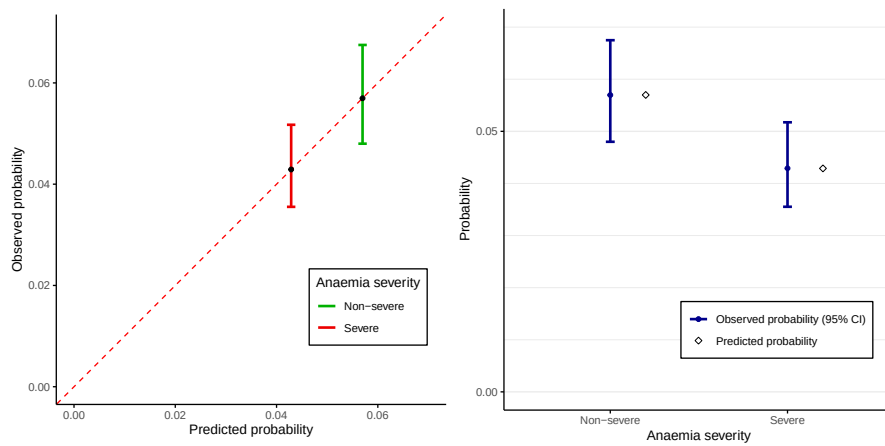


Figure C.23: Calibration plots for anaemia severity — model **including** parasite grade.

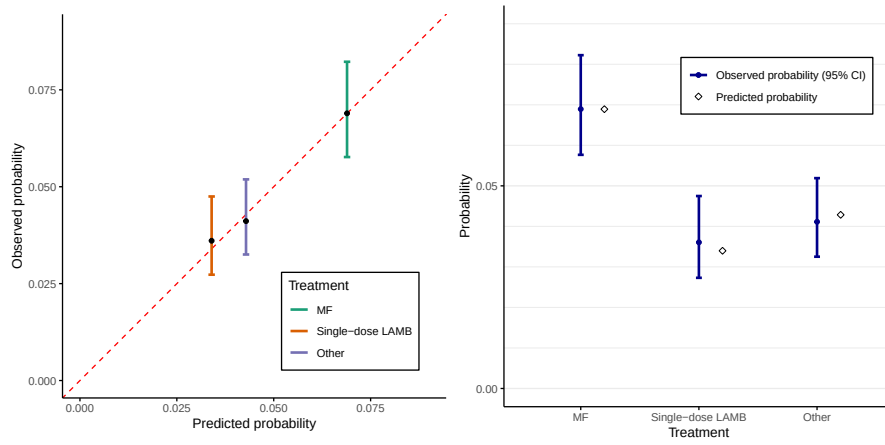


Figure C.24: Calibration plots for treatment — model **including** parasite grade.

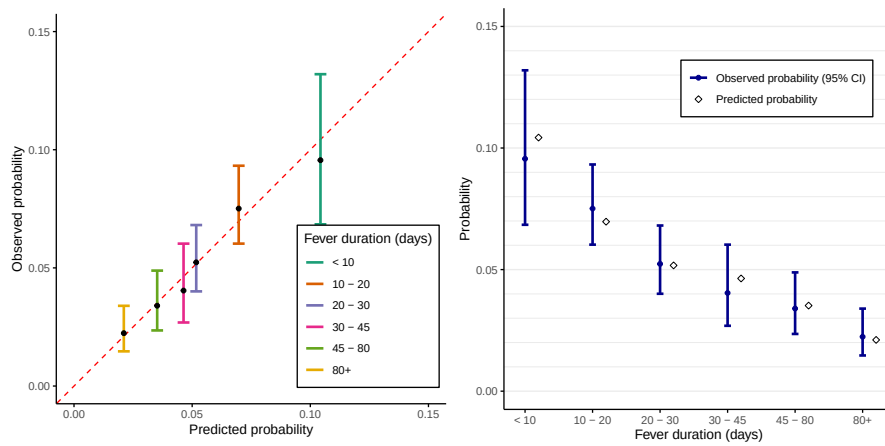


Figure C.25: Calibration plots for fever duration — model **excluding** parasite grade.

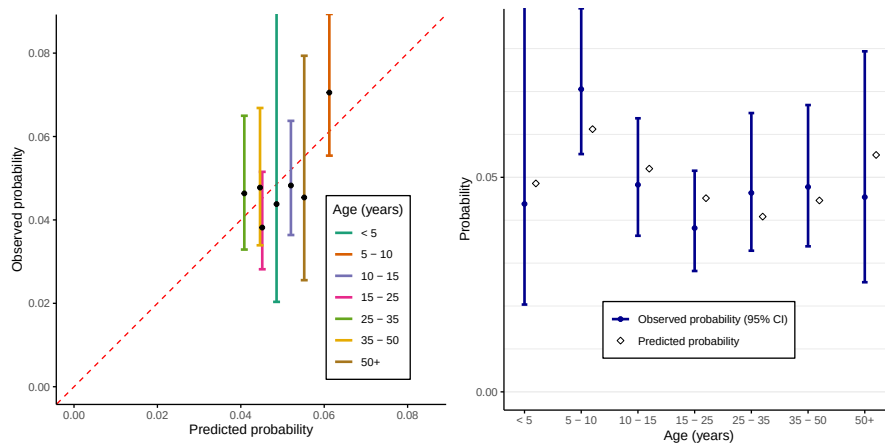


Figure C.26: Calibration plots for age — model **excluding** parasite grade.

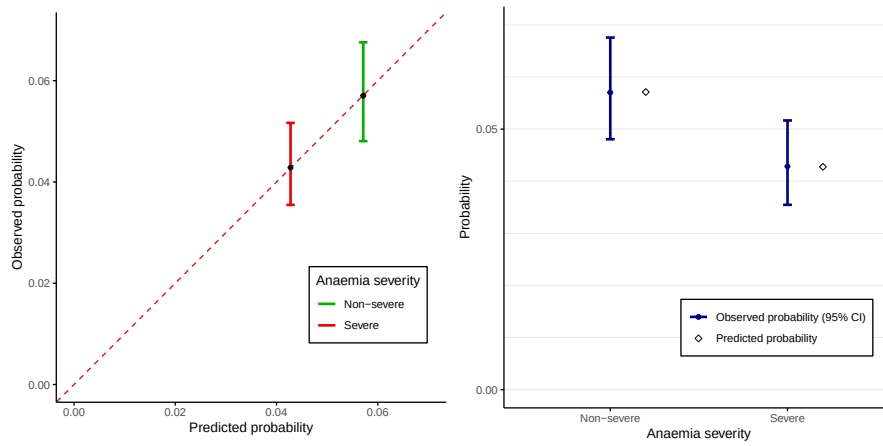


Figure C.27: Calibration plots for anaemia severity — model **excluding** parasite grade.

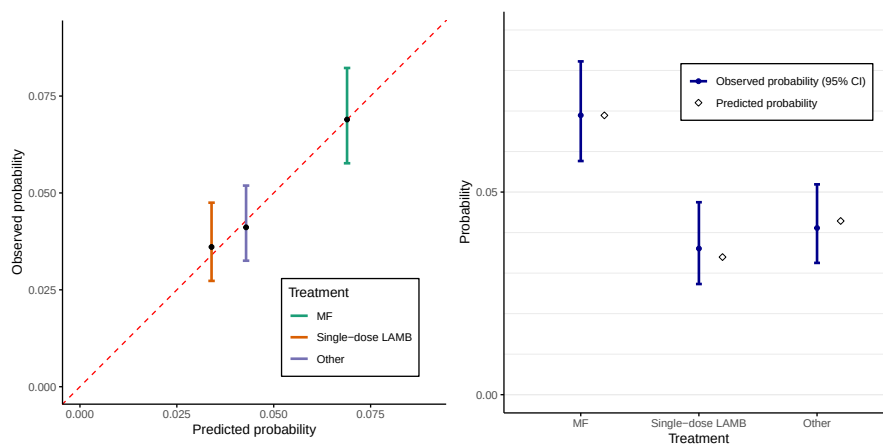


Figure C.28: Calibration plots for treatment — model **excluding** parasite grade.

D

Appendix — East Africa model results

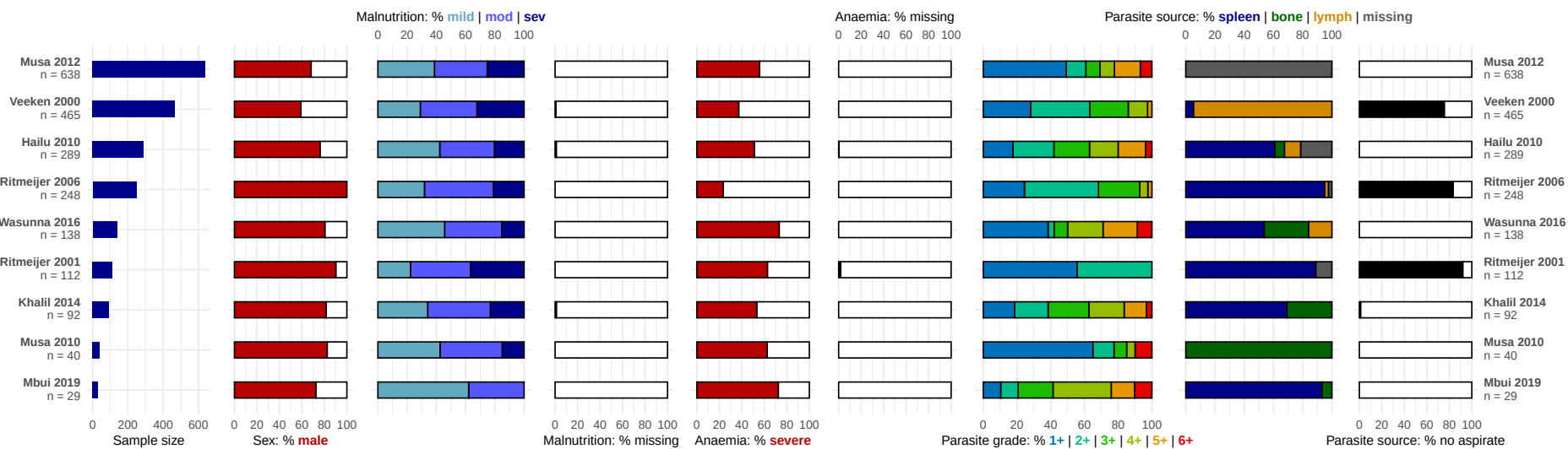


Figure D.1: Distribution of categorical predictors by study. Missing data excluded from stacked bar charts, except for parasite source, where ‘missing’ refers to patients where aspirates were performed, but aspirate source not available.

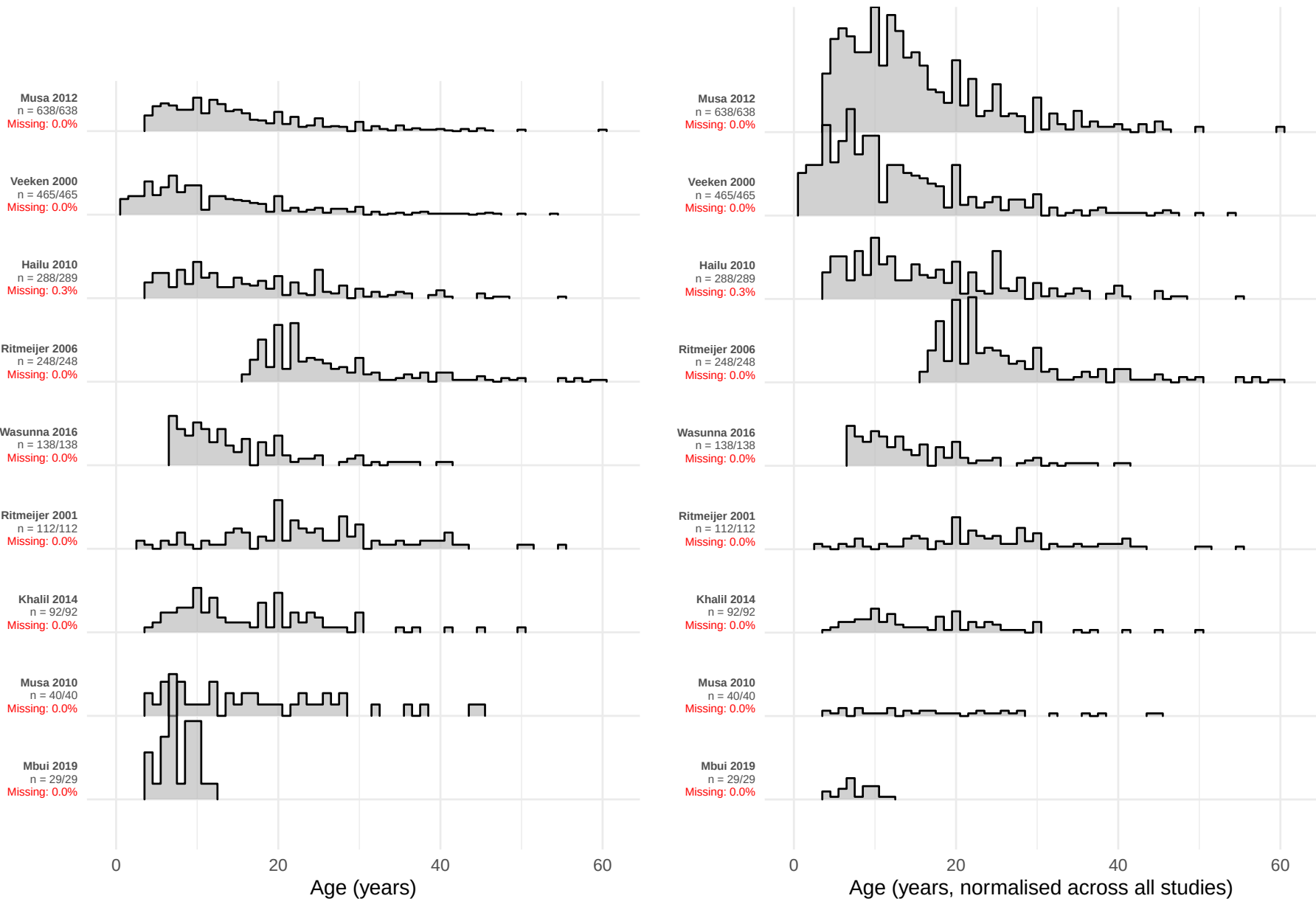


Figure D.2: Distribution of age across studies from East Africa. Missing age data described by study.

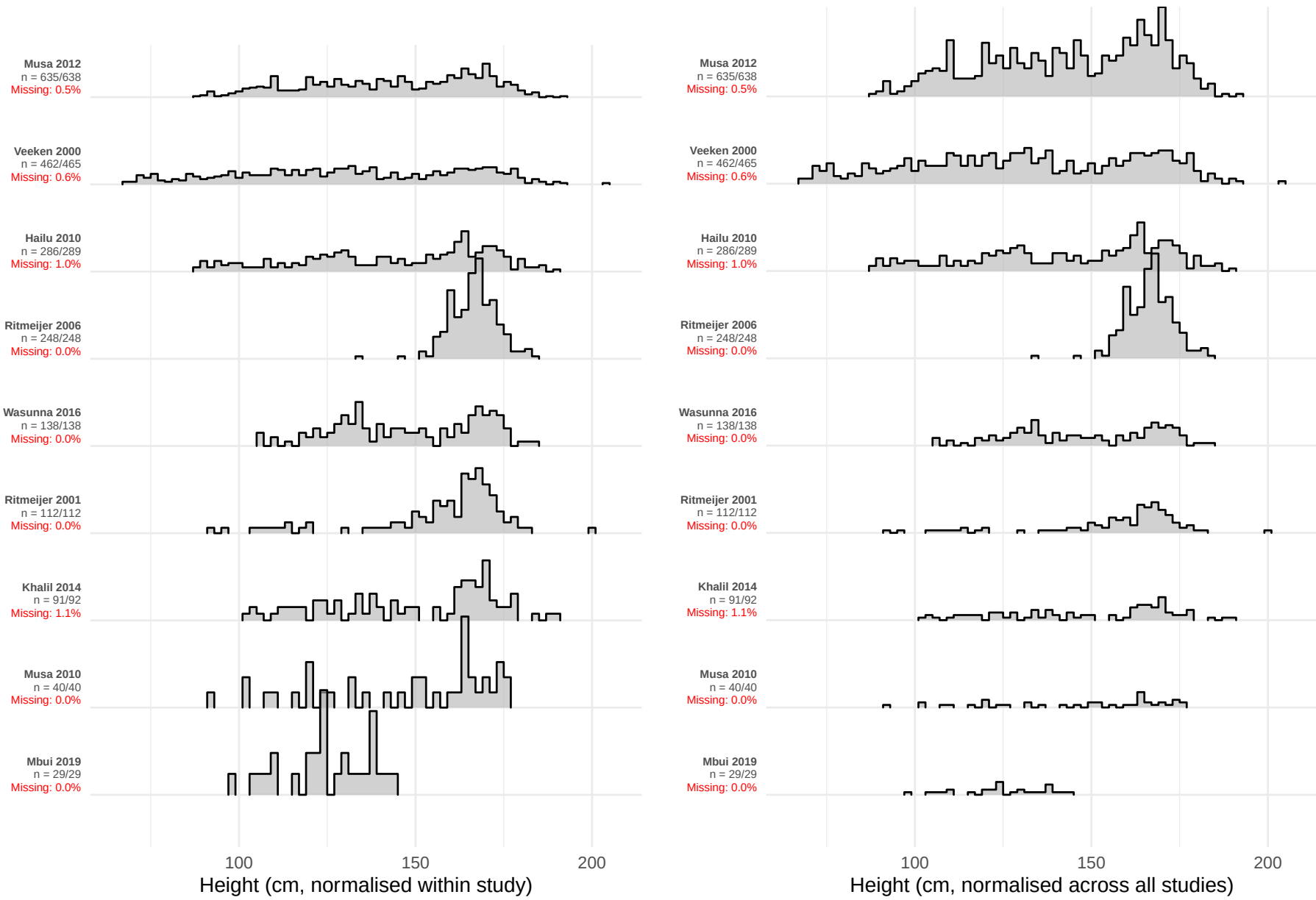


Figure D.3: Distribution of height across studies from East Africa. Missing data described by study.

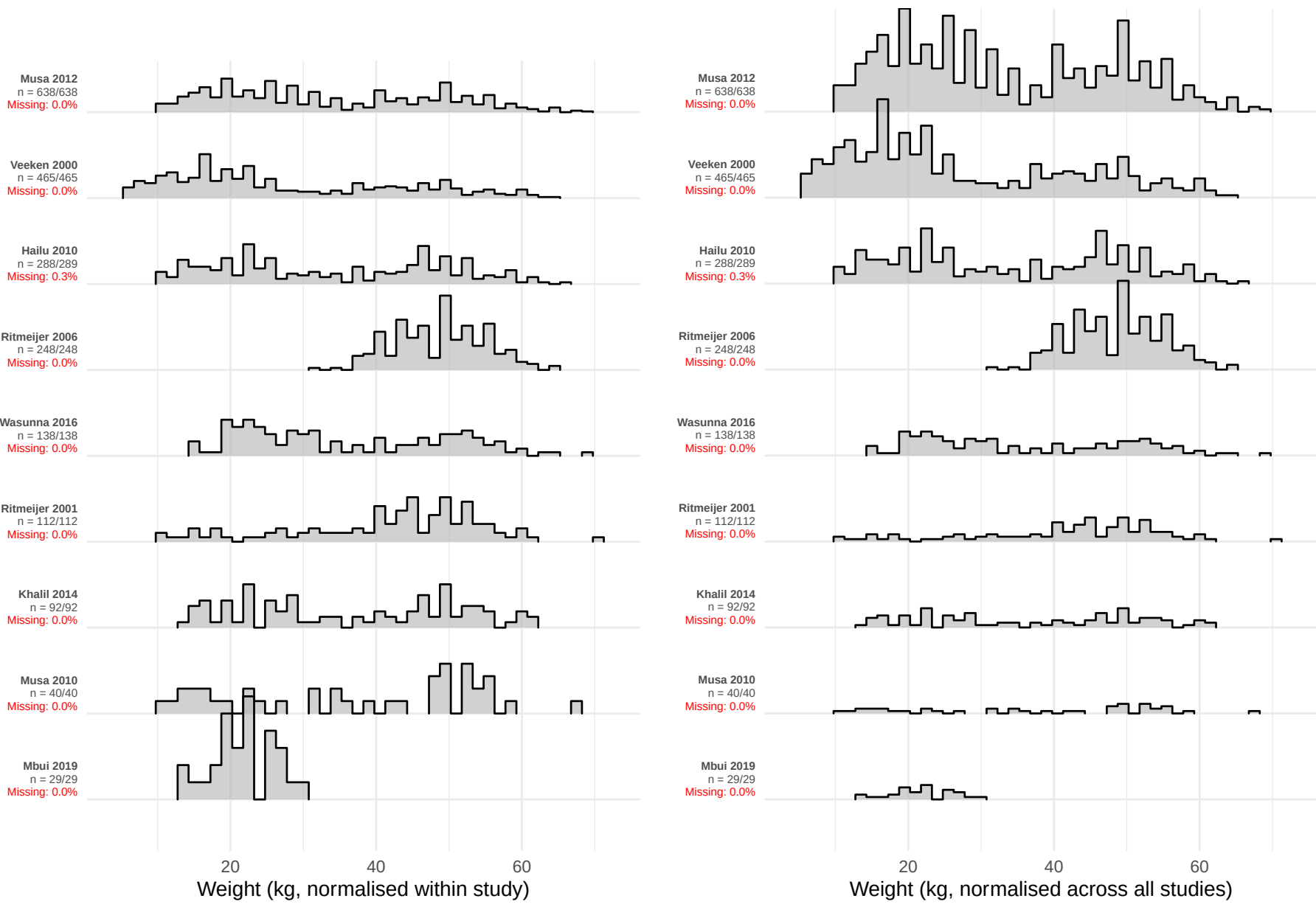


Figure D.4: Distribution of weight across studies from East Africa. Missing data described by study.

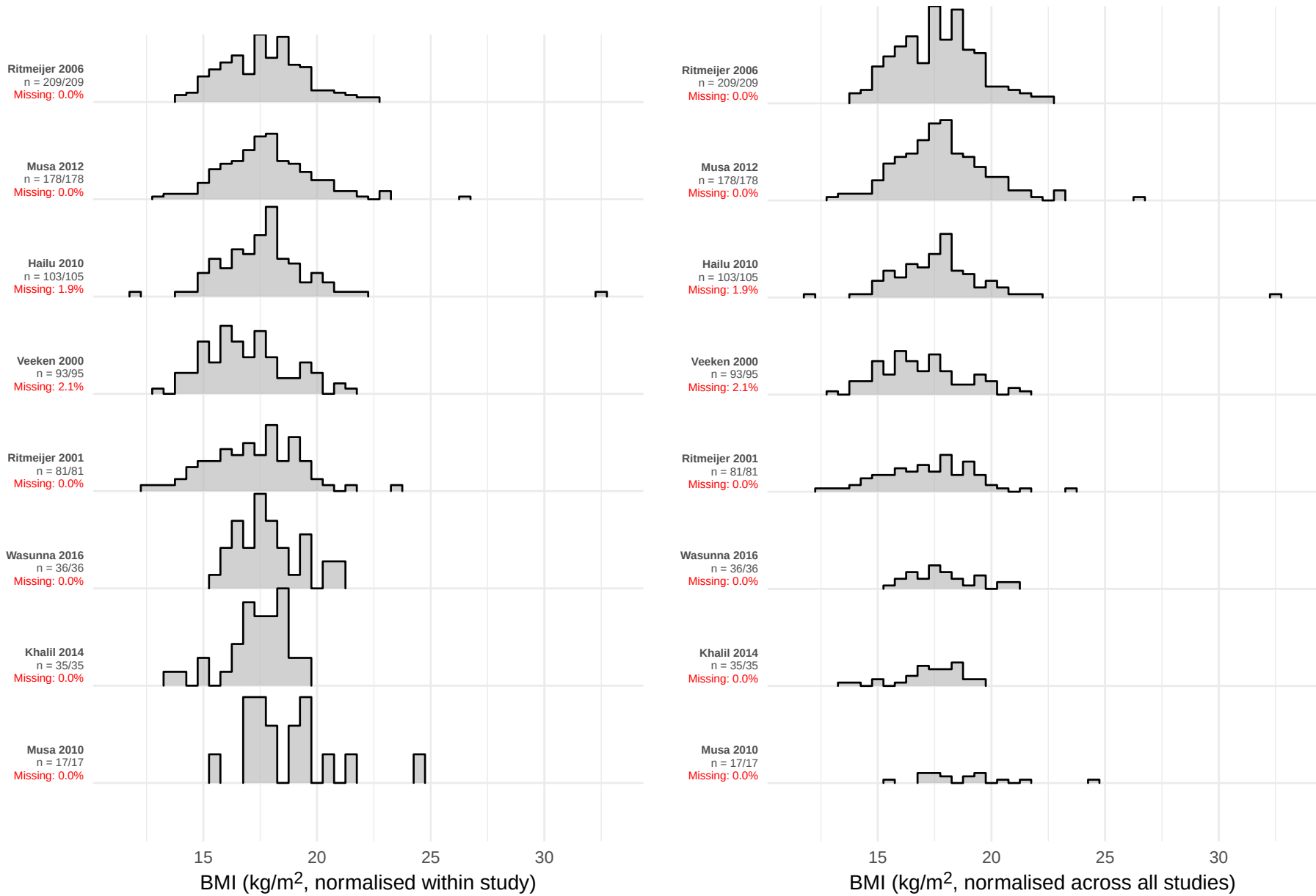


Figure D.5: Distribution of BMI across studies from East Africa. Including only participants aged 19 and over (Pandey 2017 excluded). Missing data described by study.

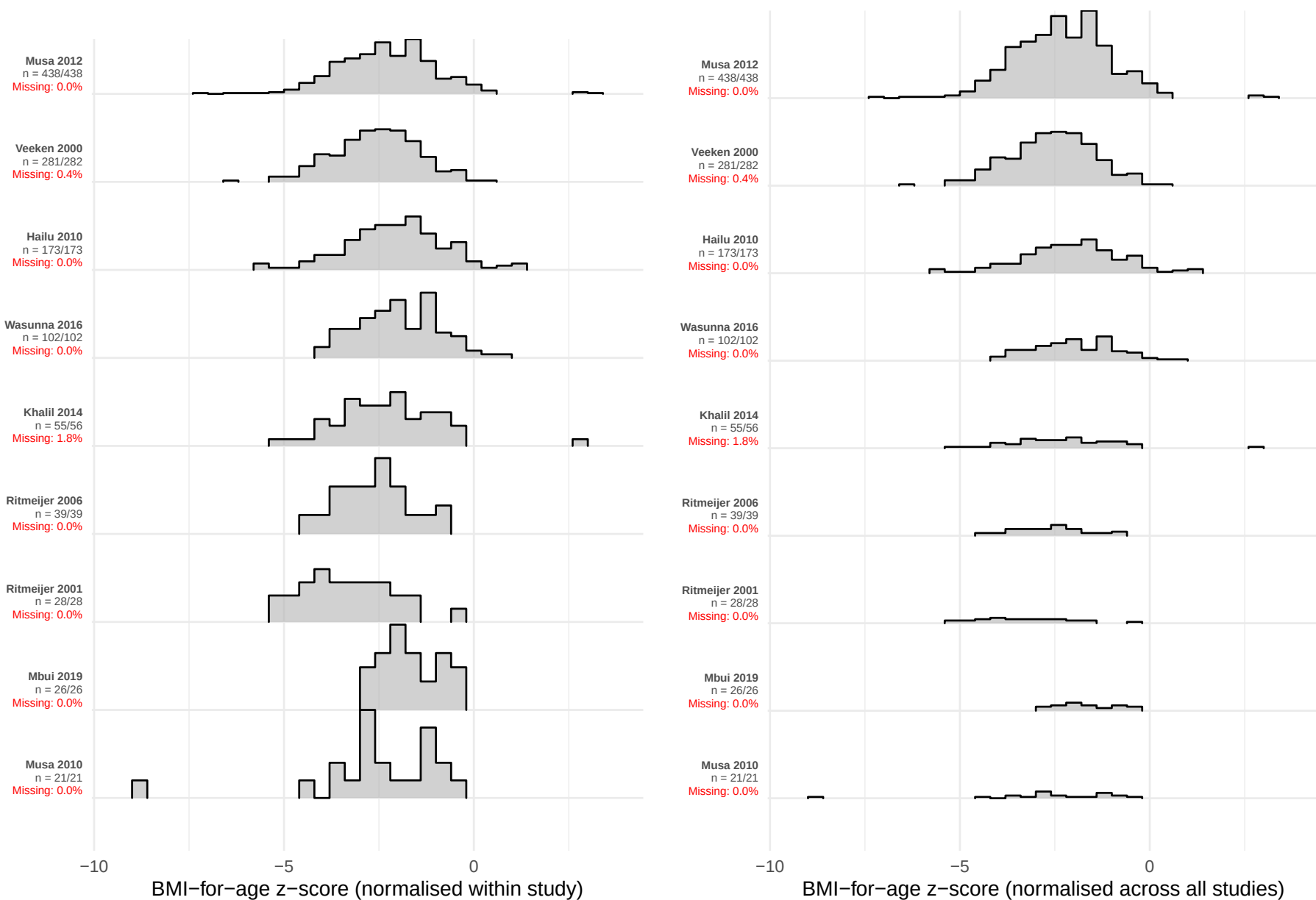


Figure D.6: Distribution of BMI-for-age z score across studies from East Africa. Including only participants aged from 5–18, inclusive. Missing data described by study.

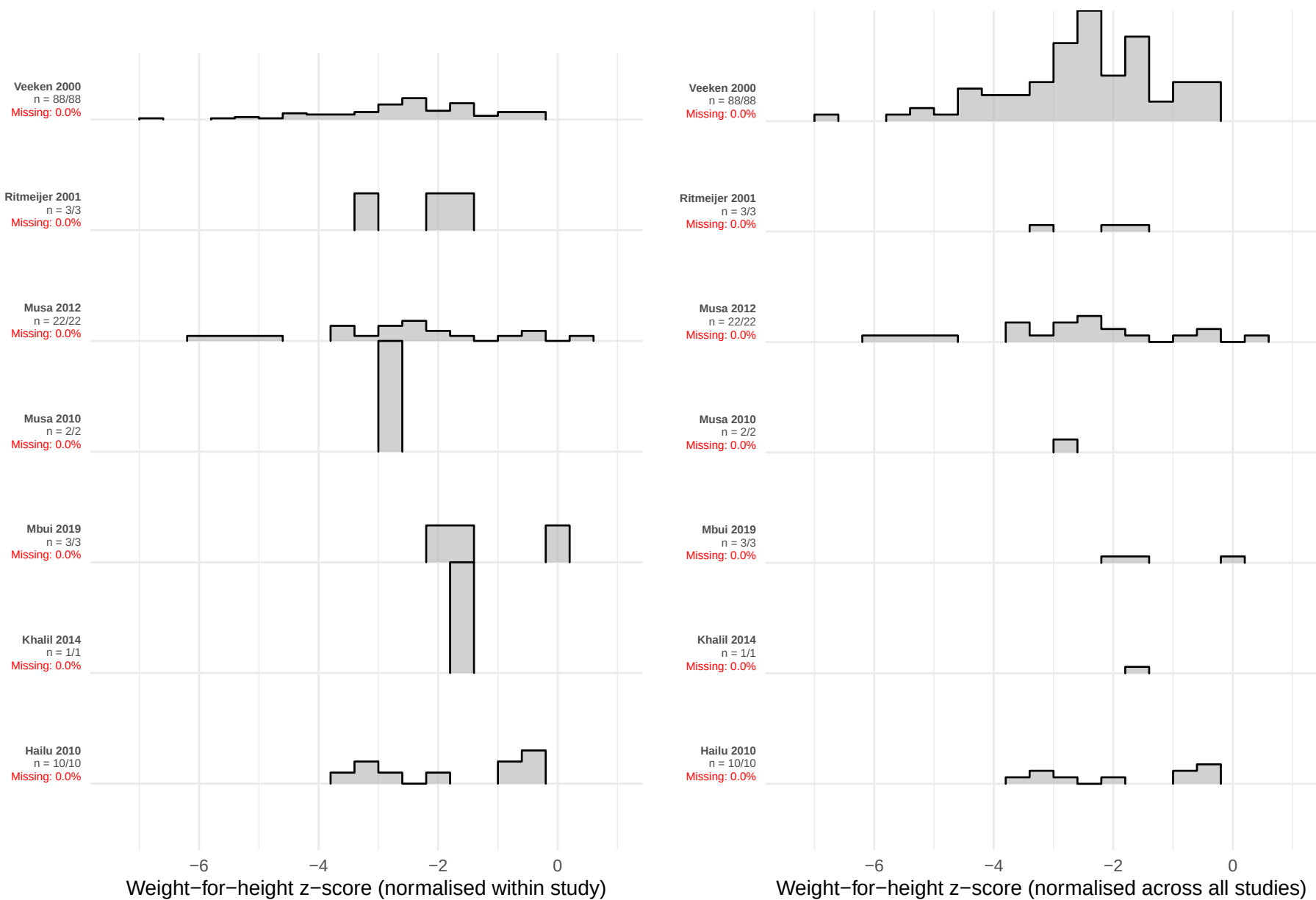


Figure D.7: Distribution of weight-for-height z score across studies from East Africa. Including only participants aged under 5. Missing data described by study.

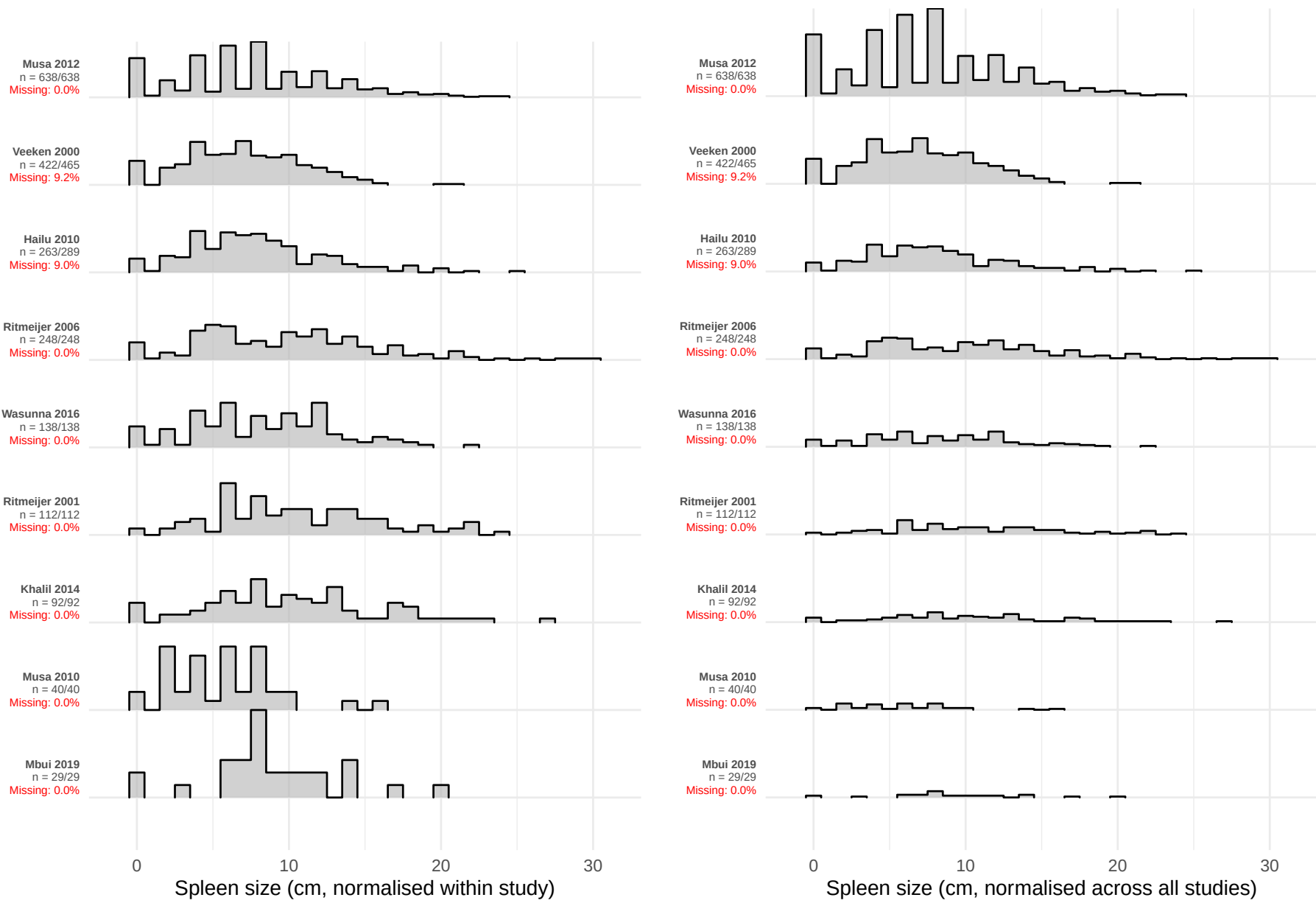


Figure D.8: Distribution of spleen size across studies from East Africa. Missing data described by study.

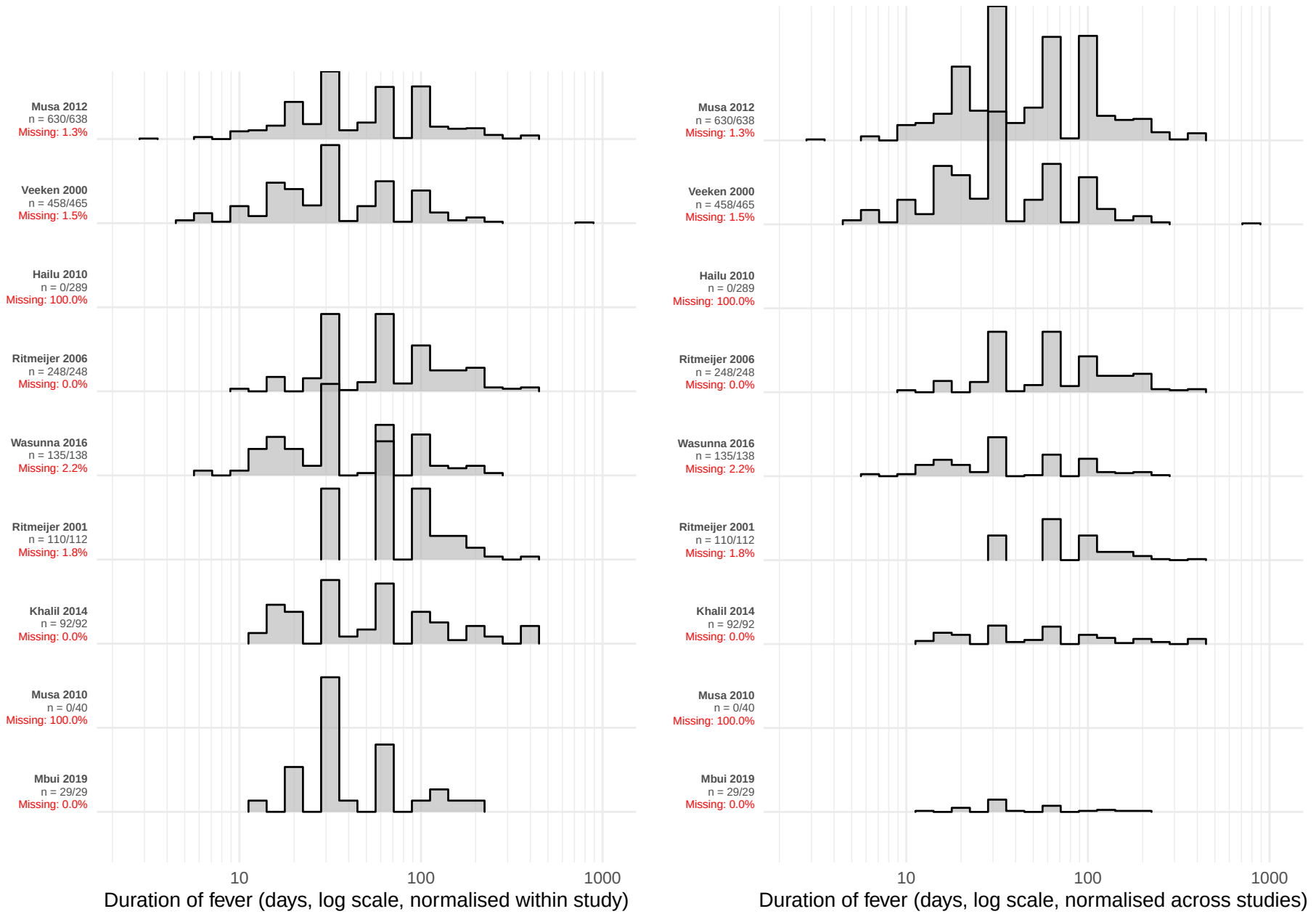


Figure D.9: Distribution of fever duration (log scale) across studies from East Africa. Missing data described by study.

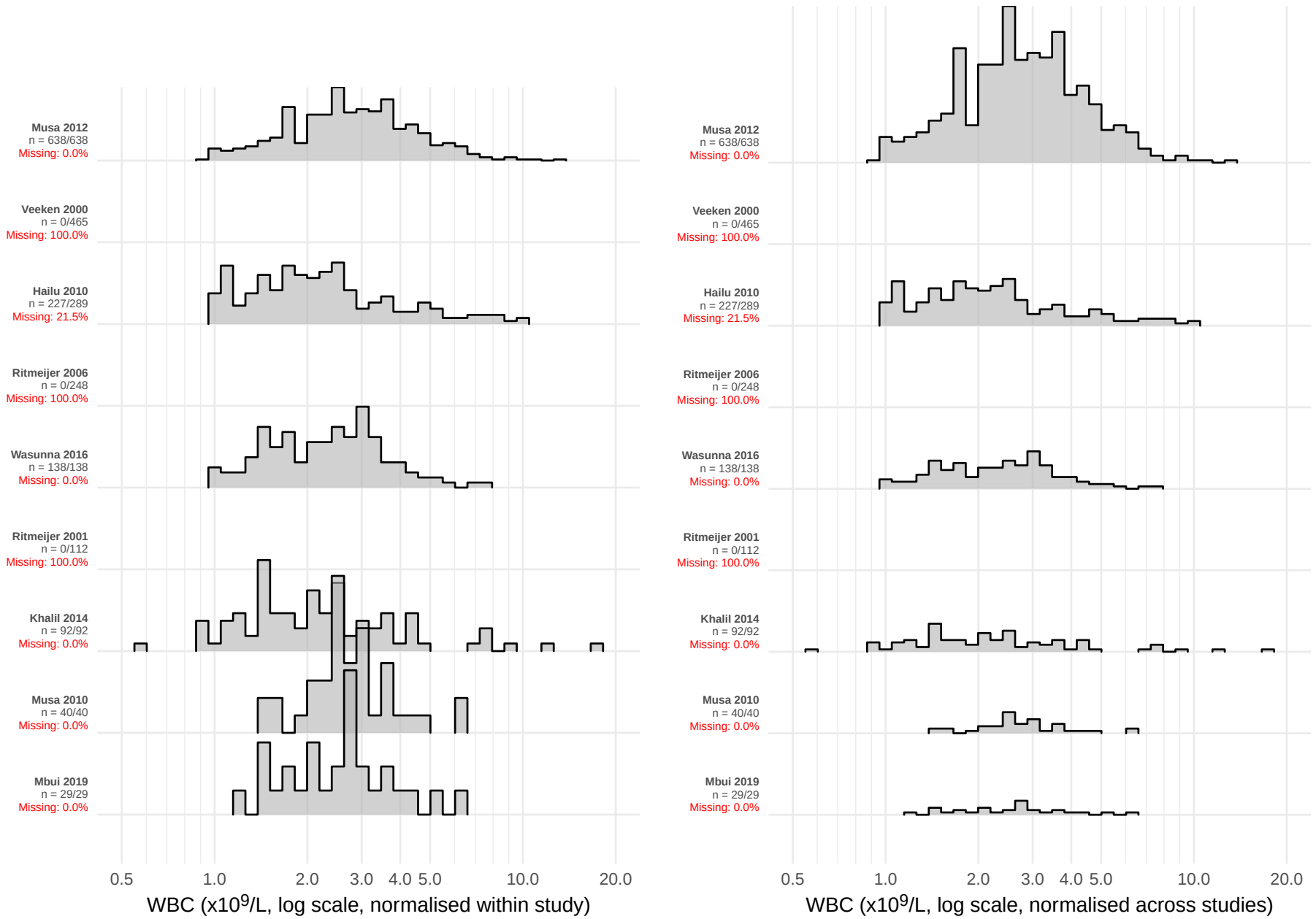


Figure D.10: Distribution of white blood cell count (log scale) across studies from East Africa. Missing data described by study.

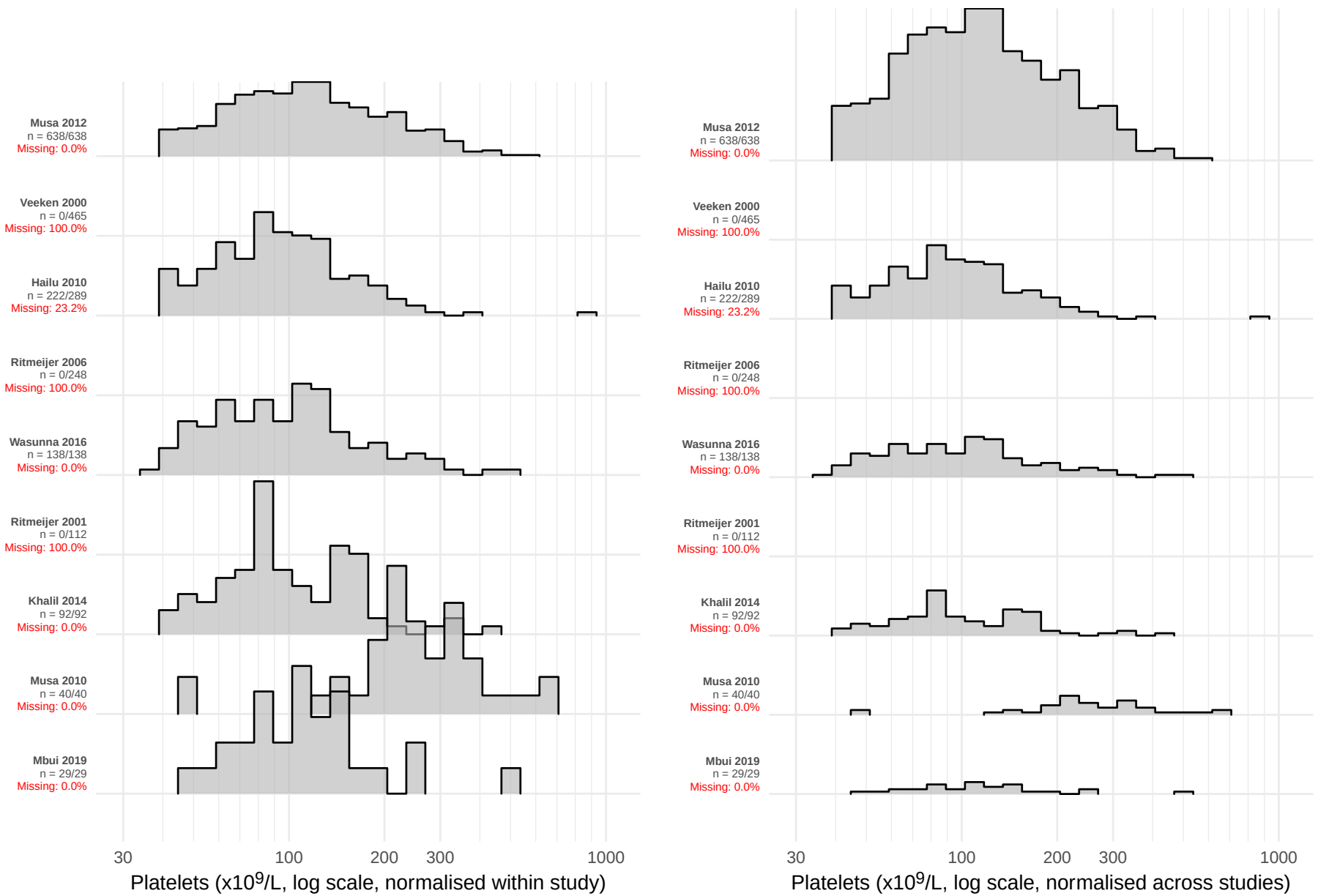


Figure D.11: Distribution of platelet count (log scale) across studies from East Africa. Missing data described by study.

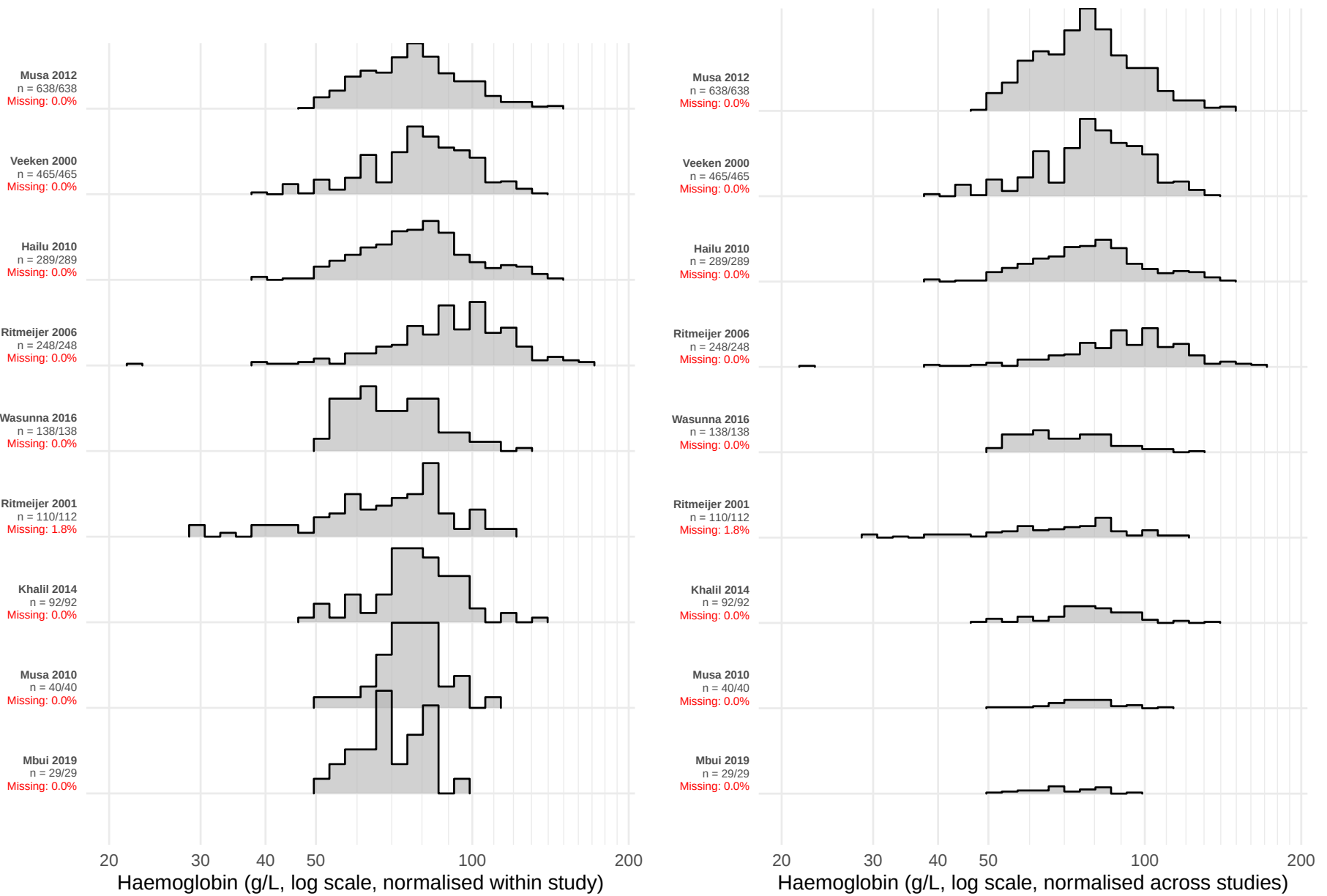


Figure D.12: Distribution of haemoglobin (log scale) across studies from East Africa. Missing data described by study.

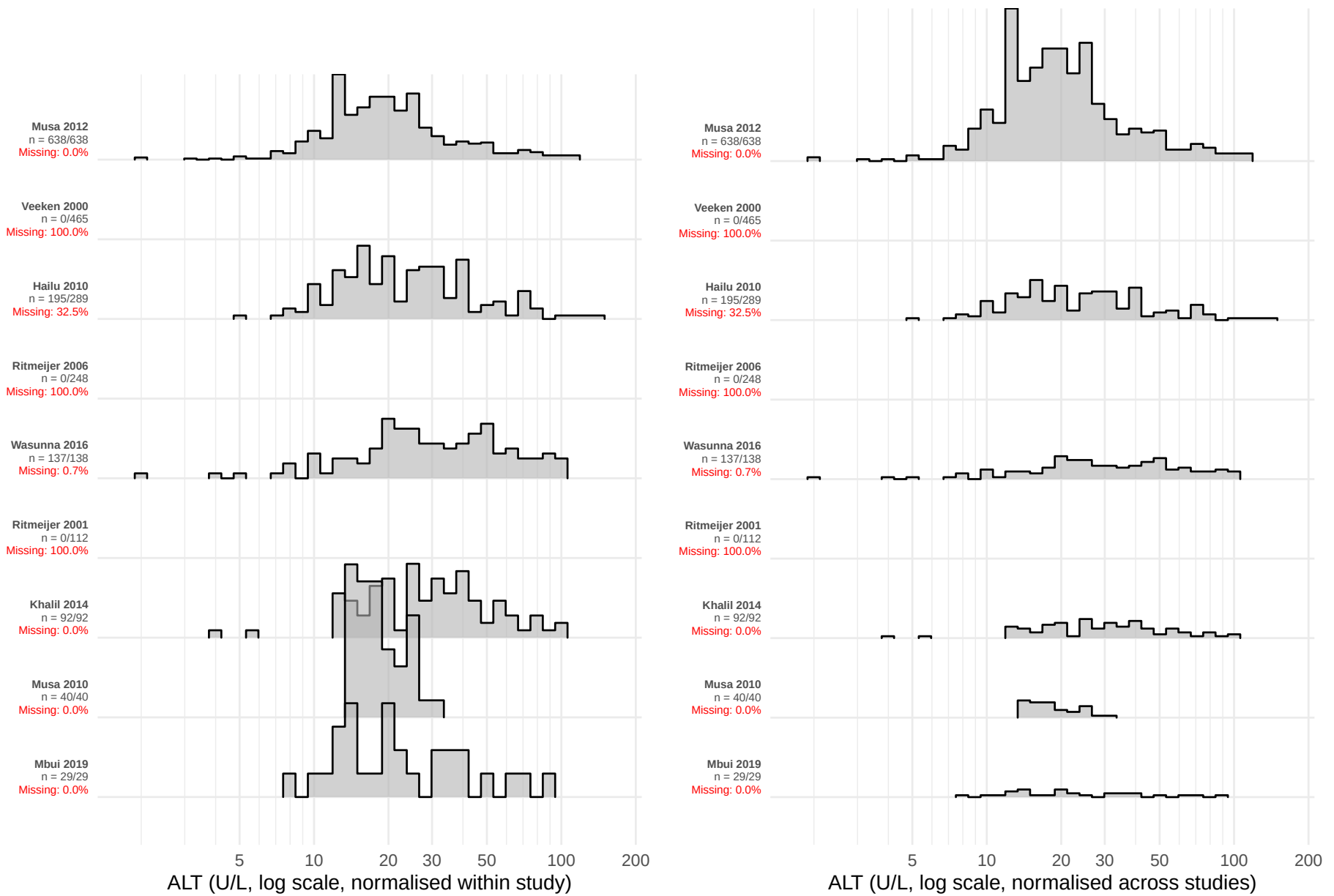


Figure D.13: Distribution of alanine transaminase (ALT, log scale) across studies from East Africa. Missing data described by study.

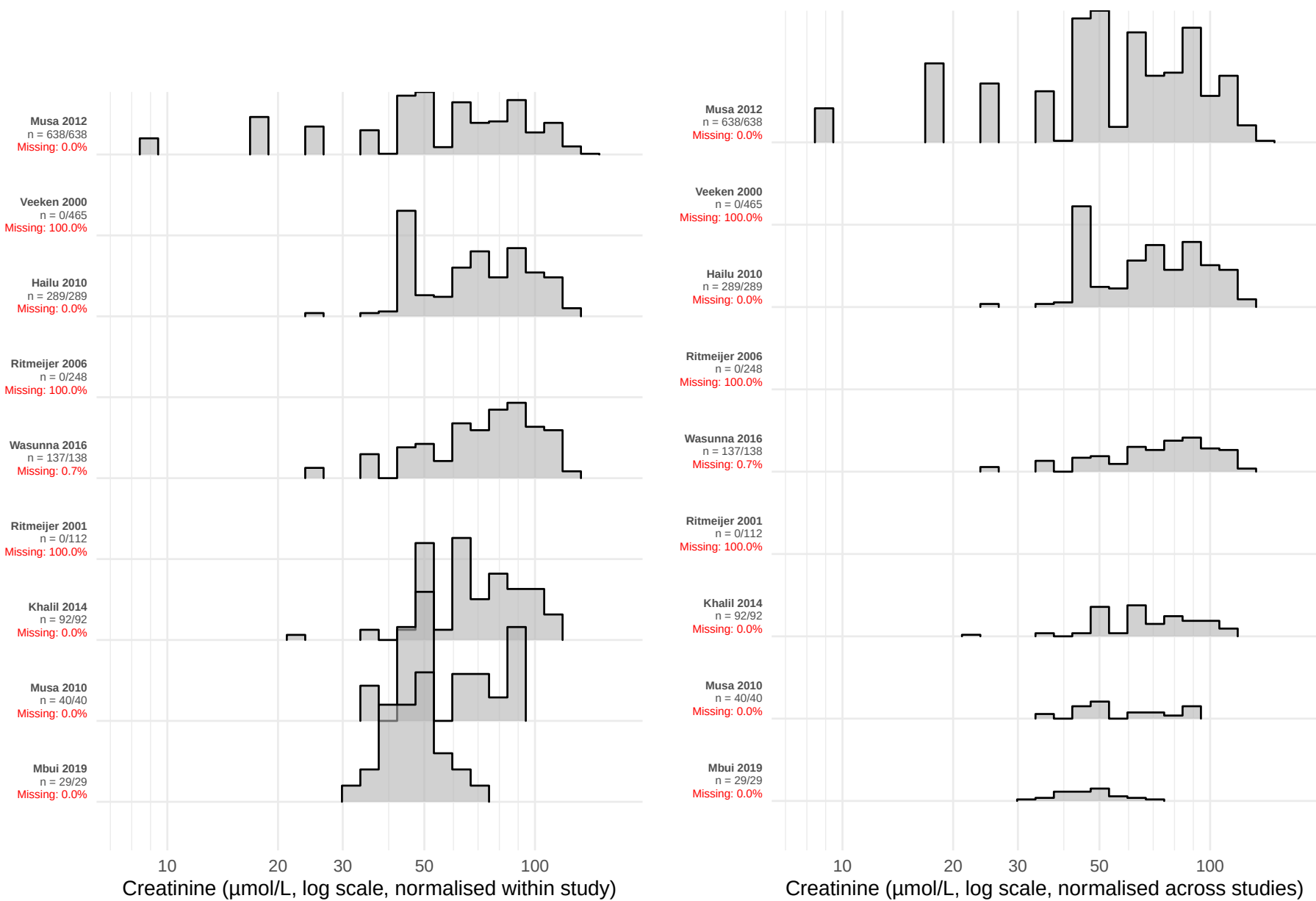


Figure D.14: Distribution of creatinine (log scale) across studies from East Africa. Missing data described by study.

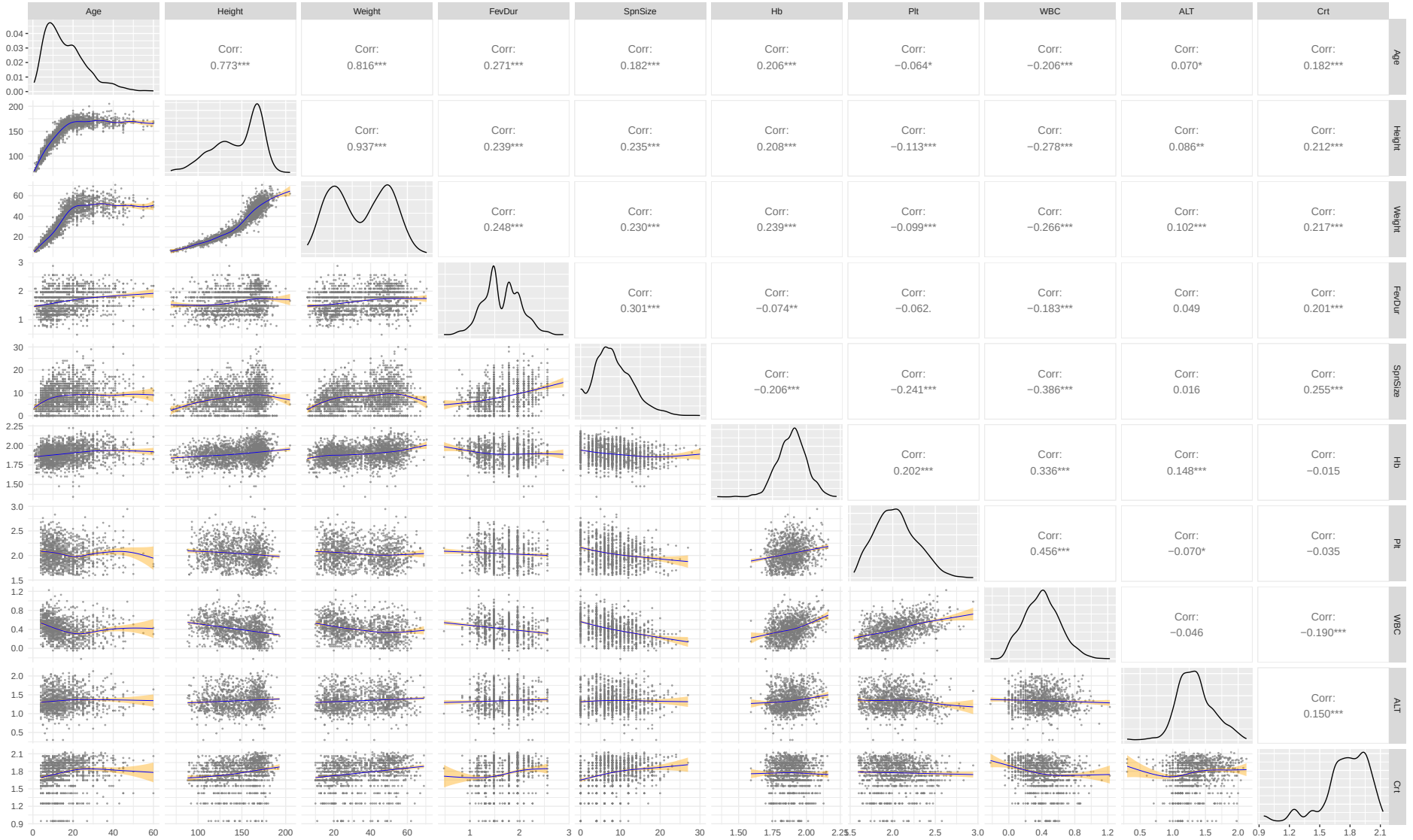


Figure D.15: Correlation between continuous variables. For scatter plots, a univariable generalised additive model is fitted (blue line) with 95% confidence interval ribbon filled (orange area). Pearson correlation coefficients are presented, ‘***’ $p < 0.001$; ‘**’ $p < 0.01$; ‘*’ $p < 0.05$; ‘.’ $p < 0.10$. Age: years; Height: cm; Weight: kg; FevDur: duration of fever, $\log_{10}(\text{days})$; SpnSize: spleen size (cm); Hb: haemoglobin, $\log_{10}(\text{g/L})$; Plt: platelets, $\log_{10}(\times 10^9/\text{L})$; WBC: white blood cells $\log_{10}(\times 10^9/\text{L})$; ALT: alanine aminotransferase, $\log_{10}(\text{U/L})$; Cr: creatinine $\log_{10}(\mu\text{mol/L})$.

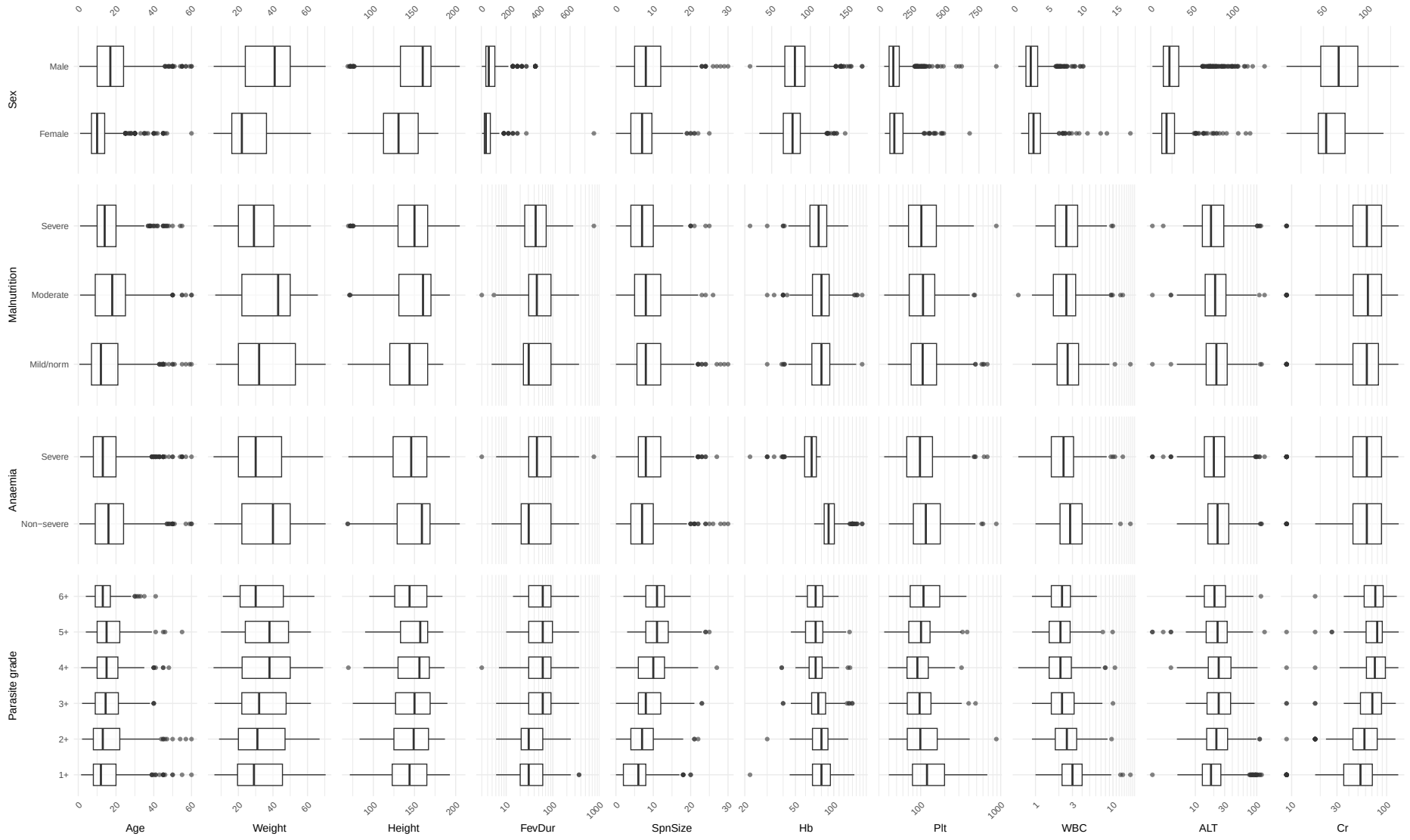


Figure D.16: Correlation between continuous and categorical variables. FevDur and laboratory tests axes transformed to log10 scale. Age: years; Height: cm; Weight: kg; FevDur: duration of fever, days; SpnSize: spleen size, cm; Hb: haemoglobin, g/L; Plt: platelets, $\times 10^9/L$; WBC: white blood cells $\times 10^9/L$; ALT: alanine aminotransferase, U/L; Cr: creatinine $\mu\text{mol/L}$.

term	estimate ¹	std.error	t statistic	df (effective)	p value
Fixed intercept	-3.7874	0.5369	-7.05	1885.50	2.43E-12
Parasite grade ²	0.3593	0.0763	4.71	1044.63	2.83E-06
Spleen size ³	-0.3335	0.1741	-1.92	1813.86	5.56E-02
Anaemia: Severe ⁴	0.5450	0.2448	2.23	1973.14	2.61E-02
Fever duration ⁵	-0.3302	0.1349	-2.45	476.32	1.47E-02
WBC count ⁶	0.2368	0.1315	1.80	419.77	7.23E-02
Malnutrition: Moderate ⁷	0.5769	0.2677	2.16	2004.45	3.13E-02
Malnutrition: Severe ⁷	0.9780	0.2860	3.42	2025.03	6.39E-04
Unadjusted intercept (Musa 2012): -3.9411					
Uniform shrinkage factor: 0.8079					
Adjusted intercepts after shrinkage: Average: -4.3182; Musa 2012: -4.4524					
Pooled τ^2 : 0.9122. Pooled ICC: 0.2169					

¹ Logit scale before uniform shrinkage.

² Continuous variable, from 1+ to 6+ as described by Chulay and Bryceson[52]. Transformed $x \mapsto x - 1$.

³ Units: centimetres. Converted to natural log scale $x \mapsto \log(x + 1)$.

⁴ Reference group: non-severe anaemia.

⁵ Units: days. Converted to natural log scale and then standardised by mean (3.550509) and standard deviation (0.8693318).

⁶ Units: $\times 10^9$ cells per litre. Converted to natural log scale and then standardised by mean (0.926133) and standard deviation (0.4911651).

⁷ Reference group: Mild/normal malnutrition.

Table D.1: Model coefficients pooled across 30 imputed datasets using Rubin’s Rules. Model **including** parasite grade. df: degrees of freedom; std.error: standard error; WBC: white blood cell count

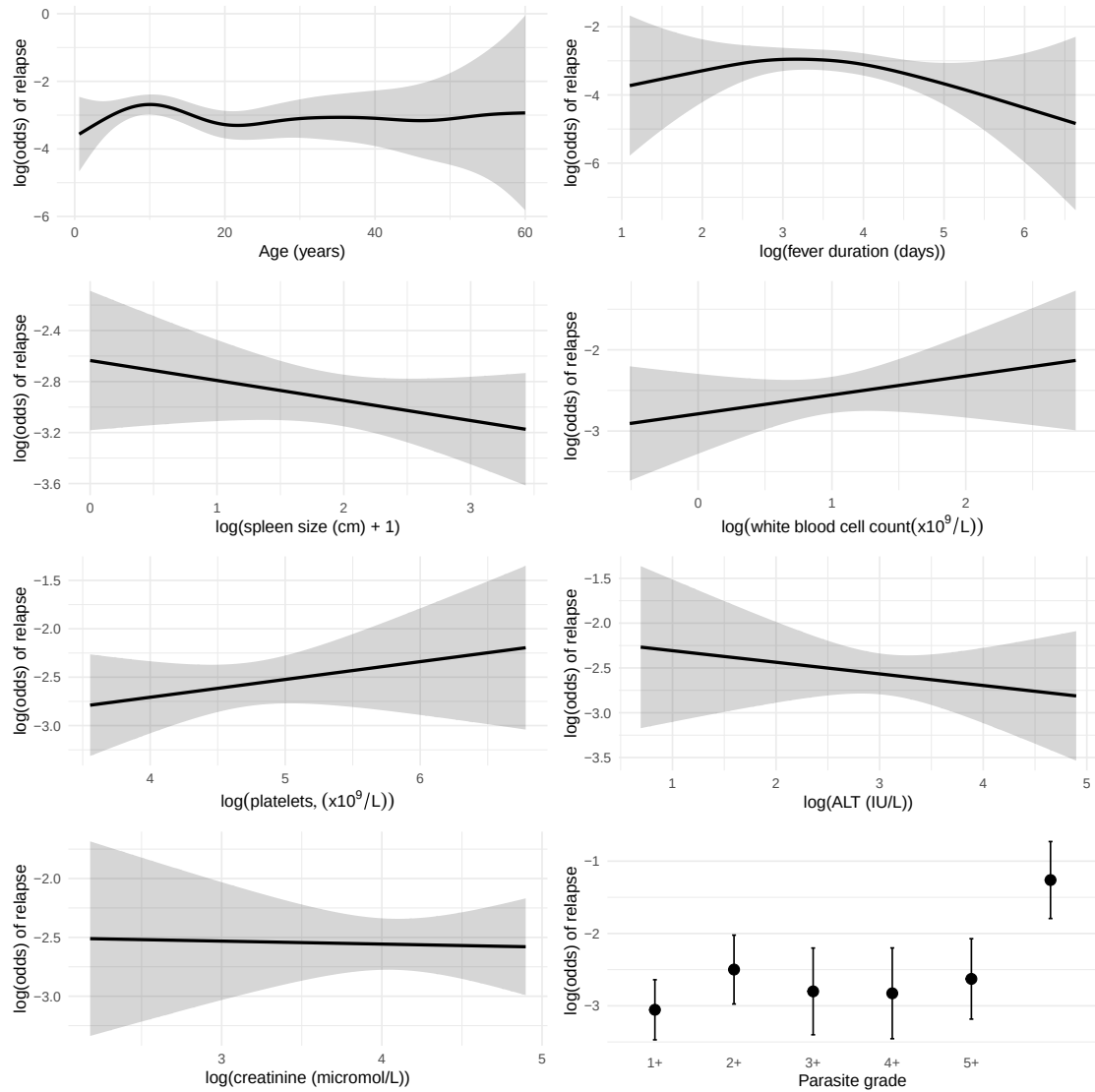


Figure D.17: Associations between transformed continuous predictors and log(odds) of relapse. For each predictor (excluding parasite grade), a univariable generalised additive model spline fit is shown, with 95% confidence intervals. For parasite grade, 95% confidence intervals are calculated for each grade using the Wilson method.

term	estimate ¹	std.error	t statistic	df (effective)	p value
Fixed intercept	-3.7674	0.4381	-8.60	2028.60	1.57E-17
Anaemia: Severe ²	0.5553	0.2365	2.35	1981.61	1.89E-02
WBC count ³	0.2254	0.1167	1.93	715.01	5.39E-02
Malnutrition: Moderate ⁴	0.5486	0.2625	2.09	2002.50	3.67E-02
Malnutrition: Severe ⁴	0.9282	0.2794	3.32	2017.72	9.08E-04
Unadjusted intercept (Musa 2012): -3.9073					
Uniform shrinkage factor: 0.7493					
Adjusted intercepts after shrinkage: Average: -4.4889; Musa 2012: -4.6065					
Pooled τ^2 : 1.040. Pooled ICC: 0.2401					

¹ Logit scale before uniform shrinkage.

² Reference group: non-severe anaemia.

³ Units: $\times 10^9$ cells per litre. Converted to natural log scale and then standardised by mean (0.926133) and standard deviation (0.4911651).

⁴ Reference group: Mild/normal malnutrition.

Table D.2: Model coefficients pooled across 30 imputed datasets using Rubin's Rules. Model **excluding** parasite grade. df: degrees of freedom; std.error: standard error; WBC: white blood cell count

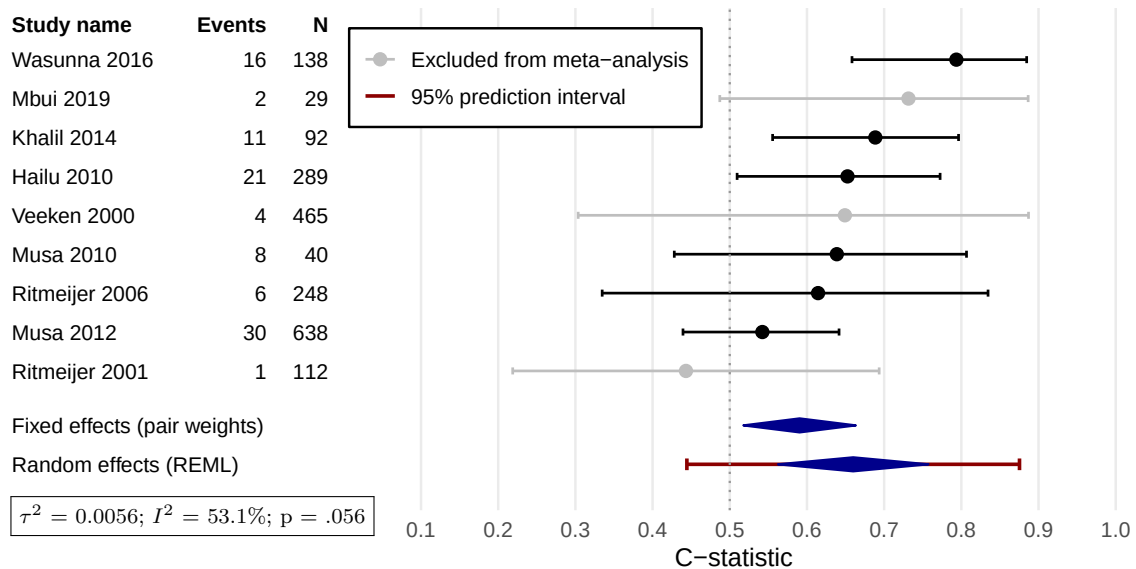


Figure D.18: Forest plot showing individual and pooled study c-statistics, for the model **excluding** parasite grade. Pooled c-statistics are presented from both fixed-effects and random-effects meta-analysis models. Pooled random-effects c-statistics and variances are estimated using restricted maximum likelihood (REML) and the Hartung-Knapp-Sidik-Jonkman method. Study-specific confidence intervals should be interpreted with caution due to small sample sizes and relapse events in some studies (see Methodology Section 3.3.7). Blue diamonds: pooled summary estimates with 95% confidence intervals. Red line: 95% prediction interval. Study ordered by c-statistic.

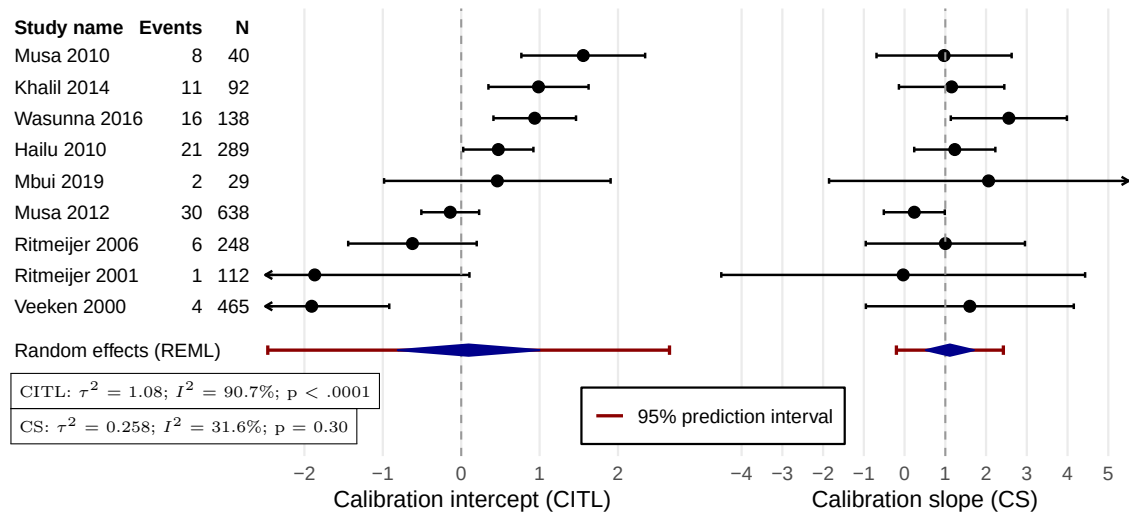


Figure D.19: Forest plots showing individual and pooled study calibration measures for the model **excluding** parasite grade. Left: calibration intercept (calibration-in-the-large, CITL); Right: calibration slope (CS). Blue diamonds: summary estimates with 95% confidence intervals.

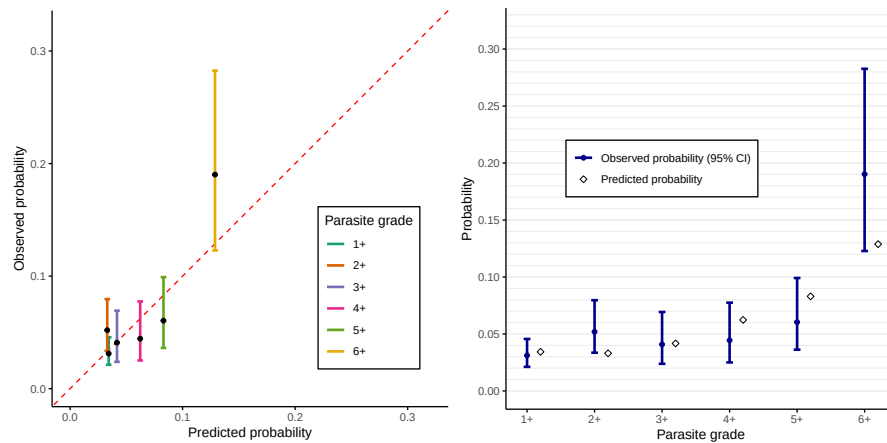


Figure D.20: Calibration plots for parasite grades — model **including** parasite grade.

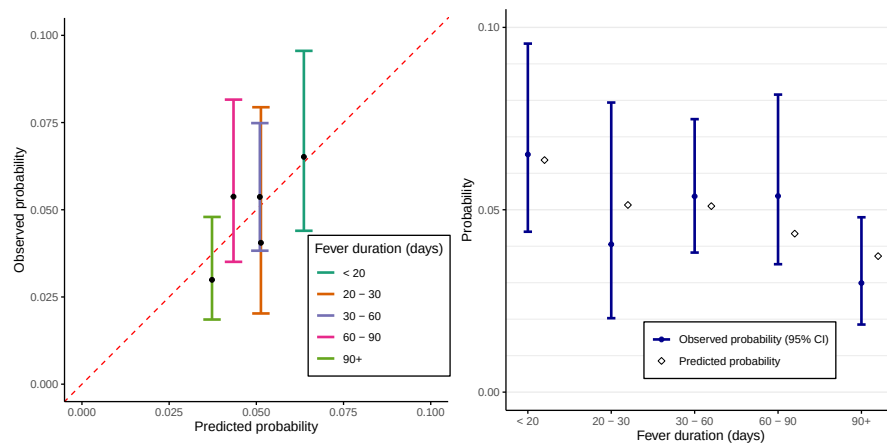


Figure D.21: Calibration plots for fever duration — model **including** parasite grade.

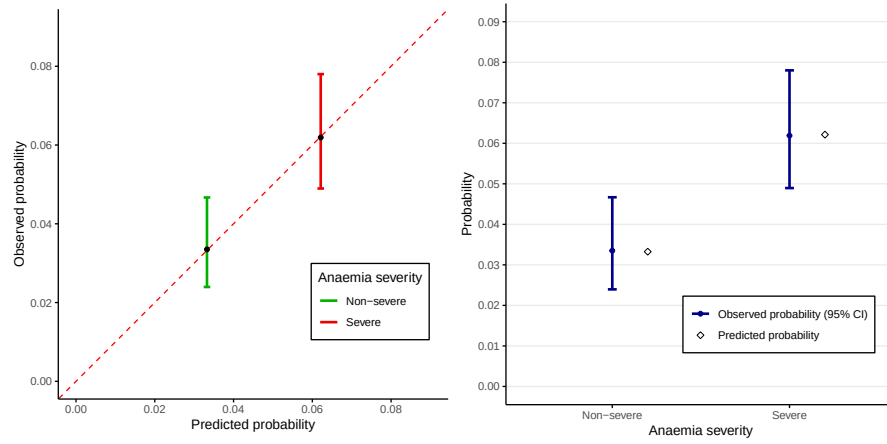


Figure D.22: Calibration plots for anaemia — model **including** parasite grade.

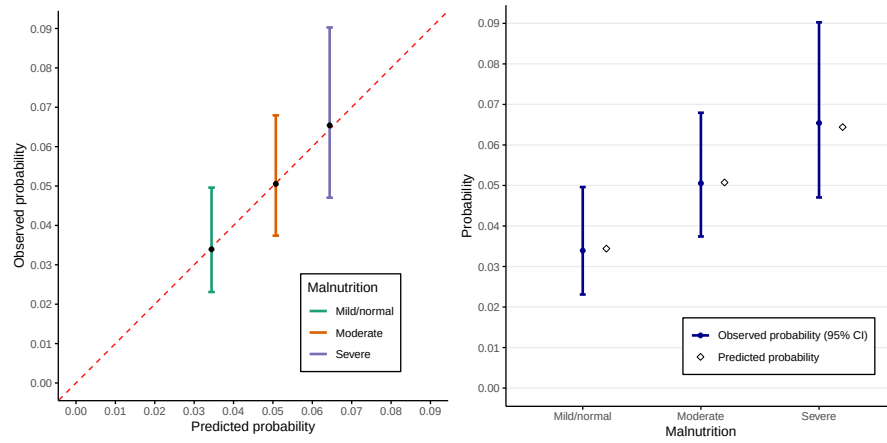


Figure D.23: Calibration plots for malnutrition — model **including** parasite grade.

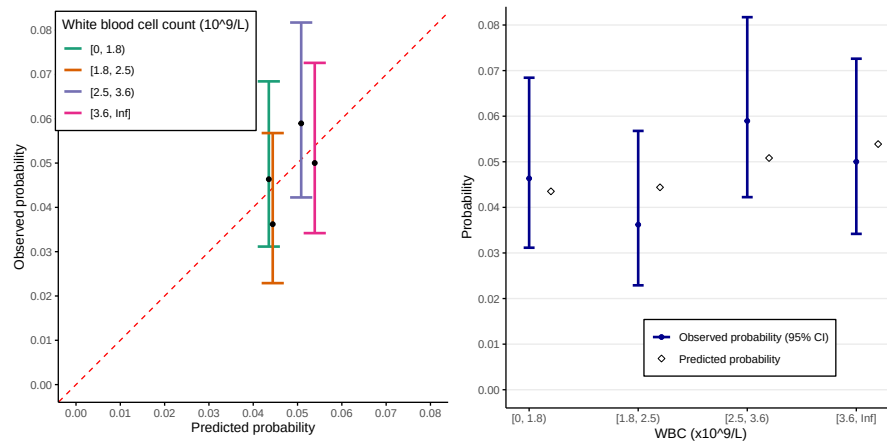


Figure D.24: Calibration plots for white blood cell count — model **including** parasite grade.

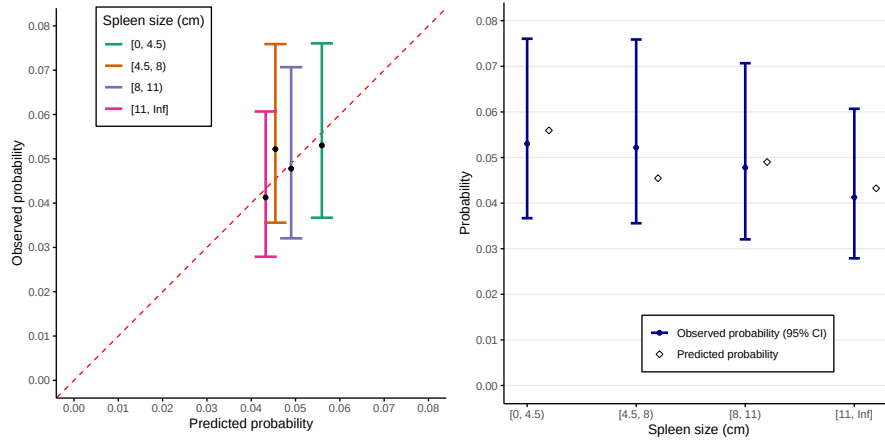


Figure D.25: Calibration plots for spleen size — model **including** parasite grade.

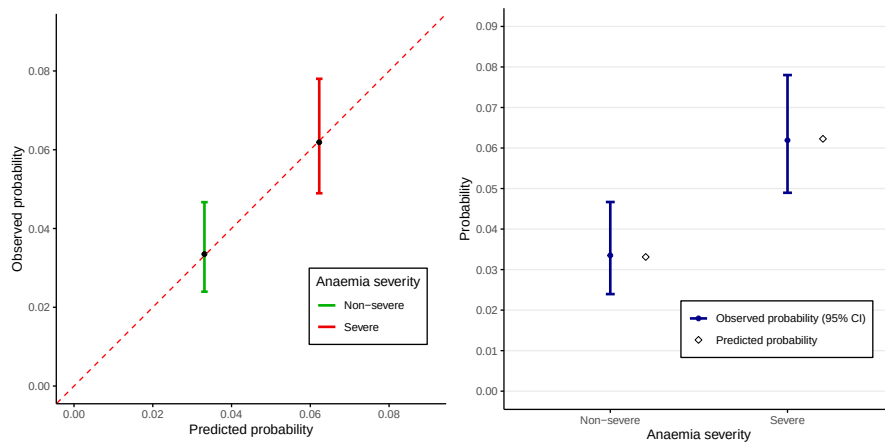


Figure D.26: Calibration plots for anaemia — model **excluding** parasite grade.

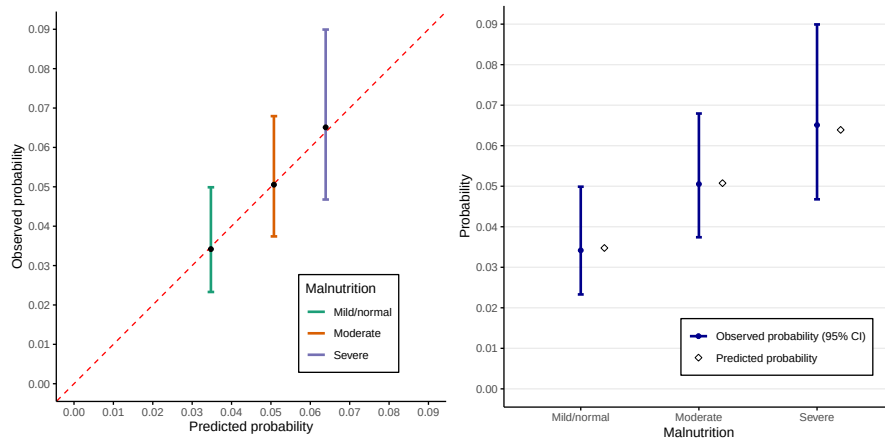


Figure D.27: Calibration plots for malnutrition — model **excluding** parasite grade.

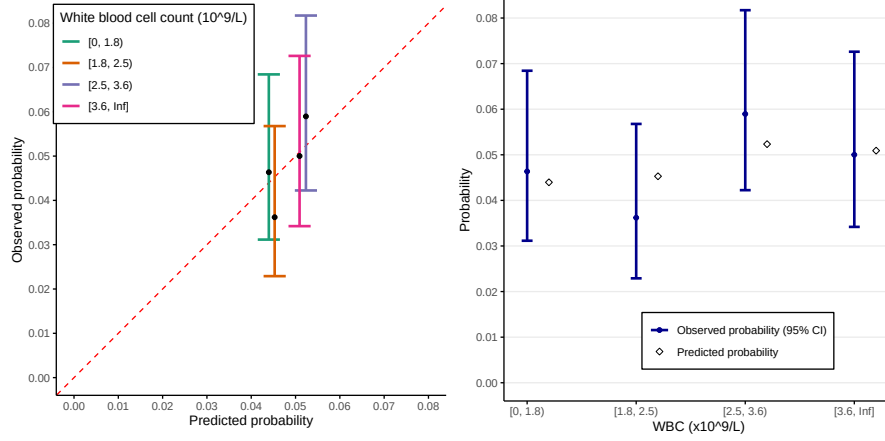


Figure D.28: Calibration plots for white blood cell count — model **excluding** parasite grade.

Predictors	Model with PG	Model without PG
Parasite grade	499¹	n/a
Malnutrition	477¹	461¹
Fever duration	390¹	176
Anaemia	298¹	361¹
Spleen size	281¹	84
WBC count	254¹	188¹
Age (linear)	205	209
Creatinine	194	101
Age (squared)	162	161
ALT	137	138
Sex	124	119
Age (cubic)	57	71
Platelets	55	53

¹ Selected in final model.

Table D.3: Frequency of predictor selection during 500 bootstrap model developments. Predictors sorted by selection frequency in the model with parasite grade. ALT: alanine aminotransferase; n/a: not applicable; PG: parasite grade; WBC: white blood cell count

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